User's Manual

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RH850/F1KH, RH850/F1KM ICUMD

User's Manual: Hardware

Renesas microcontroller
RH850 Family

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Notes for CMOS devices

(1) Voltage application waveform at input pin: Waveform distortion due to input noise or a reflected wave may cause malfunction. If the input of the CMOS device stays in the area between VIL (MAX) and VIH (MIN) due to noise, etc., the device may malfunction. Take care to prevent chattering noise from entering the device when the input level is fixed, and also in the transition period when the input level passes through the area between VIL (MAX) and VIH (MIN).

(2) Handling of unused input pins: Unconnected CMOS device inputs can be cause of malfunction. If an input pin is unconnected, it is possible that an internal input level may be generated due to noise, etc., causing malfunction. CMOS devices behave differently than Bipolar or NMOS devices. Input levels of CMOS devices must be fixed high or low by using pull-up or pull-down circuitry. Each unused pin should be connected to power supply or GND via a resistor if there is a possibility that it will be an output pin. All handling related to unused pins must be judged separately for each device and according to related specifications governing the device.

(3) Precaution against ESD: A strong electric field, when exposed to a MOS device, can cause destruction of the gate oxide and ultimately degrade the device operation. Steps must be taken to stop generation of static electricity as much as possible, and quickly dissipate it when it has occurred. Environmental control must be adequate. When it is dry, a humidifier should be used. It is recommended to avoid using insulators that easily build up static electricity. Semiconductor devices must be stored and transported in an anti-static container, static shielding bag or conductive material. All test and measurement tools including work benches and floors should be grounded. The operator should be grounded using a wrist strap. Semiconductor devices must not be touched with bare hands. Similar precautions need to be taken for PW boards with mounted semiconductor devices.

(4) Status before initialization: Power-on does not necessarily define the initial status of a MOS device. Immediately after the power source is turned ON, devices with reset functions have not yet been initialized. Hence, power-on does not guarantee output pin levels, I/O settings or contents of registers. A device is not initialized until the reset signal is received. A reset operation must be executed immediately after power-on for devices with reset functions.

(5) Power ON/OFF sequence: In the case of a device that uses different power supplies for the internal operation and external interface, as a rule, switch on the external power supply after switching on the internal power supply. When switching the power supply off, as a rule, switch off the external power supply and then the internal power supply. Use of the reverse power on/off sequences may result in the application of an overvoltage to the internal elements of the device, causing malfunction and degradation of internal elements due to the passage of an abnormal current. The correct power on/off sequence must be judged separately for each device and according to related specifications governing the device.

(6) Input of signal during power off state: Do not input signals or an I/O pull-up power supply while the device is not powered. The current injection that results from input of such a signal or I/O pull-up power supply may cause malfunction and the abnormal current that passes in the device at this time may cause degradation of internal elements. Input of signals during the power off state must be judged separately for each device and according to related specifications governing the device.

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How to Use This Manual

Readers This manual is intended for users who wish to understand the functions of the RH850/F1KH, RH850/F1KM ICUMD and design application systems using the following RH850/F1KH, RH850/F1KM ICUMD microcontrollers:

Purpose This manual is intended to give users an understanding of the hardware functions of the RH850/F1KH, RH850/F1KM ICUMD shown in the *Organization* below.

Organization This manual is divided into three parts: RH850/F1KH, RH850/F1KM ICUMD User's Manual: Hardware (this manual), RH850/F1KH, RH850/F1KM User's Manual: Hardware, and RH850G3K User's Manual: Software (R01US0125E).

RH850/F1KH, RH850/F1KM ICUMD Hardware	RH850/F1KH, RH850/F1KM Hardware	RH850G3K Software
• CPU function • Cryptographic peripherals • Peripheral functions • Flash memory programming	• Pin functions • CPU function • On-chip peripheral functions • Flash memory programming	• Overview • Processor Model • Register Reference • Exceptions and Interrupts • Memory Management • Instruction Reference • Reset • Appendix

How to read this manual It is assumed that the readers of this manual have general knowledge in the fields of electrical engineering, logic circuits, and microcontrollers.

To understand the overall functions of the ICUMD.

→Read this manual according to the Contents.

To understand the details of an instruction function

→See RH850G3K User's Manual: Software (R01US0125E) available separately.

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Conventions

Data significance: Higher digits on the left and lower digits on the right

Active low representation: $\overline{xxx}$ (overscore over pin or signal name)

Memory map address: Higher addresses on the top and lower addresses on the bottom

Note: Footnote for item marked with Note in the text

Caution: Information requiring particular attention

Remark: Supplementary information

Numeric representation: Binary ... xxxx or $xxxx_B$

Decimal ... xxxx

Hexadecimal ... $xxxx_H$

Prefix indicating power of 2 (address space, memory capacity):

K (kilo): $2^{10} = 1,024$

M (mega): $2^{20} = 1,024^2$

G (giga): $2^{30} = 1,024^3$

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Section 1 Overview

1.1 Features

The Intelligent Cryptographic Unit/Master D(ICUMD) is a hardware security module (HSM).

The features of the ICUMD are described below.

- The ICUMD integrates an Intelligent Cryptographic Unit Processor (ICUP) that controls the ICUMD sub-system and runs the user-defined security services.
- The ICUMD includes an accelerator that supports a block cipher algorithm based on the AES (Advanced Encryption Standard, FIPS PUB 197).
- The ICUMD integrates a true random number generator.
- The ICUMD can be used as a slave peripheral to run security services upon request. Also the ICUMD handles processing of other user-defined security services (for example, checking contents of memory for completeness and integrity) and operates as a master handling procedures to strengthen measures for security.
- The ICUMD has exclusive read access and programming access to a specific area of non-volatile (flash) memory which holds confidential data (keys and certificates) and highly confidential code for operating security services.

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1.1.1 Internal Block Diagram

The ICUMD consists of hardware enabled AES engine, TRNG, an ICU control register, an ICU command register, ICUP, an OS timer, a window watchdog timer, and local RAM.

The internal block diagram is shown below.

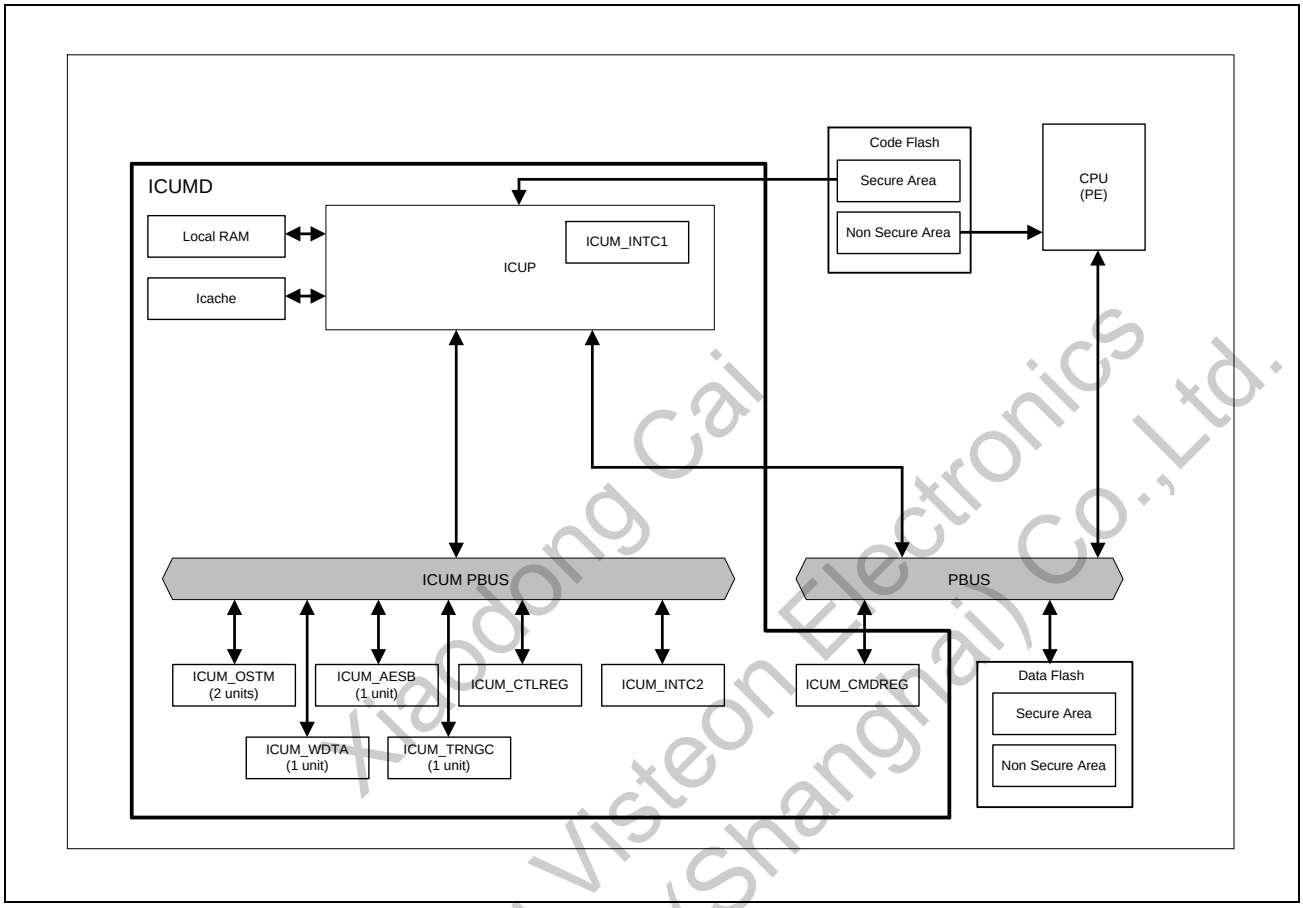


Figure 1.1 Internal Block Diagram

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Section 2 CPU System (ICUP)

This section reproduces the notes in the *RH850G3K User's Manual: Software* that concern points to be observed when using the ICUMD and provides details specific to the ICUMD that are not described in the software manual.

2.1 Overview

The ICUP controls ICUMD and executes user defined security services.

The ICUP is based on the V850E3v5-S architecture and provides the following features:

- CPU core (32-bit RISC, harvard architecture, 5-stage pipeline)
- 32-bit general-purpose registers
- Memory protection unit (MPU)
- Local RAM interface (Non wait)
- Interrupt controller (INTC)

2.1.1 Features of Architecture

2.1.1.1 Memory Management

- Memory space
  - Program area: Linear 128 MB area
  - Data area: Linear 128 MB area
- Processor protection functions
  - Memory Protection Unit (MPU)
    The MPU protects the CPU's memory space against execution or data manipulation that is not allowed for user programs (up to 8 memory protection areas can be set).
  - There is no memory protection setting check function.
- There is no memory management unit (MMU).
- Cache
  - Instruction cache 4KB-2way.

2.1.1.2 System Register Protection (SRP)

Setting the CPU operating mode to User mode enables use of the SRP (system register protection) function, which can prevent access to certain system registers.

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2.1.1.3 Functions Not Supported

The following functions are not supported by the ICUMD.

- Hardware multithreading function
- CPU virtualization support function
- Multiprocessing function

2.1.2 Clock

The ICUP uses CPUCLK_L.

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2.2 Processor Model

This section describes the items included in *Section 2* in *the RH850G3K User's Manual: Software* that require specific attention when using the ICUMD.

2.2.1 CPU Data Address and Physical Program Address Space

This section describes the CPU data address; that is, the size and addresses of the CPU address space and the physical address space.

The CPU supports the following address spaces:

- 128 MB CPU data address space
  128 MB of a 4 GB (max.) address space can be accessed by using the 32-bit general registers.
- 128 MB physical program address space
  The CPU provides 128 MB of physical address space to access instruction codes in the program memory.
  This means that up to 128 MB of external or internal memory is accessible.

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2.2.1.1 Program Space and Data Space

The figure below shows the assignment of data space and program space in the CPU address space.

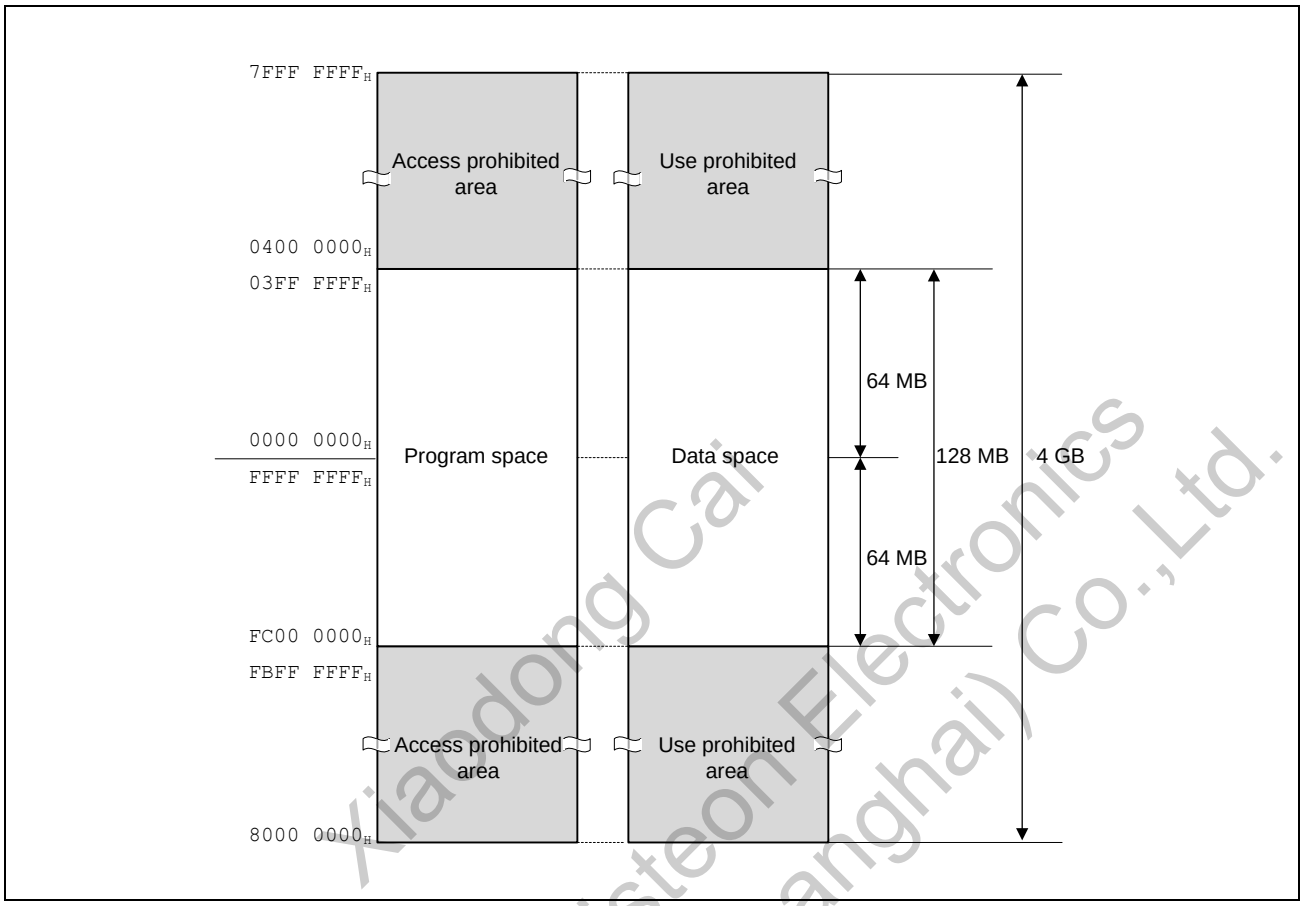


Figure 2.1 CPU Address Space

For instruction address addressing, a value whose 26th bit is sign-extended is automatically specified for the higher-order 5 bits of the register holding the instruction address. Therefore, the addressable range is 0000 0000$_H$ to 03FF FFFE$_H$ and FC00 0000$_H$ to FFFF FFFE$_H$ (and the least significant bit is always 0.) Be sure to place the instructions and the tables to be referenced using the SWITCH, CALLT, and SYSCALL instructions in this addressable address range.

2.2.2 Hardware Thread

The ICUP only supports single-thread processing. If the HALT instruction is executed, therefore, the CPU stops operating.

Also, when the SNOOZE instruction is executed, the thread that was stopped by the SNOOZE instruction starts operating again after 32 clock cycles have elapsed.

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2.3 Register Reference

This section describes the items included in *Section 3* in *the RH850G3K User's Manual: Software* that require specific attention when using the ICUMD.

2.3.1 PC — Program Counter

The PC retains the address of the instruction being executed. Bit 0 is fixed to 0, and branching to an odd number address is disabled.

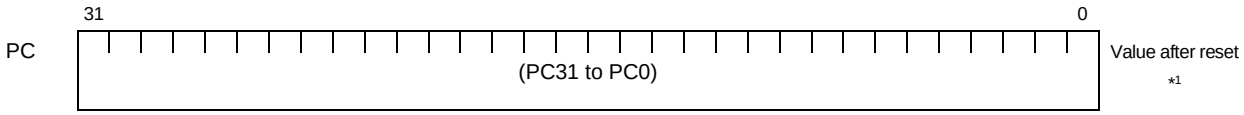


Table 2.1 PC Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 1	PC31 to PC1	These bits indicate the address of the instruction being executed.	R/W	*1
0	PC0	This bit is fixed to 0. Branching to an odd number address is disabled.	R/W	0

Note 1. The value after reset differs depending on the setting value of the reset vector. For details, see **Section 2.3.2.10, RBASE — Reset Vector Base Address**.

CAUTION

For a CPU whose instruction addressing range is 128 MB, the value resulting from sign-extension of bit 26 is automatically set to bits 31 to 27.

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2.3.2 Basic System Registers

The basic system registers are used to control the CPU status and to retain exception information.

System registers are read from or written to by using the LDSR and STSR instructions and specifying the system register number, which is made up of a register number and selection ID.

Table 2.2 Basic System Registers

Register No. (regID, selID)	Symbol	Function	Changes from the RH850G3K User's Manual: Software	Access Rights
SR0, 0	EIPC*1	Status save registers when acknowledging EI level exception	Some changes in specifications	SV
SR1, 0	EIPSW	Status save registers when acknowledging EI level exception	Some changes in specifications	SV
SR2, 0	FEPC*1	Status save registers when acknowledging FE level exception	Some changes in specifications	SV
SR3, 0	FEPSW	Status save registers when acknowledging FE level exception	Some changes in specifications	SV
SR5, 0	PSW	Program status word	Some changes in specifications	*2
SR13, 0	EIIC	EI level exception cause	No change	SV
SR14, 0	FEIC	FE level exception cause	No change	SV
SR16, 0	CTPC*1	CALLT execution status save register	Some changes in specifications	UM
SR17, 0	CTPSW	CALLT execution status save register	No change	UM
SR20, 0	CTBP*1	CALLT base pointer	Some changes in specifications	UM
SR28, 0	EIWR	EI level exception working register	No change	SV
SR29, 0	FEWR	FE level exception working register	No change	SV
SR31, 0	BSEL	Not implemented. A value of 0 is returned when read and writing is ignored.	—	SV
SR0, 1	MCFG0	Machine configuration 0	Some changes in specifications	SV
SR2, 1	RBASE*1	Reset vector base address	Some changes in specifications	SV
SR3, 1	EBASE*1	Exception handler vector address	Some changes in specifications	SV
SR4, 1	INTBP*1	Base address of the interrupt handler table	Some changes in specifications	SV
SR5, 1	MCTL	CPU control	Some changes in specifications	SV
SR6, 1	PID	Processor ID	Some changes in specifications	SV
SR11, 1	SCCFG	SYSCALL operation setting	No change	SV
SR12, 1	SCBP*1	SYSCALL base pointer	Some changes in specifications	SV
SR0, 2	HTCFG0	Thread configuration	Some changes in specifications	SV
SR6, 2	MEA	Memory error address	No change	SV
SR7, 2	ASID	Address space ID	No change	SV
SR8, 2	MEI	Memory error information	Some changes in specifications	SV

Note 1. For a CPU whose instruction addressing range is 128 MB, the value resulting from sign-extension of bit 26 is automatically set to bits 31 to 27.

Note 2. The access rights differ depending on the bit. For details, see **Table 2.7, Access Rights for PSW Register.**

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2.3.2.1 EIPC — Status Save Register when Acknowledging EI Level Exception

When an EI level exception is acknowledged, the address of the instruction that was being executed when the EI level exception occurred, or of the next instruction, is saved to the EIPC register (see *Section 4.1.3, Types of exceptions,* in *the RH850G3K User's Manual: Software*).

Because there is only one pair of EI level exception status save registers, when processing multiple exceptions, the contents of these registers must be saved by a program.

Be sure to set an even-numbered address to the EIPC register. An odd-numbered address must not be specified.

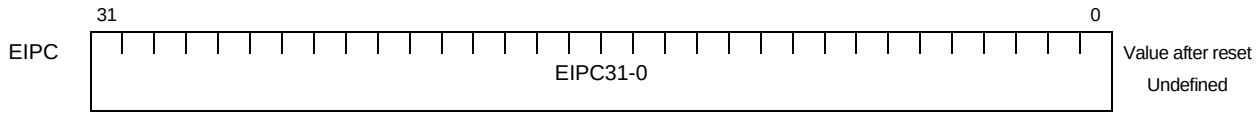


Table 2.3 EIPC Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 1	EIPC31-1	These bits indicate the PC saved when an EI level exception is acknowledged.	R/W	Undefined
0	EIPC0	This bit indicates the PC saved when an EI level exception is acknowledged. Always set this bit to 0. Even if it is set to 1, the value transferred to the PC when the EIRET instruction is executed is 0.	R/W	Undefined

CAUTION

For a CPU whose instruction addressing range is 128 MB, the value resulting from sign-extension of bit 26 is automatically set to bits 31 to 27 of EIPC.

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2.3.2.2 EIPSW — Status Save Register when Acknowledging EI Level Exception

When an EI level exception is acknowledged, the current PSW setting is saved to the EIPSW register. Because there is only one pair of EI level exception status save registers, when processing multiple exceptions, the contents of these registers must be saved by a program.

EIPSW

31	30	29										19	18		16	15	14			12	11		9	8	7	6	5	4	3	2	1	0
0	UM	0	0	0	0	0	0	0	0	0	0	0	CU2-0			EBV	0	0	0	Debug			0	NP	EP	ID	SAT	CY	OV	S	Z

Value after reset: 0000 0020H

Table 2.4 EIPSW Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31	—	(Reserved for future expansion. Be sure to clear to 0.)	R	0
30	UM	This bit stores the PSW.UM bit setting when an EI level exception is acknowledged.	R/W	0
29 to 19	—	(Reserved for future expansion. Be sure to clear to 0.)	R	0
18 to 16	CU2-0	The ICUMD does not support these bits. Be sure to clear to 0.	R	0
15	EBV	This bit stores the PSW.EBV bit setting when an EI level exception is acknowledged.	R/W	0
14 to 12	—	(Reserved for future expansion. Be sure to clear to 0.)	R	0
11 to 9	Debug	These bits store the PSW.Debug field when an EI level exception is acknowledged.	R	0
8	—	(Reserved for future expansion. Be sure to clear to 0.)	R	0
7	NP	This bit stores the PSW.NP bit setting when an EI level exception is acknowledged.	R/W	0
6	EP	This bit stores the PSW.EP bit setting when an EI level exception is acknowledged.	R/W	0
5	ID	This bit stores the PSW.ID bit setting when an EI level exception is acknowledged.	R/W	1
4	SAT	This bit stores the PSW.SAT bit setting when an EI level exception is acknowledged.	R/W	0
3	CY	This bit stores the PSW.CY bit setting when an EI level exception is acknowledged.	R/W	0
2	OV	This bit stores the PSW.OV bit setting when an EI level exception is acknowledged.	R/W	0
1	S	This bit stores the PSW.S bit setting when an EI level exception is acknowledged.	R/W	0
0	Z	This bit stores the PSW.Z bit setting when an EI level exception is acknowledged.	R/W	0

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2.3.2.3 FEPC — Status Save Register when Acknowledging FE Level Exception

When an FE level exception is acknowledged, the address of the instruction that was being executed when the FE level exception occurred, or of the next instruction, is saved to the FEPC register (see *Section 4.1.3, Types of exceptions, in the RH850G3K User's Manual: Software*).

Because there is only one pair of FE level exception status save registers, when processing multiple exceptions, the contents of these registers must be saved by a program.

Be sure to set an even-numbered address to the FEPC register. An odd-numbered address must not be specified.

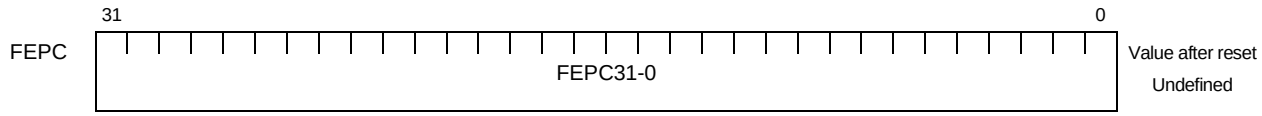


Table 2.5 FEPC Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 1	FEPC31-1	These bits indicate the PC saved when an FE level exception is acknowledged.	R/W	Undefined
0	FEPC0	This bit indicates the PC saved when an FE level exception is acknowledged. Always set this bit to 0. Even if it is set to 1, the value transferred to the PC when the FERET instruction is executed is 0.	R/W	Undefined

CAUTION

For a CPU whose instruction addressing range is 128 MB, the value resulting from sign-extension of bit 26 is automatically set to bits 31 to 27 of FEPC.

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2.3.2.4 FEPSW — Status Save Register when Acknowledging FE Level Exception

When an FE level exception is acknowledged, the current PSW setting is saved to the FEPSW register. Because there is only one pair of FE level exception status save registers, when processing multiple exceptions, the contents of these registers must be saved by a program.

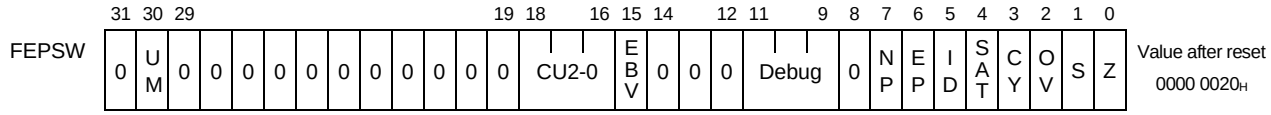


Table 2.6 FEPSW Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31	—	(Reserved for future expansion. Be sure to clear to 0.)	R	0
30	UM	This bit stores the PSW.UM bit setting when an FE level exception is acknowledged.	R/W	0
29 to 19	—	(Reserved for future expansion. Be sure to clear to 0.)	R	0
18 to 16	CU2-0	The ICUMD does not support these bits. Be sure to clear to 0.	R	0
15	EBV	This bit stores the PSW.EBV bit setting when an FE level exception is acknowledged.	R/W	0
14 to 12	—	(Reserved for future expansion. Be sure to clear to 0.)	R	0
11 to 9	Debug	These bits store the PSW.Debug field when an FE level exception is acknowledged.	R	0
8	—	(Reserved for future expansion. Be sure to clear to 0.)	R	0
7	NP	This bit stores the PSW.NP bit setting when an FE level exception is acknowledged.	R/W	0
6	EP	This bit stores the PSW.EP bit setting when an FE level exception is acknowledged.	R/W	0
5	ID	This bit stores the PSW.ID bit setting when an FE level exception is acknowledged.	R/W	1
4	SAT	This bit stores the PSW.SAT bit setting when an FE level exception is acknowledged.	R/W	0
3	CY	This bit stores the PSW.CY bit setting when an FE level exception is acknowledged.	R/W	0
2	OV	This bit stores the PSW.OV bit setting when an FE level exception is acknowledged.	R/W	0
1	S	This bit stores the PSW.S bit setting when an FE level exception is acknowledged.	R/W	0
0	Z	This bit stores the PSW.Z bit setting when an FE level exception is acknowledged.	R/W	0

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2.3.2.5 PSW — Program Status Word

PSW (program status word) is a set of flags that indicate the program status (instruction execution result) and bits that indicate the operation status of the CPU (flags are bits in the PSW that are referenced by a condition instruction (Bcond, CMOV, etc.)).

When the LDSR instruction is used to change the contents of bits 7 to 0 in this register, the changed contents become valid from the instruction following the LDSR instruction.

The access rights for the PSW register differ depending on the bit. All bits can be read, but some bits can only be written under certain conditions. See **Table 2.7** for the access rights for each bit.

Table 2.7 Access Rights for PSW Register

Bit		Access Right When Reading	Access Right When Writing
30	UM	UM	SV*1
15	EBV		SV*1
7	NP		SV*1
6	EP		SV*1
5	ID		SV*1
4	SAT		UM
3	CY		UM
2	OV		UM
1	S		UM
0	Z		UM

Note 1. The access right for the whole PSW register is UM, so the PIE exception does not occur even if the register is written by using an LDSR instruction when PSW.UM is 1. In this case, writing is ignored.

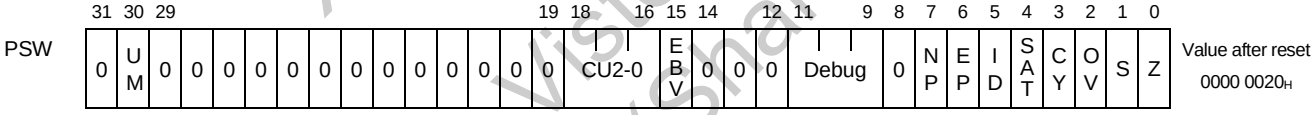


Table 2.8 PSW Register Contents (1/2)

Bit Position	Bit Name	Function	R/W	Value After Reset
31	—	(Reserved for future expansion. Be sure to clear to 0.)	R	0
30	UM	This bit indicates that the CPU is in the user mode (the UM mode). 0: Supervisor mode 1: User mode	R/W	0
29 to 19	—	(Reserved for future expansion. Be sure to clear to 0.)	R	0
18 to 16	CU2-0	The ICUMD does not support these bits. Be sure to clear to 0.	R	0
15	EBV	This bit indicates the reset vector and exception vector operation. For details, see the RBASE register in **Section 2.3.2.10, RBASE — Reset Vector Base Address** and EBASE register in **Section 2.3.2.11, EBASE — Exception Handler Vector Address**.	R/W	0
14 to 12	—	(Reserved for future expansion. Be sure to clear to 0.)	R	0
11 to 9	Debug	These bits are used in debugging function for the development tool. In normal cases, set these bits to “0”.	R	0
8	—	(Reserved for future expansion. Be sure to clear to 0.)	R	0

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Table 2.8 PSW Register Contents (2/2)

Bit Position	Bit Name	Function	R/W	Value After Reset
7	NP	This bit indicates that an FE-level exception is being processed. When an FE level exception is acknowledged, this bit is set to 1, which prohibits occurrence of multiple exceptions.*1 0: FE level exception handling is not in progress. 1: FE level exception handling is in progress.	R/W	0
6	EP	This bit indicates that an exception other than an interrupt is being serviced. It is set to 1 when the corresponding exception occurs. This bit does not affect acknowledging an exception request even when it is set to 1. 0: An interrupt is being serviced. 1: An exception other than an interrupt is being serviced.	R/W	0
5	ID	This bit indicates enabling or disabling of the acknowledgment of EI-level exceptions. It is set to 1 when an EI level exception or FE level exception is acknowledged, disabling the acknowledgment of further EI-level exception.*1 This bit is also used to disable the acknowledgment of EI level exceptions in critical sections of an ordinary program or during the servicing of interrupts. It is set to 1 when the DI instruction is executed, and cleared to 0 when the EI instruction is executed. Changes to the value of the ID bit by the EI or DI instruction will be effective from the next instruction. 0: The acknowledgment of an EI level exception is enabled. 1: The acknowledgment of an EI level exception is disabled.	R/W	1
4	SAT*2	This bit indicates that the operation result is saturated because the result of a saturated operation instruction operation has overflowed. This is a cumulative flag, so when the operation result of the saturated operation instruction becomes saturated, this bit is set to 1, but it is not cleared to 0 when the operation result for a subsequent instruction is not saturated. This bit is cleared to 0 by the LDSR instruction. This bit is neither set to 1 nor cleared to 0 when an arithmetic operation instruction is executed. 0: Not saturated 1: Saturated	R/W	0
3	CY	This bit indicates whether a carry or borrow has occurred in the operation result. 0: Carry and borrow have not occurred. 1: Carry or borrow has occurred.	R/W	0
2	OV*2	This bit indicates whether or not an overflow has occurred during an operation. 0: Overflow has not occurred. 1: Overflow has occurred.	R/W	0
1	S*2	This bit indicates whether or not the result of an operation is negative. 0: Result of operation is positive or 0. 1: Result of operation is negative.	R/W	0
0	Z	This bit indicates whether or not the result of an operation is 0. 0: Result of operation is not 0. 1: Result of operation is 0.	R/W	0

Note 1. Exceptions from certain sources are accepted regardless of the states of the ID and NP bits. For details, see *List of Exception Sources* in *the RH850G3K User's Manual: Software*.

Note 2. The operation result of the saturation processing is determined in accordance with the contents of the OV flag and S flag during a saturated operation. When only the OV flag is set to 1 during a saturated operation, the SAT flag is set to 1.

Operation Result Status	Flag Status			Operation Result after Saturation Processing
	SAT	OV	S	
Exceeded positive maximum value	1	1	0	7FFF FFFF$_H$
Exceeded negative maximum value	1	1	1	8000 0000$_H$
Positive (maximum value not exceeded)	Value prior to operation is retained.	0	0	Operation result itself
Negative (maximum value not exceeded)			1	

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2.3.2.6 CTPC — Status Save Register when Executing CALLT

When a CALLT instruction is executed, the address of the next instruction after the CALLT instruction is saved to CTPC.

Be sure to set an even-numbered address to the CTPC register. An odd-numbered address must not be specified.

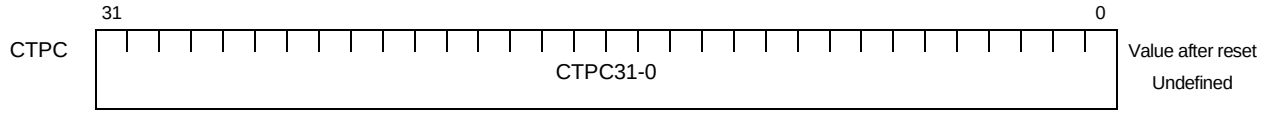


Table 2.9 CTPC Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 1	CTPC31-1	These bits indicate the PC of the instruction after the CALLT instruction.	R/W	Undefined
0	CTPC0	This bit indicates the PC of the instruction after the CALLT instruction. Always set this bit to 0. Even if it is set to 1, the value transferred to the PC when the CTRET instruction is executed is 0.	R/W	Undefined

CAUTION

For a CPU whose instruction addressing range is 128 MB, the value resulting from sign-extension of bit 26 is automatically set to bits 31 to 27 of CTPC.

2.3.2.7 CTBP — CALLT Base Pointer

The CTBP register is used to specify table addresses of the CALLT instruction and generate target addresses.

Be sure to set the CTBP register to a halfword address.

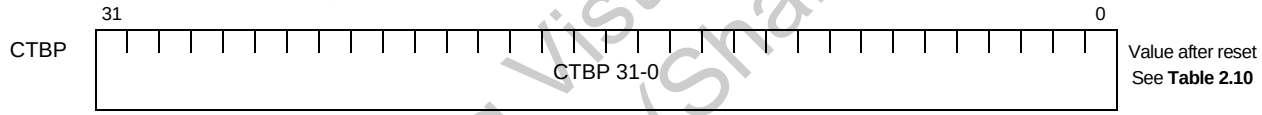


Table 2.10 CTBP Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 1	CTBP31-1	These bits indicate the base pointer address of the CALLT instruction. These bits indicate the first address in the table used by the CALLT instruction.	R/W	Undefined
0	CTBP0	This bit indicates the base pointer address of the CALLT instruction. This bit indicates the first address in the table used by the CALLT instruction. Always set this bit to 0.	R	0

CAUTION

For a CPU whose instruction addressing range is 128 MB, the value resulting from sign-extension of bit 26 is automatically set to bits 31 to 27 of CTBP.

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2.3.2.8 HTCFG0 — Thread Configuration Register

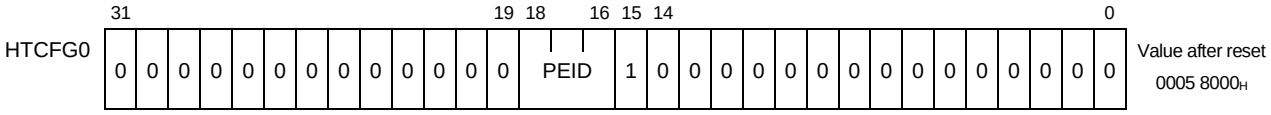


Table 2.11 HTCFG0 Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 19	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
18 to 16	PEID	These bits indicate the processor element number.	R	101_B
15	—	(When writing, always write 1. This bit is read as 1.)	R	1
14 to 0	—	(Reserved for future expansion. Be sure to set to 0.)	R	0

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2.3.2.9 MEI — Memory Error Information Register

This register is used to store information about the instruction that caused the exception when a misaligned (MAE) or memory protection (MDP) exception occurs. This information is used during emulation.

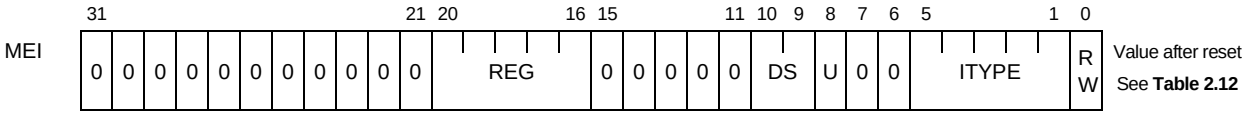


Table 2.12 MEI Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 21	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
20 to 16	REG	These bits indicate the number of the source or destination register accessed by the instruction that caused the exception. For details, see **Table 2.13**.	R/W	Undefined
15 to 11	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
10, 9	DS	These bits indicate the type of data handled by the instruction that caused the exception.*1 0: Byte (8 bits) 1: Halfword (16 bits) 2: Word (32 bits) 3: Double-word (64 bits) For details, see **Table 2.13**.	R/W	Undefined
8	U	These bits indicate the sign extension method of the instruction that caused the exception. 0: Signed 1: Unsigned For details, see **Table 2.13**.	R/W	Undefined
7, 6	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
5 to 1	ITYPE	These bits indicate the instruction that caused the exception. For details, see **Table 2.13**.	R/W	Undefined
0	RW	This bit indicates whether the operation of the instruction that caused the exception was read (Load-memory) or write (Store-memory) 0: Read (Load-memory) 1: Write (Store-memory) For details, see **Table 2.13**.	R/W	Undefined

Note 1. Even if the data is divided and access is made several times due to the specifications of the hardware, the original data type indicated by the instruction is stored.

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Table 2.13 Instructions Causing Exceptions and Values of MEI Register

Instruction	REG	DS	U	RW	ITYPE
SLD.B	dst	0 (byte)	0 (signed)	0 (read)	00000_B
SLD.BU	dst	0 (byte)	1 (unsigned)	0 (read)	00000_B
SLD.H	dst	1 (halfword)	0 (signed)	0 (read)	00000_B
SLD.HU	dst	1 (halfword)	1 (unsigned)	0 (read)	00000_B
SLD.W	dst	2 (word)	0 (signed)	0 (read)	00000_B
SST.B	src	0 (byte)	0 (signed)	1 (write)	00000_B
SST.H	src	1 (halfword)	0 (signed)	1 (write)	00000_B
SST.W	src	2 (word)	0 (signed)	1 (write)	00000_B
LD.B (disp16)	dst	0 (byte)	0 (signed)	0 (read)	00001_B
LD.BU (disp16)	dst	0 (byte)	1 (unsigned)	0 (read)	00001_B
LD.H (disp16)	dst	1 (halfword)	0 (signed)	0 (read)	00001_B
LD.HU (disp16)	dst	1 (halfword)	1 (unsigned)	0 (read)	00001_B
LD.W (disp16)	dst	2 (word)	0 (signed)	0 (read)	00001_B
ST.B (disp16)	src	0 (byte)	0 (signed)	1 (write)	00001_B
ST.H (disp16)	src	1 (halfword)	0 (signed)	1 (write)	00001_B
ST.W (disp16)	src	2 (word)	0 (signed)	1 (write)	00001_B
LD.B (disp23)	dst	0 (byte)	0 (signed)	0 (read)	00010_B
LD.BU (disp23)	dst	0 (byte)	1 (unsigned)	0 (read)	00010_B
LD.H (disp23)	dst	1 (halfword)	0 (signed)	0 (read)	00010_B
LD.HU (disp23)	dst	1 (halfword)	1 (unsigned)	0 (read)	00010_B
LD.W (disp23)	dst	2 (word)	0 (signed)	0 (read)	00010_B
ST.B (disp23)	src	0 (byte)	0 (signed)	1 (write)	00010_B
ST.H (disp23)	src	1 (halfword)	0 (signed)	1 (write)	00010_B
ST.W (disp23)	src	2 (word)	0 (signed)	1 (write)	00010_B
LD.DW (disp23)	dst	3 (double-word)	0 (signed)	0 (read)	00010_B
ST.DW (disp23)	src	3 (double-word)	0 (signed)	1 (write)	00010_B
LDL.W	dst	2 (word)	0 (signed)	0 (read)	00111_B
STC.W	src	2 (word)	0 (signed)	1 (write)	00111_B
CAXI	dst	2 (word)	1 (unsigned)	0 (read)*1	01000_B
SET1	—	0 (byte)	1 (unsigned)	0 (read)*1	01001_B
CLR1	—	0 (byte)	1 (unsigned)	0 (read)*1	01001_B
NOT1	—	0 (byte)	1 (unsigned)	0 (read)*1	01001_B
TST1	—	0 (byte)	1 (unsigned)	0 (read)	01001_B
PREPARE	—	2 (word)	1 (unsigned)	1 (write)	01100_B
DISPOSE	—	2 (word)	1 (unsigned)	0 (read)	01100_B
PUSHSP	—	2 (word)	1 (unsigned)	1 (write)	01101_B
POPSP	—	2 (word)	1 (unsigned)	0 (read)	01101_B
SWITCH	—	1 (halfword)	0 (signed)	0 (read)	10000_B
CALLT	—	1 (halfword)	1 (unsigned)	0 (read)	10001_B
SYSCALL	—	2 (word)	1 (unsigned)	0 (read)	10010_B
Interrupt (table reference)*2	—	2 (word)	1 (unsigned)	0 (read)	10101_B

Note 1. This exception occurs when the instruction executes a read access.

Note 2. An exception occurs when the table reference interrupt vector is read.

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NOTE

dst: destination register number, src: source register number.

Misaligned Exception

Access to data at an address that does not correspond to the required boundary for the unit of access leads to a misaligned exception. For example, reading or writing a word from or to an address that is not a multiple of 4 leads to an exception of this type. However, if the type is double-word, access at any word-form boundary does not lead to an exception. When MCTL.MA is 1, access proceeds without generating misaligned exceptions except double-word accesses to an address that is not a multiple of 4.

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2.3.2.10 RBASE — Reset Vector Base Address

This register indicates the reset vector address when there is a reset. If the PSW.EBV bit is 0, this vector address is also used as the exception vector address.

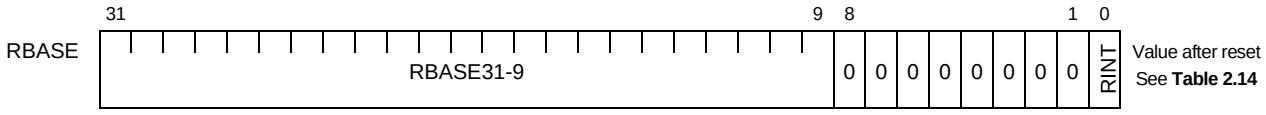


Table 2.14 RBASE Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 9	RBASE31-9	These bits indicate the reset vector when there is a reset. When PSW.EBV = 0, this address is also used as the exception vector. The RBASE8-0 bits are not assigned as names because these bits are always 0	R	*1
8 to 1	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
0	RINT	When the RINT bit is set, the offset address of the exception handler used during interrupt servicing is decremented. This bit is valid when PSW.EBV = 0.	R	0

Note 1. This is the same value as the Value After Reset for ICUMRESV[31:9]. For details about ICUMRESV, see **Section 16.8.2, OPBT9 — Option Byte 9.**

2.3.2.11 EBASE — Exception Handler Vector Address

This register indicates the exception handler vector address. This register is valid when the PSW.EBV bit is 1.



Table 2.15 EBASE Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 9	EBASE31-9	The exception handler routine address is changed to the address resulting from adding the offset address of each exception to the base address specified for this register. The EBASE8-0 bits are not assigned as names because these bits are always 0.	R/W	Undefined
8 to 1	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
0	RINT	When the RINT bit is set, the offset address of the exception handler used during interrupt servicing is decremented.	R/W	Undefined

CAUTION

For a CPU whose instruction addressing range is 128 MB, the value resulting from sign-extension of bit 26 is automatically set to bits 31 to 27 of EBASE.

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2.3.2.12 INTBP — Base Address of the Interrupt Handler Table

This register indicates the base address of the address table when the table reference method is selected as the interrupt handler address selection method.

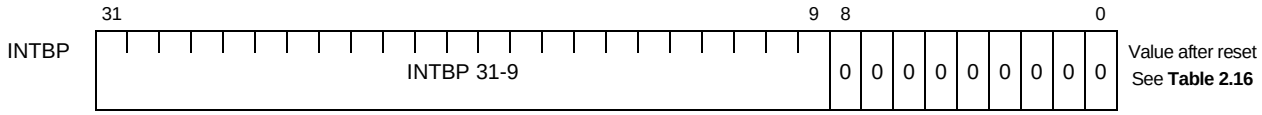


Table 2.16 INTBP Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 9	INTBP31-9	These bits indicate the base pointer address for an interrupt prescribed in the expanded specifications. The value indicated by these bits is the first address in the table used to determine the exception handler when the interrupt prescribed by the expanded specifications (EIINT0 to 31) is acknowledged. The INTBP8-0 bits are not assigned as names because these bits are always 0.	R/W	Undefined
8 to 0	—	(Reserved for future expansion. Be sure to set to 0.)	R	0

CAUTION

For a CPU whose instruction addressing range is 128 MB, the value resulting from sign-extension of bit 26 is automatically set to bits 31 to 27 of INTBP.

2.3.2.13 PID — Processor ID

The PID register retains a processor identifier that is unique to the CPU. The PID register is a read-only register.

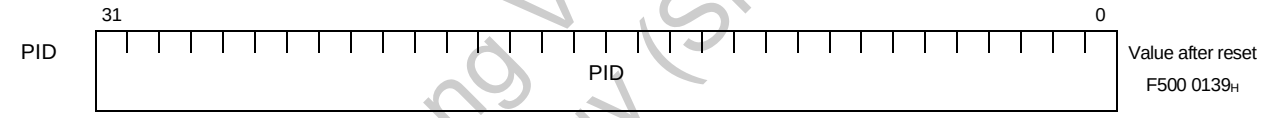


Table 2.17 PID Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 24	PID	Architecture identifier This identifier indicates the architecture of the processor.	R	$F5_H$
23 to 8		Function identifier This identifier indicates the functions of the processor. These bits indicate whether or not functions defined per bit are implemented (1: implemented, 0: not implemented). Bits 23 to 9: Reserved Bit 8: Memory protection unit (MPU) function	R	0001_H
7 to 0		Version identifier This identifier indicates the version of the processor.	R	39_H

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2.3.2.14 SCBP — SYSCALL Base Pointer

The SCBP register is used to specify a table address of the SYSCALL instruction and generate a target address. Be sure to set an appropriate value to this register before using the SYSCALL instruction.

Be sure to set a word address to the SCBP register.

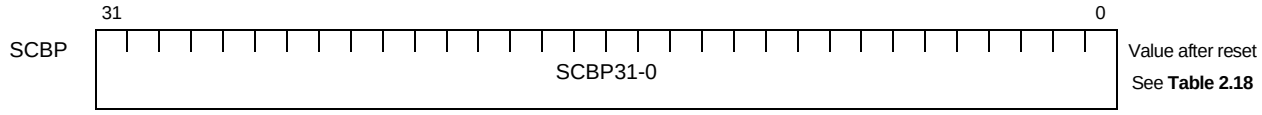


Table 2.18 SCBP Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 2	SCBP31-2	These bits indicate the base pointer address of the SYSCALL instruction. These bits indicate the first address in the table used by the SYSCALL instruction.	R/W	Undefined
1, 0	SCBP1-0	These bits indicate the base pointer address of the SYSCALL instruction. These bits indicate the first address in the table used by the SYSCALL instruction. Always set this bit to 0.	R	0

CAUTION

For a CPU whose instruction addressing range is 128 MB, the value resulting from sign-extension of bit 26 is automatically set to bits 31 to 27 of SCBP.

2.3.2.15 MCFG0 — Machine Configuration

This register indicates the CPU configuration.

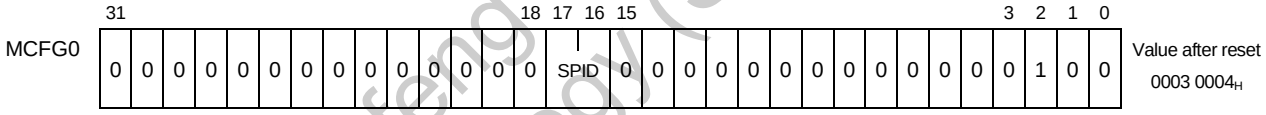


Table 2.19 MCFG0 Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 18	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
17, 16	SPID	These bits indicate the system protection number.	R	11_B
15 to 3	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
2	—	(Reserved for future expansion. Be sure to set to 1.)	R	1
1, 0	—	(Reserved for future expansion. Be sure to set to 0.)	R	0

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2.3.2.16 MCTL — Machine Control

This register is used to control the CPU.

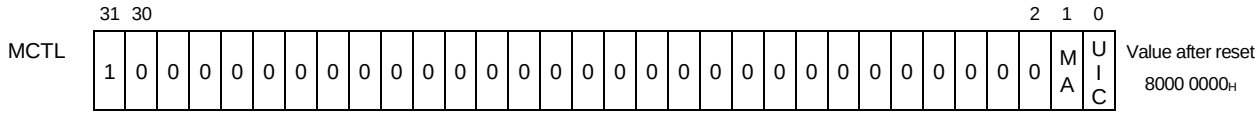


Table 2.20 MCTL Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31	—	(Reserved for future expansion. Be sure to set to 1.)	R	1
30 to 2	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
1	MA	This bit is used to control the misaligned access. 0: In the event of a misaligned access, an exception is always generated. 1: The correct operation is controlled by hardware.	R/W	0
0	UIC	This bit is used to control the interrupt enable/disable operation in the user mode. When this bit is set to 1, executing the EI/DI instruction becomes possible.	R/W	0

2.3.3 Interrupt Function Registers

Table 2.21 Interrupt Function System Registers

Register No. (regID, selID)	Symbol	Function	Changes from the RH850G3K User's Manual: Software	Access Permission
SR10, 2	ISPR	Priority of interrupt being serviced	No change	SV
SR11, 2	PMR	Interrupt priority masking	No change	SV
SR12, 2	ICSR	Interrupt control status	No change	SV
SR13, 2	INTCFG	Interrupt function setting	No change	SV

2.3.4 MPU Function Registers

Table 2.22 MPU Function System Registers (1/2)

Register No. (regID, selID)	Symbol	Function	Changes from the RH850G3K User's Manual: Software	Access Permission
SR0, 5	MPM	Memory protection operation mode setting	Some changes in specifications	SV
SR1, 5	MPRC	MPU region control	Some changes in specifications	SV
SR4, 5	MPBRGN	MPU base region number	Some changes in specifications	SV
SR5, 5	MPTRGN	MPU end region number	Some changes in specifications	SV
SR11, 5	MCR	Memory protection setting check result	No change	SV
SR0, 6	MPLA0	Protection area minimum address	Some changes in specifications	SV
SR1, 6	MPUA0	Protection area maximum address	Some changes in specifications	SV
SR2, 6	MPAT0	Protection area attribute	Some changes in specifications	SV
SR4, 6	MPLA1	Protection area minimum address	Some changes in specifications	SV
SR5, 6	MPUA1	Protection area maximum address	Some changes in specifications	SV
SR6, 6	MPAT1	Protection area attribute	Some changes in specifications	SV
SR8, 6	MPLA2	Protection area minimum address	Some changes in specifications	SV

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Table 2.22 MPU Function System Registers (2/2)

Register No. (regID, selID)	Symbol	Function	Changes from the RH850G3K User's Manual: Software	Access Permission
SR9, 6	MPUA2	Protection area maximum address	Some changes in specifications	SV
SR10, 6	MPAT2	Protection area attribute	Some changes in specifications	SV
SR12, 6	MPLA3	Protection area minimum address	Some changes in specifications	SV
SR13, 6	MPUA3	Protection area maximum address	Some changes in specifications	SV
SR14, 6	MPAT3	Protection area attribute	Some changes in specifications	SV
SR16, 6	MPLA4	Protection area minimum address	Some changes in specifications	SV
SR17, 6	MPUA4	Protection area maximum address	Some changes in specifications	SV
SR18, 6	MPAT4	Protection area attribute	Some changes in specifications	SV
SR20, 6	MPLA5	Protection area minimum address	Some changes in specifications	SV
SR21, 6	MPUA5	Protection area maximum address	Some changes in specifications	SV
SR22, 6	MPAT5	Protection area attribute	Some changes in specifications	SV
SR24, 6	MPLA6	Protection area minimum address	Some changes in specifications	SV
SR25, 6	MPUA6	Protection area maximum address	Some changes in specifications	SV
SR26, 6	MPAT6	Protection area attribute	Some changes in specifications	SV
SR28, 6	MPLA7	Protection area minimum address	Some changes in specifications	SV
SR29, 6	MPUA7	Protection area maximum address	Some changes in specifications	SV
SR30, 6	MPAT7	Protection area attribute	Some changes in specifications	SV
SR0, 7	MPLA8	Not implemented. A value of 0 is returned when read and writing is ignored.	—	SV
SR1, 7	MPUA8		—	SV
SR2, 7	MPAT8		—	SV
SR4, 7	MPLA9		—	SV
SR5, 7	MPUA9		—	SV
SR6, 7	MPAT9		—	SV
SR8, 7	MPLA10		—	SV
SR9, 7	MPUA10		—	SV
SR10, 7	MPAT10		—	SV
SR12, 7	MPLA11		—	SV
SR13, 7	MPUA11		—	SV
SR14, 7	MPAT11		—	SV
SR16, 7	MPLA12		—	SV
SR17, 7	MPUA12		—	SV
SR18, 7	MPAT12		—	SV
SR20, 7	MPLA13		—	SV
SR21, 7	MPUA13		—	SV
SR22, 7	MPAT13		—	SV
SR24, 7	MPLA14		—	SV
SR25, 7	MPUA14		—	SV
SR26, 7	MPAT14		—	SV
SR28, 7	MPLA15		—	SV
SR29, 7	MPUA15		—	SV
SR30, 7	MPAT15		—	SV

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2.3.4.1 MPM — Memory Protection Operation Mode

The memory protection mode register is used to define the basic operating state of the memory protection function.

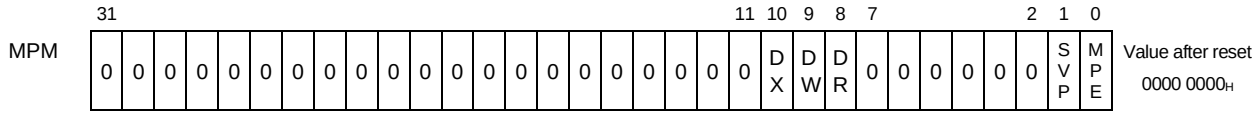


Table 2.23 MPM Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 11	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
10	DX	This bit specifies the default operation when an instruction is executed at an address that does not exist in a protection area. 0: Disable executing an instruction at an address that does not exist in a protection area. 1: Enable executing an instruction at an address that does not exist in a protection area. The setting of this bit affects the access operation when the protection areas overlap.	R/W	0
9	DW	This bit specifies the default operation when writing to an address that does not exist in a protection area. 0: Disable writing to an address that does not exist in a protection area. 1: Enable writing to an address that does not exist in a protection area. The setting of this bit affects the access operation when the protection areas overlap.	R/W	0
8	DR	This bit specifies the default operation when reading from an address that does not exist in a protection area. 0: Disable reading from an address that does not exist in a protection area. 1: Enable reading from an address that does not exist in a protection area. The setting of this bit affects the access operation when the protection areas overlap.	R/W	0
7 to 2	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
1	SVP*3	In SV mode (when PSW.UM = 0), this bit is used to specify whether to restrict access according to the SX, SW, and SR bits of the MPAT register for each protection area. *1 0: As usual, implicitly enable all access in SV mode. 1: Restrict access according to the SX, SW, and SR bits even in SV mode.*2	R/W	0
0	MPE	This bit is used to specify whether to enable or disable MPU function. 0: Disable 1: Enable	R/W	0

Note 1. When the SVP bit is set to 1, access is restricted according to the setting of each protection area even in SV mode. Therefore, specify protection areas before setting the SVP bit to prevent the access of the program itself from being restricted.

Note 2. If access is restricted in SV mode, execution of MDP exceptions or the MIP exception handling itself might not be possible depending on the settings. Be careful to specify settings so that access to the memory area necessary for the exception handler and exception handling is permitted.

Note 3. Processing the table reference for a SYSCALL instruction or for an EIINT with table reference while the SVP bit is 1 all access is enabled since the CPU is in supervisor mode.

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2.3.4.2 MPRC — MPU Region Control

Bits used to perform special memory protection function operations are located in this register.

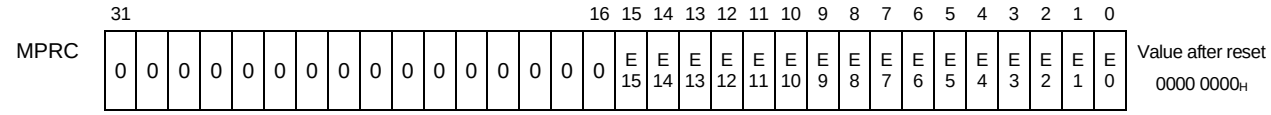


Table 2.24 MPRC Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 16	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
15 to 8	E15 to E8	The ICUMD does not support these bits. When writing, always write 0.	R	0
7 to 0	E7 to E0	These are enable bits for each of the protection areas. The value of each Ei bit reflects that of the corresponding MPATi.E bit. (i = 7 to 0).	R/W	0

2.3.4.3 MPBRGN — MPU Base Region

This register indicates the minimum usable MPU area number.

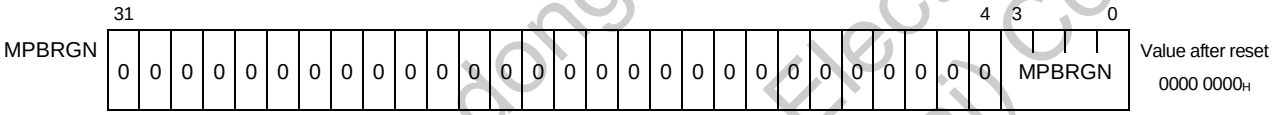


Table 2.25 MPBRGN Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 4	—	(Reserved for future expansion.)	R	0
3 to 0	MPBRGN	The ICUMD does not support these bits.	R	0000_B

2.3.4.4 MPTRGN — MPU End Region

This register indicates the maximum usable MPU area number.

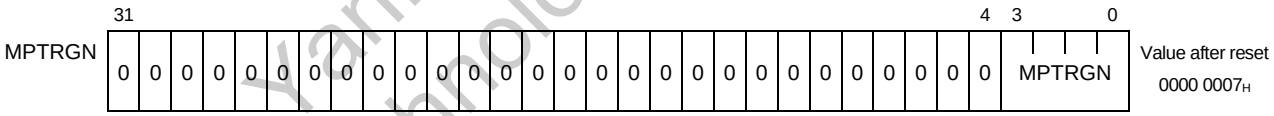


Table 2.26 MPTRGN Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 4	—	(Reserved for future expansion.)	R	0
3 to 0	MPTRGN	The ICUMD does not support these bits.	R	0111_B

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2.3.4.5 MPLAi — Protection Area Minimum Address

This register indicates the minimum address of area i (i = 0 to 7).

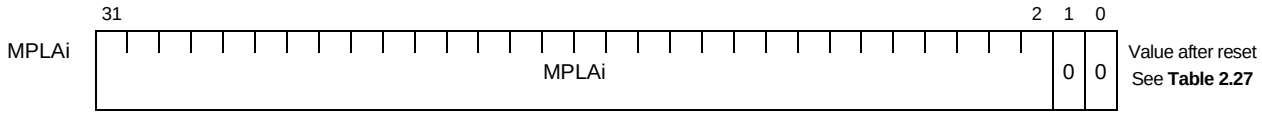


Table 2.27 MPLAi Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 2	MPLA31 to 2	These bits indicate the minimum address of area i. The MPLA1 to 0 bits are implicitly set to 0.	R/W	Undefined
1, 0	—	(Reserved for future expansion. Be sure to set to 0.)	R	0

CAUTION

For a CPU whose instruction addressing range is 128 MB, the value resulting from sign-extension of bit 26 is automatically set to bits 31 to 27 of MPLAi.

2.3.4.6 MPUAi — Protection Area Maximum Address

This register indicates the maximum address of area i (i = 0 to 7).

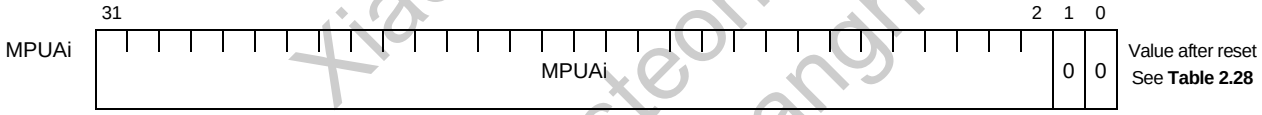


Table 2.28 MPUAi Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 2	MPUA31 to 2	These bits indicate the maximum address of area i. The MPUA1 to MPUA0 bits are implicitly set to 1.	R/W	Undefined
1, 0	—	(Reserved for future expansion. Be sure to set to 0.)	R	0

CAUTION

For a CPU whose instruction addressing range is 128 MB, the value resulting from sign-extension of bit 26 is automatically set to bits 31 to 27 of MPUAi.

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2.3.4.7 MPATi — Protection Area Attribute

This register indicates the attributes of area i (i = 0 to 7).

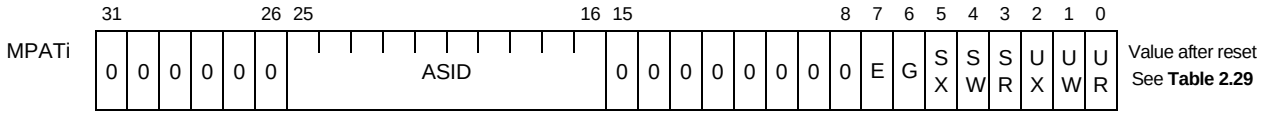


Table 2.29 MPATi Register Contents

Bit Position	Bit Name	Function	R/W	Value After Reset
31 to 26	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
25 to 16	ASID	The ICUMD does not support this bit. When writing, always write 0.	R/W	Undefined
15 to 8	—	(Reserved for future expansion. Be sure to set to 0.)	R	0
7	E	This bit indicates whether the protection area setting of area i is enabled or disabled. 0: Protection area setting of area i is disabled. 1: Protection area setting of area i is enabled.	R/W	0
6	G	If this bit is set to 1, the ASID bit is ignored under the area match condition. When writing, always write 1.	R/W	Undefined
5	SX	This bit indicates the execution privilege for the supervisor mode.*1 0: Execution is disabled. 1: Execution is enabled.	R/W	Undefined
4	SW	This bit indicates the write permission for the supervisor mode.*1 0: Writing is disabled. 1: Writing is enabled.	R/W	Undefined
3	SR	This bit indicates the read permission for the supervisor mode.*1 0: Reading is disabled. 1: Reading is enabled.	R/W	Undefined
2	UX	Indicates the execution privilege for the user mode. 0: The execution privilege is not granted to the user mode. 1: The execution privilege is granted to the user mode.	R/W	Undefined
1	UW	Enables/disables writing in the user mode. 0: The write permission is disabled for the user mode. 1: The write permission is enabled for the user mode.	R/W	Undefined
0	UR	Enables/disables reading in the user mode. 0: The read permission is disabled for the user mode. 1: The read permission is enabled for the user mode.	R/W	Undefined

Note 1. If access is restricted in SV mode, execution of MDP exceptions or the MIP exception handling itself might not be possible depending on the settings. Be careful to specify settings so that access to the memory area necessary for the exception handler and exception handling is permitted.

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2.4 Instruction Reference

Instructions for which some specifications differ are shown below.

Table 2.30 Instructions with Different Specifications

Instruction	Operation
SWITCH	When the SWITCH instruction straddles the boundary of the 128 MB space (the next address from the SWITCH instruction), the first address of the table is handled as follows: ● When the address of the SWITCH instruction is PC = 03FF FFFE$_H$, the first address in the table is calculated as (PC + 2) = 0400 0000$_H$. ● When the address of the SWITCH instruction is PC = 0400 0000$_H$ (the physical address is FC00 0000$_H$), the first address in the table is calculated as (PC + 2) = FC00 0002$_H$.
SET1, CLR1, NOT1, CAXI	If the instructions are executed on other than local RAM (self area) or on-chip peripheral I/O area (self area), atomicity is not guaranteed.

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2.5 System Error Notification Setting Registers

2.5.1 ICUM_SEGCONT — SYSERR Exception Notification Setting Register

ICUM_SEGCONT is an enable register used to set whether to enable or disable error notification of each SYSERR source. If bit 6, 4, 2, or 1 is set to 1, the function to send a SYSERR notification if an error occurs is enabled for the respective bit. If any of these bits is 0, the error flag of the corresponding bit will be set if an error occurs, but a SYSERR notification will not be sent.

When writing, always write 0 to bits 15 to 7, 5, 3, and 0.

Access: This register can be read or written in 16-bit units.

Address: FFFE E9F0$_H$

Value after reset: 0000$_H$ (Initialized by any reset sources.)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	SEG_CONT RMWE	—	SEG_CONT RAME	—	SEG_CONT EXTE	SEG_CONT FCHE	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R/W	R	R/W	R	R/W	R/W	R

Table 2.31 ICUM_SEGCONT Register Contents

Bit Position	Bit Name	Function
15 to 7	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
6	SEG_CONT RMWE	Local RAM read-modify-write error notification enable Sets whether to enable SYSERR notification for byte write or half-word write to local RAM.
5	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
4	SEG_CONT RAME	Local RAM area error notification enable Sets whether to enable SYSERR notification if an error occurs when the ICUP accesses data in the local RAM area.
3	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
2	SEG_CONT EXTE	Bus access error notification function enable Sets whether to enable SYSERR notification if an error occurs when the ICUP accesses data in an area other than the code flash and local RAM.
1	SEG_CONT FCHE	Code flash error notification enable Sets whether to enable SYSERR notification if an error occurs when the ICUP accesses data in the code flash area.
0	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.

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2.5.2 ICUM_SEGFLAG — System Error Source Storing Register

ICUM_SEGFLAG is a flag register used to store the occurrence state of each SYSERR source. When a SYSERR source occurs, the flag corresponding to the error source is set to 1. Each flag is cleared to 0 by reading the flag state of 1 and then writing 0.

In successive writing, even if 0 is written to a flag in this register from which 0 has been read, the flag will be set if an error occurs between when the flag is read and when the flag is written.

When writing, always write 0 to bit 15 to 7, 5, 3, and 0.

Access: This register can be read or written in 16-bit units.

Address: FFFE E9F2$_H$

Value after reset: 0000$_H$ (Initialized by any reset sources.)

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	SEG_FLAG RMWF	—	SEG_FLAG RAMF	—	SEG_FLAG EXTF	SEG_FLAG FCHF	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R/W	R	R/W	R	R/W	R/W	R

Table 2.32 ICUM_SEGFLAG Register Contents

Bit Position	Bit Name	Function
15 to 7	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
6	SEG_FLAG RMWF	Local RAM read-modify-write error flag This bit is set to 1 if a read-modify-write error is detected during byte write or half-word write to local RAM.
5	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
4	SEG_FLAG RAMF	Local RAM area error flag This bit is set to 1 on detection of an error when the ICUP accessed data in the local RAM area.
3	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
2	SEG_FLAG EXTF	Bus access error flag This bit is set to 1 on detection of an error when the ICUP accessed data in the area other than the code flash and local RAM area.
1	SEG_FLAG FCHF	Code flash error flag This bit is set to 1 on detection of an error when the ICUP accessed data in the code flash area.
0	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.

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2.6 Usage Notes

2.6.1 Synchronization of Store Instruction Completion and Subsequent Instruction Execution

When a control register is updated by a store instruction, there is a time lapse between when the CPU executes the store instruction and when the control register is actually updated. Therefore, if the updated content of the control register is to be used by the instruction following the store instruction, appropriate synchronization is required. The following shows the processes required to perform this synchronization.

- When the updated content of the control register is to be used by the subsequent instruction

  An example of this is, when an interrupt is enabled by executing the EI instruction after the interrupt request is cleared by accessing the control register of peripheral circuit. In this case, execute the following process.

  - Execute a store instruction to update the control registers (ST.W, etc.).
  - Perform a dummy read of the above registers (LD.W, etc.).
  - Execute SYNCP.
  - Execute the subsequent instruction (EI).

- When the updated content of the control register or memory is to be used in the instruction fetch of the subsequent instruction

  An example of this is, if an instruction is to be executed after it is written to memory. In this case, execute the following process.

  - Execute a store instruction to update the memory (ST.W, etc.).
  - Perform a dummy read of the above memory (LD.W, etc.).
  - Execute SYNCI.
  - Execute the subsequent instruction (such as a branch instruction).

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2.6.2 Avoiding Timing Hazards for System Registers

The following procedures are required to avoid timing hazards when the settings of some registers are updated.

- Instruction fetching

  When instruction fetching is carried out after the registers below have been updated:

  After updating the registers, execute an EIRET, FERET, or SYNCI instruction before instruction fetching starts.

  PSW.UM, MCFG0.SPID, ASID, and MPU-related registers (Register number: sr*, 5-7)

- Loading and storing

  When a load or store instruction for the registers below follows updating of the given register: After updating the register, execute a SYNCP instruction before executing the load or store instruction.

  ASID and MPU protection area setting registers (Register number: sr*, 6-7)

  Do not execute a load or store instruction as the instruction immediately before or after and instruction that updates the register below.

  MCTL.MA

- Interrupts

  Update the registers below while interrupts are disabled (PSW.ID = 1).

  PSW.EBV, EBASE, INTBP, ISPR, PMR, ICSR, and INTCFG

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Section 3 Address Space

3.1 ICUMD Address Space

This section describes the ICUMD memory map and specific memory space.

CAUTION

Do not access the access prohibited area, or an address to which no register has been allocated.

Table 3.1 ICUMD Address Space (F1KH-D8 8-MB 324 Pin Product)

Address	Address Space Type	Size	Access
0000 0000$_H$ to 003F FFFF$_H$	Code flash (bank A)	4 MB	CPU (PE) and ICUP
0040 0000$_H$ to 007F FFFF$_H$	Access prohibited area		
0080 0000$_H$ to 00BF FFFF$_H$	Code flash (bank B)	4 MB	
00C0 0000$_H$ to 00FF FFFF$_H$	Access prohibited area		
0100 0000$_H$ to 0100 7FFF$_H$	Code flash (Extended user area)	32 KB	
0100 8000$_H$ to FBFF FFFF$_H$	Access prohibited area		ICUP only
FC00 0000$_H$ to FDFF FFFF$_H$	Remap window area	32 MB	
FE00 0000$_H$ to FE9C FFFF$_H$	Access prohibited area		CPU (PE) and ICUP
FE9D 0000$_H$ to FE9F FFFF$_H$	Local RAM (CPU2 (PE2) area)	192 KB	
FEA0 0000$_H$ to FEBC FFFF$_H$	Access prohibited area		
FEBD 0000$_H$ to FEBF FFFF$_H$	Local RAM (CPU1 (PE1) area)	192 KB	
FEC0 0000$_H$ to FEDE FFFF$_H$	Access prohibited area		ICUP only
FEDF 0000$_H$ to FEDF FFFF$_H$	Local RAM (self area)	64 KB	
FEE0 0000$_H$ to FEEB 7FFF$_H$	Access prohibited area		CPU (PE) and ICUP
FEEB 8000$_H$ to FEEF FFFF$_H$	Global RAM A	288 KB	
FEF0 0000$_H$ to FEF0 FFFF$_H$	Retention RAM	64 KB	
FEF1 0000$_H$ to FEFB 7FFF$_H$	Access prohibited area		
FEFB 8000$_H$ to FEFF FFFF$_H$	Global RAM B	288 KB	
FF00 0000$_H$ to FF1E FFFF$_H$	Access prohibited area		
FF1F 0000$_H$ to FF1F 00FF$_H$	ICUM_CMDREG*1	32KB	CPU (PE) and ICUP
FF1F 0100$_H$ to FF1F FFFF$_H$	Access prohibited area		
FF20 0000$_H$ to FF23 FFFF$_H$	Data flash	256 KB	CPU (PE) and ICUP
FF24 0000$_H$ to FF9F FFFF$_H$	Access prohibited area		
FFA0 0000$_H$ to FFFD FFFF$_H$	On-chip peripheral I/O area	6 MB - 128 KB	
FFFE 0000$_H$ to FFFE DFFF$_H$	Access prohibited area		ICUP only
FFFE E000$_H$ to FFFE FFFF$_H$	On-chip peripheral I/O area (self area)	8 KB	
FFFF 0000$_H$ to FFFF 4FFF$_H$	Access prohibited area		CPU (PE) and ICUP
FFFF 5000$_H$ to FFFF FFFF$_H$	On-chip peripheral I/O area	44 KB	

Note 1. This area can be accessed by CPU (PE).

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Table 3.2 ICUMD Address Space (F1KM-S4 4-MB 272 Pin Product)

Address	Address Space Type	Size	Access
0000 0000_H to 003F $FFFF_H$	Code flash	4 MB	CPU (PE) and ICUP
0040 0000_H to 00FF $FFFF_H$	Access prohibited area		
0100 0000_H to 0100 $7FFF_H$	Code flash (Extended user area)	32 KB	
0100 8000_H to FBFF $FFFF_H$	Access prohibited area		ICUP only
FC00 0000_H to FDFF $FFFF_H$	Remap window area	32 MB	
FE00 0000_H to FEBB $FFFF_H$	Access prohibited area		CPU (PE) and ICUP
FEBC 0000_H to FEBF $FFFF_H$	Local RAM (CPU1 (PE1) area)	256 KB	
FEC0 0000_H to FEDE $FFFF_H$	Access prohibited area		ICUP only
FEDF 0000_H to FEDF $FFFF_H$	Local RAM (self area)	64 KB	
FEE0 0000_H to FEEE $7FFF_H$	Access prohibited area		CPU (PE) and ICUP
FEEE 8000_H to FEEF $FFFF_H$	Global RAM A	96 KB	
FEF0 0000_H to FEF0 $FFFF_H$	Retention RAM	64 KB	
FEF1 0000_H to FEFE $7FFF_H$	Access prohibited area		
FEFE 8000_H to FEFF $FFFF_H$	Global RAM B	96 KB	
FF00 0000_H to FF1E $FFFF_H$	Access prohibited area		
FF1F 0000_H to FF1F $00FF_H$	ICUM_CMDREG*1	32KB	CPU (PE) and ICUP
FF1F 0100_H to FF1F $FFFF_H$	Access prohibited area		
FF20 0000_H to FF21 $FFFF_H$	Data flash	128 KB	CPU (PE) and ICUP
FF22 0000_H to FF9F $FFFF_H$	Access prohibited area		
FFA0 0000_H to FFFD $FFFF_H$	On-chip peripheral I/O area	6 MB - 128 KB	
FFFE 0000_H to FFFE $DFFF_H$	Access prohibited area		ICUP only
FFFE $E000_H$ to FFFE $FFFF_H$	On-chip peripheral I/O area (self area)	8 KB	
FFFF 0000_H to FFFF $4FFF_H$	Access prohibited area		CPU (PE) and ICUP
FFFF 5000_H to FFFF $FFFF_H$	On-chip peripheral I/O area	44 KB	

Note 1. This area can be accessed by CPU (PE).

The size, and thus the address range, of the code flash, data flash, and RAM differs depending on the product. See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

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3.2 Address Space Viewed from ICUP

Table 3.3 and **Table 3.4** show address spaces viewed from ICUP.

3.2.1 Code Flash Area

When the ICUMD is enabled, the secure area ranges from the start address specified in the ICUMRAC bit to the most significant address in the code flash area. The most significant address differs depending on the product.

For details of the code flash area, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

3.2.2 Data Flash Area

When the ICUMD is enabled, the secure area ranges from the start address specified in the ICUMRAD bit to the most significant address in the data flash area. The most significant address differs depending on the product.

For details of the data flash area, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

3.2.3 Space in which Instructions can be Fetched

Instructions of ICUP can be fetched from the code flash area and local RAM.

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3.2.4 Data Space Accessible by ICUP

See **Table 3.3** and **Table 3.4** for the spaces accessible from ICUP.

Table 3.3 Address Space Viewed from Each Bus Master (8-MB Product)

Address	Resource	From ICUP
0000 0000$_H$ to 003F FFFF$_H$	Code flash (bank A)	✓
0040 0000$_H$ to 007F FFFF$_H$	Access prohibited area	
0080 0000$_H$ to 00BF FFFF$_H$	Code flash (bank B)	✓
00C0 0000$_H$ to 00FF FFFF$_H$	Access prohibited area	
0100 0000$_H$ to 0100 7FFF$_H$	Code flash (Extended user area)	✓
0100 8000$_H$ to FBFF FFFF$_H$	Access prohibited area	
FC00 0000$_H$ to FDFF FFFF$_H$	Remap window area	✓
FE00 0000$_H$ to FE9C FFFF$_H$	Access prohibited area	
FE9D 0000$_H$ to FE9F FFFF$_H$	Local RAM (CPU2 (PE2) area)	✓
FEA0 0000$_H$ to FEBC FFFF$_H$	Access prohibited area	
FEBD 0000$_H$ to FEBF FFFF$_H$	Local RAM (CPU1 (PE1) area)	✓
FEC0 0000$_H$ to FEDE FFFF$_H$	Access prohibited area	
FEDF 0000$_H$ to FEDF FFFF$_H$	Local RAM (self area)	✓
FEE0 0000$_H$ to FEEB 7FFF$_H$	Access prohibited area	
FEEB 8000$_H$ to FEEF FFFF$_H$	Global RAM A	✓
FEF0 0000$_H$ to FEF0 FFFF$_H$	Retention RAM	✓
FEF1 0000$_H$ to FEFB 7FFF$_H$	Access prohibited area	
FEFB 8000$_H$ to FEFF FFFF$_H$	Global RAM B	✓
FF00 0000$_H$ to FF1E FFFF$_H$	Access prohibited area	
FF1F 0000$_H$ to FF1F 00FF$_H$	ICUM_CMDREG*1	✓
FF1F 0100$_H$ to FF1F FFFF$_H$	Access prohibited area	
FF20 0000$_H$ to FF23 FFFF$_H$	Data flash	✓
FF24 0000$_H$ to FF9F FFFF$_H$	Access prohibited area	
FFA0 0000$_H$ to FFFD FFFF$_H$	On-chip peripheral I/O area	✓
FFFE 0000$_H$ to FFFE DFFF$_H$	Access prohibited area	
FFFE E000$_H$ to FFFE FFFF$_H$	On-chip peripheral I/O area (self area)	✓
FFFF 0000$_H$ to FFFF 4FFF$_H$	Access prohibited area	
FFFF 5000$_H$ to FFFF FFFF$_H$	On-chip peripheral I/O area	✓

Note 1. This area can be accessed by CPU (PE).

Note : The following color coding is used in the map above.

Fetch and data access available
Data access available
Access prohibited

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Table 3.4 Address Space Viewed from ICUP (4-MB Product)

Address	Resource	From ICUP
0000 0000_H to 003F $FFFF_H$	Code flash	✓
0040 0000_H to 00FF $FFFF_H$	Access prohibited area	
0100 0000_H to 0100 $7FFF_H$	Code flash (Extended user area)	✓
0100 8000_H to FBFF $FFFF_H$	Access prohibited area	
FC00 0000_H to FDFF $FFFF_H$	Remap window area	✓
FE00 0000_H to FEBB $FFFF_H$	Access prohibited area	
FEBC 0000_H to FEBF $FFFF_H$	Local RAM (CPU1 (PE1) area)	✓
FEC0 0000_H to FEDE $FFFF_H$	Access prohibited area	
FEDF 0000_H to FEDF $FFFF_H$	Local RAM (self area)	✓
FEE0 0000_H to FEEE $7FFF_H$	Access prohibited area	
FEEE 8000_H to FEEF $FFFF_H$	Global RAM A	✓
FEF0 0000_H to FEF0 $FFFF_H$	Retention RAM	✓
FEF1 0000_H to FEFE $7FFF_H$	Access prohibited area	
FEFE 8000_H to FEFF $FFFF_H$	Global RAM B	✓
FF00 0000_H to FF1E $FFFF_H$	Access prohibited area	
FF1F 0000_H to FF1F $00FF_H$	ICUM_CMDREG*1	✓
FF1F 0100_H to FF1F $FFFF_H$	Access prohibited area	
FF20 0000_H to FF21 $FFFF_H$	Data flash	✓
FF22 0000_H to FF9F $FFFF_H$	Access prohibited area	
FFA0 0000_H to FFFD $FFFF_H$	On-chip peripheral I/O area	✓
FFFE 0000_H to FFFE $DFFF_H$	Access prohibited area	
FFFE $E000_H$ to FFFE $FFFF_H$	On-chip peripheral I/O area (self area)	✓
FFFF 0000_H to FFFF $4FFF_H$	Access prohibited area	
FFFF 5000_H to FFFF $FFFF_H$	On-chip peripheral I/O area	✓

Note 1. This area can be accessed by CPU (PE).

Note : The following color coding is used in the map above.

Fetch and data access available
Data access available
Access prohibited

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3.3 Peripheral I/O Address Map

Table 3.5 and **Table 3.6** shows peripheral I/O address map.

Table 3.5 Peripheral I/O Address Map (self area of ICUMD)

Address	Peripheral I/O
FFFE E000$_H$ to FFFE E03F$_H$	ICUM_OSTM0
FFFE E040$_H$ to FFFE E07F$_H$	ICUM_OSTM1
FFFE E080$_H$ to FFFE E08F$_H$	ICUM_WDTA0
FFFE E090$_H$ to FFFE E09F$_H$	ICUM_PROT0
FFFE E0A0$_H$ to FFFE E0BF$_H$	ICUM_TRNGC0
FFFE E0C0$_H$ to FFFE E0FF$_H$	Access prohibited area
FFFE E100$_H$ to FFFE E17F$_H$	ICUM_AESB0
FFFE E180$_H$ to FFFE E1FF$_H$	Access prohibited area
FFFE E200$_H$ to FFFE E2BF$_H$	ICUM_CTLREG
FFFE E2C0$_H$ to FFFE E2FF$_H$	Access prohibited area
FFFE E300$_H$ to FFFE E3FF$_H$	Local RAM ECC (ICUP)
FFFE E400$_H$ to FFFE E4FF$_H$	Cache ECC (ICUP)
FFFE E500$_H$ to FFFE E5FF$_H$	ICUM Flash Cache CTL
FFFE E600$_H$ to FFFE E9EF$_H$	Access prohibited area
FFFE E9F0$_H$ to FFFE E9FF$_H$	ICUM_SEG
FFFE EA00$_H$ to FFFE EBFF$_H$	ICUM_INTC1
FFFE EC00$_H$ to FFFE ECBF$_H$	Access prohibited area
FFFE ECC0$_H$ to FFFE ECCF$_H$	ICUM_FENMI
FFFE ECD0$_H$ to FFFE ECDF$_H$	ICUM_FEINT
FFFE ECE0$_H$ to FFFE ECFF$_H$	Access prohibited area
FFFE ED00$_H$ to FFFE ED1F$_H$	P-Bus guard (ICUM_PBG00)
FFFE ED20$_H$ to FFFE ED2F$_H$	ERRSLV (ICUM_PBG00)
FFFE ED30$_H$ to FFFE ED3F$_H$	Access prohibited area
FFFE ED40$_H$ to FFFE ED5F$_H$	P-Bus guard (ICUM_PBG01)
FFFE ED60$_H$ to FFFE ED6F$_H$	ERRSLV (ICUM_PBG01)
FFFE ED70$_H$ to FFFE ED7F$_H$	Access prohibited area
FFFE ED80$_H$ to FFFE ED87$_H$	ERRSLV (Code flash guard)
FFFE ED88$_H$ to FFFE ED9F$_H$	Access prohibited area
FFFE EDA0$_H$ to FFFE EDAF$_H$	ERRSLV (Data flash guard)
FFFE EDB0$_H$ to FFFE EDBF$_H$	Access prohibited area
FFFE EDC0$_H$ to FFFE EDFF$_H$	Data flash ECC (ICUP)
FFFE EE00$_H$ to FFFE EF53$_H$	Code flash ECC (ICUP)
FFFE EF54$_H$ to FFFE EFFF$_H$	Access prohibited area
FFFE F000$_H$ to FFFE FFFF$_H$	ICUM_INTC2

Table 3.6 Peripheral I/O Address Map (shared area between CPU (PE) and ICUP)

Address	Peripheral Group	Peripheral I/O
FF1F 0000$_H$ to FF1F 00FF$_H$	2	ICUM_CMDREG

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3.3.1 Remap Area

For access to all address areas, the ICUMD has a remapping function for address translation during access to specific address ranges. The destination area for access by the remapping function is called the Remap area and this area is specified via the ICUM_SICREMAP register.

For access to the Remap area, set the higher-order 7 bits of the area to be remapped in the ICUM_SICREMAP register and then access the area by accessing the range from FC00 0000_H to FDFF $FFFF_H$ (32 MB).

The Remap area takes up a 32 MB range. Access to a different 32 MB Remap area requires a new setting in the ICUM_SICREMAP register.

If an exception is acknowledged while remapping is in use and the exception handler changes ICUM_SICREMAP register, the exception handler must include saving and restoring the value of ICUM_SICREMAP register. The processing following return from the exception handler will not proceed normally if restoring the ICUM_SICREMAP register is not executed correctly.

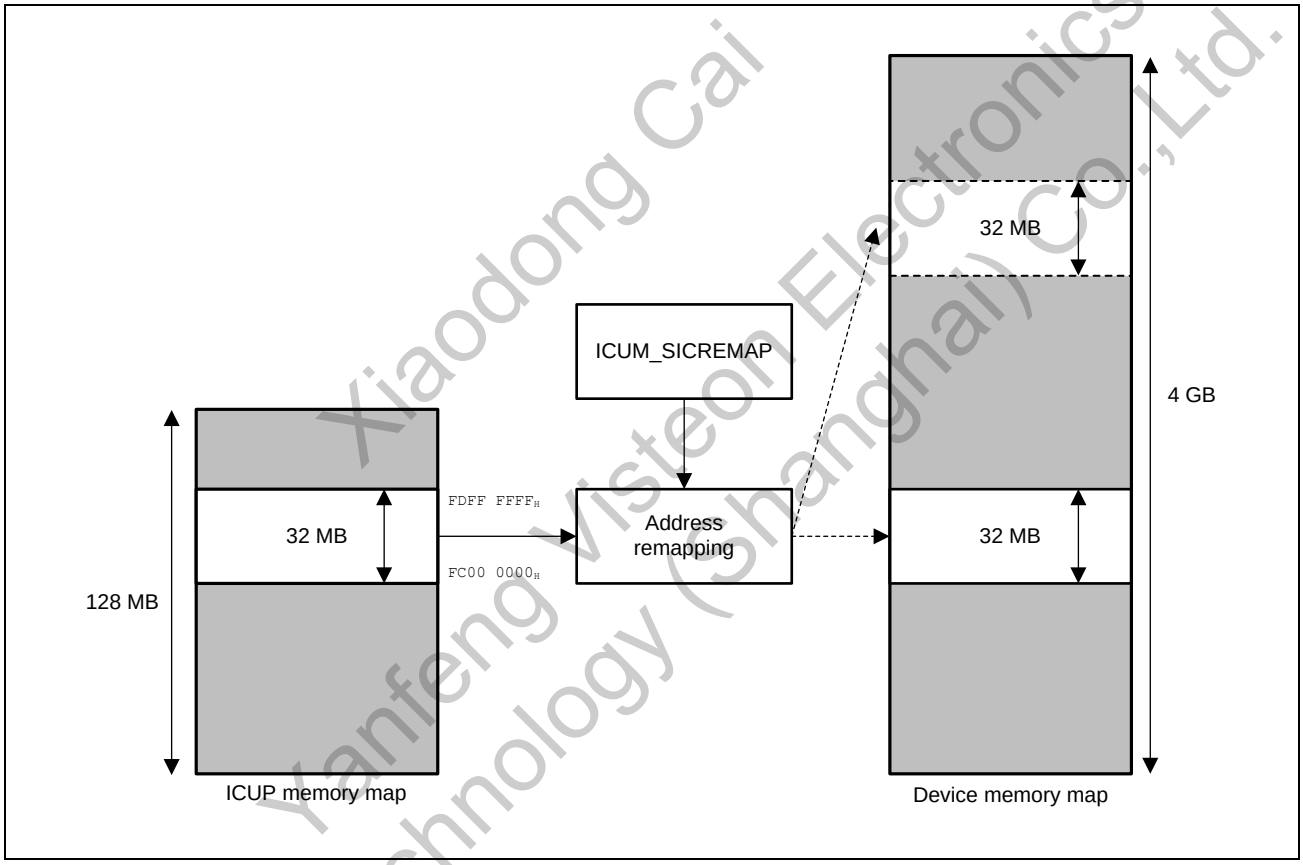


Figure 3.1 Remap Function Configuration

CAUTION

The memory protection settings implemented by the MPU function are applied to the settings for the remapped window area.

These settings therefore need to be reconfigured according to the area to be accessed.

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Section 4 Write-Protection Registers

This section describes the write-protection registers used in the ICUMD.

4.1 Overview

For details of the write-protection register functions, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

4.1.1 Registers to Be Write-Protected

The registers that are protected by the write-protection registers are listed below.

Table 4.1 Write-Protected Registers

Protection Target	Protected Register	Protected Register		Protection Cluster
		Command	Status	
ICUM_CTLREG	ICUM_SICREMAP	ICUM_PROT0PCMD	ICUM_PROT0PS	ICUMD control protection cluster
	ICUM_UPGPS			
	ICUM_IDAS			
	ICUM_STOP			

4.2 Registers

4.2.1 List of Registers

The following table lists the write-protection control registers.

For details, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

The only differences from the write protection register incorporated in the RH850/F1KH, RH850/F1KM are the register symbols.

For the ICUM_PROT0PCMD register, see the section on the "PROTCMDn Register", and for the ICUM_PROT0PS register, see the section on the "PROTSn Register".

Table 4.2 List of Registers

Register Name	Symbol	Address
ICUMD control protection cluster		
Protection command register 0	ICUM_PROT0PCMD	FFFE E090$_H$
Protection status register 0	ICUM_PROT0PS	FFFE E094$_H$

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Section 5 Operating Modes

The behaviors of the ICUMD in each operating mode of the RH850/F1KH, RH850/F1KM are listed below. For other details, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

- Normal operating mode: The ICUMD operates normally.
- User boot mode: The ICUMD operates normally.
- Serial programming mode: The ICUMD stops operating.
- Boundary scan mode: The ICUMD stops operating.

5.1 Registers

5.1.1 List of Registers

Table 5.1 List of Registers

Register Name	Symbol	Address
User boot mode status register	ICUM_UBMDSTS	FFF8 3508$_H$

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5.1.2 ICUM_UBMDSTS — User Boot Mode Status Register

This register indicates the Operating Mode of the device. The status of the mode pin is taken in this register when an external reset or power-on reset is generated.

Access: This register is a read-only register that can be read in 32-bit units.

Address: FFF8 3508$_H$

Value after reset: 0000 0001$_H$ / 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	UB MODE
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1/0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 5.2 ICUM_UBMDSTS Register Contents

Bit Position	Bit Name	Function
31 to 1	—	Reserved When read, the value after reset is returned.
0	UBMODE	This bit indicates a user boot mode. 0:Not user boot mode 1:User boot mode

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Section 6 Exception/Interrupts

6.1 Features

See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*. When referring to this manual, "IC<name>" should be handled as "ICUM_IC<name>".

6.2 ICUMD Interrupt Sources

6.2.1 ICUMD Interrupt Sources

6.2.1.1 ICUMD FE Level Non-Maskable Interrupts

(1) Priority
See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

(2) Return PC
Return or recovery from an FE non-maskable interrupt is not possible.

(3) Status Register
See **Section 6.3.4, ICUM_FNC — FE Level NMI Status Register**.

(4) Return Instruction
None

Table 6.1 FE Level Non-Maskable Interrupt Requests

Interrupt			Interrupt Request		Unit	Priority Group	Exception Source code	Handler address 00000...
	Control Register							
Symbol	Name	Address	Name	Source				
FENMI	ICUM_FNC	FFFE EA78$_H$	EXTTNMI	NMI pin	Port	*1	0E0$_H$	0E0$_H$
			EXTWDTA0NMI	WDTA0 FENMI interrupt	WDTA0			
			ICUWDTA0NMI	ICUM_WDTA0 FENMI interrupt	ICUM_WDTA0			

Note 1. See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

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6.2.1.2 ICUMD FE Level Maskable Interrupts

(1) Priority
See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

(2) Return PC
The return PC returned from an interrupt handling routine by the FERET instruction is the PC from when the program was suspended (current PC).

(3) Status Register
See **Section 6.3.5, ICUM_FIC — FE Level Maskable Interrupt Status Register**.

(4) Return Instruction
FERET

Table 6.2 FE Level Maskable Interrupt Requests

Interrupt			Interrupt Request		Unit	Priority Group	Exception Source code	Handler address 00000...
	Control Register							
Symbol	Name	Address	Name	Source				
FEINT	ICUM_FIC	FFFE EA7A$_H$	INTEXTLVIL	LVI voltage detection (falling)	LVI	*1	0F0$_H$	0F0$_H$
			INTEXTLVIH	LVI voltage detection (rising)	LVI			
			INTEXTSTPRQ	STOP mode request of standby controller	Standby controller			
			INTICUECCCRAM	Cache RAM 1-bit ECC error or 2-bit ECC error interrupt	RAM			
			INTICUECCLRAM	Local RAM 1-bit ECC error or 2-bit ECC error interrupt	RAM			
			INTICUECCFLI0	Code flash 1-bit ECC error or 2-bit ECC error interrupt	Code flash			
			INTICUECCEEP0	Data flash 1-bit ECC error or 2-bit ECC error interrupt	Data flash			
			INTICUOSTM0_FE*2	ICUM_OSTM0 interrupt	ICUM_OSTM0			
			INTICUOSTM1_FE*3	ICUM_OSTM1 interrupt	ICUM_OSTM1			

Note 1. See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

Note 2. As INTICUOSTM0 is also assigned to EIINT, using them simultaneously is prohibited.

Note 3. As INTICUOSTM1 is also assigned to EIINT, using them simultaneously is prohibited.

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6.2.1.3 ICUMD EI Level Maskable Interrupts

(1) Interrupt Naming Rules

See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*. When referring to this manual, "IC <name>" should be handled as "ICUM_IC<name>".

(2) Priority

See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

(3) Return PC

The return PC returned from an interrupt handling routine by the EIRET instruction is the PC from when the program was suspended (current PC).

(4) Control Register

EI Level Maskable Interrupt Control Register

See **Section 6.3.2, ICUM_ICxxx — EI Level Interrupt Control Registers**.

(5) Return Instruction

EIRET instruction

(6) Configuration

EI-level maskable interrupts are controlled by the two controllers, INTC1 and INTC2. The interrupts are supported on a total 48 channels (RH850/F1KM-S4) / 52 channels (RH850/F1KH-D8) with a cascade connection of INTC1 and INTC2.

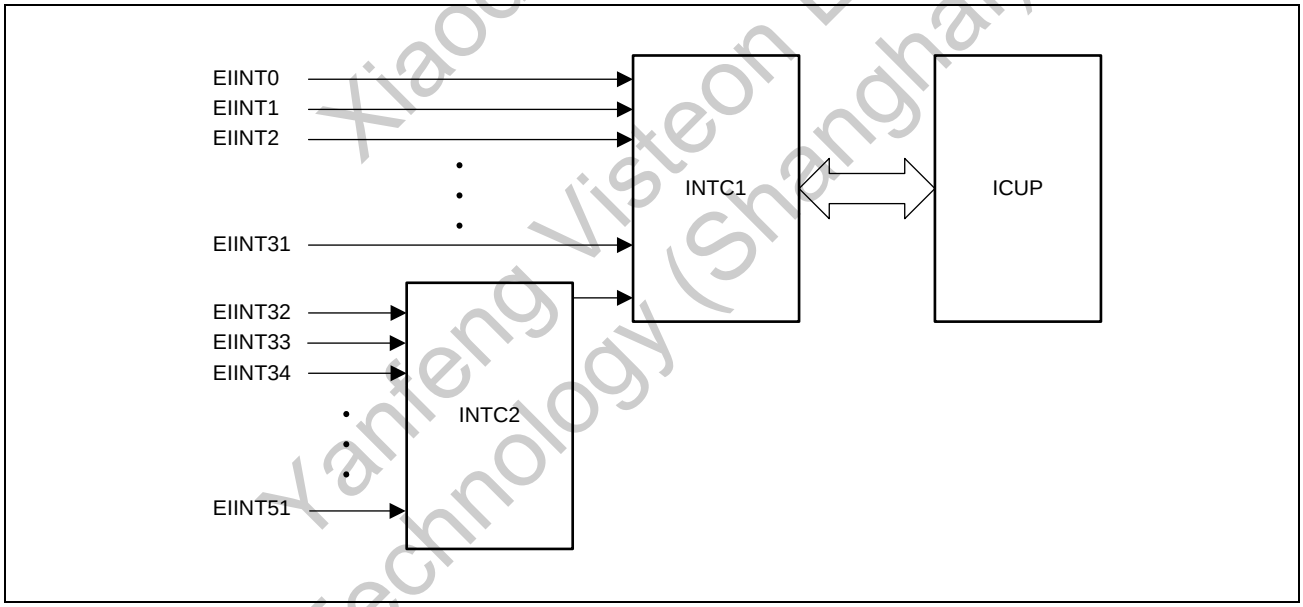


Figure 6.1 Configuration Diagram of EI Level Maskable Interrupt

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Table 6.3 lists EI Level Maskable Interrupts.

Table 6.3 EI Level Maskable Interrupt Sources (1/2)

Interrupt			Interrupt Request				Exception Source code	Handler Address (Offset)*3		
	Control Register							Direct Jumping to an Address		
Channel No.	Name	Address	Name	Source	Unit	Detection Type*1		RINT = 0	RINT = 1	Reference to a Table*2
0	ICUM_ICPES	FFFE EA00$_H$	INTPES	CPU (PE) interrupt request	ICUM_CMDREG	Level	1000$_H$	*4	*5	+000$_H$
1	ICUM_ICPE	FFFE EA02$_H$	INTPE	CPU (PE) interrupt request	ICUM_CMDREG	Level	1001$_H$			+004$_H$
2	Reserved	FFFE EA04$_H$					1002$_H$			+008$_H$
3	Reserved	FFFE EA06$_H$					1003$_H$			+00C$_H$
4	ICUM_ICICUAESB0RD	FFFE EA08$_H$	INTICUAESB0RD	Completion of AES encryption or decryption	ICUM_AESB0	Edge	1004$_H$			+010$_H$
5	Reserved	FFFE EA0A$_H$					1005$_H$			+014$_H$
6	ICUM_ICICUTRNGC0	FFFE EA0C$_H$	INTICUTRNGC0	ICUM_TRNGC0 output data ready	ICUM_TRNGC0	Edge	1006$_H$			+018$_H$
7	ICUM_ICICUOSTM0	FFFE EA0E$_H$	INTICUOSTM0	ICUM_OSTM0 overflow	ICUM_OSTM0	Edge	1007$_H$			+01C$_H$
8	ICUM_ICICUOSTM1	FFFE EA10$_H$	INTICUOSTM1	ICUM_OSTM1 overflow	ICUM_OSTM1	Edge	1008$_H$			+020$_H$
9	ICUM_ICICUWDTA0	FFFE EA12$_H$	INTICUWDTA0	ICUM_WDTA0 counter 75%	ICUM_WDTA0	Edge	1009$_H$			+024$_H$
10	ICUM_ICWDTA0	FFFE EA14$_H$	INTEXTWDTA0	WDTA0 counter 75%	WDTA0	Edge	100A$_H$			+028$_H$
11	Reserved	FFFE EA16$_H$					100B$_H$			+02C$_H$
12	ICUM_ICFACI0ERR	FFFE EA18$_H$	INTFACI0ERR	Flash sequencer error	Flash	Level	100C$_H$			+030$_H$
13	ICUM_ICFACI0ENDNM	FFFE EA1A$_H$	INTFACI0ENDNM	Flash sequencer end	Flash	Edge	100D$_H$			+034$_H$
14	ICUM_ICFICUIF0CFRQ	FFFE EA1C$_H$	INTFICUIF0CFRQ	Code flash programming request	Flash	Edge	100E$_H$			+038$_H$
15	ICUM_ICFICUIF0DFRQ	FFFE EA1E$_H$	INTFICUIF0DFRQ	Data flash programming request	Flash	Edge	100F$_H$			+03C$_H$
16	ICUM_ICFICUIF0ILA	FFFE EA20$_H$	INTFICUIF0ILA	Flash violation access notice	Flash	Edge	1010$_H$			+040$_H$
17	Reserved	FFFE EA22$_H$					1011$_H$			+044$_H$
18	Reserved	FFFE EA24$_H$					1012$_H$			+048$_H$
19	Reserved	FFFE EA26$_H$					1013$_H$			+04C$_H$
20	Reserved	FFFE EA28$_H$					1014$_H$			+050$_H$
21	Reserved	FFFE EA2A$_H$					1015$_H$			+054$_H$
22	Reserved	FFFE EA2C$_H$					1016$_H$			+058$_H$
23	ICUM_ICAPBGERR	FFFE EA2E$_H$	INTAPBGERR	Violation of access to ICUM_PBG	ICUM_PBG	Edge	1017$_H$			+05C$_H$
24	Reserved	FFFE EA30$_H$					1018$_H$			+060$_H$
25	Reserved	FFFE EA32$_H$					1019$_H$			+064$_H$
26	ICUM_ICECM0CFGERR	FFFE EA34$_H$	INTECM0CFGERR	Code flash guard error	CFG	Edge	101A$_H$			+068$_H$
27	ICUM_ICECM0DFGERR	FFFE EA36$_H$	INTECM0DFGERR	Data flash guard error	DFG	Edge	101B$_H$			+06C$_H$
28	ICUM_ICEXTDCU20CDRQ	FFFE EA38$_H$	INTEXTDCU20CDRQ	Authentication challenge data request for ICUP	Debug	Edge	101C$_H$			+070$_H$
29	ICUM_ICEXTDCU20RDRQ	FFFE EA3A$_H$	INTEXTDCU20RDRQ	Authentication response data comparison request for ICUP/CPU (PE)	Debug	Edge	101D$_H$			+074$_H$
30	ICUM_ICEXTDCU20CDRQPE	FFFE EA3C$_H$	INTEXTDCU20CDRQPE	Authentication challenge data request for CPU (PE)	Debug	Edge	101E$_H$			+078$_H$
31	Reserved	FFFE EA3E$_H$					101F$_H$			+07C$_H$

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Table 6.3 ICUMD EI Level Maskable Interrupt Sources (2/2)

Interrupt			Interrupt Request				Exception Source code	Handler Address (Offset)*3		
								Direct Jumping to an Address		
Channel No.	Control Register		Name	Source	Unit	Detection Type*1		RINT = 0	RINT = 1	Reference to a Table*2
	Name	Address								
32	Reserved	FFFE EA40$_H$					1020$_H$	*4	*5	+080$_H$
33	Reserved	FFFE EA42$_H$					1021$_H$			+084$_H$
34	Reserved	FFFE EA44$_H$					1022$_H$			+088$_H$
35	Reserved	FFFE EA46$_H$					1023$_H$			+08C$_H$
35	Reserved	FFFE EA48$_H$					1024$_H$			+090$_H$
37	Reserved	FFFE EA4A$_H$					1025$_H$			+094$_H$
38	Reserved	FFFE EA4C$_H$					1026$_H$			+098$_H$
39	Reserved	FFFE EA4E$_H$					1027$_H$			+09C$_H$
40	ICUM_ICEXTDMA0	FFFE F050$_H$	INTEXTDMA0	DMA00 transfer completion	PDMA0	Edge	1028$_H$			+0A0$_H$
41	ICUM_ICEXTDMA2	FFFE F052$_H$	INTEXTDMA2	DMA02 transfer completion	PDMA0	Edge	1029$_H$			+0A4$_H$
42	ICUM_ICEXTDMA4	FFFE F054$_H$	INTEXTDMA4	DMA04 transfer completion	PDMA0	Edge	102A$_H$			+0A8$_H$
43	ICUM_ICEXTDMA6	FFFE F056$_H$	INTEXTDMA6	DMA06 transfer completion	PDMA0	Edge	102B$_H$			+0AC$_H$
44	ICUM_ICEXTDMA8	FFFE F058$_H$	INTEXTDMA8	DMA10 transfer completion	PDMA0	Edge	102C$_H$			+0B0$_H$
45	ICUM_ICEXTDMA10	FFFE F05A$_H$	INTEXTDMA10	DMA12 transfer completion	PDMA0	Edge	102D$_H$			+0B4$_H$
46	ICUM_ICEXTDMA12	FFFE F05C$_H$	INTEXTDMA12	DMA14 transfer completion	PDMA0	Edge	102E$_H$			+0B8$_H$
47	ICUM_ICEXTDMA14	FFFE F05E$_H$	INTEXTDMA14	DMA16 transfer completion	PDMA0	Edge	102F$_H$			+0BC$_H$
48	ICUM_ICEXTDMA32*6	FFFE F060$_H$	INTEXTDMA32	DMA00 transfer completion	PDMA1	Edge	1030$_H$			+0C0$_H$
49	ICUM_ICEXTDMA34*6	FFFE F062$_H$	INTEXTDMA34	DMA02 transfer completion	PDMA1	Edge	1031$_H$			+0C4$_H$
50	ICUM_ICEXTDMA36*6	FFFE F064$_H$	INTEXTDMA36	DMA04 transfer completion	PDMA1	Edge	1032$_H$			+0C6$_H$
51	ICUM_ICEXTDMA38*6	FFFE F066$_H$	INTEXTDMA38	DMA06 transfer completion	PDMA1	Edge	1033$_H$			+0C8$_H$

Note 1. This indicates whether an interrupt source is detected at the level or edge. This also affects the value after reset of an EI level interrupt control register. For details, see **Section 6.3.2, ICUM_ICxxx — EI Level Interrupt Control Registers**.

Note 2. The table reference method uses a table for reading an exception handler address for each interrupt channel and it extracts the handler address by referencing to the table. The position of the table is determined by the following formula.
Exception handler address read position = INTBP register + channel no. × 4 bytes

Note 3. For details, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

Note 4. Irrespective of interrupt channels, an offset address is determined in the range from +100$_H$ to 170$_H$ according to the priority order (0 to 7).

Note 5. Irrespective of the priority order, offset addresses are uniformly +100$_H$.

Note 6. These interrupts are supported only in RH850/F1KH-D8.

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6.2.1.4 Inter-PE Interrupts

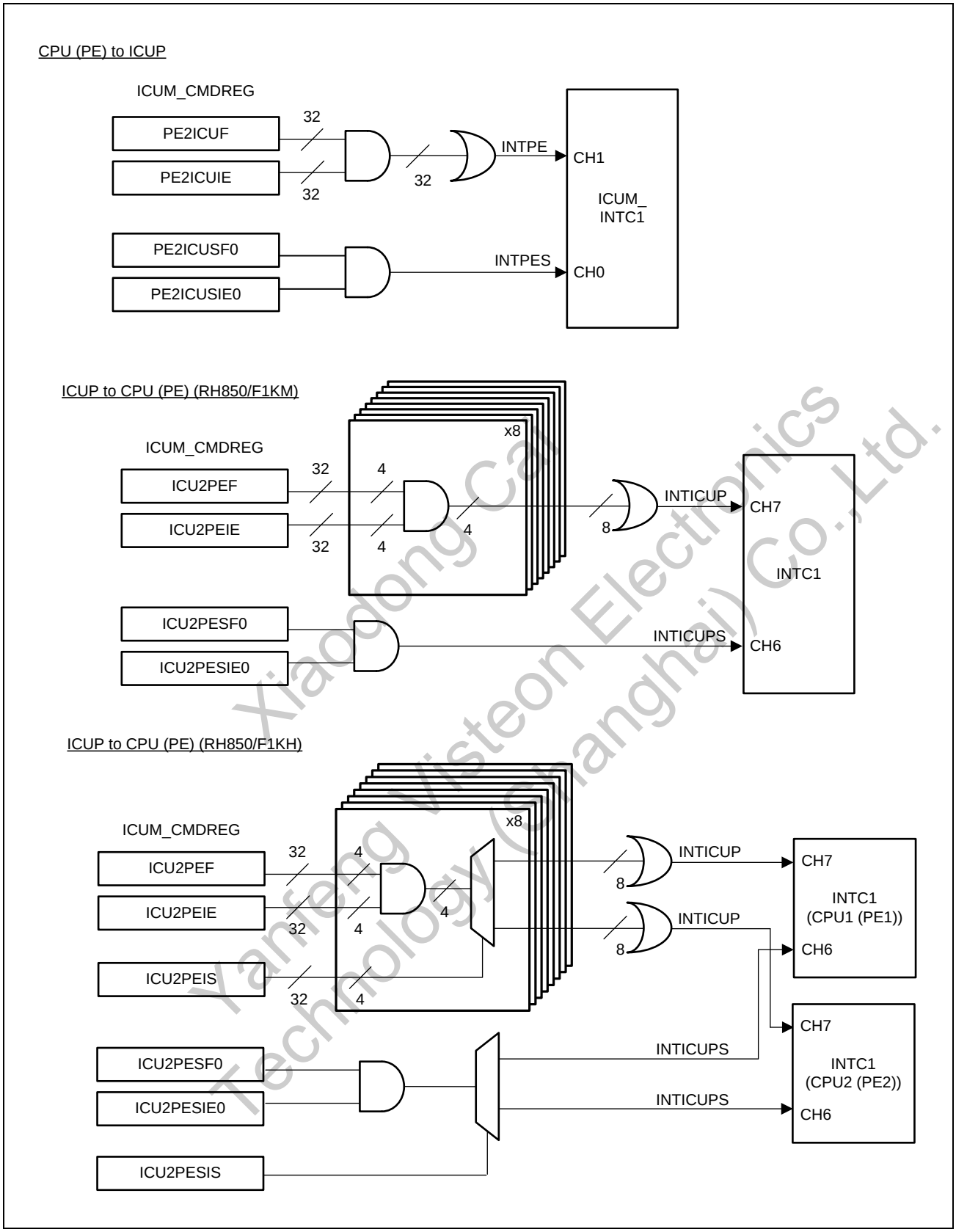
CPU (PE) can generate interrupt signal to ch0, 1 of the ICUM_INTC1 EIINT in ICUP via command register. The ICUP can generate interrupt signals to CPU (PE) via command register.

The interrupt signals from CPU (PE) to ICUP and the interrupt signals from ICUP to CPU (PE) are generated based on ICUM_CMDREG's setting.

Priority channel interrupt signals are added as expanded function from ICUMD. For details, see **Section 10, ICUMD Control Register (ICUM_CTLREG)**.

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Figure 6.2 Inter-PE Interrupt Block Diagram

6.2.2 ICUM FE Level Non-Maskable Interrupt Sources

6.2.2.1 List of Registers

The following table lists the ICUM FE Level Non-Maskable Interrupt Sources registers.

Table 6.4 List of Registers

Register Name	Symbol	Address
FENMI factor register	ICUM_FENMIF	FFFE ECC0$_H$
FENMI factor clear register	ICUM_FENMIFC	FFFE ECC8$_H$

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6.2.2.2 ICUM_FENMIF — FENMI Factor Register

This register contains information about which source has generated the FE level non-maskable interrupt (ICUM_FENMI). This register is initialized by any reset.

Access: This register is a read-only register that can be read in 32-bit units.

Address: FFFE ECC0$_H$

Value after reset: 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	ICUM_WDTA0NMIF	—	—	—	—	—	—	WDTA0NMIF	TNMIF
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 6.5 ICUM_FENMIF Register Contents

Bit Position	Bit Name	Function
31 to 9	—	Reserved When read, the value after reset is returned.
8	ICUM_WDTA0NMIF	ICUM_WDTA0NMI flag 0: No ICUM_WDTA0NMI occurred 1: ICUM_WDTA0NMI has occurred
7 to 2	—	Reserved When read, the value after reset is returned.
1	WDTA0NMIF	WDTA0NMI flag 0: No WDTA0NMI occurred 1: WDTA0NMI has occurred
0	TNMIF	Input signal flag from the NMI pin 0: No TNMI occurred 1: TNMI has occurred

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6.2.2.3 ICUM_FENMIFC — FENMI Factor Clear Register

This register clears the FE level non-maskable interrupt flags of the ICUM_FENMIF register.

Access: This register is a write-only register that can be written in 32-bit units.

Address: FFFE ECC8$_H$

Value after reset: 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	ICUM_WDTA0NMIFC	—	—	—	—	—	—	WDTA0NMIFC	TNMIFC
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	W	R	R	R	R	R	R	W	W

Table 6.6 ICUM_FENMIFC Register Contents

Bit Position	Bit Name	Function
31 to 9	—	Reserved When writing, write the value after reset.
8	ICUM_WDTA0NMIFC	ICUM_WDTA0NMIF flag clear 0: — 1: Clear
7 to 2	—	Reserved When writing, write the value after reset.
1	WDTA0NMIFC	WDTA0NMIF flag clear 0: — 1: Clear
0	TNMIFC	TNMIF flag clear 0: — 1: Clear

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6.2.3 ICUM FE Level Maskable Interrupt Sources

6.2.3.1 List of Registers

The following table lists the ICUM FE Level Maskable Interrupt Sources registers.

Table 6.7 List of Registers

Register Name	Symbol	Address
FEINT factor register	ICUM_FEINTF	FFFE ECD0$_H$
FEINT factor mask register	ICUM_FEINTFMSK	FFFE ECD4$_H$
FEINT factor clear register	ICUM_FEINTFC	FFFE ECD8$_H$

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6.2.3.2 ICUM_FEINTF — FEINT Factor Register

This register contains information about which source has generated the FE level maskable interrupt (ICUM_FEINT). This register is initialized by any reset.

Access: This register is a read-only register that can be read in 32-bit units.

Address: FFFE ECD0$_H$

Value after reset: 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LVIHFEIF	LVILFEIF
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	ICUM_OSTM1FEIF	ICUM_OSTM0FEIF	ICUM_STPRQFEIF	—	—	—	ICUM_ECCCRAMFEIF	ICUM_ECCLRAMFEIF	ICUM_ECCEEP0FEIF	ICUM_ECCFLI0FEIF
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 6.8 ICUM_FEINTF Register Contents (1/2)

Bit Position	Bit Name	Function
31 to 18	—	Reserved When read, the value after reset is returned.
17	LVIHFEIF	INTEXTLVIH interrupt occurrence 0: No interrupt occurred 1: Interrupt has occurred
16	LVILFEIF	INTEXTLVIL interrupt occurrence 0: No interrupt occurred 1: Interrupt has occurred
15 to 10	—	Reserved When read, the value after reset is returned.
9	ICUM_OSTM1FEIF	INTICUOSTM1_FE interrupt occurrence 0: No interrupt occurred 1: Interrupt has occurred
8	ICUM_OSTM0FEIF	INTICUOSTM0_FE interrupt occurrence 0: No interrupt occurred 1: Interrupt has occurred
7	ICUM_STPRQFEIF	INTEXTSTPRQ interrupt occurrence 0: No interrupt occurred 1: Interrupt has occurred
6 to 4	—	Reserved When read, the value after reset is returned.
3	ICUM_ECCCRAMFEIF	INTICUECCCRAM interrupt occurrence 0: No interrupt occurred 1: Interrupt has occurred
2	ICUM_ECCLRAMFEIF	INTICUECCLRAM interrupt occurrence 0: No interrupt occurred 1: Interrupt has occurred

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Table 6.8 ICUM_FEINTF Register Contents (2/2)

Bit Position	Bit Name	Function
1	ICUM_ECCEEP0FEIF	INTICUECCEEP0 interrupt occurrence 0: No interrupt occurred 1: Interrupt has occurred
0	ICUM_ECCFLI0FEIF	INTICUECCFLI0 interrupt occurrence 0: No interrupt occurred 1: Interrupt has occurred

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6.2.3.3 ICUM_FEINTFMSK — FEINT Factor Mask Register

This register masks the FE level maskable interrupt (ICUM_FEINT). This register is initialized by any reset.

Access: This register can be read or written in 32-bit units.

Address: FFFE ECD4$_H$

Value after reset: FFFF FFFF$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LVIHFEIFMSK	LVILFEIFMSK
Value after reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	ICUM_OSTM1FEIFMSK	ICUM_OSTM0FEIFMSK	ICUM_STPRQFEIFMSK	—	—	—	ICUM_ECCCRAMFEIFMSK	ICUM_ECCLRAMFEIFMSK	ICUM_ECCEEP0FEIFMSK	ICUM_ECCFLI0FEIFMSK
Value after reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
R/W	R	R	R	R	R	R	R/W	R/W	R/W	R	R	R	R/W	R/W	R/W	R/W

Table 6.9 ICUM_FEINTFMSK Register Contents (1/2)

Bit Position	Bit Name	Function
31 to 18	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
17	LVIHFEIFMSK	INTEXTLVIH interrupt mask 0: Not masked 1: Masked
16	LVILFEIFMSK	INTEXTLVIL interrupt mask 0: Not masked 1: Masked
15 to 10	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
9	ICUM_OSTM1FEIFMSK	INTICUOSTM1_FE interrupt mask 0: Not masked 1: Masked
8	ICUM_OSTM0FEIFMSK	INTICUOSTM0_FE interrupt mask 0: Not masked 1: Masked
7	ICUM_STPRQFEIFMSK	INTEXTSTPRQ interrupt mask 0: Not masked 1: Masked
6 to 4	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
3	ICUM_ECCCRAMFEIFMSK	INTICUECCCRAM interrupt mask 0: Not masked 1: Masked
2	ICUM_ECCLRAMFEIFMSK	INTICUECCLRAM interrupt mask 0: Not masked 1: Masked

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Table 6.9 ICUM_FEINTFMSK Register Contents (2/2)

Bit Position	Bit Name	Function
1	ICUM_ECCEEP0FEIFMSK	INTICUECCEEP0 interrupt mask 0: Not masked 1: Masked
0	ICUM_ECCFLI0FEIFMSK	INTICUECCFLI0 interrupt mask 0: Not masked 1: Masked

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6.2.3.4 ICUM_FEINTFC — FEINT Factor Clear Register

This register clears the bits of the FEINT factor register (ICUM_FEINTF).

Access: This register is a write-only register that can be written in 32-bit units.

Address: FFFE ECD8$_H$

Value after reset: 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LVIHFEIFC	LVILFEIFC
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	W	W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	ICUM_OSTM1FEIFC	ICUM_OSTM0FEIFC	ICUM_STPRQFEIFC	—	—	—	ICUM_ECCCRAMFEIFC	ICUM_ECCLRAMFEIFC	ICUM_ECCEEP0FEIFC	ICUM_ECCFLI0FEIFC
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	W	W	W	R	R	R	W	W	W	W

Table 6.10 ICUM_FEINTFC Register Contents (1/2)

Bit Position	Bit Name	Function
31 to 18	—	Reserved When writing, write the value after reset.
17	LVIHFEIFC	LVIHFEIF flag clear 0: — 1: Clear
16	LVILFEIFC	LVILFEIF flag clear 0: — 1: Clear
15 to 10	—	Reserved When writing, write the value after reset.
9	ICUM_OSTM1FEIFC	ICUM_OSTM1FEIF flag clear 0: — 1: Clear
8	ICUM_OSTM0FEIFC	ICUM_OSTM0FEIF flag clear 0: — 1: Clear
7	ICUM_STPRQFEIFC	ICUM_STPRQFEIF flag clear 0: — 1: Clear
6 to 4	—	Reserved When writing, write the value after reset.
3	ICUM_ECCCRAMFEIFC	ICUM_ECCCRAMFEIF flag clear 0: — 1: Clear
2	ICUM_ECCLRAMFEIFC	ICUM_ECCLRAMFEIF flag clear 0: — 1: Clear

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Table 6.10 ICUM_FEINTFC Register Contents (2/2)

Bit Position	Bit Name	Function
1	ICUM_ECCEEP0FEIFC	ICUM_ECCEEP0FEIF flag clear 0: — 1: Clear
0	ICUM_ECCFLI0FEIFC	ICUM_ECCFLI0FEIF flag clear 0: — 1: Clear

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6.3 Interrupt Controller Control Registers

Writing to the ICUM_ICxxx, ICUM_IMRm (m = 0, 1), ICUM_FNC, and ICUM_FIC registers is enabled only in supervisor mode (PSW.UM = 0).

6.3.1 List of Registers

The following table lists the Interrupt Controller Control registers.

Table 6.11 List of Registers

Register Name	Symbol	Address
EI level interrupt control registers	ICUM_ICxxx	See **Table 6.3**
EI level interrupt mask registers	ICUM_IMR0	FFFE EAF0$_H$
	ICUM_IMR1	FFFE F404$_H$
FE level NMI status register	ICUM_FNC	FFFE EA78$_H$
FE level maskable interrupt status register	ICUM_FIC	FFFE EA7A$_H$

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6.3.2 ICUM_ICxxx — EI Level Interrupt Control Registers

These registers, each of which is for a channel of EI level maskable interrupt (INT), are used to set the conditions to control each channel. This register is initialized by any reset.

Access: This register can be read or written in 16-bit units.

Address: See **Table 6.3, EI Level Maskable Interrupt Sources.**

Value after reset: 0087_H (edge detection), 8087_H (level detection)*1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	CTxxx	—	—	RFxxx	—	—	—	—	MKxxx	TBxxx	—	—	—	P2xxx	P1xxx	P0xxx
Value after reset	0/1*1	0	0	0	0	0	0	0	1	0	0	0	0	1	1	1
R/W	R	R	R	R/W	R	R	R	R	R/W	R/W	R	R	R	R/W	R/W	R/W

Note 1. The value after reset differs depending on the detection type of a given interrupt (edge detection: 0, level detection: 1). For details, see detection types in **Table 6.3, EI Level Maskable Interrupt Sources**.

Table 6.12 ICUM_ICxxx Register Contents (1/2)

Bit Position	Bit Name	Function
15	CTxxx	This bit indicates an interrupt channel type. This bit is read only. 0: Detection in synchronization with an edge is currently selected. 1: Detection of the high level is currently selected.
14, 13	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
12	RFxxx	This is an interrupt request flag. The RFxxx bit can be written from a program. Setting the RFxxx bit to 1 generates EI level maskable interrupt n (INTn), just as when an interrupt request is acknowledged. 0: No interrupt request is made. 1: Interrupt request is made. **Input Interface** / **Operation** Detection in synchronization with an edge (CTxxx = 0) — This bit is automatically cleared when an interrupt request is acknowledged by the CPU core. It can be set and cleared by software. Detection of the high level(CTxxx = 1) — This bit cannot be set or cleared by software. It is read only. It is not cleared when an interrupt request is acknowledged by the CPU core.
11 to 8	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
7	MKxxx	This is an interrupt request mask bit. When the MKxxx bit is set, interrupt requests from the channel are masked and, therefore, not issued to the CPU core. The interrupt pending status is not reflected in the ICSR.PMEI bit for any channels that are masked. When the interrupt request from the channel is masked with MKxxx = 1, the RFxxx still reflects the interrupt request to the channel and can be polled in software. When the MKxxx bit is cleared, interrupt requests from the channel are issued to the CPU core for subsequent processing. The state of the MKxxx bit is also reflected in the corresponding ICUM_IMRm register. 0: Enables interrupt processing 1: Disables interrupt processing

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Table 6.12 ICUM_ICxxx Register Contents (2/2)

Bit Position	Bit Name	Function
6	TBxxx	This bit is used to select the way to determine the interrupt vector 0: Direct jumping to an address determined from the level of priority 1: Referencing a table For details on the way to determine interrupt vector, see the *RH850G3K Users' Manual: Software*.
5 to 3	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
2 to 0	P2xxx to P0xxx	These bits specify the interrupt priority as one of 8 levels, with 0 as the highest and 7 as the lowest. When multiple EI level-interrupt requests are made simultaneously, the interrupt from the source with the highest priority setting in these bits is selected and conveyed to the CPU core for servicing first. When the P2xxx to P0xxx bits specify the same priority level for simultaneously occurring interrupt requests, the source with the lower channel number takes priority. This order is fixed.

CAUTION

Do not access the ICxxx registers of interrupt channels that are not listed in **Table 6.3, EI Level Maskable Interrupt Sources.**

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6.3.3 ICUM_IMRm — EI Level Interrupt Mask Register m (m = 0, 1)

These registers are a collection of the MKxxx bits of the ICUM_ICxxx registers. Each bit of ICUM_IMRm reflects the setting of the corresponding MKxxx bit. This setting for ICUM_IMRm is reflected in the corresponding MKxxx bit. This register is initialized by any reset.

Access: This register can be read or written in 32-bit units.

ICUM_IMRmH and ICUM_IMRmL can be read or written in 16-bit units.

ICUM_IMRmHH, ICUM_IMRmHL, ICUM_IMRmLH, and ICUM_IMRmLL can be read or written in 8- or 1-bit units.

Address: ICUM_IMR0: FFFE EAF0$_H$

ICUM_IMR0L: FFFE EAF0$_H$

ICUM_IMR0H: FFFE EAF2$_H$

ICUM_IMR0LL: FFFE EAF0$_H$

ICUM_IMR0LH: FFFE EAF1$_H$

ICUM_IMR0HL: FFFE EAF2$_H$

ICUM_IMR0HH: FFFE EAF3$_H$

ICUM_IMR1: FFFE F404$_H$

ICUM_IMR1L: FFFE F404$_H$

ICUM_IMR1H: FFFE F406$_H$*1

ICUM_IMR1LH: FFFE F405$_H$

ICUM_IMR1HL: FFFE F406$_H$*1

Value after reset: FFFF FFFF$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICUM_IMRmEIMK (m × 32 + 31)	ICUM_IMRmEIMK (m × 32 + 30)	ICUM_IMRmEIMK (m × 32 + 29)	ICUM_IMRmEIMK (m × 32 + 28)	ICUM_IMRmEIMK (m × 32 + 27)	ICUM_IMRmEIMK (m × 32 + 26)	ICUM_IMRmEIMK (m × 32 + 25)	ICUM_IMRmEIMK (m × 32 + 24)	ICUM_IMRmEIMK (m × 32 + 23)	ICUM_IMRmEIMK (m × 32 + 22)	ICUM_IMRmEIMK (m × 32 + 21)	ICUM_IMRmEIMK (m × 32 + 20)	ICUM_IMRmEIMK (m × 32 + 19)	ICUM_IMRmEIMK (m × 32 + 18)	ICUM_IMRmEIMK (m × 32 + 17)	ICUM_IMRmEIMK (m × 32 + 16)
Value after reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICUM_IMRmEIMK (m × 32 + 15)	ICUM_IMRmEIMK (m × 32 + 14)	ICUM_IMRmEIMK (m × 32 + 13)	ICUM_IMRmEIMK (m × 32 + 12)	ICUM_IMRmEIMK (m × 32 + 11)	ICUM_IMRmEIMK (m × 32 + 10)	ICUM_IMRmEIMK (m × 32 + 9)	ICUM_IMRmEIMK (m × 32 + 8)	ICUM_IMRmEIMK (m × 32 + 7)	ICUM_IMRmEIMK (m × 32 + 6)	ICUM_IMRmEIMK (m × 32 + 5)	ICUM_IMRmEIMK (m × 32 + 4)	ICUM_IMRmEIMK (m × 32 + 3)	ICUM_IMRmEIMK (m × 32 + 2)	ICUM_IMRmEIMK (m × 32 + 1)	ICUM_IMRmEIMK (m × 32 + 0)
Value after reset	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Table 6.13 ICUM_IMRm Register Contents (m = 0, 1)

Bit Position	Bit Name	Function
31 to 0	ICUM_IMRmEIMK (m × 32 + 31) to ICUM_IMRmEIMK (m × 32 + 0)	These are interrupt mask bits for EI level maskable interrupt (INT) channels 0 to 47. 0: Enables interrupt handling 1: Disables interrupt handling

Note 1. This register is supported only in RH850/F1KH-D8.

CAUTION

MKxxx bits that correspond to channels that are not listed in **Table 6.3, EI Level Maskable Interrupt Sources** must be set to 1.

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6.3.4 ICUM_FNC — FE Level NMI Status Register

This register indicates the status of an FE level non-maskable interrupt (ICUM_FENMI).

Access: This register is a read-only register that can be read in 16-bit units.

Address: FFFE EA78$_H$

Value after reset: 0000$_H$

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	FNRF	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 6.14 ICUM_FNC Register Contents

Bit Position	Bit Name	Function
15 to 13	—	Reserved When read, the value after reset is returned.
12	FNRF	Interrupt request flag 0: No interrupt request occurred 1: Interrupt request occurred This bit is automatically cleared when an FE level NMI interrupt request is acknowledged by the ICUP core.
11 to 0	—	Reserved When read, the value after reset is returned.

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6.3.5 ICUM_FIC — FE Level Maskable Interrupt Status Register

This register indicates the status of an FE level maskable interrupt (ICUM_FEINT).

Access: This register is a read-only register that can be read in 16-bit units.

Address: FFFE EA7A$_H$

Value after reset: 0000$_H$

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	FIRF	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 6.15 ICUM_FIC Register Contents

Bit Position	Bit Name	Function
15 to 13	—	Reserved When read, the value after reset is returned.
12	FIRF	Interrupt request flag 0: No interrupt request occurred 1: Interrupt request occurred
11 to 0	—	Reserved When read, the value after reset is returned.

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Section 7 Resets

7.1 Overview

Several system reset functions are provided in order to initialize CPU core and peripheral functions, as well as their associated registers.

When the ICUMD is used, a reset can be triggered by the following source in addition to the reset sources of the products:

- Overflow of watchdog timer (ICUM_WDTA0RES)

7.1.1 Reset Flag

To identify a reset source, the ICUMD includes two registers with a flag for each reset source.

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7.2 Registers

This section describes registers related to the reset controller, ICUM_WDTA0RES.

7.2.1 Overview of Reset Controller Registers

The reset controller is controlled and operated by the following registers.

Table 7.1 List of Registers

Module Name	Register Name	Symbol	Address
RESCTL	**General reset flag registers**		
	Reset factor register	RESF	FFF8 0760$_H$
	Reset factor clear register	RESFC	FFF8 0768$_H$
	Redundant reset factor register	RESFR	FFF8 0860$_H$
	Redundant reset factor clear register	RESFCR	FFF8 0868$_H$

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7.2.2 Details of General Reset Flag Registers

7.2.2.1 RESF — Reset Factor Register

This register contains information about which type of resets occurred since the last power-on clear reset. This register is initialized by a power-up reset PURES.

Each reset condition sets the corresponding flag in the register.

For the functions of the bits other than RESF12, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

Access: This register is a read-only register that can be read in 32-bit units.

Address: FFF8 0760$_H$

Value after reset: 0000 0300$_H$ / 0000 0200$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	RESF14	—	RESF12	RESF11*1	RESF10	RESF9	RESF8	RESF7	RESF6	RESF5	RESF4	RESF3	RESF2	RESF1	RESF0
Value after reset	0	0	0	0	0	0	1	1/0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 7.2 RESF Register Contents

Bit Position	Bit Name	Function
12	RESF12	ICUM_WDTA0 reset flag 0: No reset has occurred 1: Reset has occurred

Note 1. This bit is supported only in RH850/F1KH.

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7.2.2.2 RESFC — Reset Factor Clear Register

This register clears the reset flags of the RESF register.

For the functions of the bits other than RESFC12, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

Access: This register is a write-only register that can be written in 32-bit units.

Address: FFF8 0768$_H$

Value after reset: Undefined

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	RESFC14	—	RESFC12	RESFC11*1	RESFC10	RESFC9	RESFC8	RESFC7	RESFC6	RESFC5	RESFC4	RESFC3	RESFC2	RESFC1	RESFC0
Value after reset	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
R/W	R	W	R	W	W	W	W	W	W	W	W	W	W	W	W	W

Table 7.3 RESFC Register Contents

Bit Position	Bit Name	Function
12	RESFC12	ICUM_WDTA0 reset flag clear 0: Do not clear reset flag 1: Clear reset flag

Note 1. This bit is supported only in RH850/F1KH.

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7.2.2.3 RESFR — Redundant Reset Factor Register

This register is a duplicate of the reset factor flag register. This register is initialized by a power-up reset PURES.

In accordance with the setting conditions for each bit in the reset factor flag register, the same bits are set in this register.

For the functions of the bits other than RESFR12, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

Access: This register is a read-only register that can be read in 32-bit units.

Address: FFF8 0860$_H$

Value after reset: 0000 0300$_H$ / 0000 0200$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	RESFR14	—	RESFR12	RESFR11*1	RESFR10	RESFR9	RESFR8	RESFR7	RESFR6	RESFR5	RESFR4	RESFR3	RESFR2	RESFR1	RESFR0
Value after reset	0	0	0	0	0	0	1	1/0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 7.4 RESFR Register Contents

Bit Position	Bit Name	Function
12	RESFR12	ICUM_WDTA0 reset flag 0: No reset has occurred 1: Reset has occurred

Note 1. This bit is supported only in RH850/F1KH.

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7.2.2.4 RESFCR — Redundant Reset Factor Clear Register

This register clears the reset flags in the RESFR register.

For the functions of the bits other than RESFCR12, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

Access: This register is a write-only register that can be written in 32-bit units.

Address: FFF8 0868$_H$

Value after reset: Undefined

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	RESFCR14	—	RESFCR12	RESFCR11*1	RESFCR10	RESFCR9	RESFCR8	RESFCR7	RESFCR6	RESFCR5	RESFCR4	RESFCR3	RESFCR2	RESFCR1	RESFCR0
Value after reset	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
R/W	R	W	R	W	W	W	W	W	W	W	W	W	W	W	W	W

Table 7.5 RESFCR Register Contents

Bit Position	Bit Name	Function
12	RESFCR12	ICUM_WDTA0 reset flag clear 0: Do not clear reset flag 1: Clear reset flag

Note 1. This bit is supported only in RH850/F1KH.

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7.3 Functional Description

This section describes the reset function of the watchdog timer.

For details of resets other than the watchdog timer reset, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

7.3.1 Watchdog Timer (ICUM_WDTA) Reset

The watchdog timer can be configured to generate a reset if the overflow time expires. After a watchdog timer (ICUM_WDTA0) reset is generated, the corresponding watchdog timer reset flag (the RESF.RESF12 and RESFR.RESFR12 bits for ICUM_WDTA0RES) is set.

After that, bits RESF.RESF12 and RESFR.RESFR12 are not cleared automatically, even if ICUM_WDTA0RES enters the inactive state. Bits RESF.RESF12 and RESFR.RESFR12 are cleared as described below:

- Setting the RESFC.RESFC12 bit to 1 clears the RESF.RESF12 bit.
- Setting the RESFCR.RESFCR12 bit to 1 clears the RESFR.RESFR12 bit.

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Section 8 Stand-By Controller (STBC)

This section contains a generic description of the stand-by controller.

The first part in this section describes the stand-by functions of the ICUMD. The remainder of the section describes the stand-by functions of the RH850/F1KH, RH850/F1KM that affect the ICUMD.

8.1 Stand-By Functions of ICUMD

8.1.1 Functions

8.1.1.1 Types of Stand-By Mode

This ICUMD supports the following stand-by modes:

- STOP mode

  STOP mode stops the clocks in the ICUMD. STOP mode is entered after a stop request INTEXTSTPRQ is acknowledged and the STPRQEN bit in the ICUM_STOP register is set to 1.

8.1.1.2 Wake-Up Control

The wake-up sources are shared with those of the RH850/F1KH, RH850/F1KM.

For details, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

8.1.1.3 Transition to Stand-By Mode

The figure below shows transitions between RUN mode and the stand-by modes.

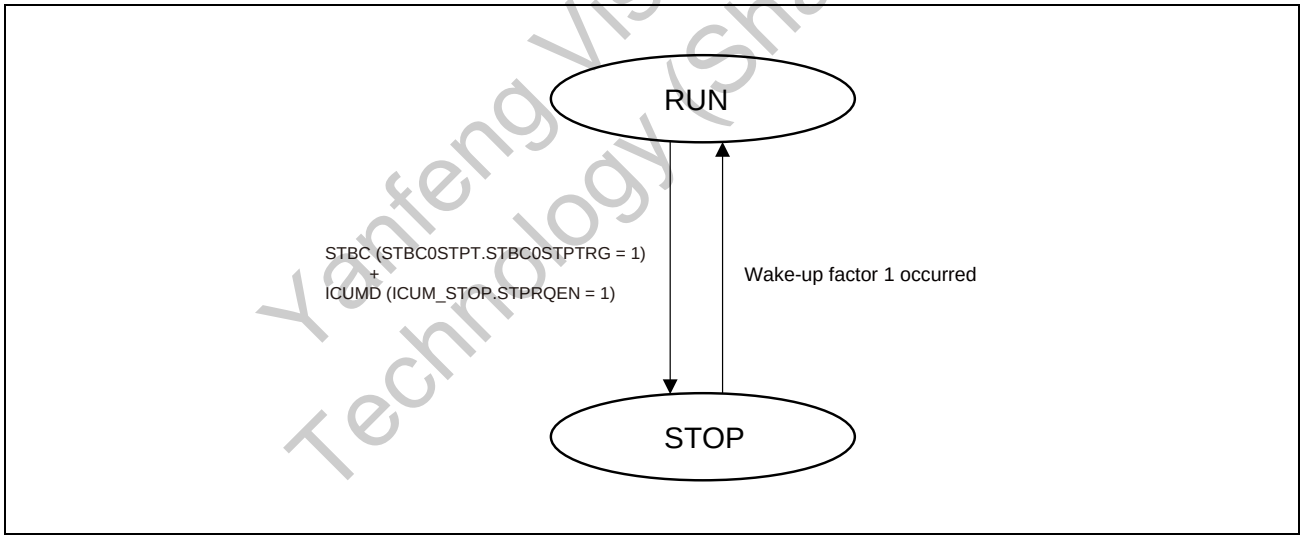


Figure 8.1 Transition of ICUMD Modes

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8.1.2 Registers

8.1.2.1 BOOTCTRL — CPU Start Control Register

This register is used to control operation of the CPU (PE).

Access: This register can be read or written in 32-bit units.

Address: FFC5 8000H

Value after reset: 0000 0020H

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	CPU2 EN*1	CPU1 EN	—
Value after reset	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R/W	R/W	R

Table 8.1 BOOTCTRL Register Contents

Bit Position	Bit Name	Function
31 to 3	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
2	CPU2EN*2	CPU2 (PE2) operation/stop setting 0: CPU2 (PE2) stops 1: CPU2 (PE2) operates*3 Clearing this bit to 0 is prohibited.
1	CPU1EN	CPU1 (PE1) operation/stop setting 0: CPU1 (PE1) stops 1: CPU1 (PE1) operates Clearing this bit to 0 is prohibited.
0	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.

Note 1. This bit is only supported for RH850/F1KH.

Note 2. This bit setting is valid regardless of the value of PE2BOOTEN (bit 3 of option byte 0) when ICUMD is activated. For further details about option byte 0, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

Note 3. It cannot be written when PE2EN (bit 2 of option byte 0) is "0". For further details about option byte 0, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

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8.2 RH850/F1KH, RH850/F1KM Stand-By Functions

8.2.1 Functions

8.2.1.1 Types of Stand-By Mode

- STOP mode

  In STOP mode, the clock supply to a certain domain can be stopped.

  STOP mode is entered when the STBC0STPT.STBC0STPTRG bit is set to 1, and ICUMD sets the ICUM_STOP.STPRQEN bit to 1 upon acknowledgement of a stop request INTEXTSTPRQ.

- DeepSTOP Mode

  To further reduce power consumption, the power supply of the Isolated area (ISO area) can be stopped.

  DeepSTOP mode is entered when the STBC0PSC.STBC0DISTRG bit is set to 1 and the ICUMD sets the ICUM_STOP.STPRQEN bit to 1 upon acknowledgement of a stop request INTEXTSTPRQ.

  When the RH850/F1KH, RH850/F1KM transitions from RUN mode to DeepSTOP mode, the ICUMD transitions modes in the following order: RUN → STOP → Power OFF.

For further details about stand-by modes, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

8.2.1.2 Wake-up Control

See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

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8.2.1.3 Transition to Stand-By Mode

The figures below show a state transition chart.

- ICUM-RUN indicates the state in which CPU1 (PE1) / CPU2 (PE2) is stopped and only ICUP is operating.

  CPU1 (PE1) / CPU2 (PE2) must be started by software processing.

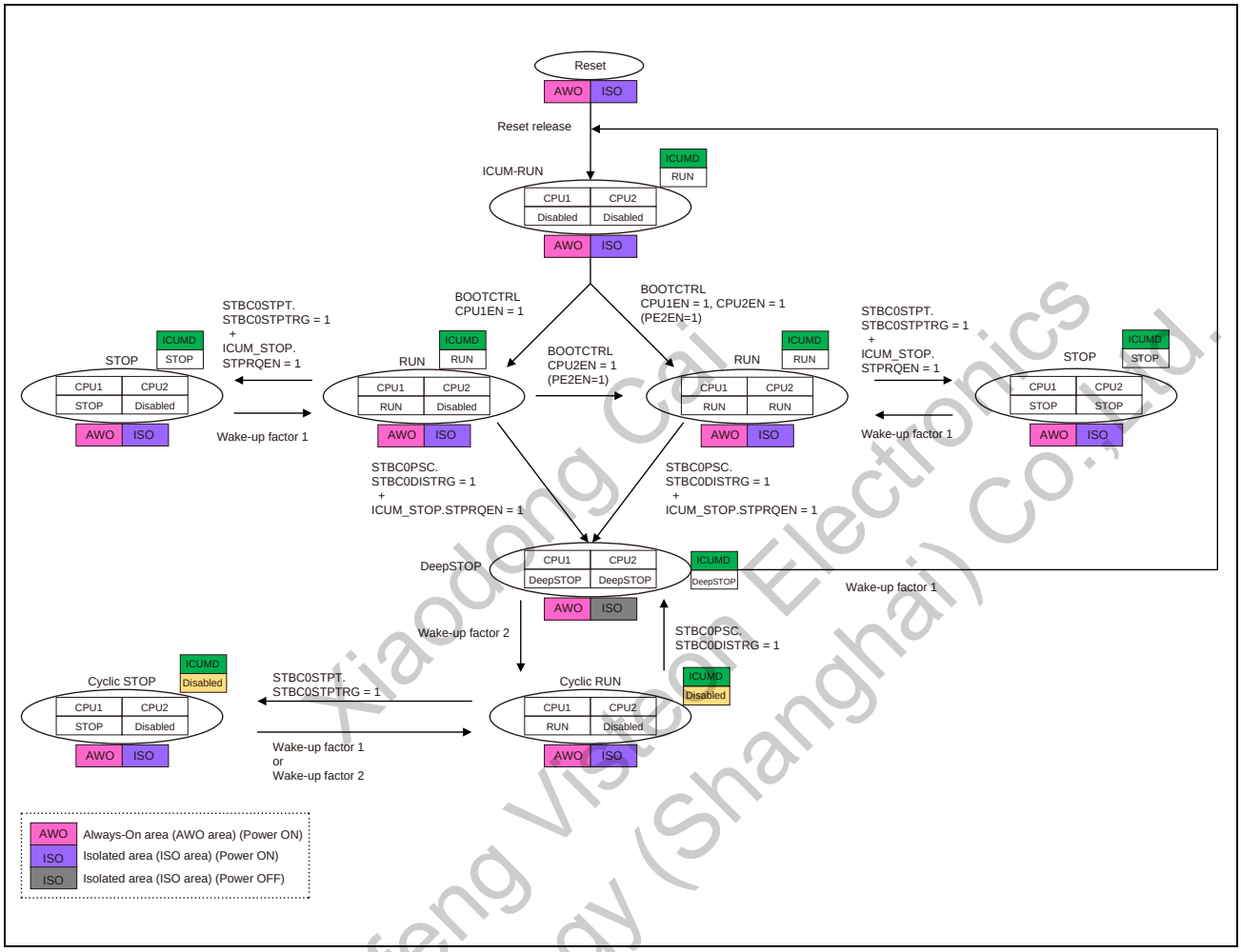


Figure 8.2 RH850/F1KH Mode Transition

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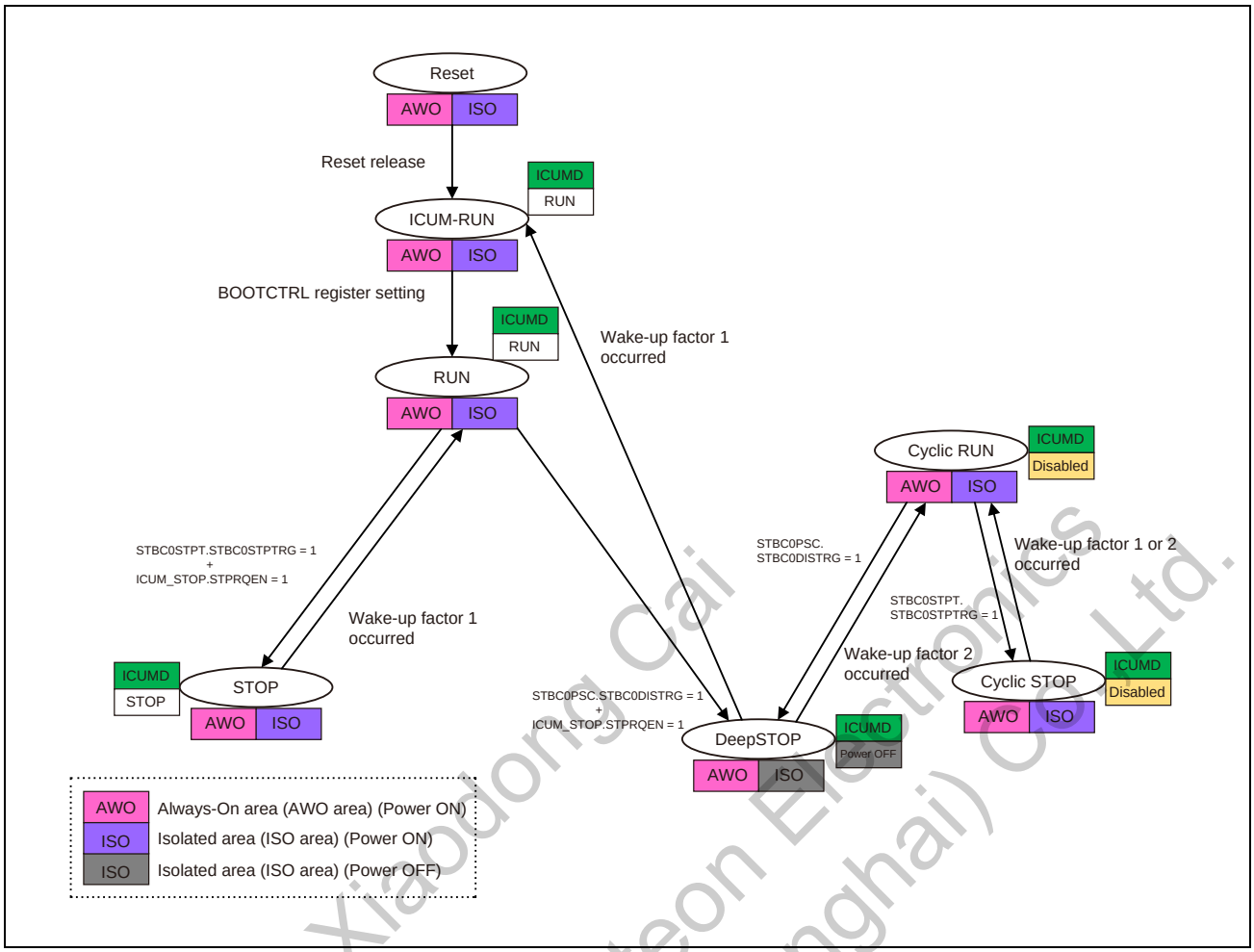


Figure 8.3 RH850/F1KM Mode Transition

8.2.2 Registers

8.2.2.1 List of Registers

See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

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8.2.3 Modes

This section describes the procedure for transitioning to STOP mode and DeepSTOP mode.

8.2.3.1 STOP Mode

In STOP mode, the clock supply to the Always-On area (AWO area) and Isolated area (ISO area) is stopped. It is therefore necessary to stop all of the peripheral functions to which the clock supply is to be stopped before transitioning to STOP mode.

Procedure for transitioning to Stand-By

The procedure for transiting to STOP mode (example) is shown below.

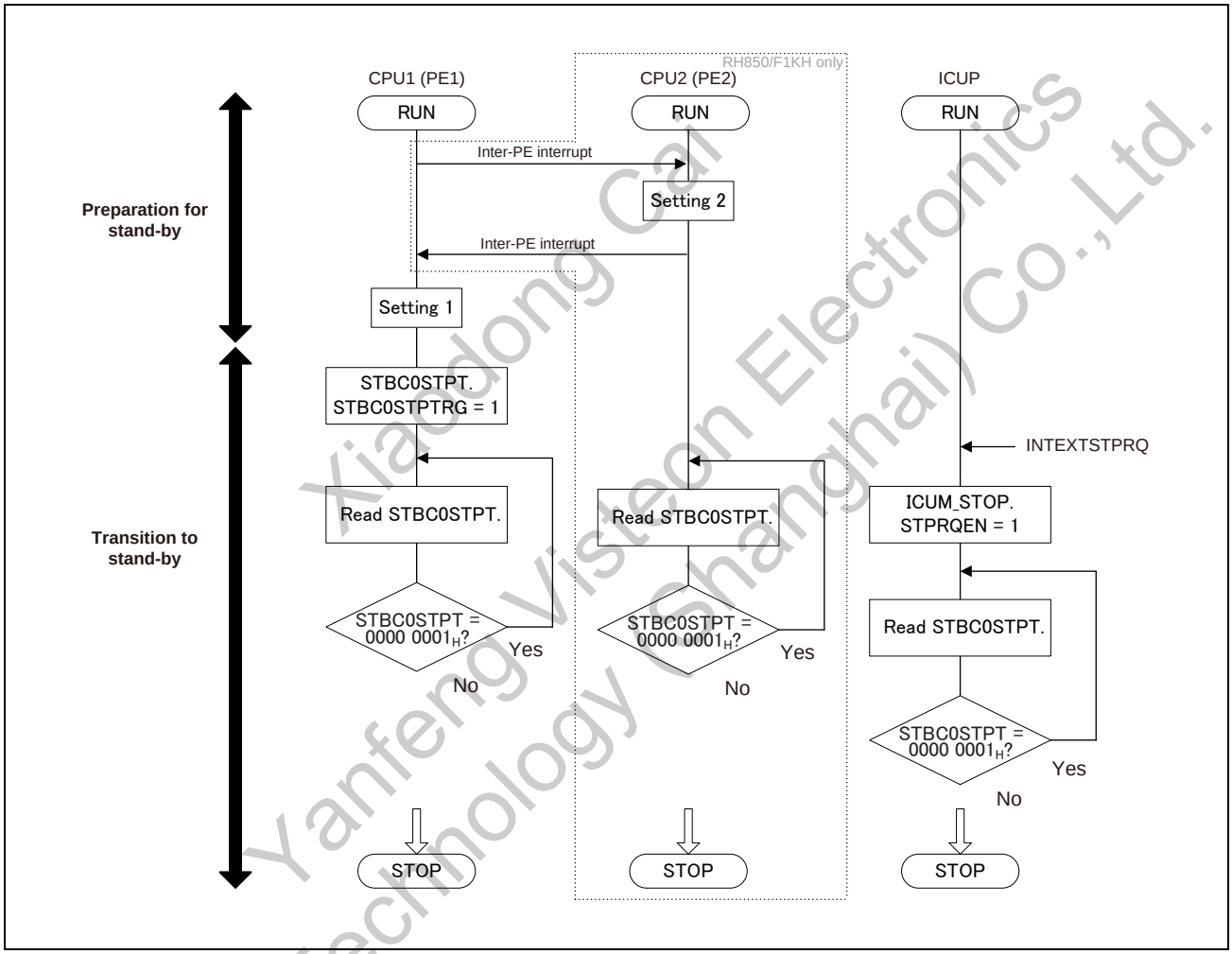


Figure 8.4 Example of Transition to STOP Mode

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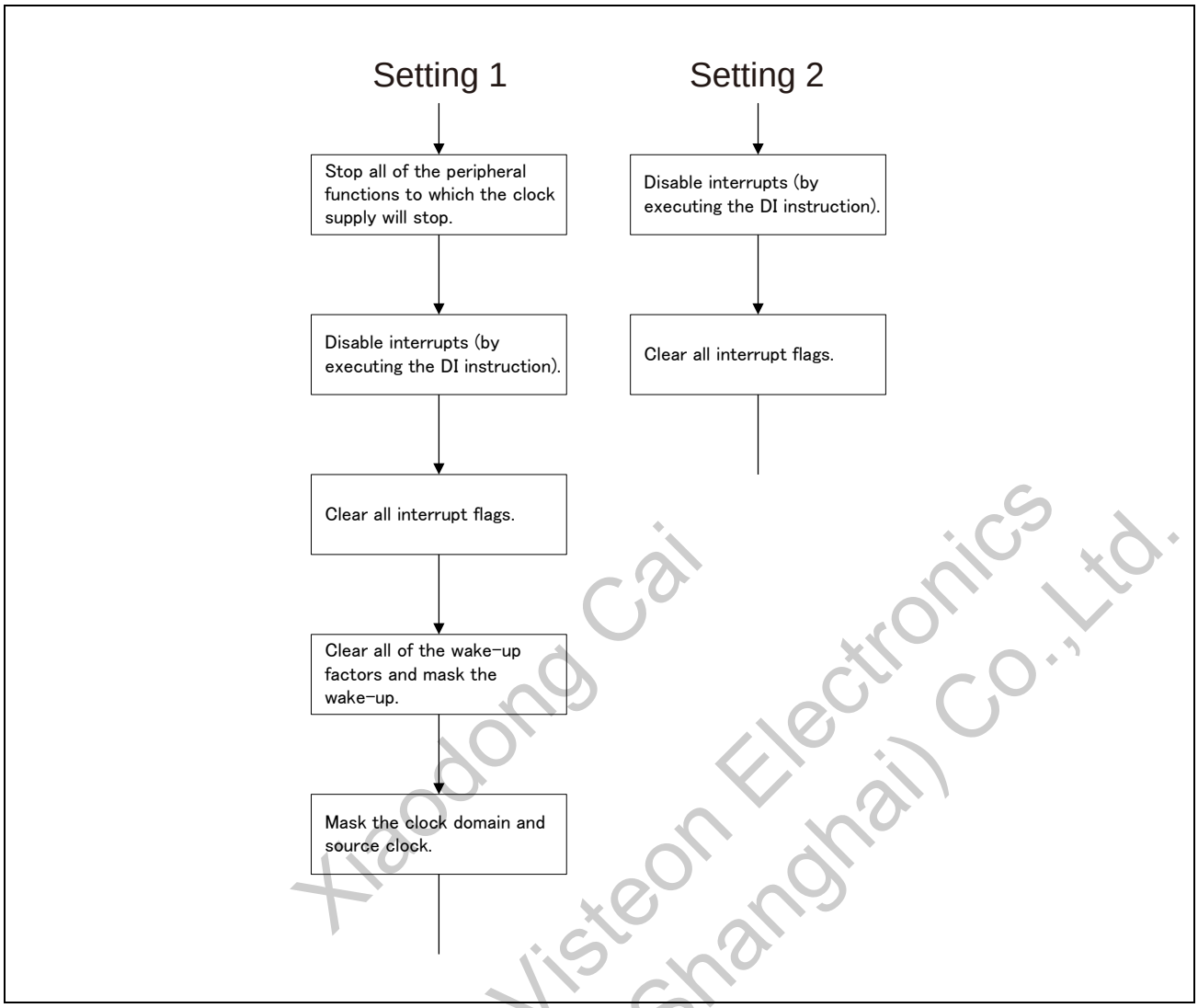


Figure 8.5 Example of STOP Mode Transition Settings

End of Stand-By

STOP mode is terminated by the occurrence of a wake-up event.

Wake-Up Handling

See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

After wake-up from STOP mode, the system transitions to RUN mode.

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8.2.3.2 DeepSTOP Mode

Procedure for transitioning to Stand-By

The procedure for transiting to DeepSTOP mode (example) is shown below.

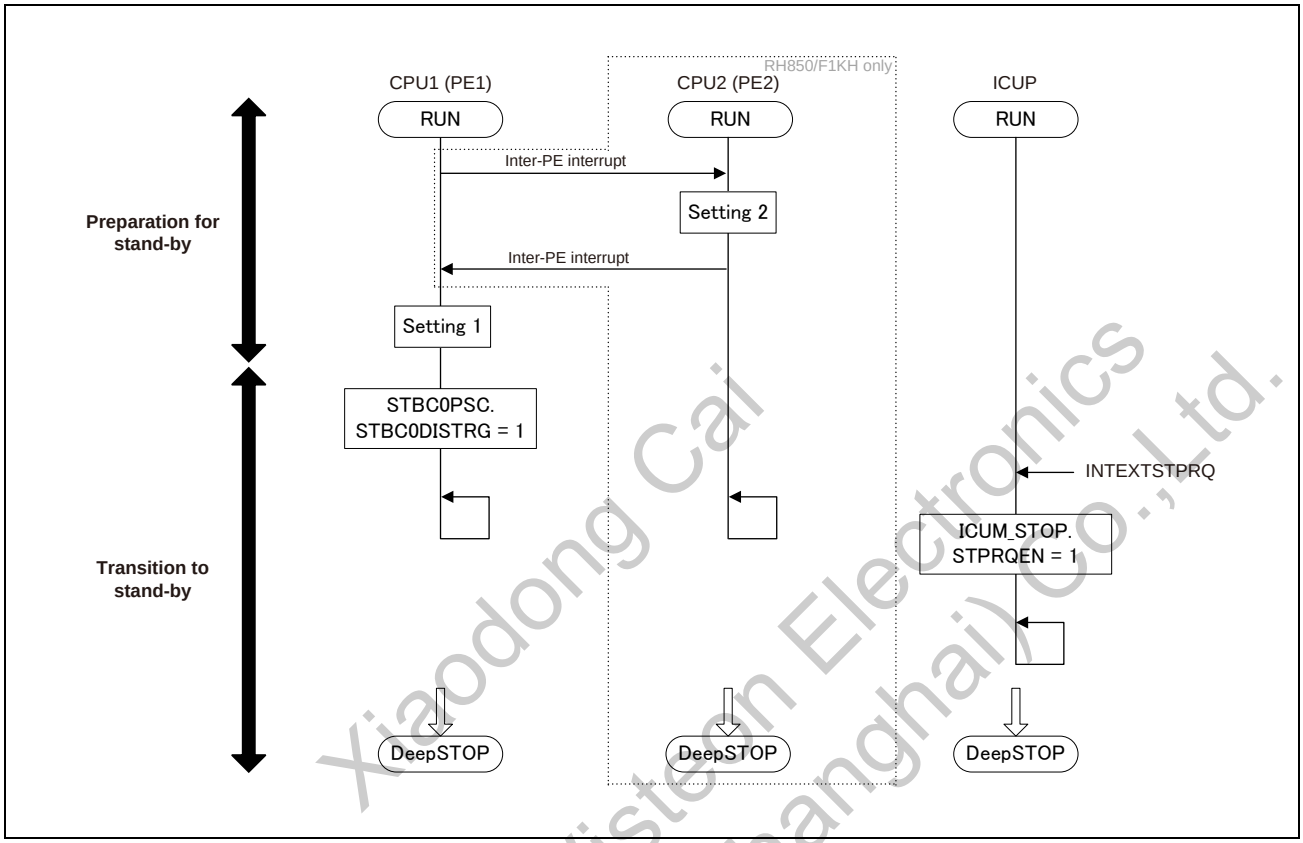


Figure 8.6 Example of Transition to DeepSTOP Mode

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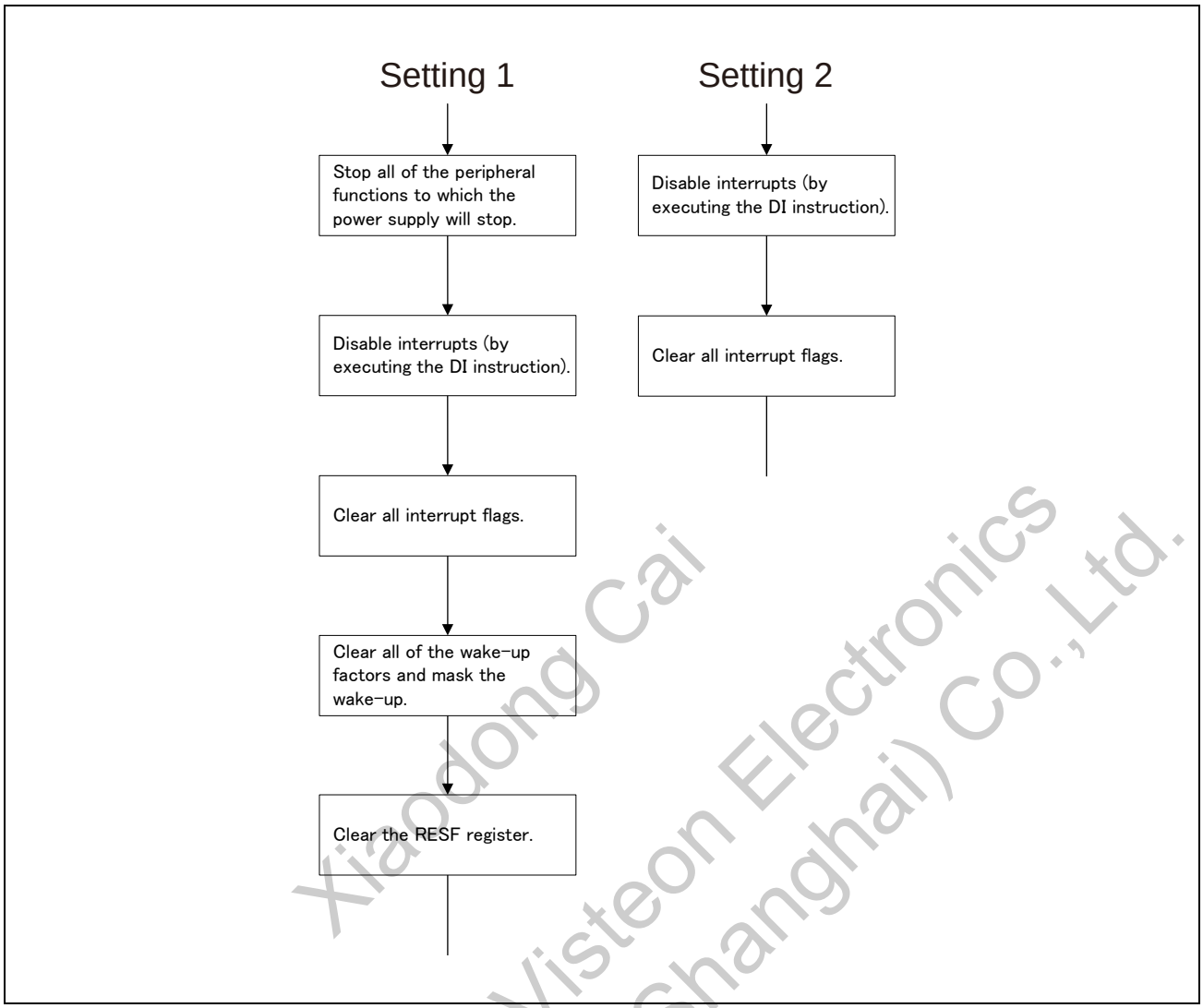


Figure 8.7 Example of DeepSTOP Mode Transition Settings

End of Stand-By

DeepSTOP mode is terminated by the occurrence of a wake-up event.

Wake-Up Handling

See the *RH850/F1KH, RH850/F1KM User's manual: Hardware*.

After wake-up, the system transitions to ICUM-RUN mode.

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Section 9 Cryptographic Peripherals

9.1 AES (ICUM_AESB0)

The ICUM_AESB0 is an accelerator for fast cipher processing based on AES algorithm.

9.1.1 Features of ICUM_AESB0

9.1.1.1 Number of Units and Channels

ICUMD has one ICUM_AESB0 channel.

9.1.1.2 Register Base Address

The ICUM_AESB0 base address is shown in the table below.

Table 9.1 Register Base Address

Base Address Name	Base Address
<ICUM_AESB0_base>	FFFE $E100_H$

9.1.1.3 Clock Supply

The ICUM_AESB0 clock supply is shown in the table below.

Table 9.2 Clock Supply

Unit Name	Unit Clock Name	Supply Clock Name
ICUM_AESB0	PCLK	CPUCLK_L

9.1.1.4 Interrupt Requests

The ICUM_AESB0 interrupt request is listed in the table below.

Table 9.3 Interrupt Requests

Unit Interrupt Signal	Description	Interrupt Number	DMA Trigger Number
ICUM_AESB0			
INTICUAESB0RD	Completion of ICUM_AESB0 encryption or decryption	4	—

9.1.1.5 Reset Sources

The ICUM_AESB0 reset sources are listed in the table below. ICUM_AESB0 is initialized by these reset sources.

Table 9.4 Reset Sources

Unit Name	Reset Source
ICUM_AESB0	All reset sources (ISORES)

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9.1.2 Overview

9.1.2.1 Functional Overview

The ICUM_AESB0 supports encryption and decryption based on AES algorithm. The engine conforms FIPS PUB 197. Features are listed as shown below.

- Block size: 128-bit
- Key size: 128-bit
- The ICUM_AESB0 supports the following block cipher modes
  - Electronic Codebook (ECB) mode following NIST SP800-38A
  - Cipher Block Chaining (CBC) mode following NIST SP800-38A
  - Counter (CTR) mode following NIST SP800-38A
  - Cipher Feedback 128 (CFB-128) mode following NIST SP800-38A
  - Output Feedback (OFB) mode following NIST SP800-38A
  - Galois Counter Mode (GCM) following NIST SP800-38D[*2]
  - XEX-TCB-CTS (XTS) mode following IEEE 1619-2007
  - Counter with CBC-MAC (CCM) mode following NIST SP800-38C[*1*2]
- The ICUM_AESB0 supports the following message authentication algorithm
  - Cipher-based MAC (CMAC) following NIST SP800-38B
- The ICUM_AESB0 supports the following compression function
  - Miyaguchi-Preneel (MP) following Secure Hardware Extension Functional Specification Version1.1
- The ICUM_AESB0 supports three execution contexts to allow fast switching between three different block cipher modes.
- Key is selected from key registers (KEY_i [i = 0 to 2]), individual secret keys (ROM_KEY_i [i = 0 to 1 for context #0 or #1, i = 0 to 2 for context #2]), and imported key by using secret key (ROM_KEY_2).
- Support Pseudo Random Number Generator (PRNG) following AIS 20.

Note 1. The data length of associated data supported by ICUM_AESB0 is as follows.
0 ≤ length of associated data < 2^16 - 2^8 [byte]

Note 2. CCM and GCM output decrypted data regardless of the result from the tag verification. If the verification result is failed, the whole memory area to store the decryption result shall be cleared in software.

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9.1.2.2 Block Diagram

Figure 9.1 is a high-level view of the ICUM_AESB0.

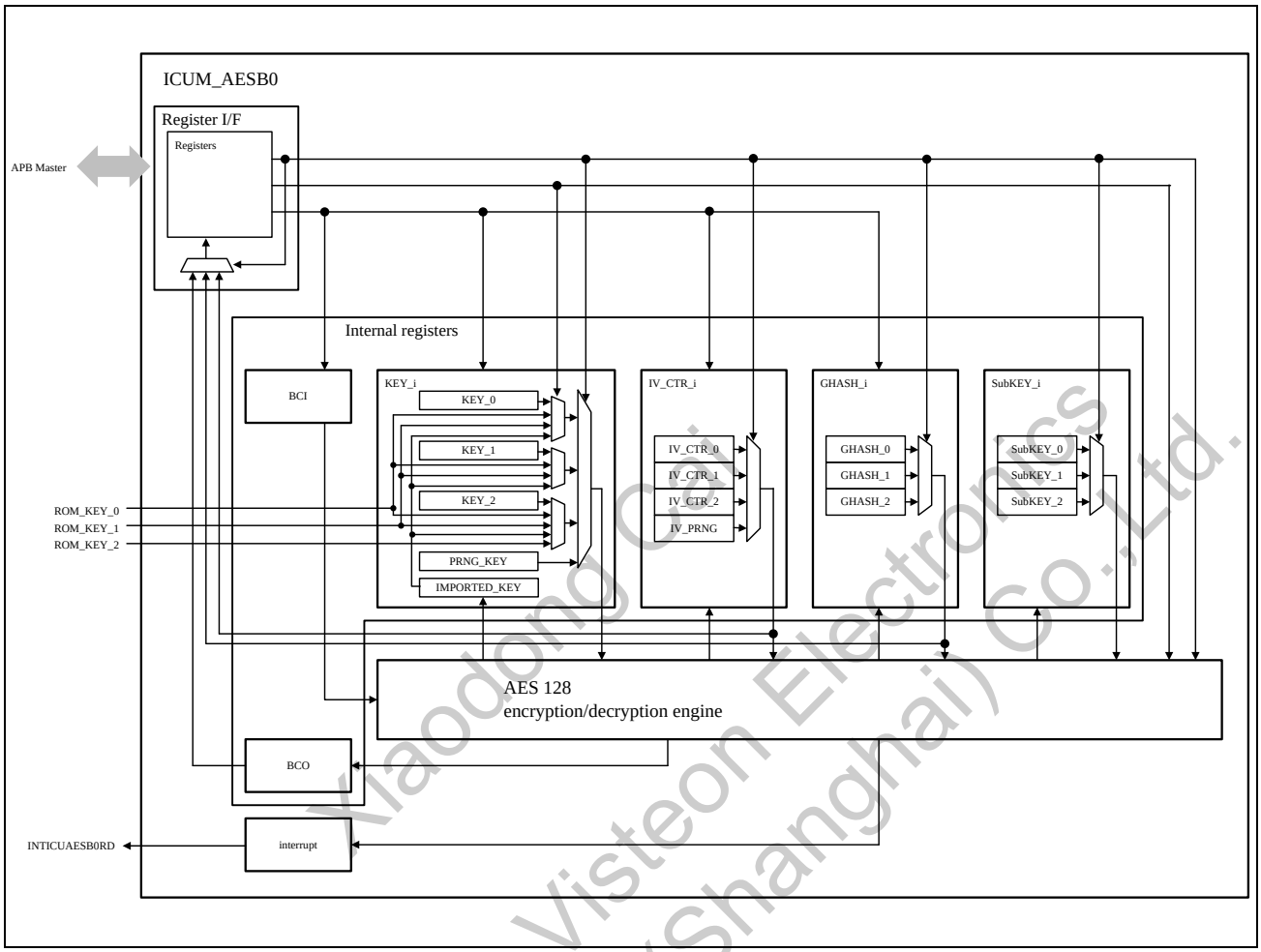


Figure 9.1 Block Diagram of the ICUM_AESB0

The ICUM_AESB0 consists of the following internal registers to perform the AES calculation:

- Block Cipher Input register (BCI)
- Block Cipher Output register (BCO)
- Write only Key register (KEY_i [i = 0 to 2], PRNG_KEY)
- Key register for imported key (IMPORTED_KEY)
- Initial vector or counter register (IV_CTR_i [i = 0 to 2])
- Random seed for PRNG (IV_PRNG)
- GHASH value of GCM mode (GHASH_i [i = 0 to 2])
- Subkey register (SubKEY_i [i = 0 to 2])

These internal registers can be accessed via ICUM_AESB0IDAT, ICUM_AESB0ODAT, ICUM_AESB0DIDi [i = 0 to 3], or ICUM_AESB0DODi [i = 0 to 3].

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9.1.2.3 Read Only Secret Keys

Three read-only secret keys (ROM_KEY_i [i = 0 to 2]) are available within the ICUM_AESB0.

Each read-only secret key has the following characteristics:

- Size: 128-bit
- Random data value not derived from any device data and not stored or printed on any kind of support
- Programmed by Renesas during production before shipment
- Not erasable nor updatable by any self-programming operation
- Can be read only by the ICUM_AESB0. ICUP cannot read ROM_KEY_i [i = 0 to 2].

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9.1.3 Registers

9.1.3.1 List of Registers

The ICUM_AESB0 registers are listed in the table below.

Table 9.5 List of Registers

Module Name	Register	Symbol	Address
ICUM_AESB0	AES control register	ICUM_AESB0CTRL	<ICUM_AESB0_base> + 0000_H
	AES data control register	ICUM_AESB0DCTL	<ICUM_AESB0_base> + 0004_H
	AES context control register 0	ICUM_AESB0CCTL0	<ICUM_AESB0_base> + 0008_H
	AES context control register 1	ICUM_AESB0CCTL1	<ICUM_AESB0_base> + $000C_H$
	AES status register	ICUM_AESB0STR	<ICUM_AESB0_base> + 0010_H
	AES status clear register	ICUM_AESB0STC	<ICUM_AESB0_base> + 0014_H
	AES data input register	ICUM_AESB0IDAT	<ICUM_AESB0_base> + 0020_H
	AES data output register	ICUM_AESB0ODAT	<ICUM_AESB0_base> + 0024_H
	AES direct data input register 0	ICUM_AESB0DID0	<ICUM_AESB0_base> + 0030_H
	AES direct data input register 1	ICUM_AESB0DID1	<ICUM_AESB0_base> + 0034_H
	AES direct data input register 2	ICUM_AESB0DID2	<ICUM_AESB0_base> + 0038_H
	AES direct data input register 3	ICUM_AESB0DID3	<ICUM_AESB0_base> + $003C_H$
	AES direct data output register 0	ICUM_AESB0DOD0	<ICUM_AESB0_base> + 0040_H
	AES direct data output register 1	ICUM_AESB0DOD1	<ICUM_AESB0_base> + 0044_H
	AES direct data output register 2	ICUM_AESB0DOD2	<ICUM_AESB0_base> + 0048_H
	AES direct data output register 3	ICUM_AESB0DOD3	<ICUM_AESB0_base> + $004C_H$
	AES version register	ICUM_AESB0VER	<ICUM_AESB0_base> + $007C_H$

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9.1.3.2 ICUM_AESB0CTRL — AES Control Register

This register controls the ICUM_AESB0 module.

Access: ICUM_AESB0CTRL can be read or written in 32-bit units.
ICUM_AESB0CTRLL and ICUM_AESB0CTRLH can be read or written in 16-bit units.
ICUM_AESB0CTRLLL, ICUM_AESB0CTRLLH, ICUM_AESB0CTRLHL, and ICUM_AESB0CTRLHH can be read or written in 8-bit units.

Address: ICUM_AESB0CTRL: <ICUM_AESB0_base> + 0000_H
ICUM_AESB0CTRLL: <ICUM_AESB0_base> + 0000_H
ICUM_AESB0CTRLH: < ICUM_AESB0_base > + 0002_H
ICUM_AESB0CTRLLL: <ICUM_AESB0_base> + 0000_H
ICUM_AESB0CTRLLH: < ICUM_AESB0_base > + 0001_H
ICUM_AESB0CTRLHL: <ICUM_AESB0_base> + 0002_H
ICUM_AESB0CTRLHH: < ICUM_AESB0_base > + 0003_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	ODATSRC[1:0]		—	—	IDATDEST[1:0]		—	—	—	—	—	CTX[2:0]		
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R/W	R/W	R	R	R/W	R/W	R	R	R	R	R	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	BYTE_ ORDER	DATRE GSW	—	—	—	—	RST3	RST2	RST1	RST0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R/W	R/W	R	R	R	R	R/W	R/W	R/W	R/W

Table 9.6 ICUM_AESB0CTRL Register Contents (1/2)

Bit Position	Bit Name	Function
31 to 30	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
29 to 28	ODATSRC[1:0]	Output data source selection bit Combined with the CTX[2:0] bits to specify the source register of output data to be read from ICUM_AESB0ODAT/ICUM_AESB0DODi [i = 0 to 3]. See **Table 9.7** for details. 00: Block Cipher Output register (BCO) 01: Setting Prohibited 10: Initialization vector/counter register (IV_CTR_i [i = 0 to 2]) 11: GHASH register (GHASH_i [i = 0 to 2])
27 to 26	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
25 to 24	IDATDEST[1:0]	Input data destination selection bit Combined with the CTX[2:0] bits to specify the destination register in which data from ICUM_AESB0IDAT/ICUM_AESB0DIDi [i = 0 to 3] is to be stored. See **Table 9.8** for details. 00: Block Cipher Input register (BCI) 01: KEY register (KEY_i [i = 0 to 2]) 10: Initialization vector/counter register (IV_CTR_i [i = 0 to 2]) 11: GHASH register (GHASH_i [i = 0 to 2])
23 to 19	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.

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Table 9.6 ICUM_AESB0CTRL Register Contents (2/2)

Bit Position	Bit Name	Function
18 to 16	CTX[2:0]	Context select bit Specify the context number. When context #3 is selected, the block mode is always set to “OFB encryption mode.” 000: Context #0 001: Context #1 010: Context #2 011 to 110: Prohibited 111: Context #3 (PRNG)
15 to 10	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
9	BYTE_ORDER	Byte order selection bit Specify whether byte order conversion to input and output data is applied. 0: Apply byte order conversion 1: Do not apply byte order conversion. This bit must be set before accessing input data register and output data register.
8	DATREGSW	Data register selection bit Specify the registers to be used for data input and output. 0: Use ICUM_AESB0IDAT/ICUM_AESB0ODAT 1: Use ICUM_AESB0DIDi/ICUM_AESB0DODi [i = 0 to 3] This bit must not be changed during 1 block AES operation and data input-output sequence. If changed, input/output data value and calculation result of the ICUM_AESB0 are not guaranteed.
7 to 4	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
3	RST3	Context #3 soft reset bit 0: Keep the current value. 1: Initialize PRNG_KEY and IV_PRNG. This bit is automatically cleared to 0 by hardware in 1 cycle.
2	RST2	Context #2 soft reset bit 0: Keep the current value. 1: Initialize KEY_2, IV_CTR_2, GHASH_2, and SubKEY_2. This bit is automatically cleared to 0 by hardware in 1 cycle.
1	RST1	Context #1 soft reset bit 0: Keep the current value. 1: Initialize KEY_1, IV_CTR_1, GHASH_1, and SubKEY_1. This bit is automatically cleared to 0 by hardware in 1 cycle.
0	RST0	Context #0 soft reset bit 0: Keep the current value. 1: Initialize KEY_0, IV_CTR_0, GHASH_0, and SubKEY_0. This bit is automatically cleared to 0 by hardware in 1 cycle.

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Table 9.7 indicates the truth table for source register selection of output data.

Table 9.7 Truth Table for Source Setting of Output Data

ODATSRC[1:0]		CTX[2:0]			
b29	**b28**	**b18**	**b17**	**b16**	**Source Register**
0	0	0	0	0	BCO (Context #0)
0	0	0	0	1	BCO (Context #1)
0	0	0	1	0	BCO (Context #2)
0	0	1	1	1	BCO (PRNG)
0	1	0	0	0	Reserved
0	1	0	0	1	Reserved
0	1	0	1	0	Reserved
0	1	1	1	1	Reserved
1	0	0	0	0	IV_CTR_0
1	0	0	0	1	IV_CTR_1
1	0	0	1	0	IV_CTR_2
1	0	1	1	1	IV_PRNG
1	1	0	0	0	GHASH_0
1	1	0	0	1	GHASH_1
1	1	0	1	0	GHASH_2
1	1	1	1	1	Reserved
Except above					Reserved

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Table 9.8 indicates the truth table for destination register selection of input data.

Table 9.8 Truth Table for Destination Setting of Input Data

IDATDEST[1:0]		CTX[2:0]			
b25	**b24**	**b18**	**b17**	**b16**	**Destination Register**
0	0	0	0	0	BCI (Context #0)
0	0	0	0	1	BCI (Context #1)
0	0	0	1	0	BCI (Context #2)
0	0	1	1	1	BCI (PRNG)
0	1	0	0	0	KEY_0
0	1	0	0	1	KEY_1
0	1	0	1	0	KEY_2
0	1	1	1	1	PRNG_KEY
1	0	0	0	0	IV_CTR_0
1	0	0	0	1	IV_CTR_1
1	0	0	1	0	IV_CTR_2
1	0	1	1	1	IV_PRNG
1	1	0	0	0	GHASH_0
1	1	0	0	1	GHASH_1
1	1	0	1	0	GHASH_2
1	1	1	1	1	Reserved
Except above					Reserved

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9.1.3.3 ICUM_AESB0DCTL — AES Data Control Register

This register specifies data attribute used for the ICUM_AESB0 calculation.

Access: ICUM_AESB0DCTL can be read or written in 32-bit units.
ICUM_AESB0DCTLL and ICUM_AESB0DCTLH can be read or written in 16-bit units.
ICUM_AESB0DCTLLL, ICUM_AESB0DCTLLH, and ICUM_AESB0DCTLHL can be read or written in 8-bit units.

Address: ICUM_AESB0DCTL: <ICUM_AESB0_base> + 0004_H
ICUM_AESB0DCTLL: <ICUM_AESB0_base> + 0004_H
ICUM_AESB0DCTLH: <ICUM_AESB0_base> + 0006_H
ICUM_AESB0DCTLLL: <ICUM_AESB0_base> + 0004_H
ICUM_AESB0DCTLLH: <ICUM_AESB0_base> + 0005_H
ICUM_AESB0DCTLHL: <ICUM_AESB0_base> + 0006_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	ATTR[3:0]			
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R/W	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	LAST	REM[6:0]							—	—	—	—	—	IVSEL	CONT	NEW_KEY
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R	R	R	R	R	R/W	R/W	R/W

Table 9.9 ICUM_AESB0DCTL Register Contents (1/2)

Bit Position	Bit Name	Function
31 to 20	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
19 to 16	ATTR[3:0]	Input data attribute bit Specify the attributes of the next input block data. These bits are enabled when the block cipher mode is set to GCM, XTS, CCM, or CMAC. See **Table 9.10** for details. When the block cipher mode is other than GCM, XTS, CCM, and CMAC, write 0.
15	LAST	Last block bit Specify whether the next input block data is the last block of the stream data. This bit is enabled when the block cipher mode is set to GCM, XTS, or CMAC. 0: Except last block 1: Last block
14 to 8	REM[6:0]	Number of valid bits in the last block Indicate the number of valid bits counted from MSB of the input block data when the LAST bit is set to 1. These bits are enabled when the block cipher mode is set to GCM, XTS, or CMAC. 0000000: 128-bit 0000001: 1-bit 0000010: 2-bit : 1111111: 127-bit Regardless of the setting value, the ICUM_AESB0 does not execute zero padding to data input register. SW must pad 0 to invalid bits. When the LAST bit is set to 0, setting these bits are invalid.

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Table 9.9 ICUM_AESB0DCTL Register Contents (2/2)

Bit Position	Bit Name	Function
7 to 3	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
2	IVSEL	IV_CTR_i selection bit Specify whether IV_CTR_i [i = 0 to 2] is used or not. 0: Do not use IV_CTR_i value 1: Use IV_CTR_i value This bit is used for the configuration whether the result of the GHASH function is applied for the AES operation in GCM mode. See **Figure 9.5** to **Figure 9.11** to configure this bit in GCM mode.
1	CONT	Operation continuation bit Specify whether the next operation is the resumption of a suspended operation. If resumption is specified, IV_CTR_i, GHASH_i, and SubKEY_i (i: context number) is used as an intermediate value. 0: The next operation is a new or continuous operation. 1: The next operation is the resumption of a suspended operation.
0	NEW_KEY	Key update specification bit 0: Use the key data in the AES core that is used in the preceding operation. 1: Update the key data in the AES core with the key data output from the KEY block.

Table 9.10 shows input data attribute for each block cipher mode.

Table 9.10 Input Data Attribution

	ATTR[3:0]				
Block Cipher Mode	**b19**	**b18**	**b17**	**b16**	**Input Data Attribute**
GCM mode	0	0	0	0	Input data for GHASH function
	0	0	0	1	Input data for GCTR function
	0	0	1	0	Input data for tag generation Upper 64-bit: Bit length of AAD(Additional authenticated data) Lower 64-bit: Bit length of ciphertext
XTS mode	0	0	0	0	Plaintext/ciphertext
	0	0	0	1	Tweak value
	0	0	1	0	Next to last block when the decryption operation is in progress and the length of the last block is 127 or shorter.
CCM mode	0	0	0	1	Input data for parameter B_0 used as a format function.
	0	0	1	0	Encryption : Prohibited Decryption : Ciphertext
	0	0	1	1	Input data for associated data A used as a format function.
	0	1	0	0	Plaintext
	1	0	0	0	Encryption : Input data for a dummy T XOR $MSB_{Tlen}(S_0)$ Decryption : Prohibited
	1	1	0	1	Encryption : Prohibited Decryption : Input data for T XOR $MSB_{Tlen}(S_0)$
CMAC	0	0	0	0	Valid bit number of input data is not 0.
	0	0	0	1	Valid bit number of input data is 0.

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9.1.3.4 ICUM_AESB0CCTL0 —AES Context Control Register 0

This register sets configuration for context #0 and context #1.

Access: ICUM_AESB0CCTL0 can be read or written in 32-bit units.
ICUM_AESB0CCTL0L and ICUM_AESB0CCTL0H can be read or written in 16-bit units.

Address: ICUM_AESB0CCTL0: <ICUM_AESB0_base> + 0008_H
ICUM_AESB0CCTL0L: <ICUM_AESB0_base> + 0008_H
ICUM_AESB0CCTL0H: <ICUM_AESB0_base> + $000A_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	K1SEL[2:0]			—	—	—	ENCMODE1	MOD1[3:0]			
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R/W	R/W	R/W	R	R	R	R/W	R/W	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	K0SEL[2:0]			—	—	—	ENCMODE0	MOD0[3:0]			
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R/W	R/W	R/W	R	R	R	R/W	R/W	R/W	R/W	R/W

Table 9.11 ICUM_AESB0CCTL0 Register Contents (1/2)

Bit Position	Bit Name	Function
31 to 27	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
26 to 24	K1SEL[2:0]	Select key data for context #1 000: ICUM_AESB0IDAT/ICUM_AESB0DIDi [i = 0 to 3] 001: ROM_KEY_0 010: ROM_KEY_1 011: IMPORTED_KEY 100 to 111: Prohibited
23 to 21	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
20	ENCMODE1	Specify the selection of encryption/decryption for context #1. 0 : Encryption 1 : Decryption
19 to 16	MOD1[3:0]	Specify the block cipher mode for context #1. See **Table 9.12** for details.
15 to 11	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
10 to 8	K0SEL[2:0]	Select key data for context #0 000: ICUM_AESB0IDAT/ICUM_AESB0DIDi [i = 0 to 3] 001: ROM_KEY_0 010: ROM_KEY_1 011: IMPORTED_KEY 100 to 111: Prohibited

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Table 9.11 ICUM_AESB0CCTL0 Register Contents (2/2)

Bit Position	Bit Name	Function
7 to 5	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
4	ENCMODE0	Specify the selection of encryption/decryption for context #0. 0 : Encryption 1 : Decryption
3 to 0	MOD0[3:0]	Specify the block cipher mode for context #0. See **Table 9.12** for details.

Table 9.12 Block Cipher Mode

MODi[3:0] [i = 0 to 2]	Block Cipher Mode
0_H	ECB
1_H	CBC
2_H	CTR
3_H	CFB-128
4_H	OFB
5_H	GCM
6_H	XTS
7_H	CCM
8_H	CMAC
9_H	Miyaguchi-Preneel
Except above	Prohibited

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9.1.3.5 ICUM_AESB0CCTL1 — AES Context Control Register 1

This register sets configuration for context #2.

Access: ICUM_AESB0CCTL1 can be read or written in 32-bit units.
ICUM_AESB0CCTL1L can be read or written in 16-bit units.

Address: ICUM_AESB0CCTL1: <ICUM_AESB0_base> + $000C_H$
ICUM_AESB0CCTL1L: <ICUM_AESB0_base> + $000C_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	K2SEL[2:0]			—	—	—	ENCMODE2	MOD2[3:0]			
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R/W	R/W	R/W	R	R	R	R/W	R/W	R/W	R/W	R/W

Table 9.13 ICUM_AESB0CCTL1 Register Contents

Bit Position	Bit Name	Function
31 to 11	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
10 to 8	K2SEL[2:0]	Select key data for context #2 000: ICUM_AESB0IDAT/ICUM_AESB0DIDi [i = 0 to 3] 001: ROM_KEY_0 010: ROM_KEY_1 011: IMPORTED_KEY 100: ROM_KEY_2 101 to 111: Prohibited When ROM_KEY_2 is selected, block cipher mode is ECB mode regardless of b3 to b0 setting. When encryption is selected, calculation result is loaded to BCO. When decryption is selected, calculation result is loaded to IMPORTED_KEY.
7 to 5	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
4	ENCMODE2	Specify the selection of encryption/decryption for context #2. 0 : Encryption 1 : Decryption
3 to 0	MOD2[3:0]	Specify the block cipher mode for context #2. See **Table 9.12** for details.

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9.1.3.6 ICUM_AESB0STR — AES Status Register

This register indicates ICUM_AESB0 status.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_AESB0_base> + 0010_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	GHSTS	—	—	—	—	—	—	—	STS
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 9.14 ICUM_AESB0STR Register Contents

Bit Position	Bit Name	Function
31 to 9	—	Reserved When read, the value after reset is returned.
8	GHSTS	GHASH status bit Indicate whether GHASH calculation is finished when GCM mode encryption is selected and ICUM_AESB0DCTL.ATTR[3:0] = 1_H. 0: Hash value on GHASH_CTR register is invalid. 1: Hash value on GHASH_CTR register is valid. [Set condition] GHASH calculation is finished when GCM mode encryption is selected and ICUM_AESB0DCTL.ATTR[3:0] is set to 1_H. [Clear condition] ICUM_AESB0STC.GHSTS_C is set to 1. Next AES calculation is started. Reset is asserted.
7 to 1	—	Reserved When read, the value after reset is returned.
0	STS	Status bit Indicate the ICUM_AESB0 status. 0: ICUM_AESB0 is in idle state. 1: ICUM_AESB0 is running. [Set condition] AES calculation is started except GCM encryption and ICUM_AESB0DCTL.ATTR[3:0] = 1_H. [Clear condition] AES calculation is finished. Reset is asserted.

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9.1.3.7 ICUM_AESB0STC — AES Status Clear Register

This register clears ICUM_AESB0STR.

Access: This register is a write-only register that can be written in 32-bit units.

Address: <ICUM_AESB0_base> + 0014_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	GHSTS_C	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	W	R	R	R	R	R	R	R	R

Table 9.15 ICUM_AESB0STC Register Contents

Bit Position	Bit Name	Function
31 to 9	—	Reserved When writing, write the value after reset.
8	GHSTS_C	GHASH status clear bit Specify whether clear ICUM_AESB0STR.GHSTS or not. 0: Do not clear ICUM_AESB0STR.GHSTS. 1: Clear ICUM_AESB0STR.GHSTS.
7 to 0	—	Reserved When writing, write the value after reset.

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9.1.3.8 ICUM_AESB0IDAT — AES Data Input Register

This register is FIFO type data input register.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_AESB0_base> + 0020_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	AES0IDAT[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	AES0IDAT[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Table 9.16 ICUM_AESB0IDAT Register Contents

Bit Position	Bit Name	Function
31 to 0	AES0IDAT[31:0]	FIFO type input data The destination of the loaded data is specified by ICUM_AESB0CTRL.IDATDEST. See **Section 9.1.5, Data Format** for the data format. When ICUM_AESB0CTRL.DATREGSW = 1, writing to this register is invalid and read value is 0. When the destination of the data is set to BCI and write access to this register is performed 4 times, ICUM_AESB0 starts calculation. Increment condition and clear condition of writing counts are specified below. When writing counts are cleared, write access must be performed 4 times to start AES calculation. [Clear condition of writing counts] ICUM_AESB0CTRL, ICUM_AESB0DCTL, ICUM_AESB0CCTL0, or ICUM_AESB0CCTL1 is written. [Increment condition of writing counts] ICUM_AESB0IDAT is written when ICUM_AESB0CTRL.DATREGSW = 0.

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9.1.3.9 ICUM_AESB0ODAT — AES Data Output Register

This register is FIFO type data output register.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_AESB0_base> + 0024_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	AES0ODAT[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	AES0ODAT[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 9.17 ICUM_AESB0ODAT Register Contents

Bit Position	Bit Name	Function
31 to 0	AES0ODAT[31:0]	FIFO type output data The reference source of this data is specified by ICUM_AESB0CTRL.ODATSRC. See **Section 9.1.5, Data Format** for the data format. 32bit data is read in response to read access times from 128 bit reference source data. Increment condition and clear condition of reading counts are speficied below. [Clear condition of reading counts] ICUM_AESB0CTRL, ICUM_AESB0DCTL, ICUM_AESB0CCTL0, or ICUM_AESB0CCTL1 is written. [Increment condition of reading counts] ICUM_AESB0ODAT is read when ICUM_AESB0CTRL.DATREGSW = 0. In order to clear a data which is stored in BCO registers, write 00_B to ICUM_AESB0CTRL.ODATSRC and perform dummy read from ICUM_AESB0ODAT 4 times when ICUM_AESB0CTRL.DATREGSW = 0.

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9.1.3.10 ICUM_AESB0DIDi — AES Direct Data Input Register i (i = 0 to 3)

These registers are data input registers.

Access: This register can be read or written in 32-bit units.

Address: ICUM_AESB0DID0: <ICUM_AESB0_base> + 0030_H

ICUM_AESB0DID1: <ICUM_AESB0_base> + 0034_H

ICUM_AESB0DID2: <ICUM_AESB0_base> + 0038_H

ICUM_AESB0DID3: <ICUM_AESB0_base> + $003C_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	AES0DIDi[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	AES0DIDi[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Table 9.18 ICUM_AESB0DIDi Register Contents

Bit Position	Bit Name	Function
31 to 0	AES0DIDi[31:0]	Direct input data The destination of the loaded data is specified by ICUM_AESB0CTRL.IDATDEST. See **Section 9.1.5, Data Format** for the data format. When ICUM_AESB0CTRL.DATREGSW = 0, writing to this register is invalid and read value is 0. When the destination of the data is set to BCI and write access to ICUM_AESB0DIDi is performed 4 times, ICUM_AESB0 starts calculation. Increment condition and clear condition of writing counts are speficied below. When writing counts are cleared, write access must be performed 4 times to start AES calculation. [Clear condition of writing counts] ICUM_AESB0CTRL, ICUM_AESB0DCTL, ICUM_AESB0CCTL0, or ICUM_AESB0CCTL1 is written. [Increment condition of writing counts] ICUM_AESB0DIDi [i = 0 to 3] is written when ICUM_AESB0CTRL.DATREGSW = 1. If i is same as previous access, writing counts are incremented.

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9.1.3.11 ICUM_AESB0DODi — AES Direct Data Output Register i (i = 0 to 3)

These registers are data input registers.

Access: This register is a read-only register that can be read in 32-bit units.

Address: ICUM_AESB0DOD0: <ICUM_AESB0_base> + 0040_H

ICUM_AESB0DOD1: <ICUM_AESB0_base> + 0044_H

ICUM_AESB0DOD2: <ICUM_AESB0_base> + 0048_H

ICUM_AESB0DOD3: <ICUM_AESB0_base> + $004C_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	AES0DODi[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	AES0DODi[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 9.19 ICUM_AESB0DODi Register Contents

Bit Position	Bit Name	Function
31 to 0	AES0DODi [31:0]	Direct output data The reference source of this data is specified by ICUM_AESB0CTRL.ODATSRC. See **Section 9.1.5, Data Format** for the data format. When ICUM_AESB0CTRL.DATREGSW = 0, read value is 0. In order to clear a data which is stored in BCO registers, write 00_B to ICUM_AESB0CTRL.ODATSRC and perform dummy read from ICUM_AESB0DODi (I = 0 to 3) when ICUM_AESB0CTRL.DATREGSW = 1.

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9.1.3.12 ICUM_AESB0VER — AES Version Register

This register indicates AES unit version.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_AESB0_base> + $007C_H$

Value after reset: $0000\ 0001_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	AESVER[3:0]			
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 9.20 ICUM_AESB0VER Register Contents

Bit Position	Bit Name	Function
31 to 4	—	Reserved When read, the value after reset is returned.
3 to 0	AESVER[3:0]	AES unit version bits Indicate AES unit version 0001: AES unit is the ICUM_AESB0

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9.1.4 Operation

9.1.4.1 Operation Flow

(1) ECB, CBC, OFB, CFB-128, and CTR

The operating procedure for ECB, CBC, OFB, CFB-128, and CTR is shown in **Figure 9.2**.

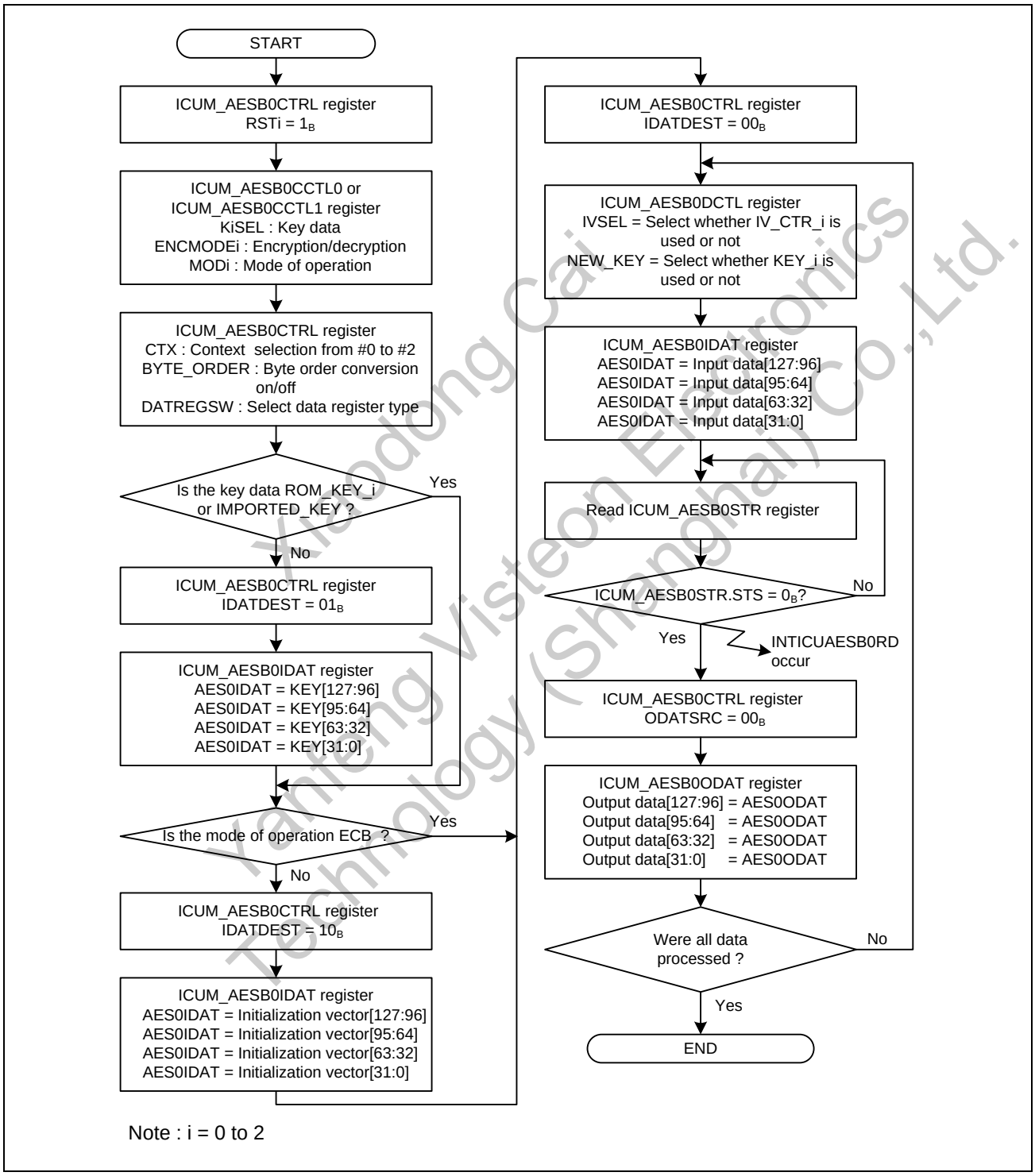


Figure 9.2 Operating procedure for ECB, CBC, OFB, CFB-128, and CTR

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(2) GCM

GCM mode requires several steps and AES operation is performed with a different configuration in each step. The input and output data in each step is shown in **Figure 9.3** and **Figure 9.4**.

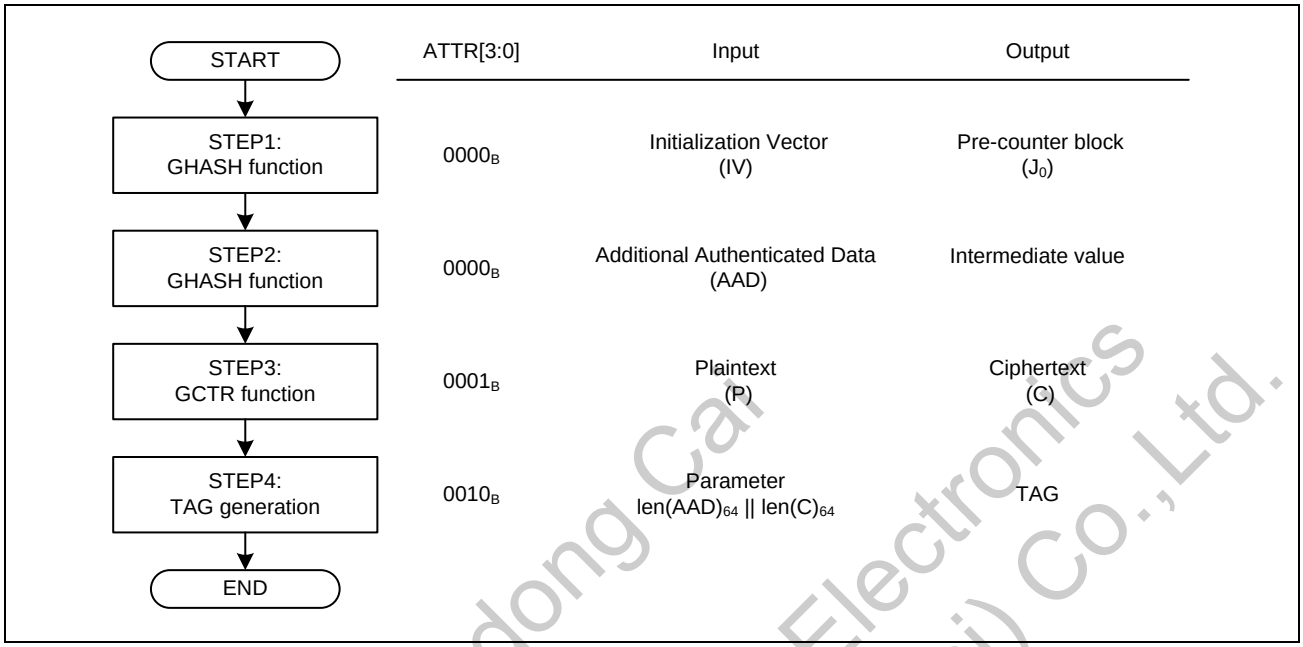


Figure 9.3 The configuration of ATTR and input/output data in each step (Encryption).

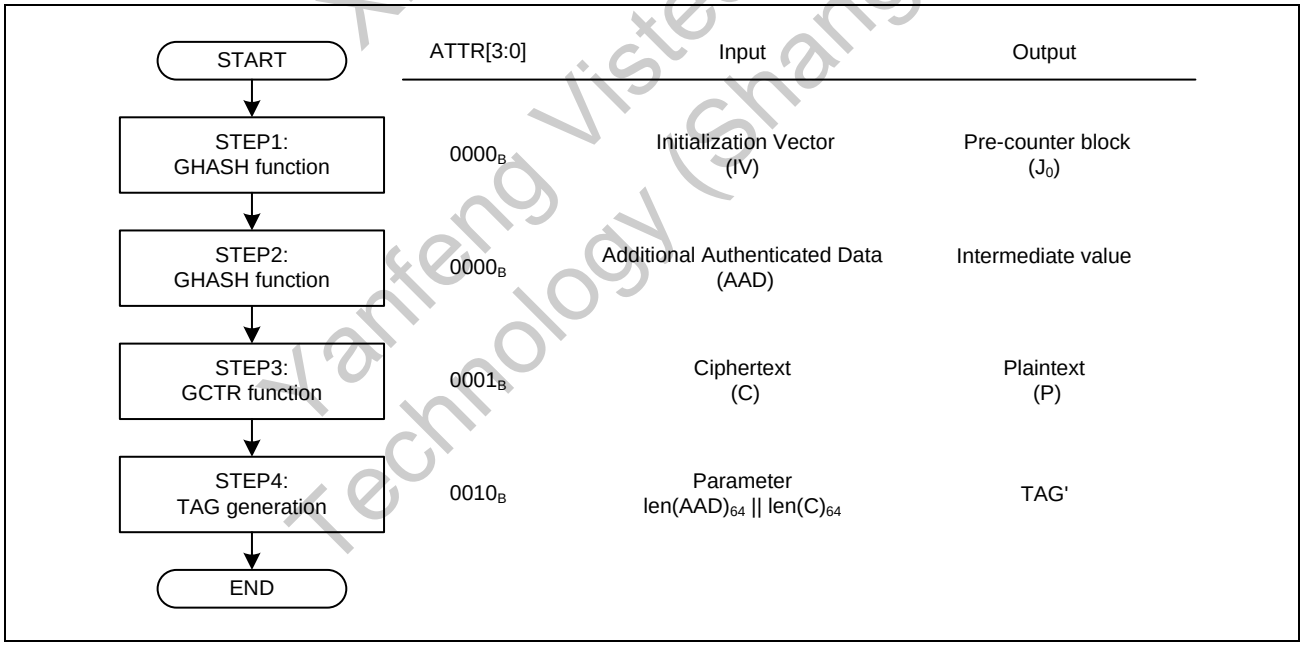


Figure 9.4 The configuration of ATTR and input/output data in each step (Decryption).

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The operating procedure for GCM is shown in **Figure 9.5** to **Figure 9.11**.

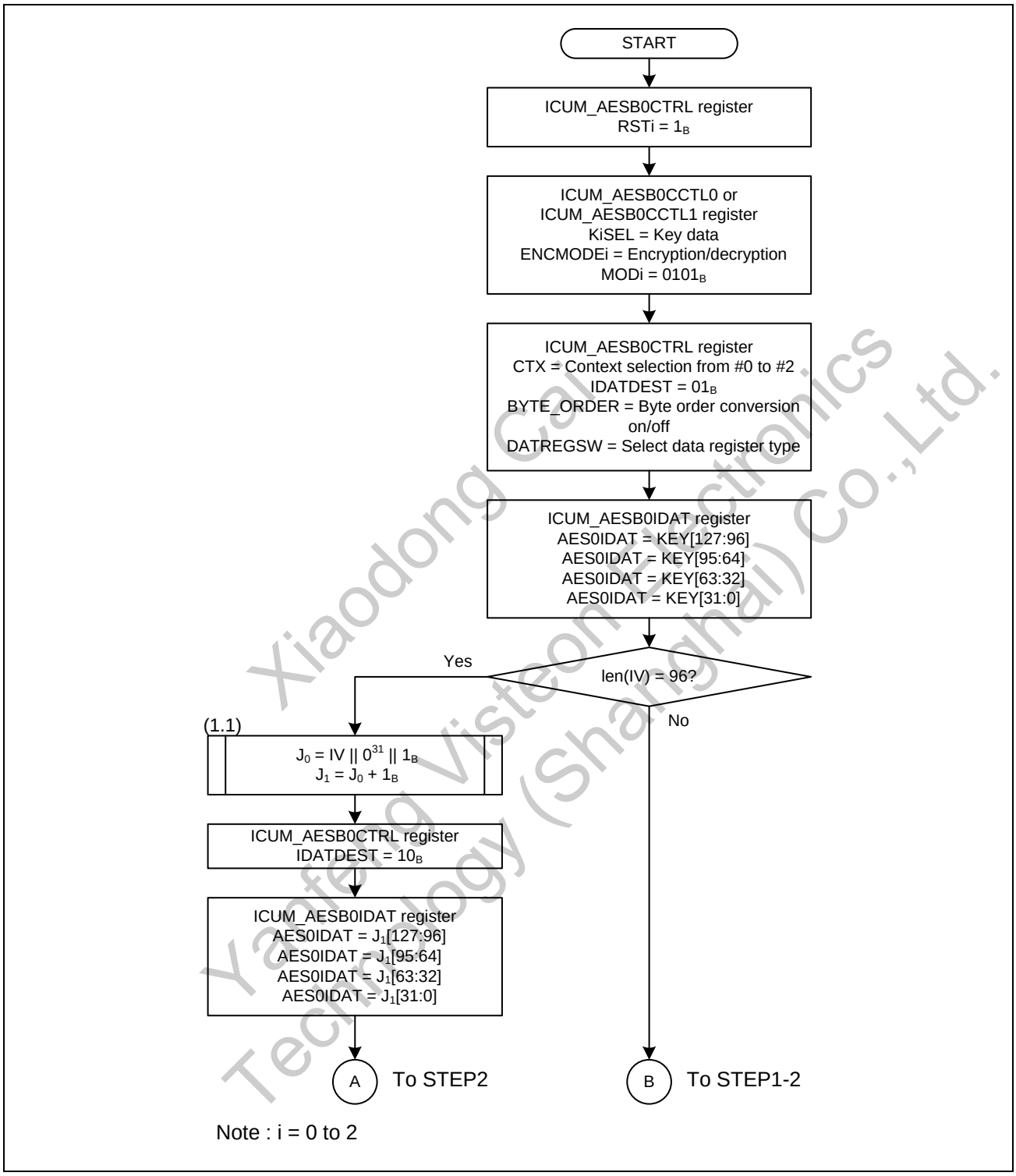


Figure 9.5 Operating procedure for GCM (STEP1-1)

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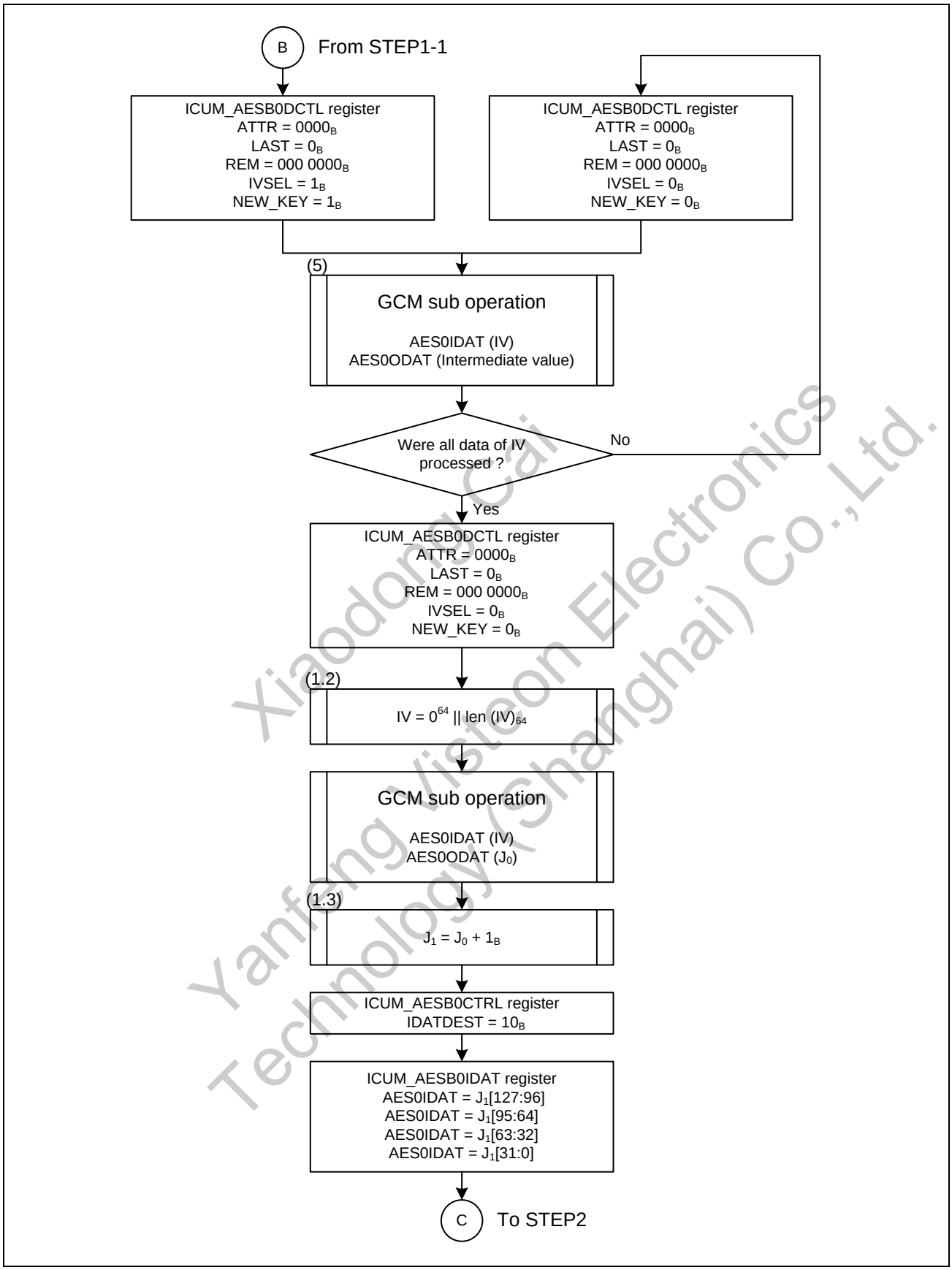


Figure 9.6 Operating procedure for GCM (STEP1-2)

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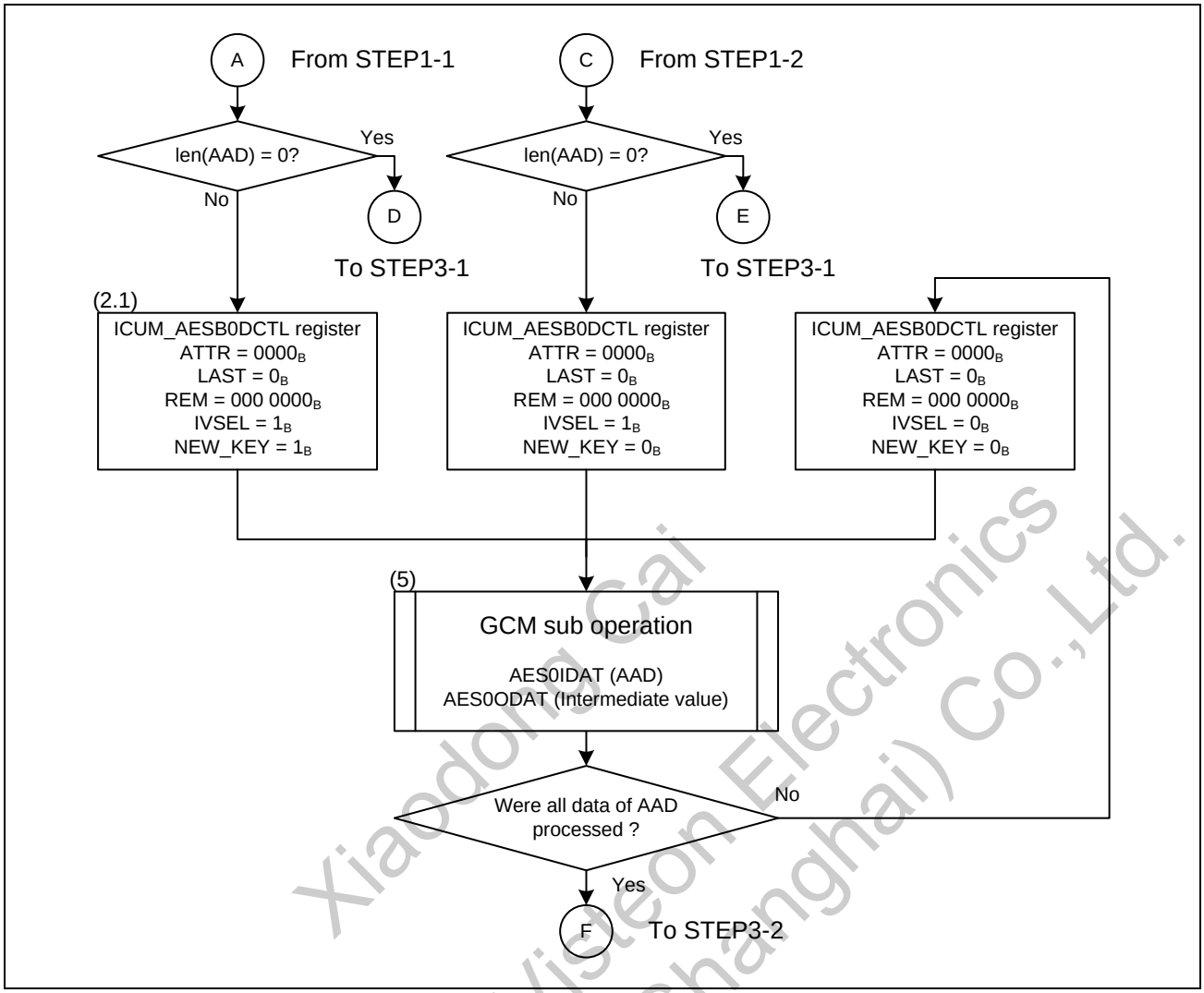
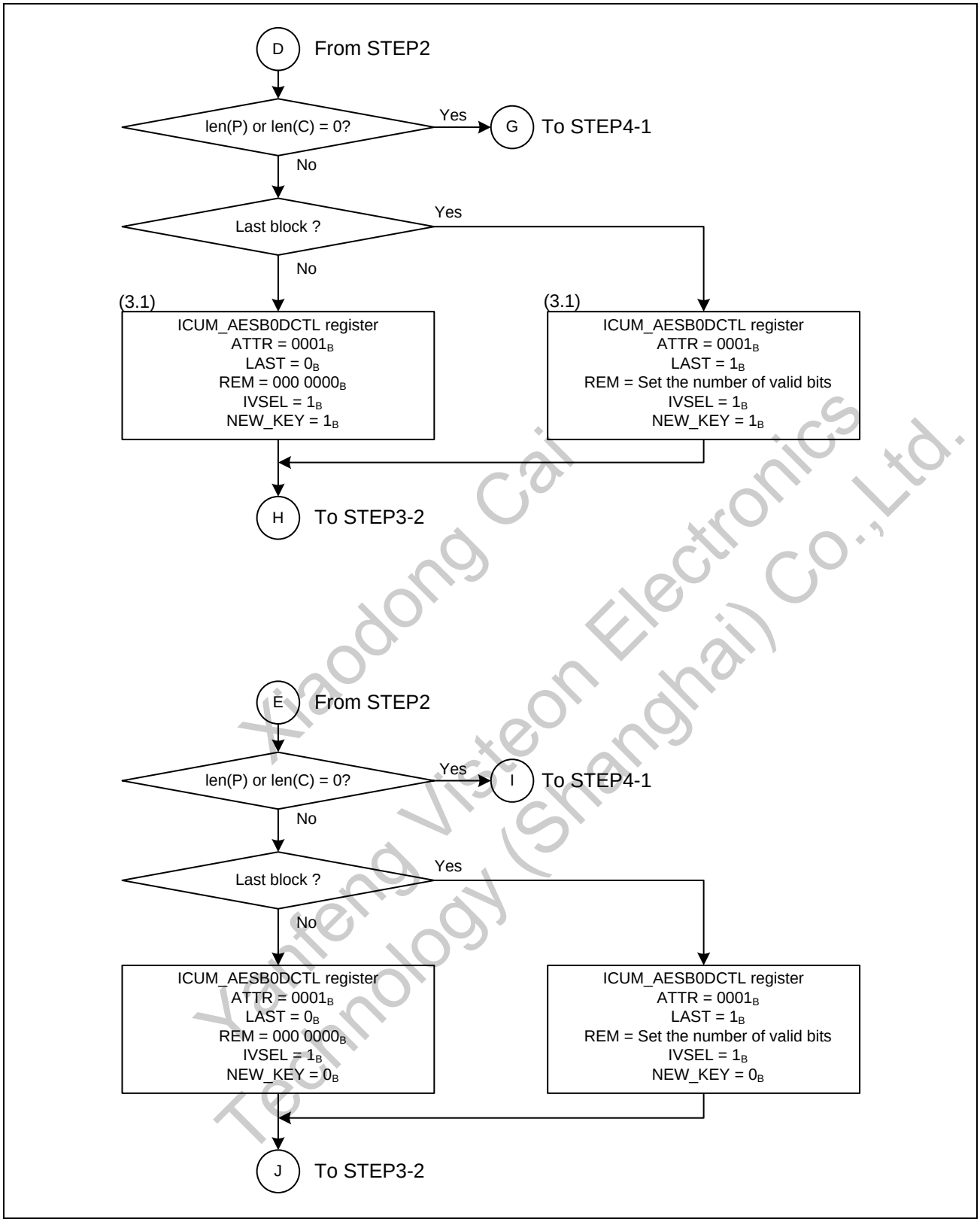


Figure 9.7 Operating procedure for GCM (STEP2)

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Figure 9.8 Operating procedure for GCM (STEP3-1)

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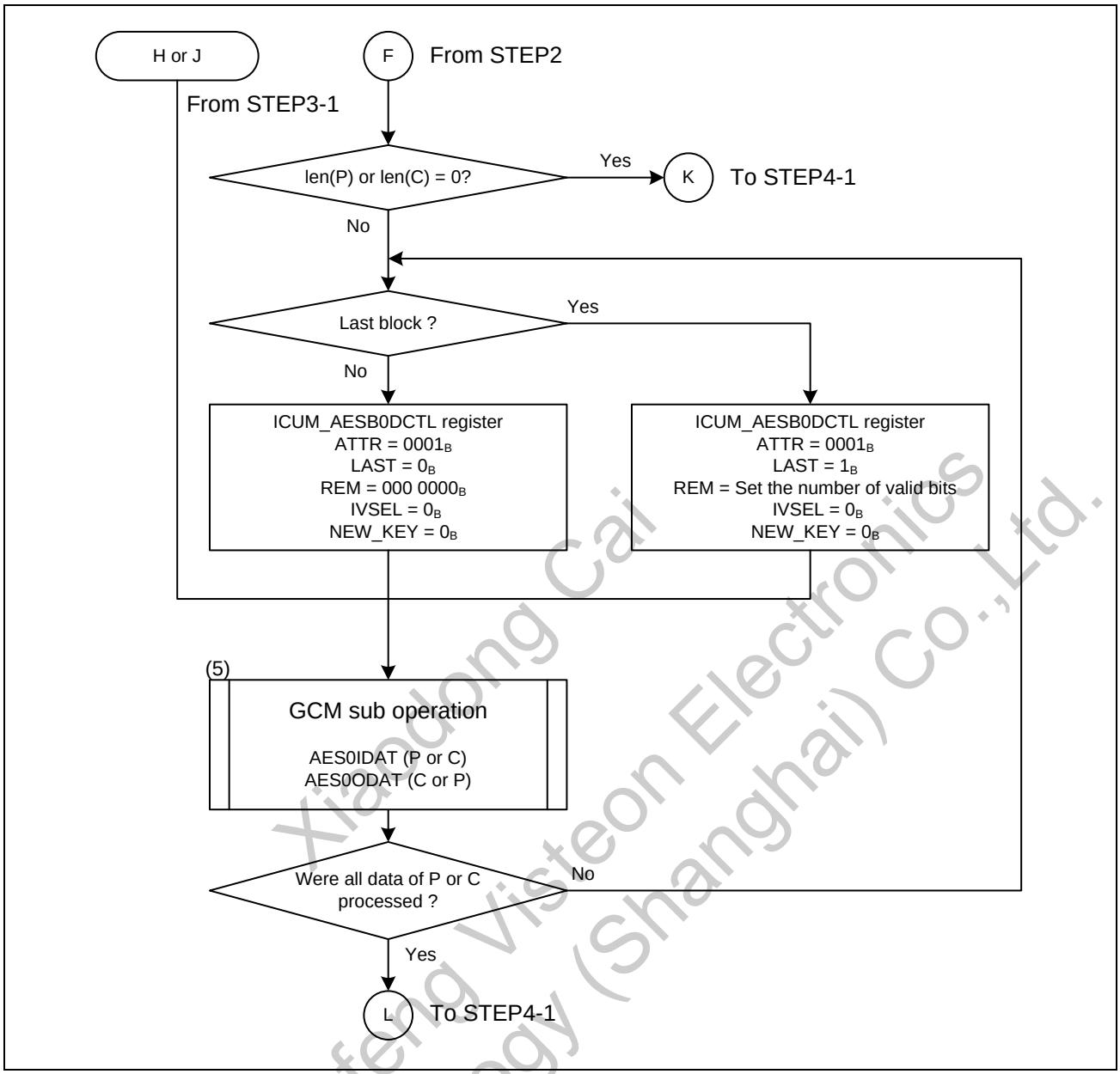


Figure 9.9 Operating procedure for GCM (STEP3-2)

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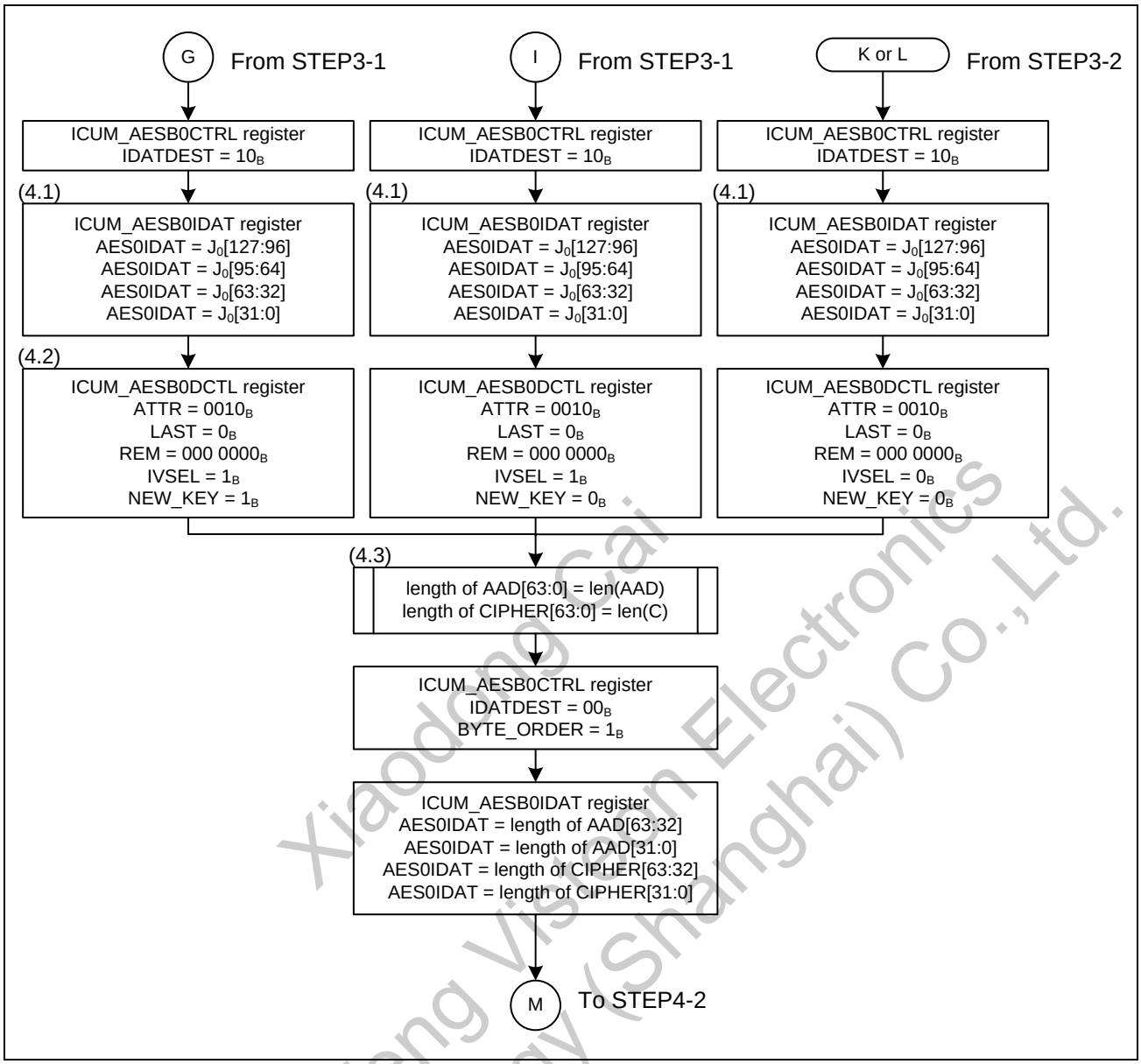


Figure 9.10 Operating procedure for GCM (STEP4-1)

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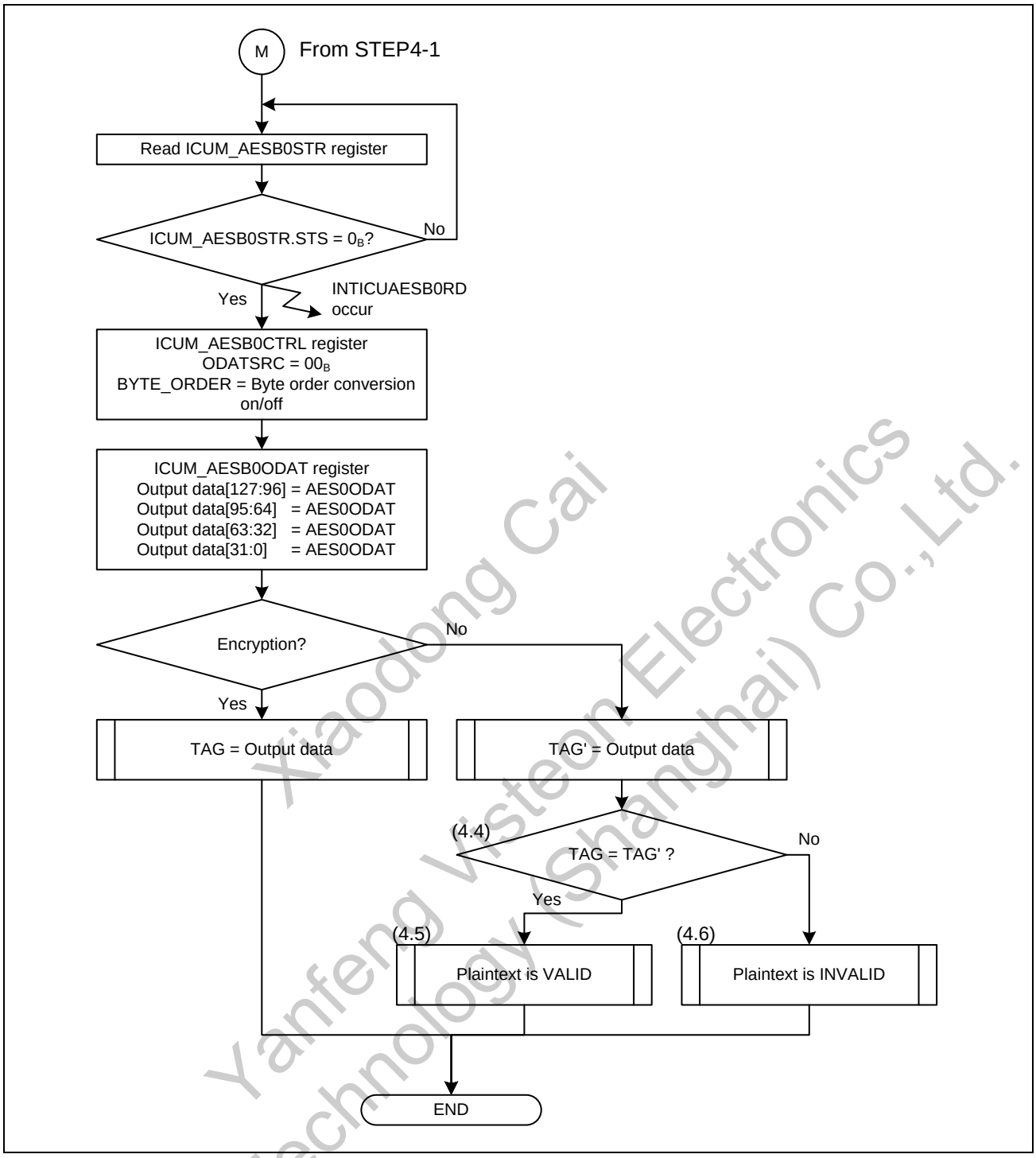


Figure 9.11 Operating procedure for in GCM (STEP4-2)

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Figure 9.12 shows the sub operation in GCM. This procedure is referred to "GCM sub operation" in **Figure 9.5** to **Figure 9.11**.

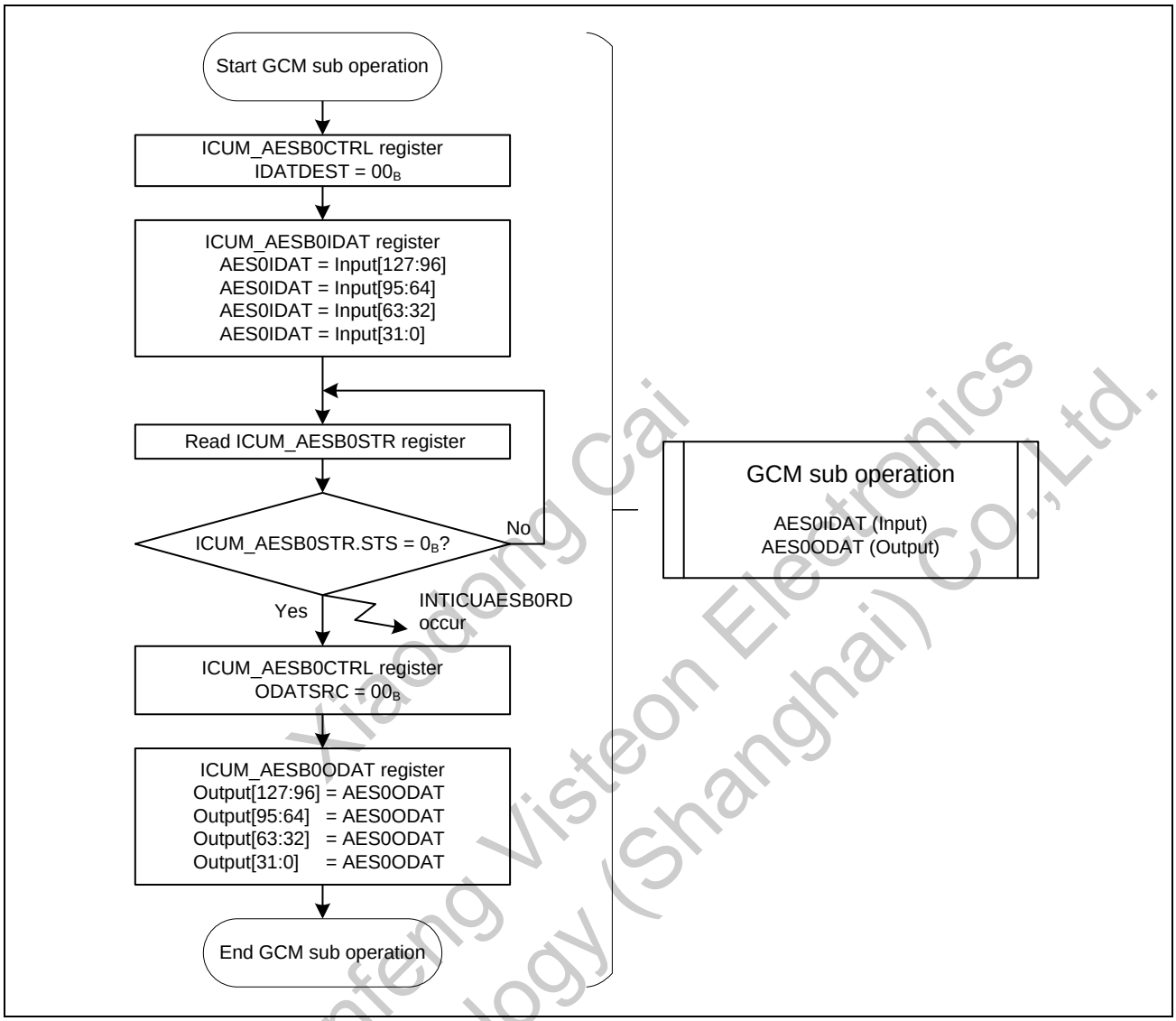


Figure 9.12 Sub operation in GCM

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[Supplement]

Given below is a brief description of the steps shown in **Figure 9.5** to **Figure 9.12**.

(1.1) Compute the J_1 from IV by software.

The value to be set in this step is obtained by the formula $J_1 = J_0 + 1_B$.

Keep J_0 in software because it is used as an input data for STEP4.

(1.2) Compute IV in this step by software.

The value to be set in this step is obtained by the formula $0^{64} \| [len(IV)]_{64}$.

(1.3) Compute the J_1 from J_0 by software.

The formula is $J_1 = J_0 + 1_B$ such that only the least significant 32-bit is incremented. The carry for 33rd bit must be ignored in this operation.

Keep J_0 computed in this phase in software because it is used as an input data for STEP4.

(2.1) Set ICUM_AESB0DCTL.NEW_KEY = 1_B for calculation to operate in a background.

(3.1) Set ICUM_AESB0DCTL.NEW_KEY = 1_B for calculation to operate in a background.

(4.1) Set J_0 which is used in STEP1 as input data.

(4.2) Set ICUM_AESB0DCTL.NEW_KEY = 1_B for calculation to operate in a background.

(4.3) Compute "length of AAD[63:0]" and "length of CIPHER[63:0]" by using len(AAD) and len(C) in software.

(4.4) Compare TAG and TAG' in software when decryption process. TAG is Authentication Tag for comparision.

(4.5) Plaintext obtained in STEP3 is valid.

(4.6) Plaintext obtained in STEP3 is invalid.

(5) Refer to **Figure 9.12**.

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(3) XTS

The operating procedure for XTS is shown in **Figure 9.13** and **Figure 9.14**.

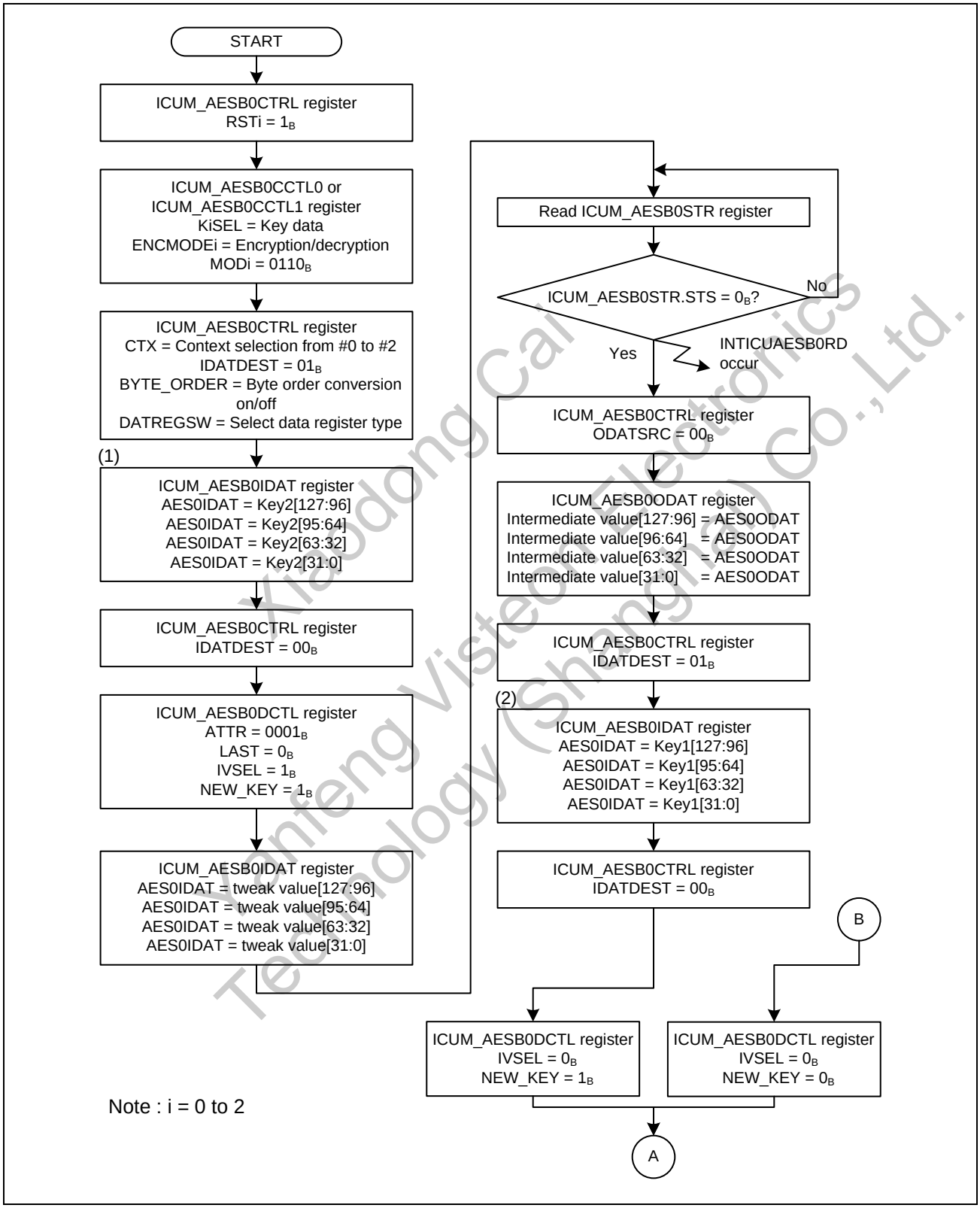
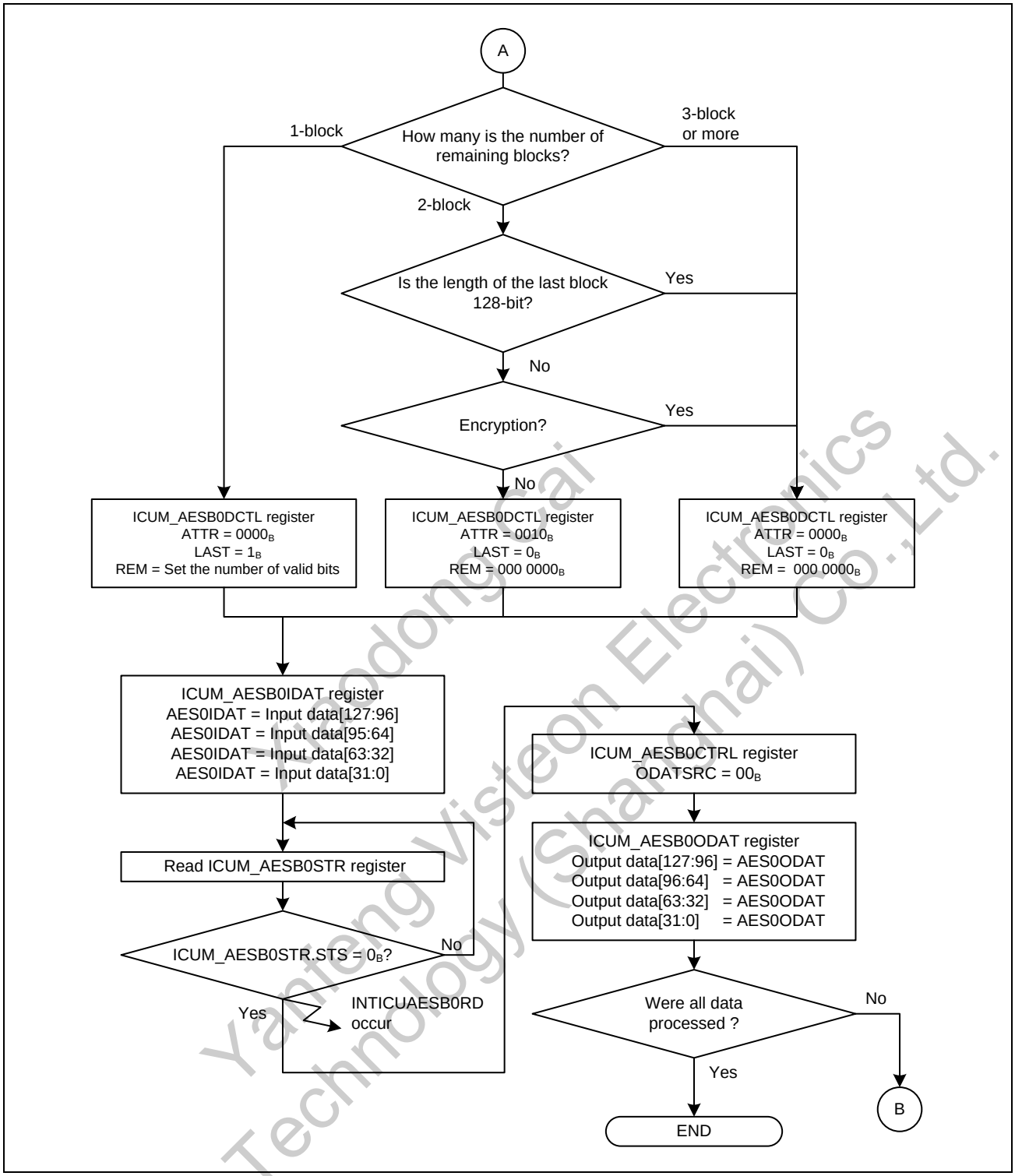


Figure 9.13 Operating procedure for XTS (1 of 2)

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Figure 9.14 Operating procedure for XTS (2 of 2)

[Supplement]

Given below is a brief description of the steps shown in **Figure 9.13** to **Figure 9.14**.

(1), (2) The secret key of XTS mode consists of a concatenation of the two keys of equal length Key1 and Key2 such that: Key = Key1 | Key2. Set Key2 during (1) and Key1 during (2) in **Figure 9.13**.

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(4) CCM

CCM mode requires several steps and AES operation is performed with a different configuration in each step. The input and output data in each step is shown in **Figure 9.15** and **Figure 9.16**.

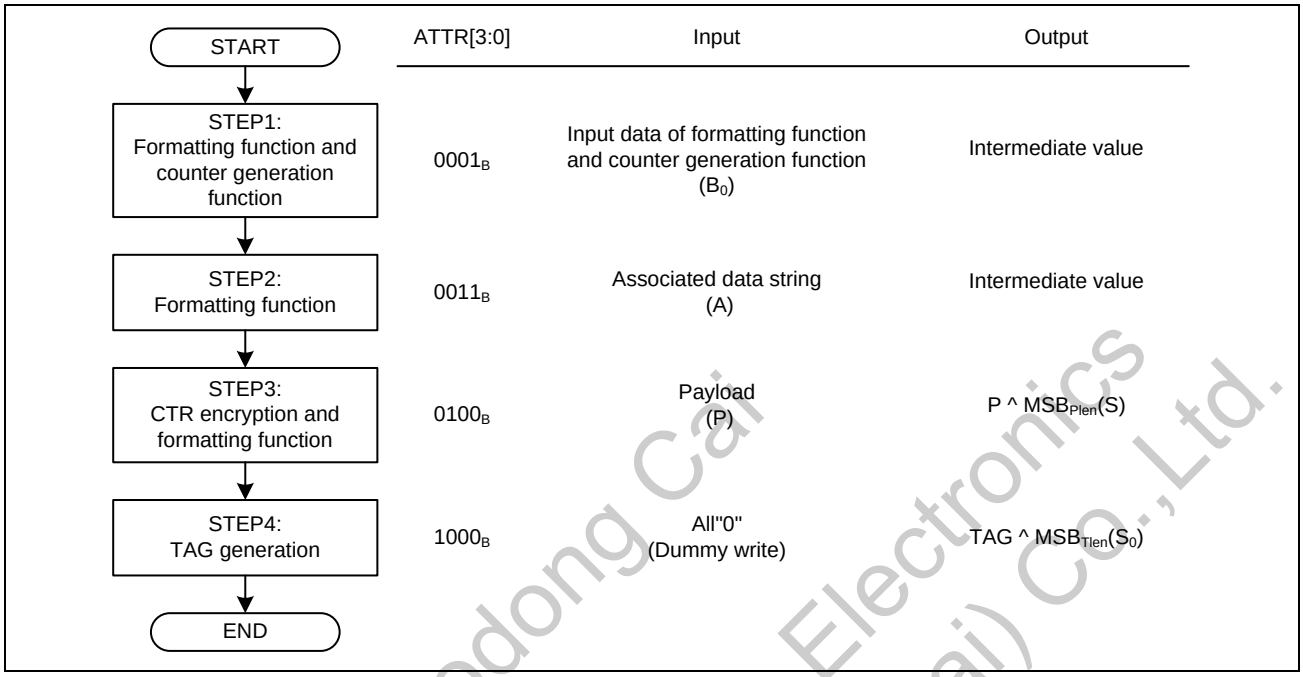


Figure 9.15 The configuration of ATTR and input/output data in each step (Encryption).

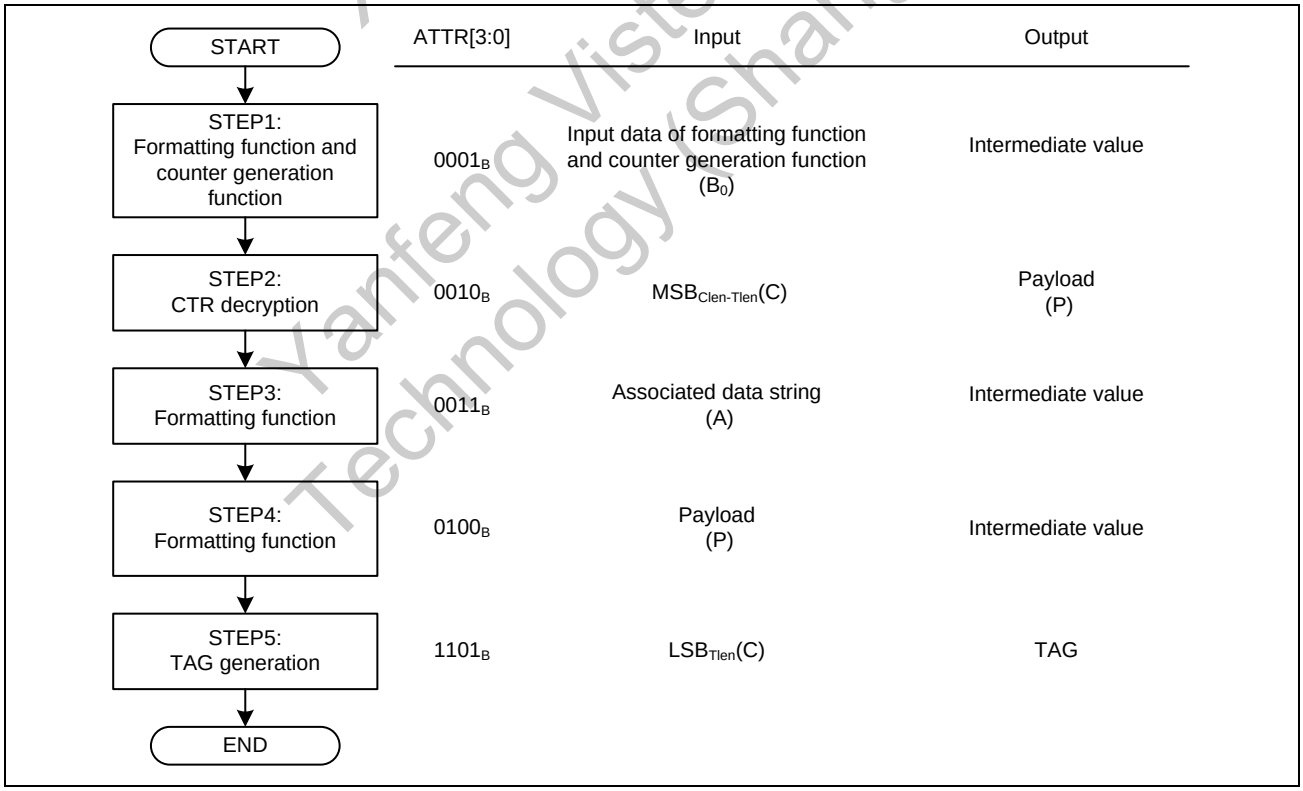


Figure 9.16 The configuration of ATTR and input/output data in each step (Decryption).

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The format of B_0 for input data in STEP1 is shown in **Table 9.21**.

Table 9.21 Format of Input data B_0

Bit Position	Symbol	Function
127 to 115	—	Reserved Please fix these bits to “0”. Otherwise, we do not ensure the computational result.
114 to 96	Alen	Bit length to indicate the associated data Set the bit length of the associated data such that 0 ≤ length of associated data < 8×(2^16 - 2^8) [bit].Write 000_B to B_0[98:96].
95 to 87	—	Reserved Please fix these bits to “0”. Otherwise, we do not ensure the computational result.
86 to 84	Nlen	Bits which interpret the length of N 000: N length is set as 56 bits 001: N length is set as 64 bits 010: N length is set as 72 bits 011: N length is set as 80 bits 100: N length is set as 88 bits 101: N length is set as 96 bits 110: N length is set as 104 bits 111: Prohibited
83	—	Reserved Please fix these bits to “0”. Otherwise, we do not ensure the computational result.
82 to 80	Tlen	Bits which interpret the length of tag. 000: Tag length is set as 32 bits 001: Tag length is set as 48 bits 010: Tag length is set as 64 bits 011: Tag length is set as 80 bits 100: Tag length is set as 96 bits 101: Tag length is set as 112 bits 110: Tag length is set as 128 bits 111: Prohibited
79 to 67	—	Reserved Please fix these bits to “0”. Otherwise, we do not ensure the computational result.
66 to 0	Plen	Bits which indicate the length of plaintext. This length is also same as the “length of ciphertext” – “length of tag”. Set multiple of 8, not less than 0 and less than 8*2^(120-56) ($=8_0000_0000_0000_0000_H$). Write 000_B to B_0[2:0].

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The operating procedure for CCM is shown in **Figure 9.17** to **Figure 9.25**.

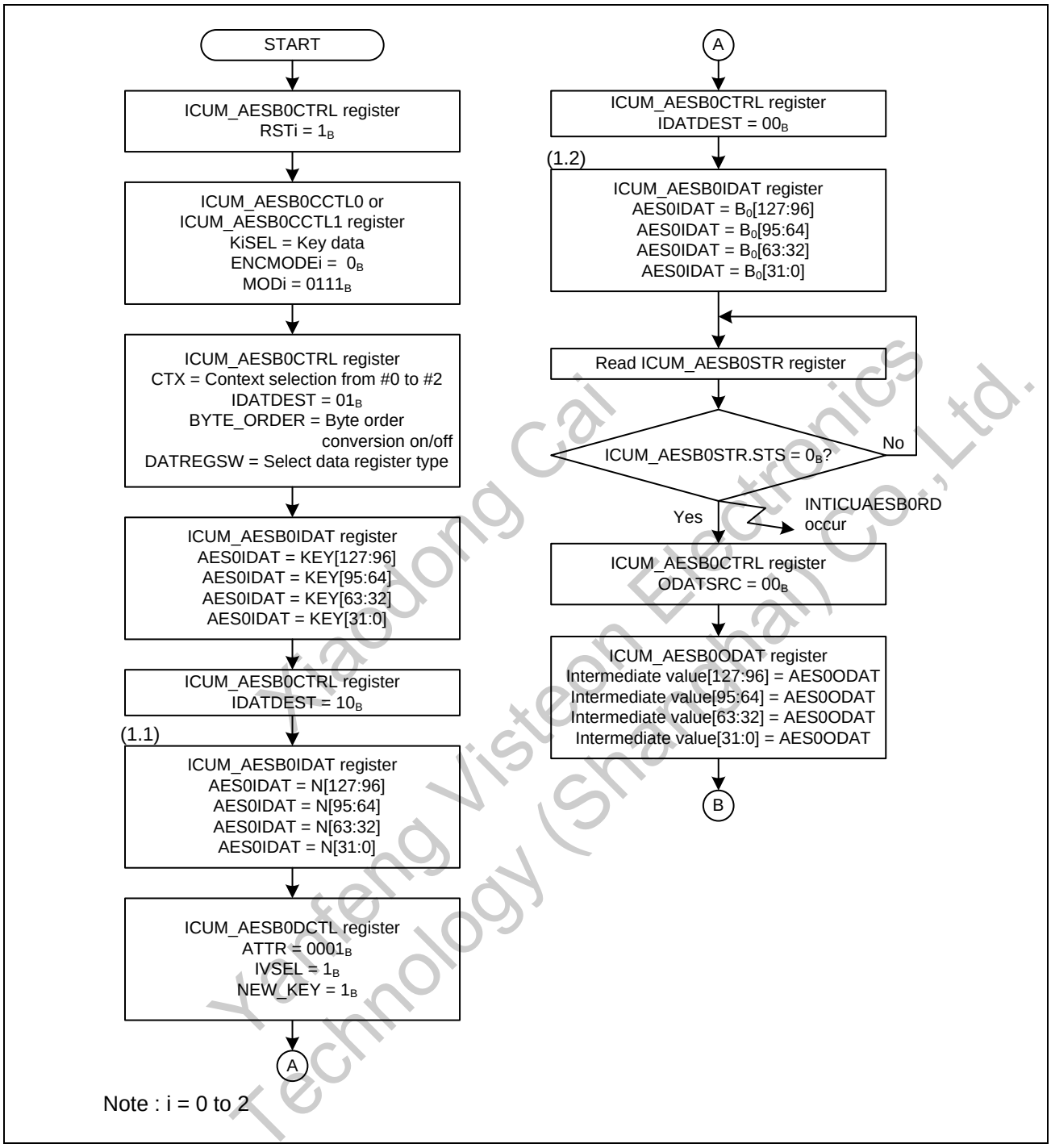


Figure 9.17 Operating procedure for encryption in CCM (STEP1)

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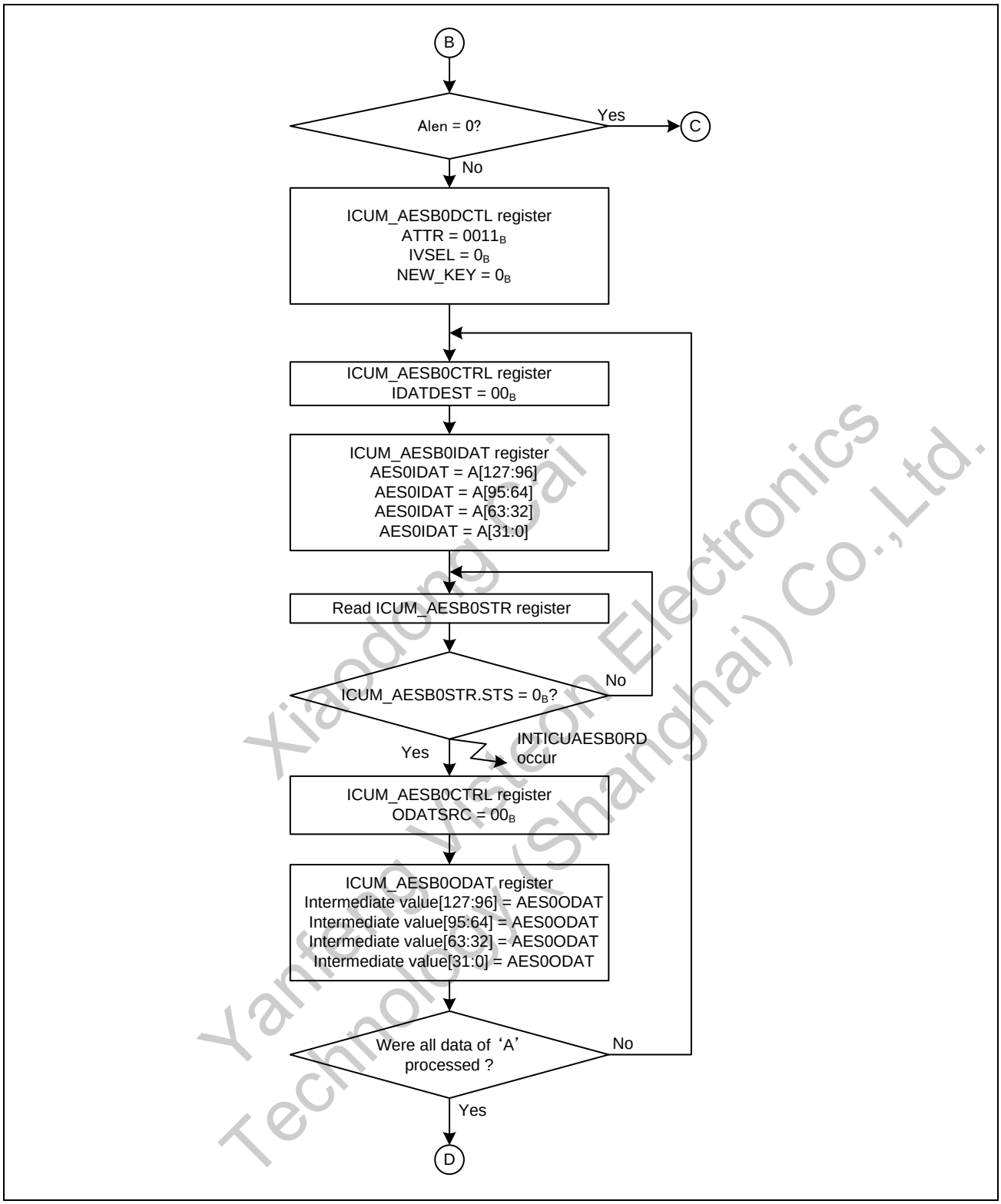


Figure 9.18 Operating procedure for encryption in CCM (STEP2)

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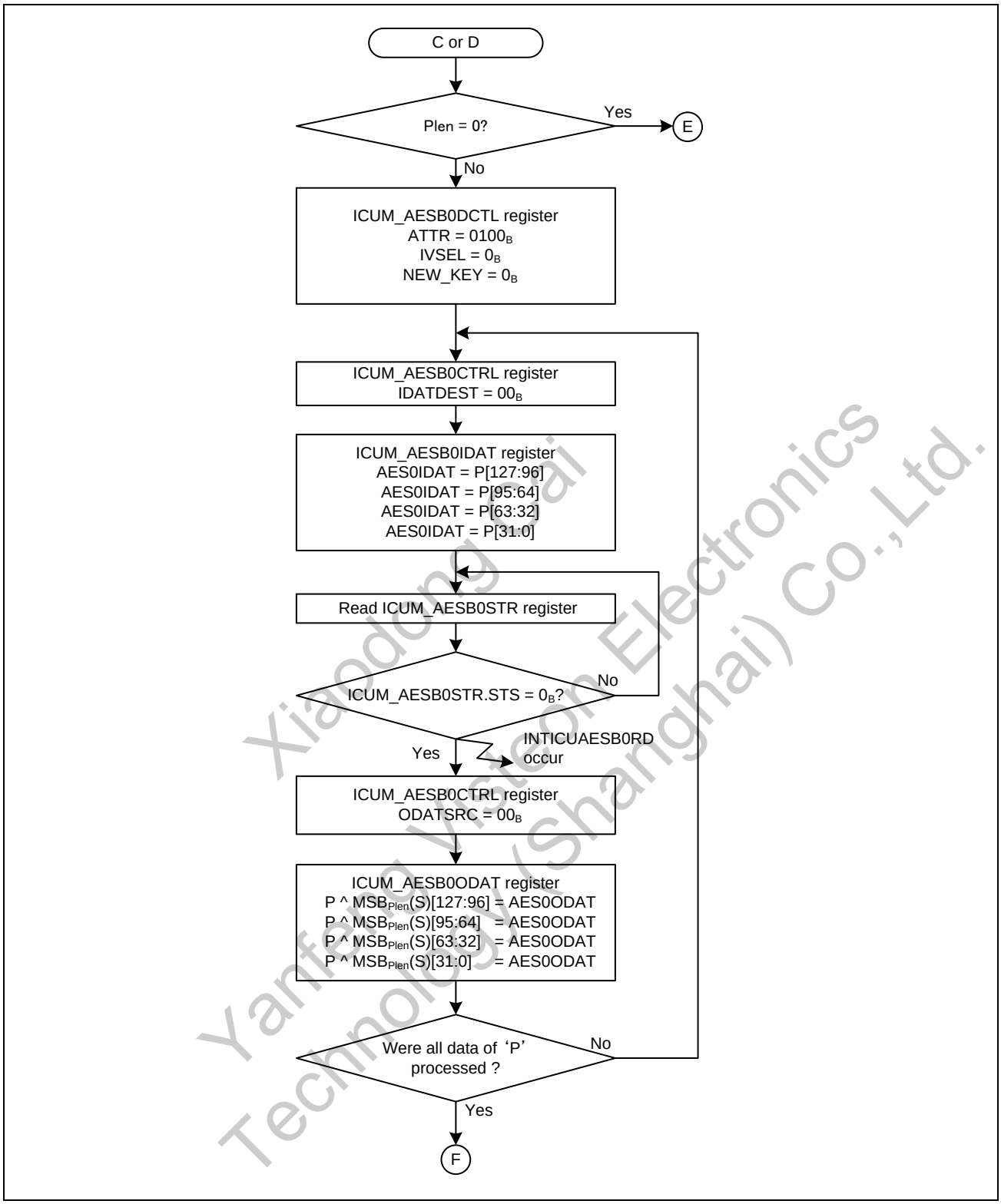


Figure 9.19 Operating procedure for encryption in CCM (STEP3)

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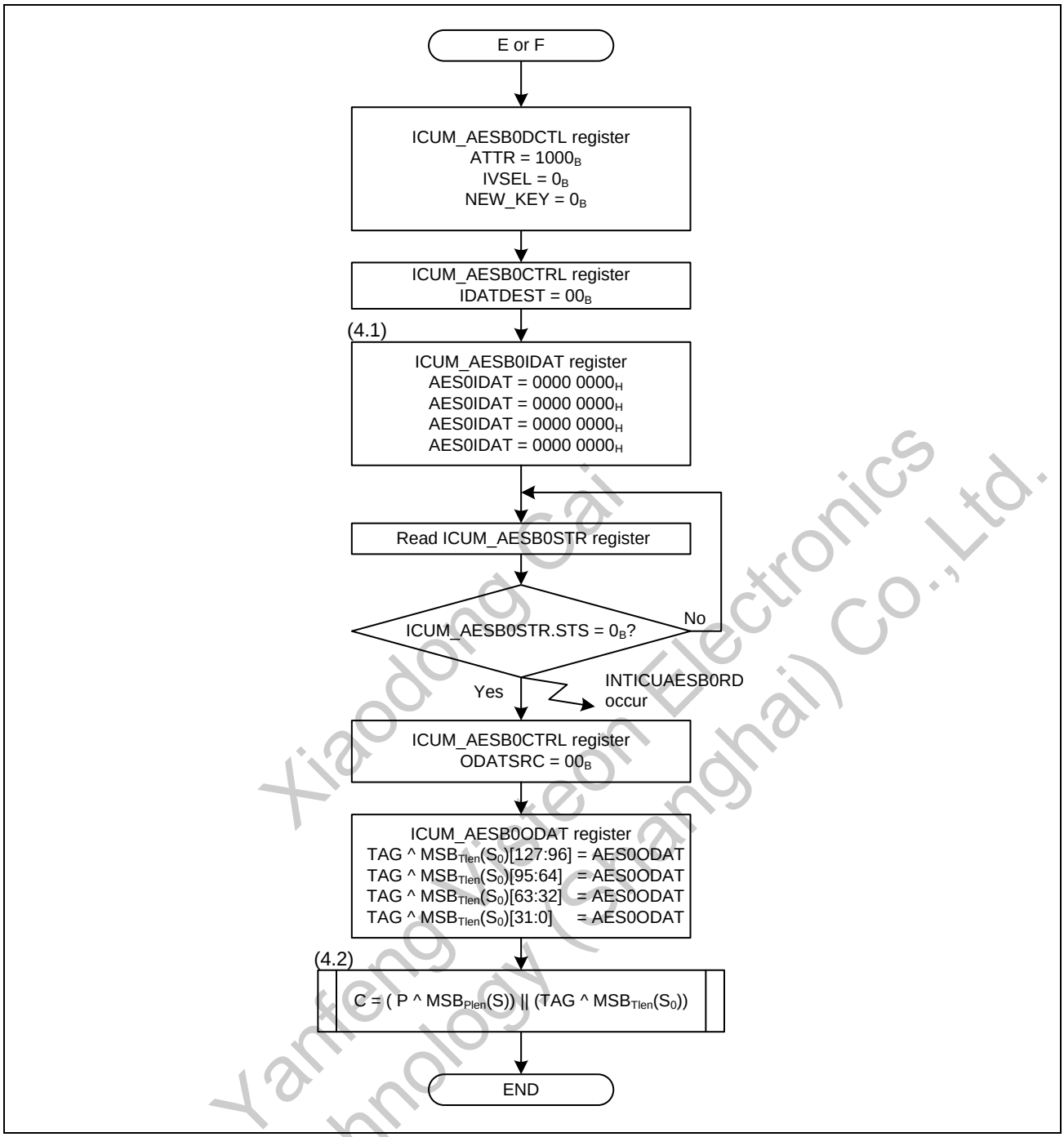


Figure 9.20 Operating procedure for encryption in CCM (STEP4)

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[Supplement]

Given below is a brief description of the steps shown in **Figure 9.17** to **Figure 9.20**.

(1.1) The MSBs of nonce N must be padded with 0 if N is less than 128-bit. When N = $123456789ABC_H$, the nonce N is modified as follows.

N[127:96] = $0000\ 0000_H$
N[95:64] = $0000\ 0000_H$
N[63:32] = $0000\ 1234_H$
N[31:0] = $5678\ 9ABC_H$

(1.2) Refer **Table 9.21** for the details about B_0

(4.1) Input all '0' as a dummy data to compute "T^$MSB_{Tlen}(S_0)$"

(4.2) C is a concatenation of "P^$MSB_{Plen}(S)$" given in STEP3 and "T^$MSB_{Tlen}(S_0)$" computed in STEP4.

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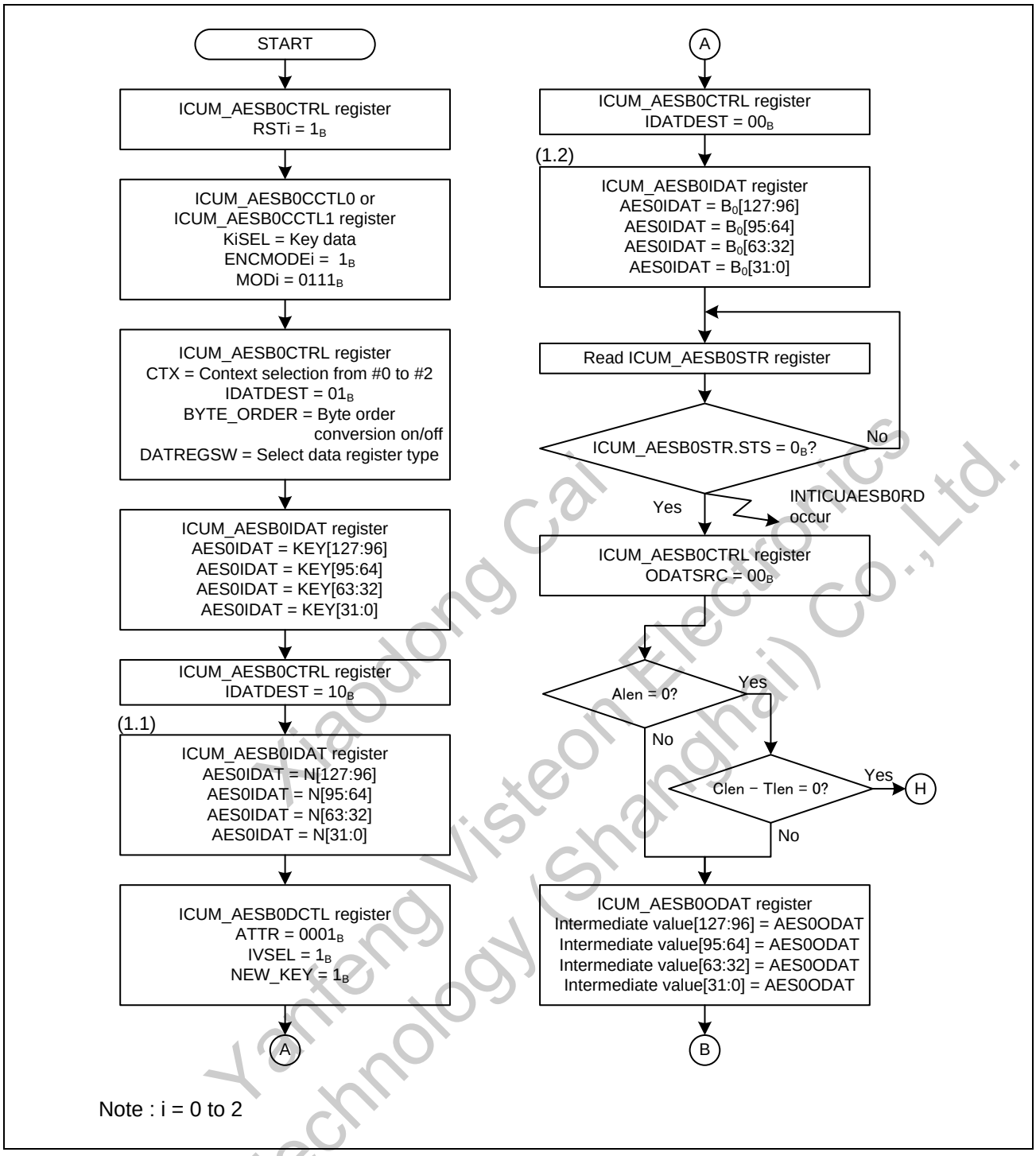


Figure 9.21 Operating procedure for decryption in CCM (STEP1)

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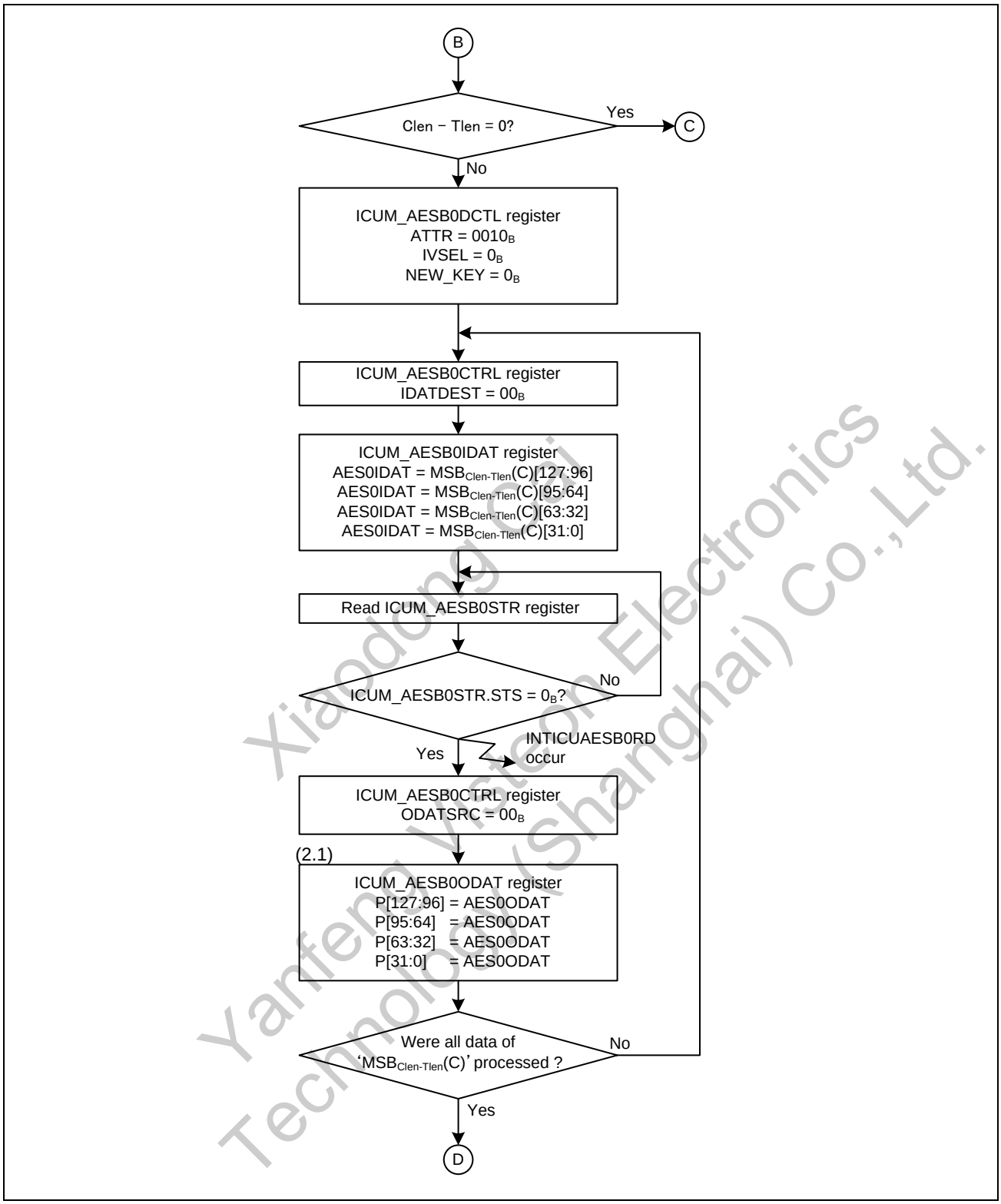


Figure 9.22 Operating procedure for decryption in CCM (STEP2)

Xiaodong Li
Yanfeng Virtech Electronics Technology (Shanghai) Co.,Ltd.

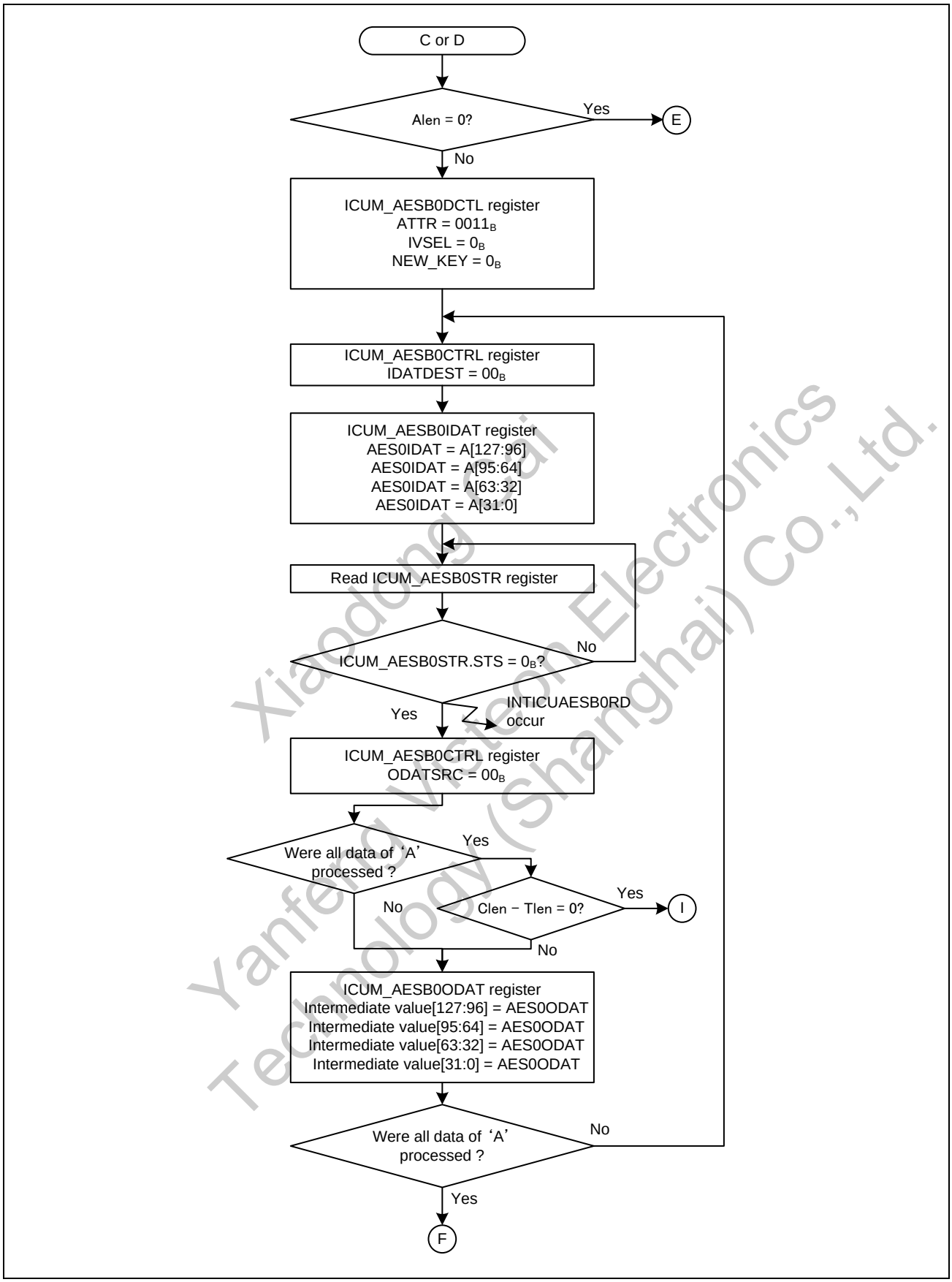


Figure 9.23 Operating procedure for decryption in CCM (STEP3)

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Yanfeng Vision Electronics
Technology (Shanghai) Co.,Ltd.

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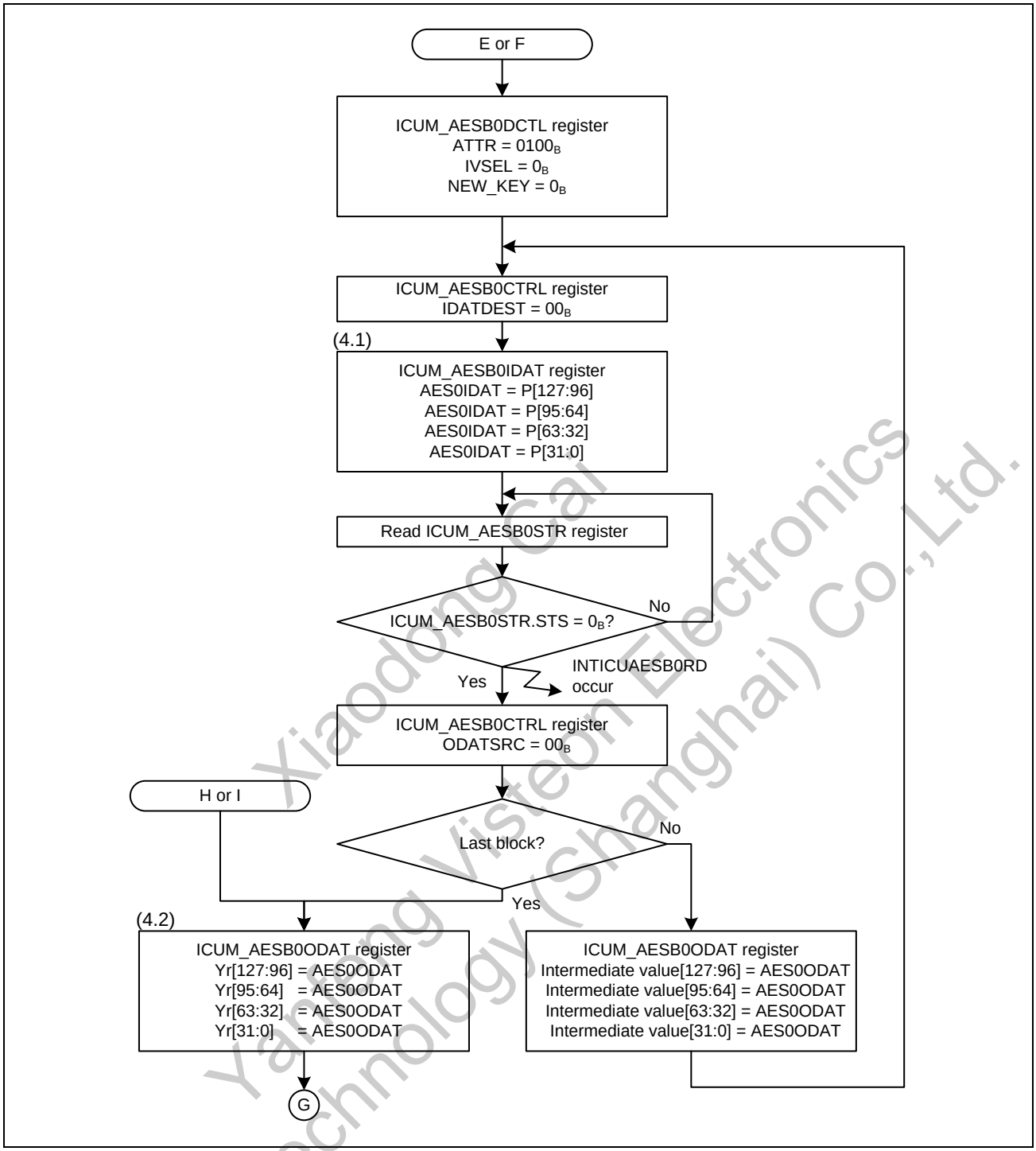


Figure 9.24 Operating procedure for decryption in CCM (STEP4)

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Yangzhou Visteon Electronics
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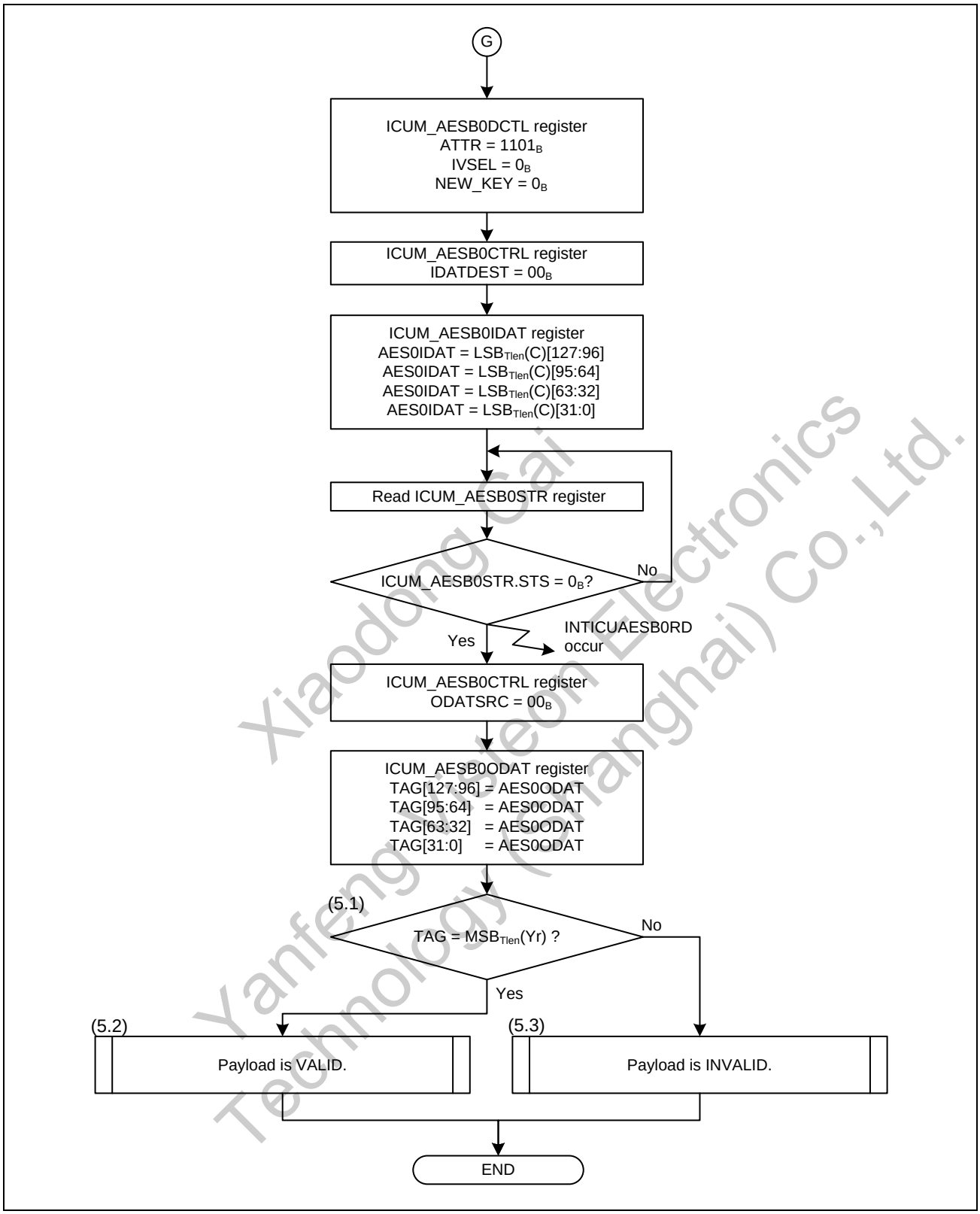


Figure 9.25 Operating procedure for decryption in CCM (STEP5)

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[Supplement]

Given below is a brief description of the steps shown in **Figure 9.21** to **Figure 9.25**.

(1.1) The MSBs of nonce N must be padded with 0 if N is less than 128-bit. When N = $123456789ABC_H$, the nonce N is modified as follows.

N[127:96] = $0000\ 0000_H$
N[95:64] = $0000\ 0000_H$
N[63:32] = $0000\ 1234_H$
N[31:0] = $5678\ 9ABC_H$

(1.2) Refer **Table 9.21** for the details about B_0

(2.1) Keep P computed in this phase in software because it is used as an input data for (4.1) in STEP4.

(4.1) P inserted in this phase is came from (2.1) in STEP2.

(4.2) Keep Yr computed in this phase in software because it is used as an input data for (5,1) in STEP5.

(5.1) Compare T obtained in STEP5 and Yr computed at (4.2) in STEP4 in software.

(5.2) Payload obtained at (2.1) in STEP2 is valid.

(5.3) Payload obtained at (2.1) in STEP2 is invalid.

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(5) CMAC

The operating procedure for CMAC is shown in **Figure 9.26** to **Figure 9.27**.

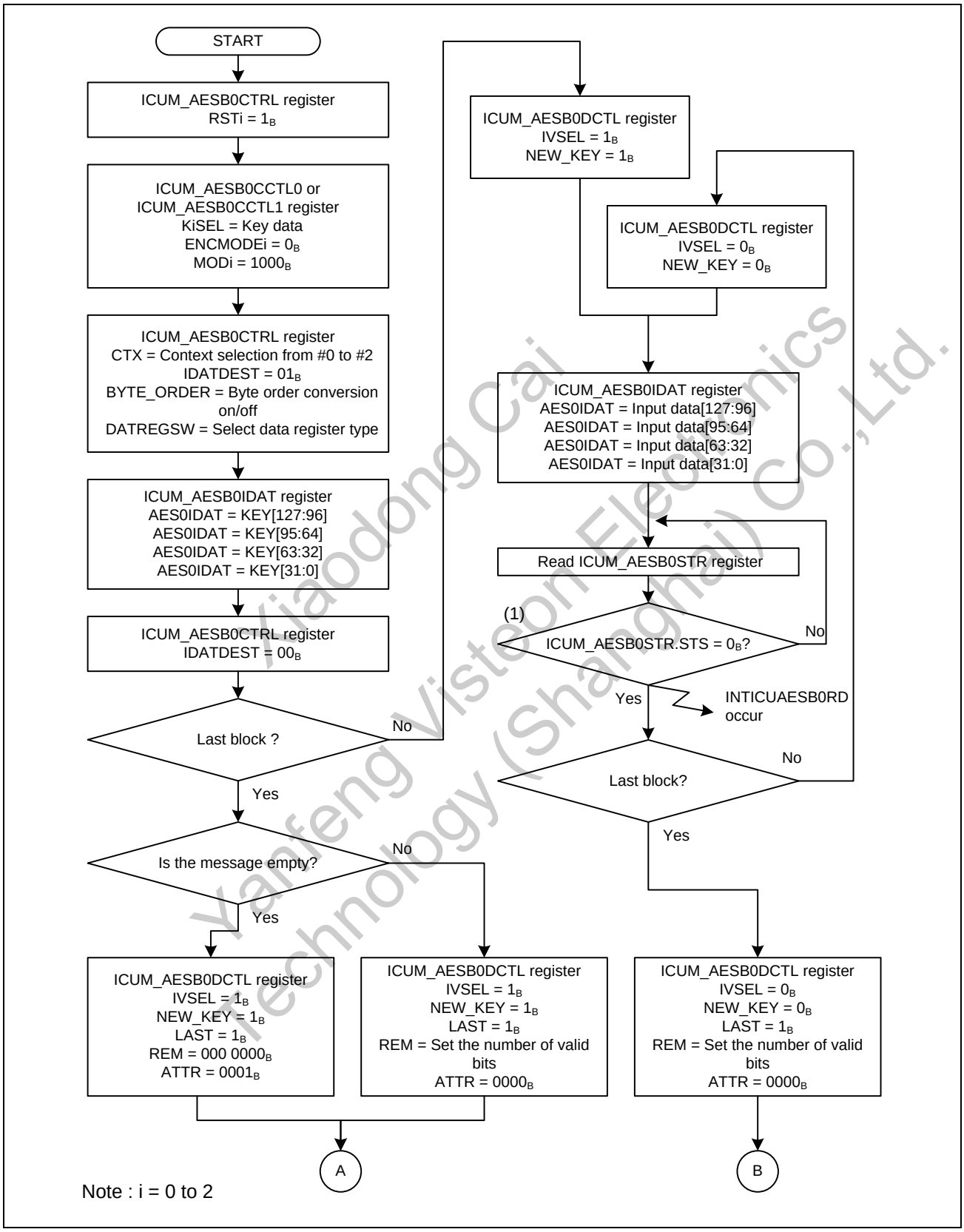


Figure 9.26 Operating procedure for CMAC (1 of 2)

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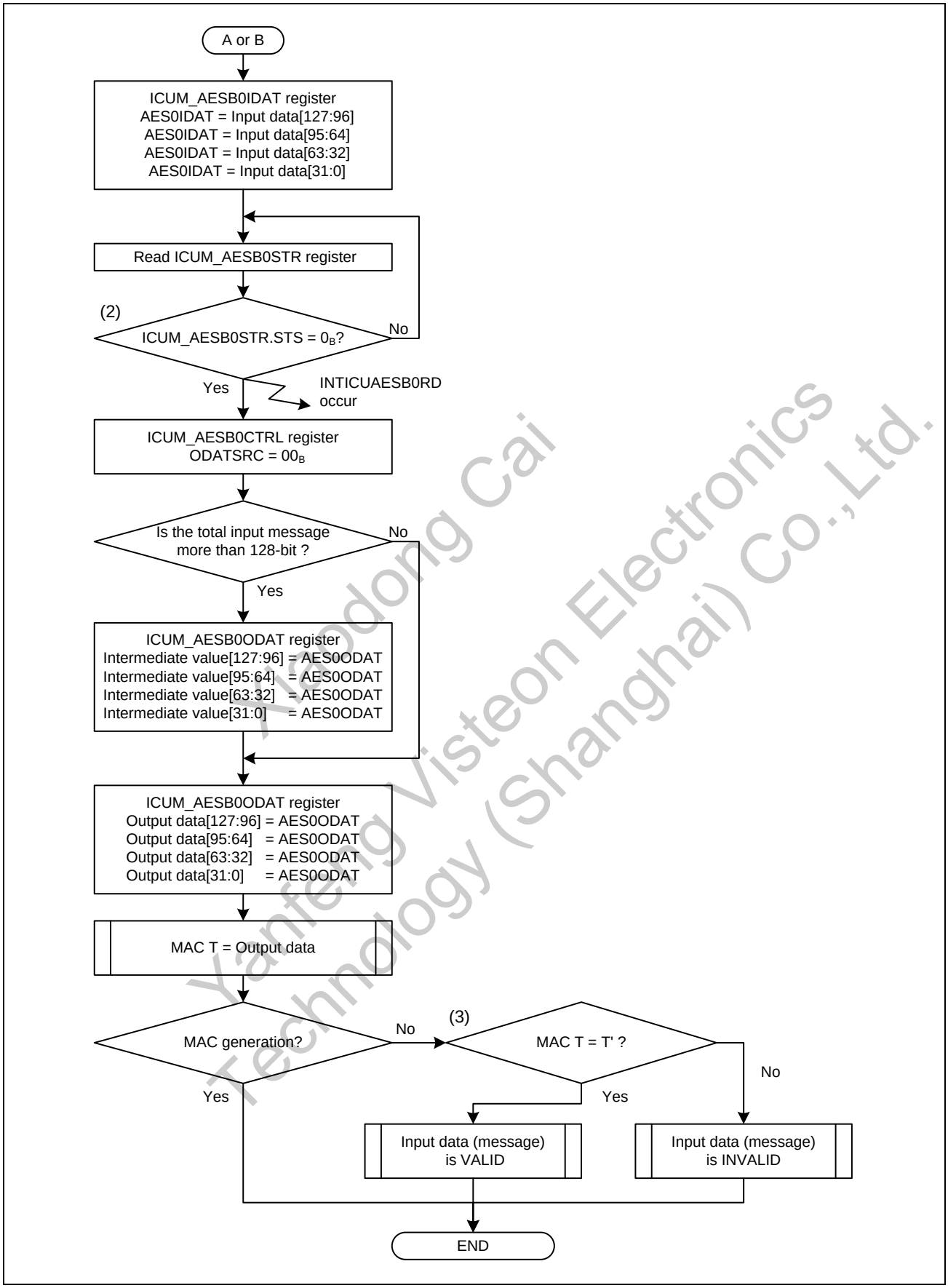


Figure 9.27 Operating procedure for CMAC (2 of 2)

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[Supplement]

Given below is a brief description of the steps shown in **Figure 9.26** to **Figure 9.27**.

(1), (2) If the input data is greater than second block, INTICUAESB0RD does not occur.

(3) Compare MAC T and T' until the bit length of T' in software when MAC verification. T' is the MAC value for comparision.

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(6) Miyaguchi Preneel

The operating procedure for Miyaguchi Preneel is shown in **Figure 9.28**.

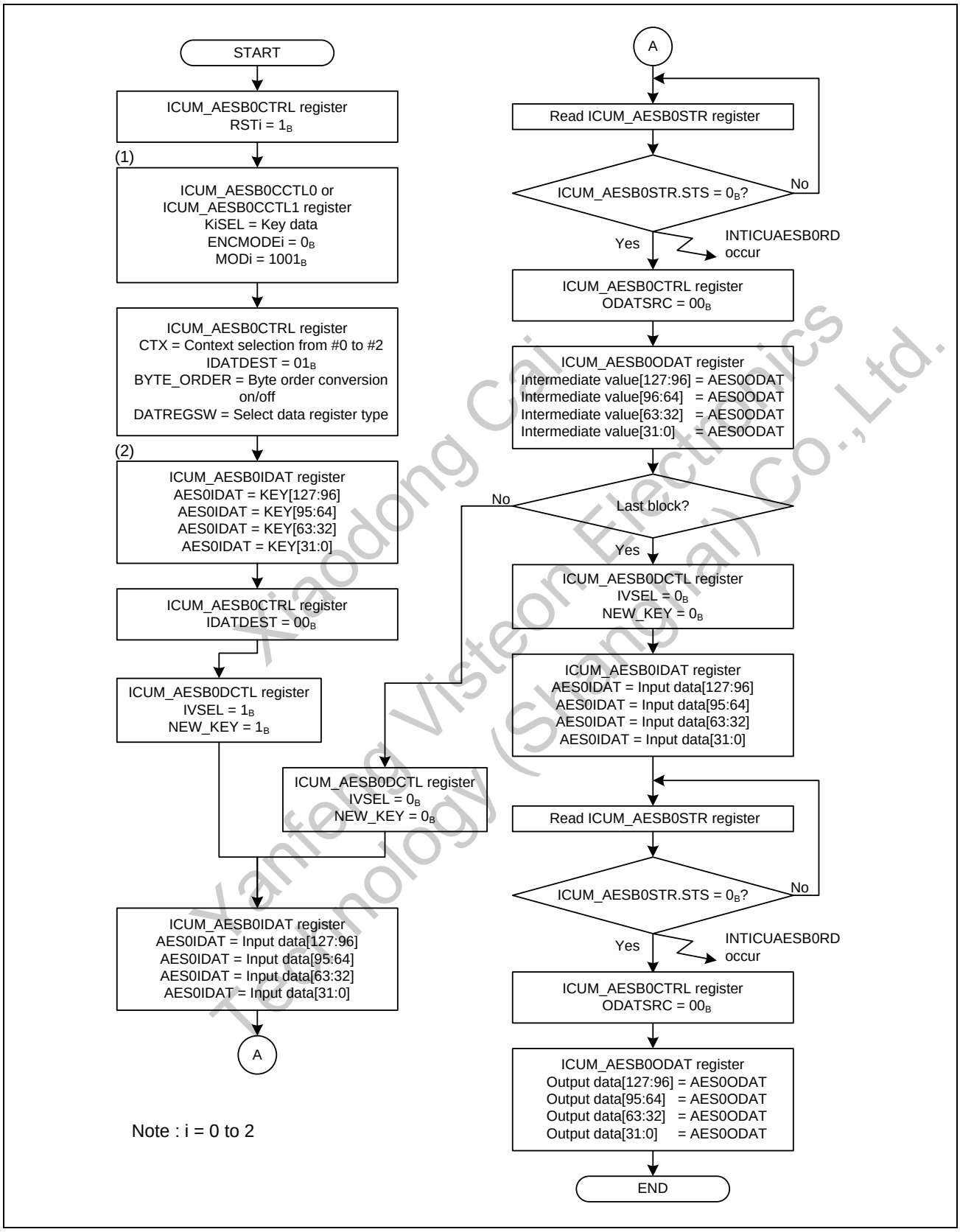


Figure 9.28 Operating procedure for Miyaguchi Preneel

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[Supplement]

Given below is a brief description of the steps shown in **Figure 9.28**.

(1) Select a key data that can input all "0".

(2) Write all "0" to KEY[127:0].

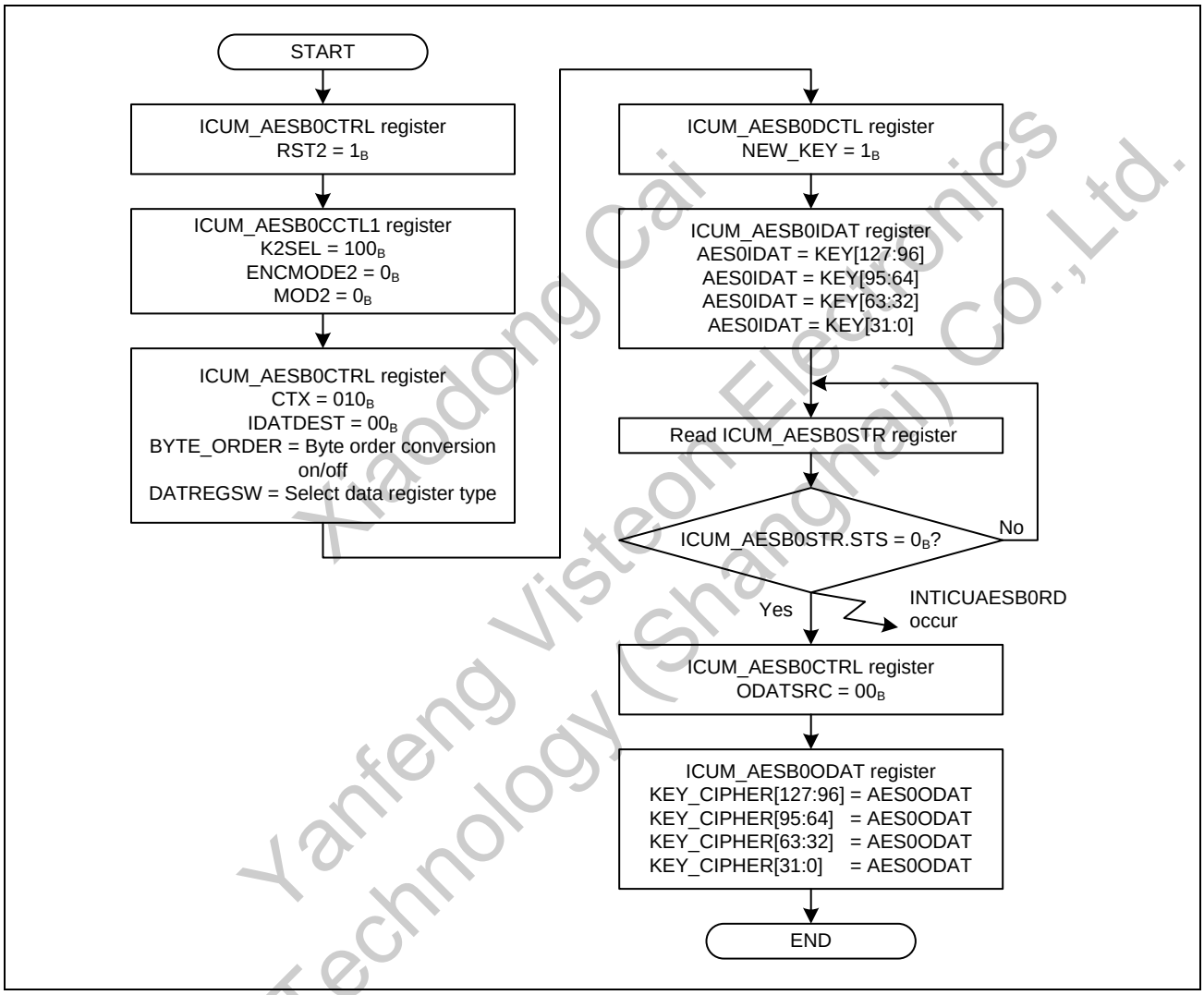
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(7) Cipher Key Output

Shown in **Figure 9.29** below is the flow of encryption processing in which the key data set up in ICUM_AESB0IDAT is encrypted with ROM_KEY_2 and read out of ICUM_AESB0ODAT.

- Block cipher mode: ECB encryption (ECB mode is automatically set up)
- Context number: CTX[2:0] =010_B
- Key data to be used: ROM_KEY_2
- Key data for encryption: KEY[127:0]
- Encrypted key data to be output: KEY_CIPHER[127:0]



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Figure 9.29 Operation procedure for cipher key output

(8) Encryption Key Input

Shown in **Figure 9.30** below is the flow of encryption processing in which the cipher key set up in ICUM_AESB0IDAT is decrypted with ROM_KEY_2 and stored in IMPORTED_KEY.

- Block cipher mode: Decryption (ECB mode is automatically set up)
- Context number: CTX[2:0] = 010_B
- Key data to be used: ROM_KEY_2
- Cipher key to be taken in : KEY_CIPHER[127:0]

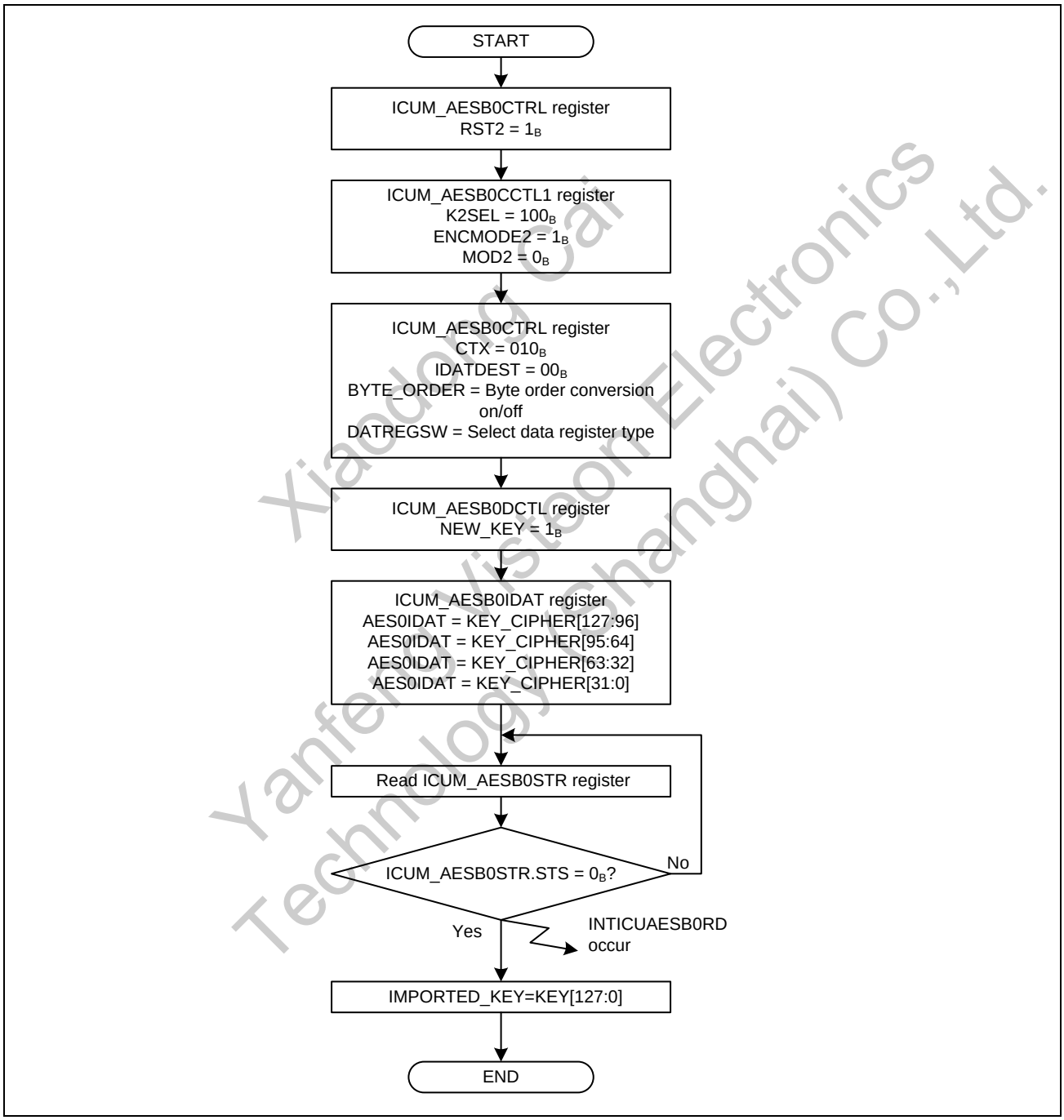


Figure 9.30 Operating procedure for encryption Key

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(9) Pseudo Random Number Generation (PRNG)

The flow for pseudo random number generation is shown in **Figure 9.31**.

- Block cipher mode: Automatically set up (OFB encryption mode)
- Context number: CTX[2:0] = 011_B
- Key data to be used: KEY[127:0] (set up via ICUM_AESB0IDAT)
- Initialization vector: random seed[127:0]
- Pseudo random number: PRNG[127:0]

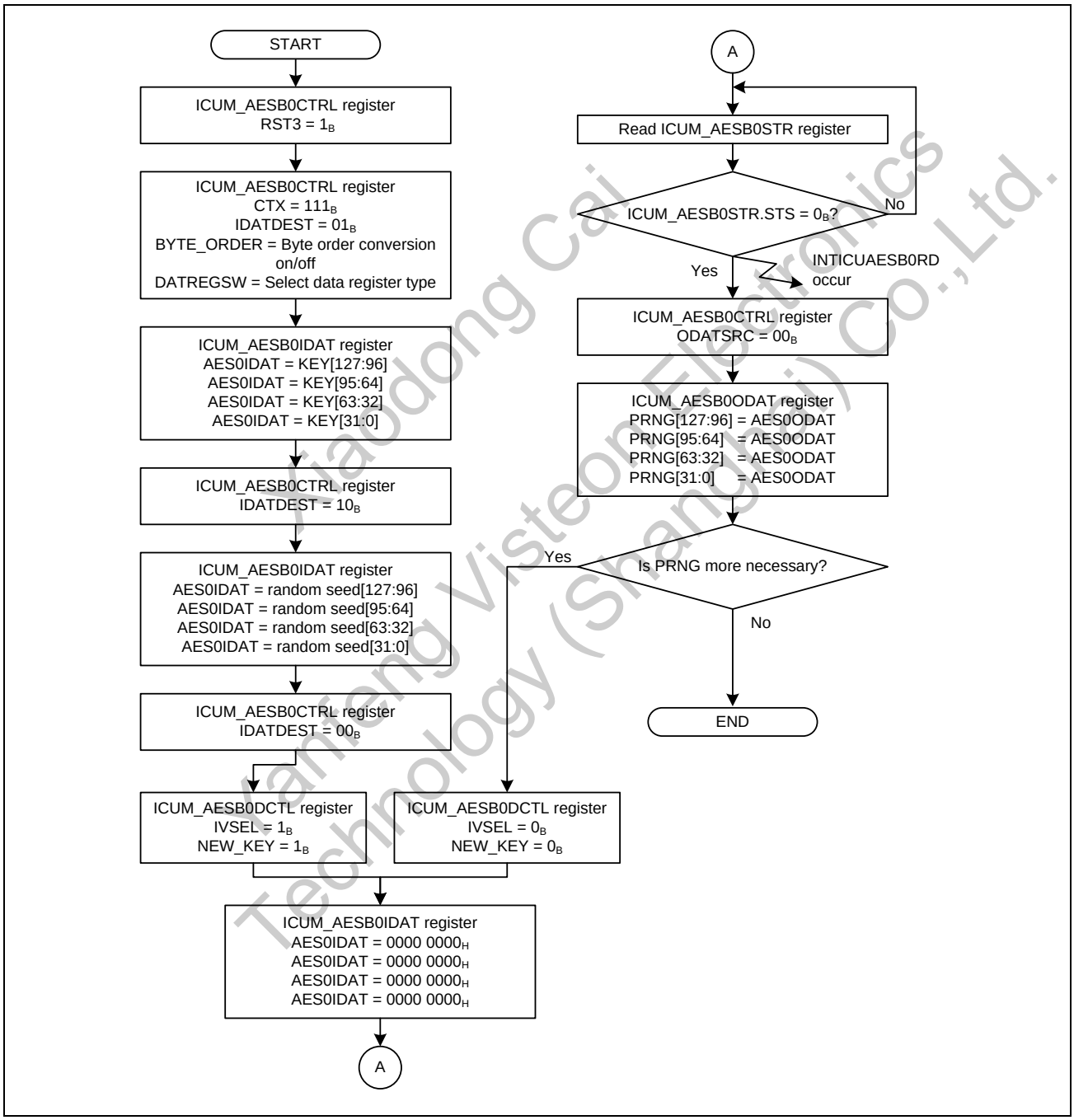


Figure 9.31 Pseudo Random Number Generation Flow

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[Supplement]

Generation of pseudo random number is accomplished by the AES using the true random number in TRNG, as the value after reset. The setting of the AES for pseudo random number generation are given below.

- Context #3 is used.
- PRNG_KEY: Defined value or random seed[*1]
- IV_PRNG: Random seed[*1]
- BCI: 0
- Block cipher mode: OFB encryption mode (automatically set up by specifying context #3.)

Note 1. The random seed refers to the output value of TRNG.

Pseudo random numbers are generated using a dedicated context. During this processing, the encryption/decryption operations for the other contexts are maintained.

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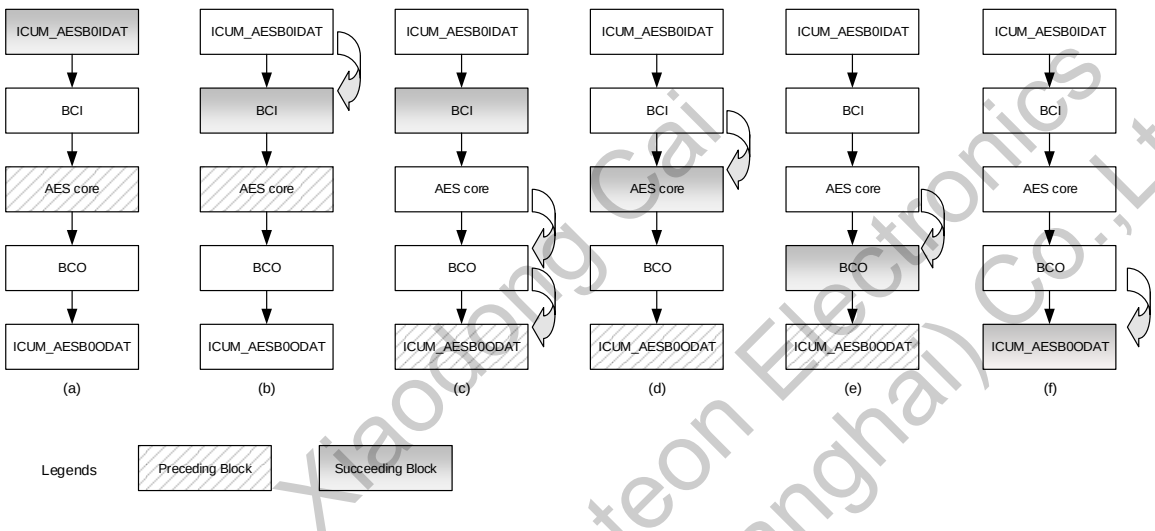
9.1.4.2 Buffering Function for Input/Output Data

The ICUM_AESB0 has a buffering function for input/output data.

The ICUM_AESB0 has one block (128-bit) buffer on the writing side. When the next input block data is written into BCI during encryption/decryption processing, it is retained in BCI until the processing is completed. The processing of this data starts after the completion of the current encryption/decryption processing.

The ICUM_AESB0 also has two-block (256-bit) buffer on the reading side, so that the results of processing on the next block data are not lost even when the encryption/decryption processing on that block data terminates while the current block data is being read.

The outline of buffering processing is shown in **Figure 9.32**.



Explanation
(a) While AES core operates encryption/decryption processing of preceding block, SW writes succeeding block to ICUM_AESB0IDAT.
(b) While AES core operates encryption/decryption processing of preceding block, 128bit succeeding block data is transferred to BCI.
(c) After AES core completes encryption/decryption processing of preceding block, calculation result of preceding block is transferred to BCO and ICUM_AESB0ODAT.
(d) Succeeding block data is transferred to AES core from BCI. AES core starts encryption/decryption processing of succeeding block.
(e) After AES core completes encryption/decryption processing of succeeding block, calculation result of succeeding block is transferred to BCO.
(f) After calculation result of preceding block is read from ICUM_AESB0ODAT, calculation result of succeeding block is transferred to ICUM_AESB0ODAT.

Figure 9.32 Outline of Buffering Processing

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9.1.4.3 Suspension and Resumption of Encryption/Decryption processing

The ICUM_AESB0 implements the functions to suspend and resume encryption/decryption processing by reading and storing the intermediate value from the AES core via ICUM_AESB0ODAT or ICUM_AESB0DODi and setting it up via ICUM_AESB0IDAT or ICUM_AESB0DIDi at a later time. Table 9-22 shows the target register to be read/written with the suspension for each block cipher mode.

Table 9.22 Target register to be read/written with the suspension for each block cipher mode.

Block cipher mode	Suspension and resumption processing	Register to be read	Register to be written
ECB	Support	-	-
CBC	Support	IV	IV
CTR	Support	IV	IV
CFB	Support	IV	IV
OFB	Support	IV	IV
GCM	Support	IV, GHASH	IV, GHASH
XTS	Support*1	IV	IV
CCM	Not support*1	IV	IV
CMAC	Support	IV	IV
Miyaguchi Preneel	Support	IV	IV

Note 1. Refer to 9.1.6.6 for detail.

The ICUM_AESB0 is able to process another stream data while the encryption/decryption processing on the current stream data is suspended (after reading its intermediate value) and resume the encryption/ decryption processing on the suspended stream data (by rewriting its intermediate value).

Operation examples are given in the following two cases:

- Suspension and resumption in the same context (common key)
- Suspension and resumption in the same context

Table 9.23 Suspension and Resumption in the Same Context (Common Key)

	Data Stream	
—	#1	#2
Block cipher mode	CFB encryption	
Context number	CTX[2:0] = 000_B	
Key	KEY[127:0](Set up via ICUM_AESB0IDAT)	
Initialization vector	IV_#1 [127:0]	IV_#2 [127:0]
Input data	PLAIN_#1 [127:0] (plain text)	PLAIN_#2 [127:0] (plain text)
Output data	CIHPER_#1 [127:0] (cipher text)	CIPHER_#2 [127:0] (cipher text)
Intermediate value	IVresume_#1	—

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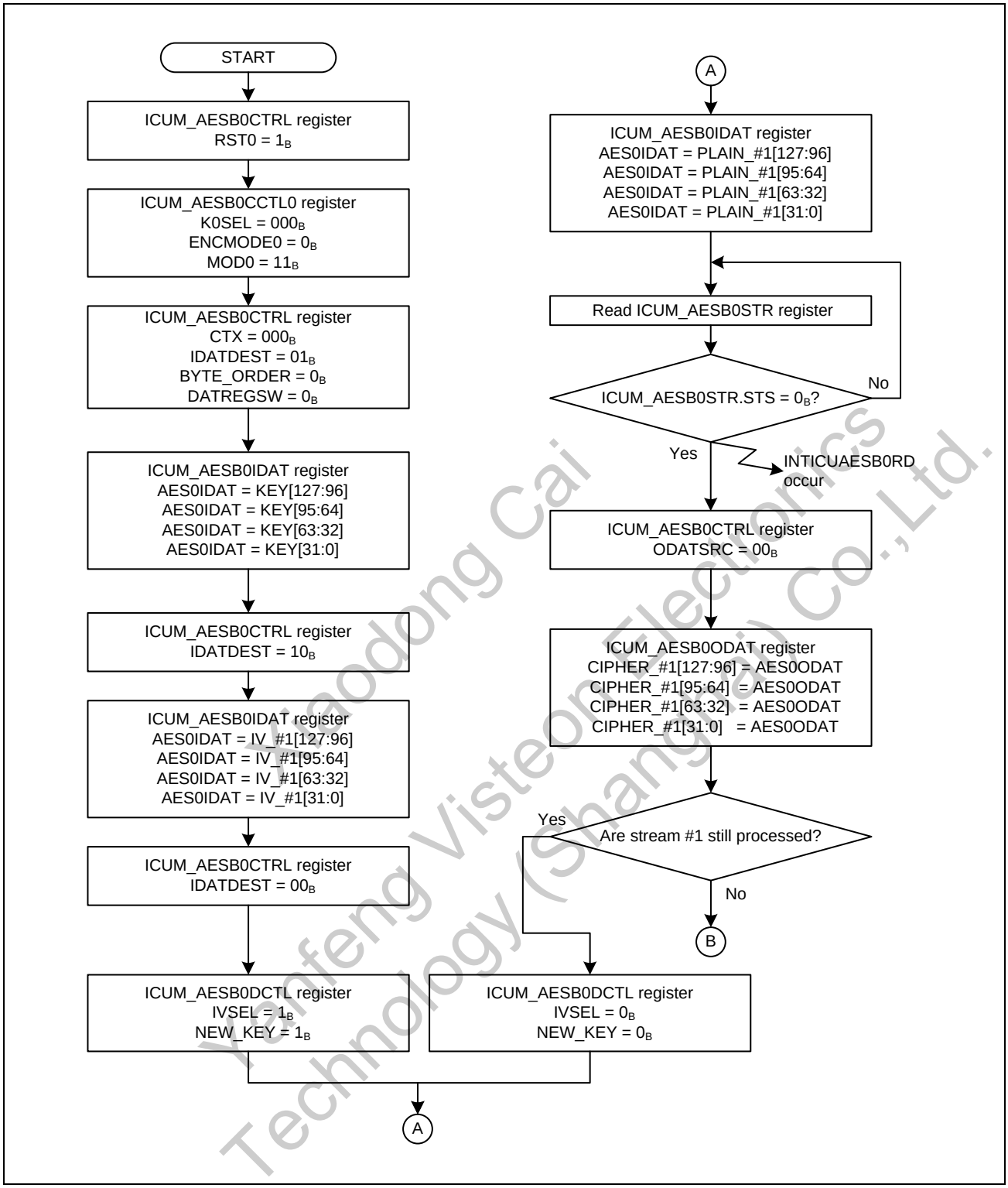
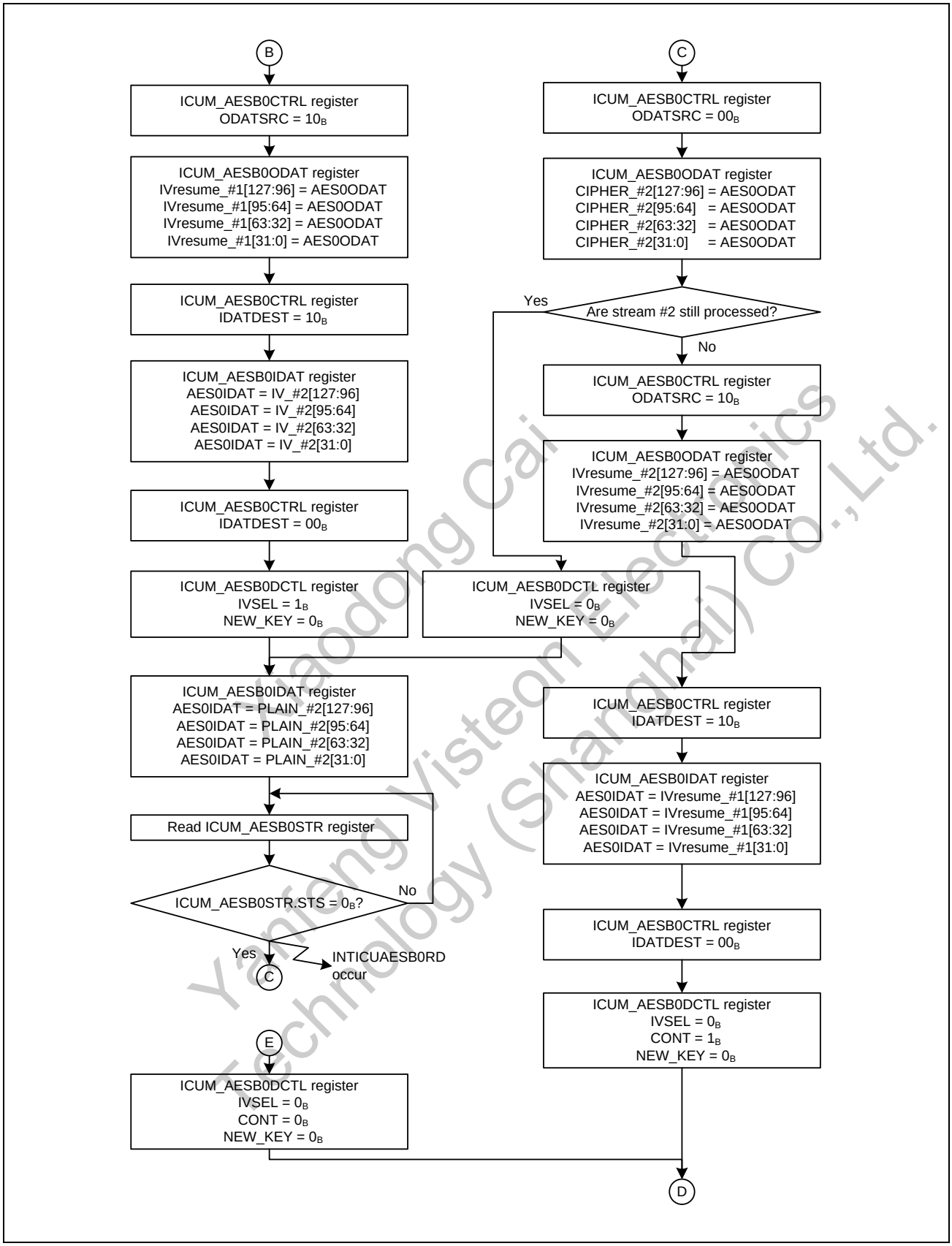


Figure 9.33 Flow for Suspension and Resumption in the Same Context (Common Key) (1 of 3)

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Figure 9.34 Flow for Suspension and Resumption in the Same Context (Common Key) (2 of 3)

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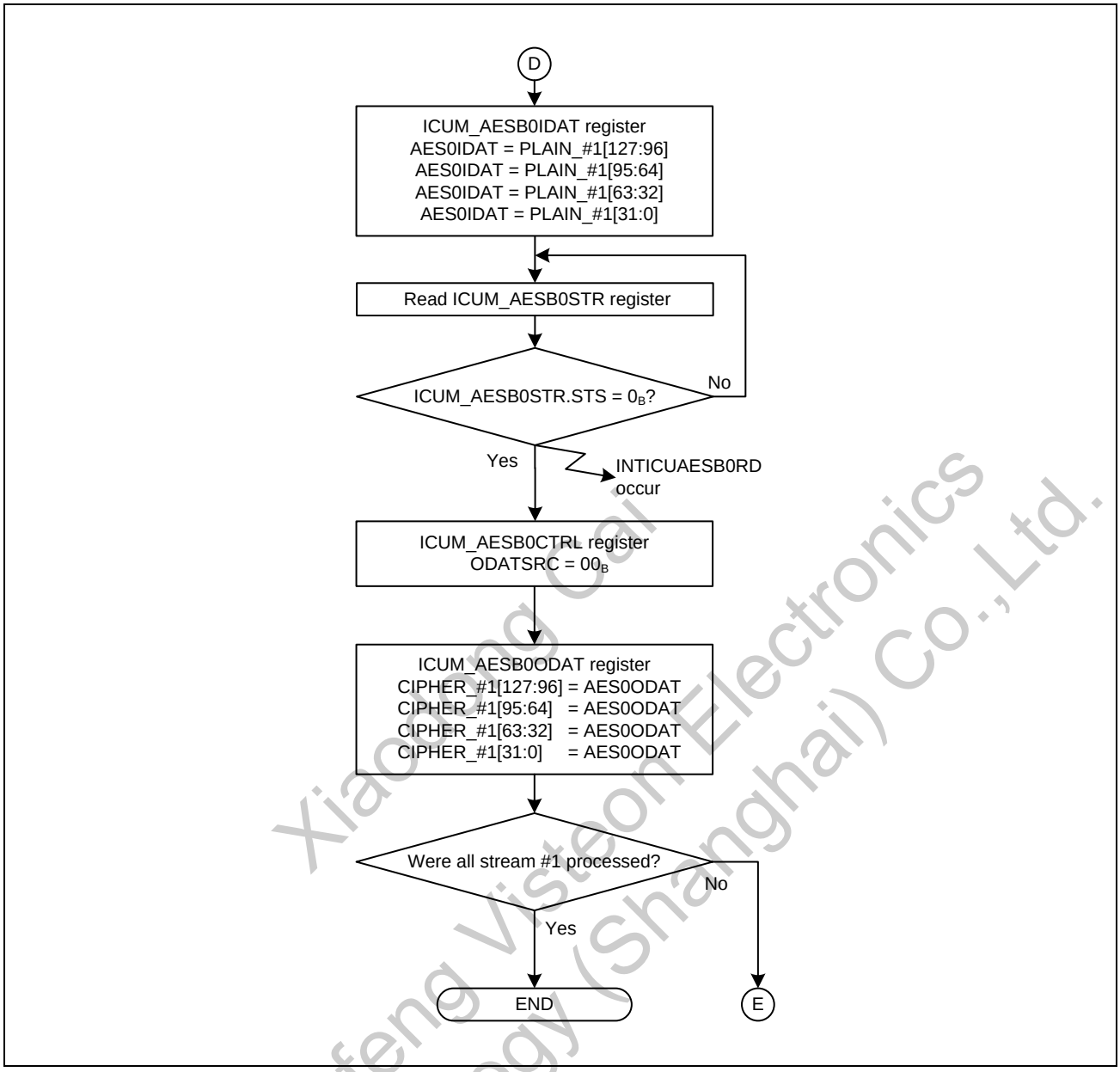


Figure 9.35 Flow for Suspension and Resumption in the Same Context (Common Key) (3 of 3)

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Table 9.24 Suspension and Resumption in the Same Context

	Data Stream	
—	#1	#2
Block cipher mode	CFB encryption	CFB encryption
Context number	CTX[2:0] = 000_B	
Key	KEY_#1 [127:0] (Set up via ICUM_AESB0IDAT)	KEY_#2 [127:0] (Set up via ICUM_AESB0IDAT)
Initialization vector	IV_#1 [127:0]	IV_#2 [127:0]
Input data	PLAIN_#1 [127:0] (plain text)	PLAIN_#2 [127:0] (plain text)
Output data	CIHPER_#1 [127:0] (cipher text)	CIPHER_#2 [127:0] (cipher text)
Intermediate value	IVresume_#1	—

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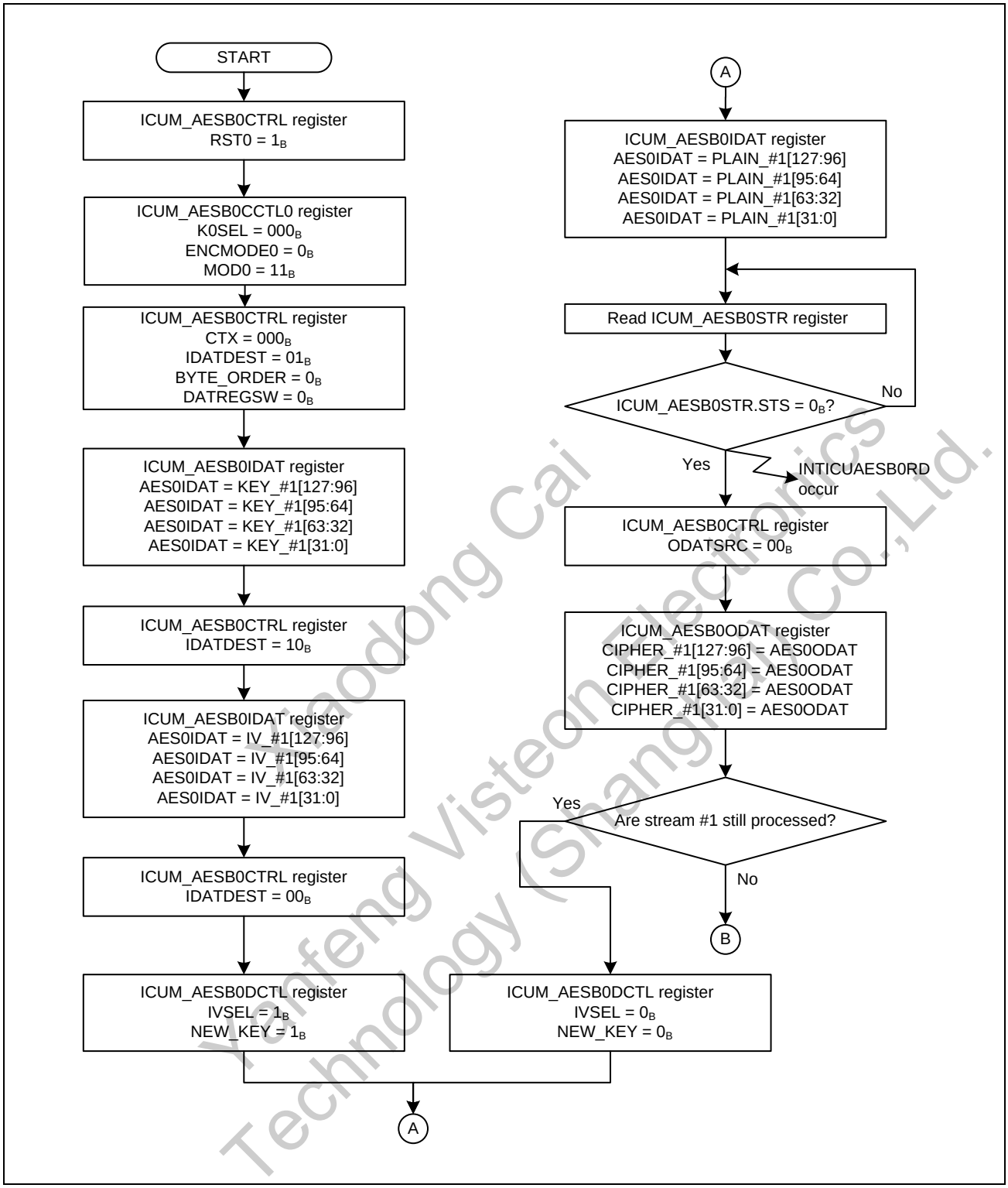


Figure 9.36 Flow for Suspension and Resumption in the Same Context (1 of 3)

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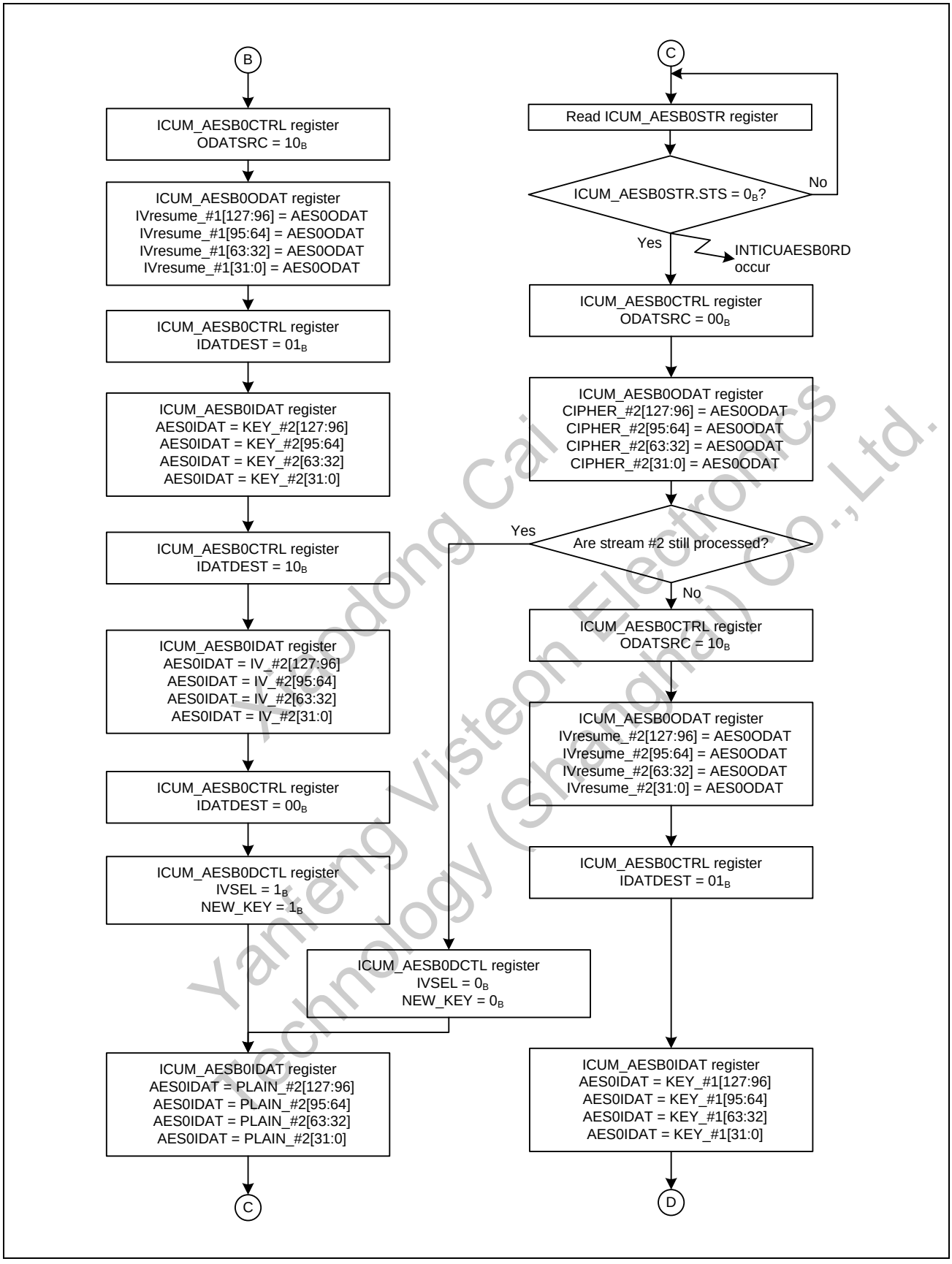


Figure 9.37 Flow for Suspension and Resumption in the Same Context (2 of 3)

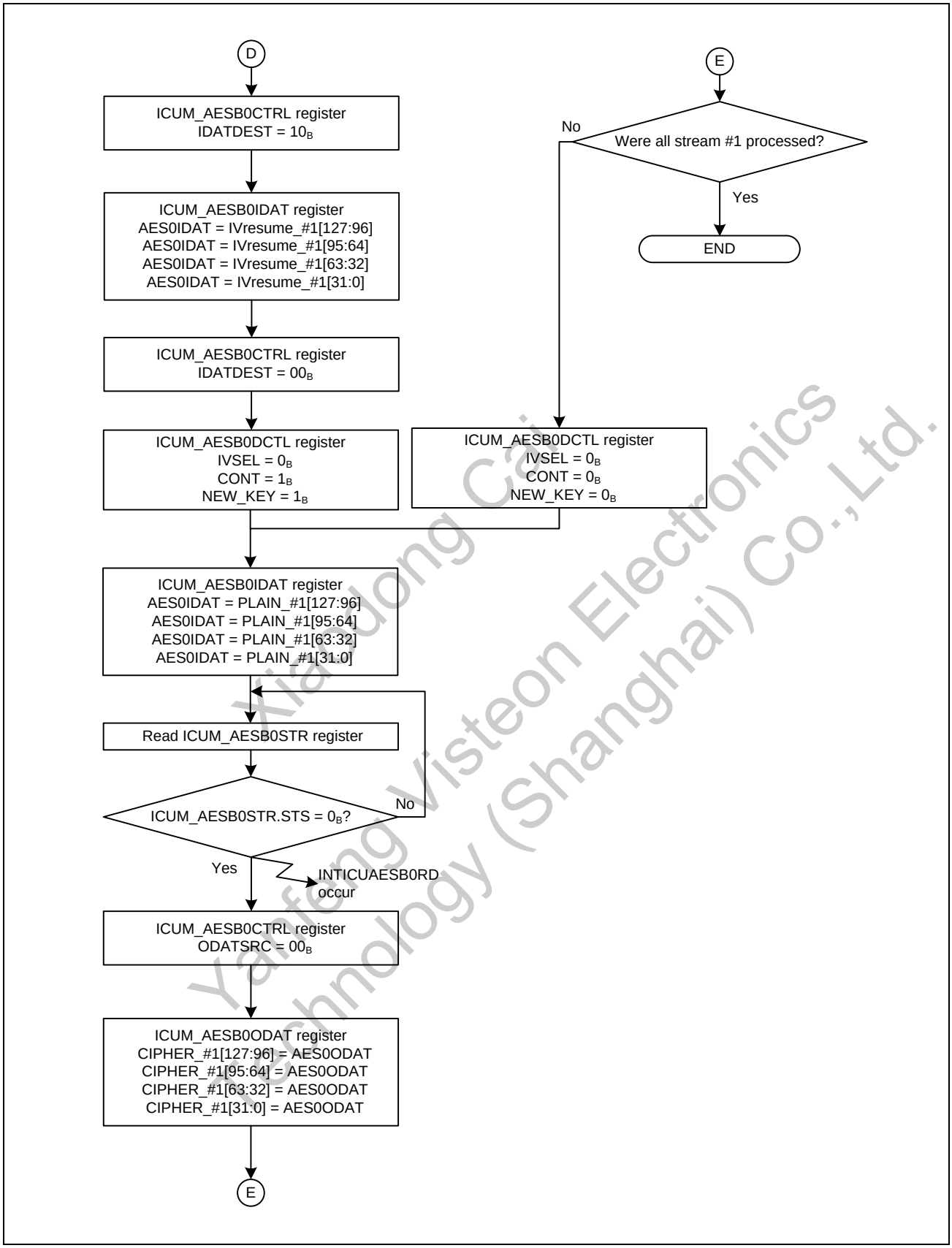


Figure 9.38 Flow for Suspension and Resumption in the Same Context (3 of 3)

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9.1.5 Data Format

The ICUM_AESB0 handles the data in each 128-bit internally. The data is set by accessing ICUM_AESB0IDAT/ICUM_AESB0DIDi or read by accessing ICUM_AESB0ODAT/ICUM_AESB0DODi. When writing to ICUM_AESB0IDAT/ICUM_AESB0DIDi or reading from ICUM_AESB0ODAT/ICUM_AESB0DODi, ordering of 32-bit unit in 128-bit and ordering of bytes in 32-bit vary depending on the data type.

When writing the intermediate value (e.g. length of AAD or length of Cipher in GCM) to ICUM_AESB0IDAT, 1st write must be most significant 32-bit. When writing the stream data to ICUM_AESB0DIDi, most significant 32-bit must be written to ICUM_AESB0DID0. ICUM_AESB0CTRL.BYTE_ORDER must be set to 1.

When writing the data (e.g. key data, initialization vector/counter value, encrypted or decrypted data), ordering of 32-bit unit in 128-bit and ordering of bytes in 32-bit vary depending on endianness of data which allocates on the memory.

- Data allocates in big endian format.

  Set 0 to the bit of ICUM_AESB0CTRL.BYTE_ORDER to convert byte ordering. Then write to ICUM_AESB0IDAT/ICUM_AESB0DIDi. Shuffling of 32-bit data by software is not needed. **Figure 9.39** indicates the writing procedure in case of big endian format data.

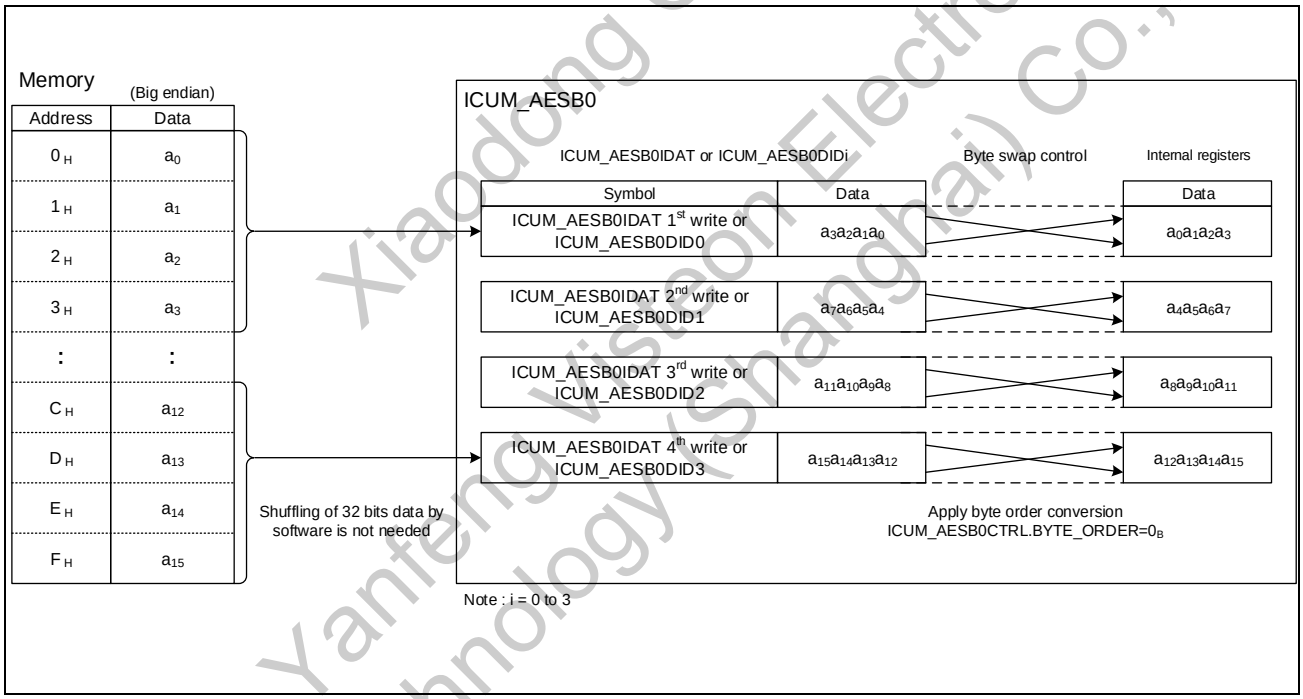


Figure 9.39 Data allocates in big endian format

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- Data allocates in little endian format.

  Set 1 to the bit of ICUM_AESB0CTRL.BYTE_ORDER. Then write to ICUM_AESB0IDAT/ICUM_AESB0DIDi. Shuffling of 32-bit data by software is needed. **Figure 9.40** indicates the writing procedure in case of little endian format data.

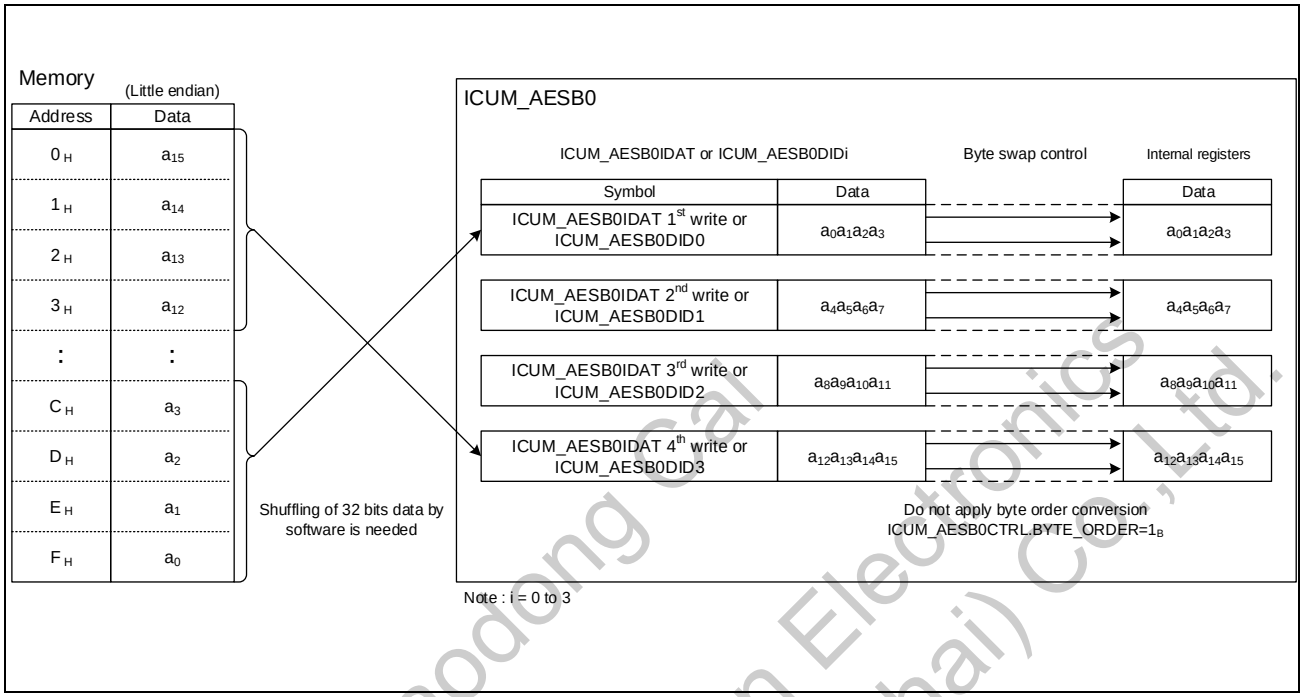


Figure 9.40 Data allocates in little endian format

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9.1.6 Notes of the Use of ICUM_AESB0

9.1.6.1 Writing Data into ICUM_AESB0IDAT

Do not write accesses to the ICUM_AESB0CTRL, ICUM_AESB0DCTL, ICUM_AESB0CCTL0, or ICUM_AESB0CCTL1 register during four times of write accesses to the ICUM_AESB0IDAT register, that is, to set a block of the 128-bit data.

Writing counts for ICUM_AESB0IDAT is cleared whenever a write access is made to these registers. In such a case, therefore, rewrite the whole 128-bit starting at the first 32-bit.

9.1.6.2 Reading Data from ICUM_AESB0ODAT

Do not write accesses to the ICUM_AESB0CTRL, ICUM_AESB0DCTL, ICUM_AESB0CCTL0, or ICUM_AESB0CCTL1 register during four times of read accesses to the ICUM_AESB0ODAT register.

Reading counts for ICUM_AESB0ODAT is cleared whenever a write access is made to these registers. In such a case, the first 32-bit of 128-bit is output when the register is read.

9.1.6.3 Writing IV_CTR_i (i: 0 to 2) or IV_PRNG during Encryption/Decryption Processing

No write shall be made into IV_CTR_i or IV_PRNG during encryption/decryption processing in context #i. Any write attempts shall be made after encryption/decryption processing in context #i is completed.

9.1.6.4 Decryption Operation in XTS Mode

If the bit length of the last block is found to be less than 128-bit during decryption processing in XTS mode, the last 2 blocks of the output data (plain text) are generated in reverse order. Adequate care should be exercised when handling the output data results.

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9.1.6.5 Buffering Function for Input/Output Data

No more than 2 blocks of data can be retained including that which is being processed among the BCI, AES core, and BCO. Reads and writes shall be performed in such a manner that the number of retained data blocks will never exceed this value.

9.1.6.6 Suspension and Resumption of Encryption/Decryption Processing

The following care shall be exercised with respect to the suspension and resumption of encryption/decryption processing:

- If the bit length of the last block that is being subjected to XTS mode decryption processing is less than 128 bits, no suspension must occur between the processing on the last block and that on the preceding block.
- When CCM is selected as Block Cipher Mode, suspension and resumption functions are prohibited. If another sequence starts while one CCM sequence is processing, CCM sequence cannot be resumed.

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9.2 True Random Number Generator (ICUM_TRNGC0)

The ICUM_TRNGC0 is a true random number generator.

9.2.1 Features of ICUM_TRNGC0

9.2.1.1 Number of Units and Channels

ICUMD has one ICUM_TRNGC0 channel.

9.2.1.2 Register Base Address

The ICUM_TRNGC0 base addresses are listed in the table below.

Table 9.25 Register Base Address

Base Address Name	Base Address
<ICUM_TRNGC0_base>	FFFE $E0A0_H$

9.2.1.3 Clock Supply

The ICUM_TRNGC0 clock supply is listed in the table below.

Table 9.26 Clock Supply

Unit Name	Unit Clock Name	Supply Clock Name
ICUM_TRNGC0	PCLK	CPUCLK_L

9.2.1.4 Interrupt Requests

The ICUM_TRNGC0 interrupt requests are listed in the table below.

Table 9.27 Interrupt Requests

Unit Interrupt Signal	Description	Interrupt Number	DMA Trigger Number
ICUM_TRNGC0			
INTICUTRNGC0	Random number generation complete	6	—

9.2.1.5 Reset Sources

The ICUM_TRNGC0 reset sources are listed in the table below. The ICUM_TRNGC0 is initialized by these reset sources.

Table 9.28 Reset Sources

Unit Name	Reset Source
ICUM_TRNGC0	All reset sources (ISORES)

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9.2.2 Overview

9.2.2.1 Functional overview

The ICUM_TRNGC0 has the following functions.

- 32-bit length random number generation.
- 128-bit length random number generation in buffer mode
- Conforming AIS (Application notes Interpretation of the Scheme) 31
- Self-diagnostic test function
  - Fault detection of the random number source
  - Online test function based on AIS31
- Serial random number generation
- Continuous random number generation
- Random number generation complete interrupt

9.2.2.2 Pseudo-Random Number Generation (PRNG)

A high throughput pseudo-random number generation (PRNG) is implemented using the ICUM_AESB0 in combination with the ICUM_TRNGC0 to get a true random seed.

The ICUM_AESB0 is configured as follows:

- [PRNG_KEY] = (predefined value or random seed)
- [IV_PRNG] = (random seed)
- [BCI] = 0
- Block cipher mode: OFB encryption

The 128-bit random seeds described in the above configuration are read from the ICUM_TRNGC0.

Pseudo random values are read-out from the BCO register in the ICUM_AESB0 by setting the CTX[2:0] bits to Context #3 (leaving 3 contexts available for other block cipher operations).

The generated output from this PRNG is compliant with AIS20 (Class K2 high).

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9.2.3 Registers

9.2.3.1 List of Registers

The ICUM_TRNGC0 registers are listed in the table below.

Table 9.29 List of Registers

Module Name	Register Name	Symbol	Address
ICUM_TRNGC0	TRNG status register	ICUM_TRNGC0SR	<ICUM_TRNGC0_base> + 0000_H
	TRNG status clear register	ICUM_TRNGC0SCLR	<ICUM_TRNGC0_base> + 0004_H
	TRNG control register	ICUM_TRNGC0CR	<ICUM_TRNGC0_base> + 0008_H
	TRNG random number result register	ICUM_TRNGC0RNRR	<ICUM_TRNGC0_base> + $000C_H$
	TRNG version register	ICUM_TRNGC0VER	<ICUM_TRNGC0_base> + $001C_H$

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9.2.3.2 ICUM_TRNGC0SR — TRNG Status Register

This registers indicates ICUM_TRNGC0 status.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_TRNGC0_base> + 0000_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	RCHKERR	RCHKS	RRDY
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 9.30 ICUM_TRNGC0SR Register Contents (1/2)

Bit Position	Bit Name	Function
31 to 3	—	Reserved When read, the value after reset is returned.
2	RCHKERR	TRNG self-diagnostic test error status bit Indicates self-diagnostic test result. 0: Self-diagnostic test is not failed. 1: Self-diagnostic test is failed. [Cleaning condition] ICUM_TRNGC0SCLR.RCHKCLR is set to 1. [Setting condition] Self-diagnostic test is failed.
1	RCHKS	TRNG self-diagnostic test status bit Indicates self-diagnostic test status. 0: Self-diagnostic test is not finished. 1: Self-diagnostic test is finished. [Cleaning condition] ICUM_TRNGC0CR.GE is set to 0. ICUM_TRNGC0SCLR.RCHKCLR is set to 1. [Setting condition] Self-diagnostic test is finished.

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Table 9.30 ICUM_TRNGC0SR Register Contents (2/2)

Bit Position	Bit Name	Function
0	RRDY	Random number ready bit Indicates whether the random number generated by the TRNG has been written into ICUM_TRNGC0RNRR. 0: No random number is stored in ICUM_TRNGC0RNRR. 1: A random number is stored in ICUM_TRNGC0RNRR. [Clearing conditions] A 32-bit random number is read when ICUM_TRNGC0CR.MOD is set to 0. A 128-bit random number is read when ICUM_TRNGC0CR.MOD is set to 1. 0 is written into ICUM_TRNGC0CR.GE. 1 is written into ICUM_TRNGC0SCLR.RRDYCLR. [Setting conditions] A 32-bit random number data is stored in ICUM_TRNGC0RNRR and ICUM_TRNGC0CR.MOD is set to 0. A 128-bit random number data is stored in ICUM_TRNGC0RNRR and ICUM_TRNGC0CR.MOD is set to 1.

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9.2.3.3 ICUM_TRNGC0SCLR — TRNG Status Clear Register

This register clears the status bits of ICUM_TRNGC0SR.

Access: This register is a write-only register that can be written in 32-bit units.

Address: <ICUM_TRNGC0_base> + 0004_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RCHKCLR	RRDYCLR
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	W	W

Table 9.31 ICUM_TRNGC0SCLR Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When writing, write the value after reset.
1	RCHKCLR	TRNG self-diagnostic test error status clear bit Writing 1 to this bit clears ICUM_TRNGC0SR.RCHKS and ICUM_TRNGC0SR.RCHKERR.
0	RRDYCLR	Random number ready clear bit Writing 1 to this bit clears ICUM_TRNGC0SR.RRDY.

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9.2.3.4 ICUM_TRNGC0CR — TRNG Control Register

This registers controls generation of random numbers.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_TRNGC0_base> + 0008_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	RCMSEL	RCHKE	MOD	GE
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R/W	R/W	R/W	R/W

Table 9.32 ICUM_TRNGC0CR Register Contents

Bit Position	Bit Name	Function
31 to 4	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
3	RCMSEL	TRNG self-diagnostic test mode selection bit Specifies evaluation criteria of the online test function as usage of generated random number. 0: AIS31 class P1 criteria Sample usage: Challenge-response protocol 1: AIS31 class P2 criteria Sample usage: Seed for pseudo random number generation of key generation During TRNG operation (GE = 1), writing to this bit is prohibited.
2	RCHKE	TRNG self-diagnostic test enable bit Specifies whether execute the self-diagnostics testing or not. 0: Self-diagnostic testing is disable 1: Self-diagnostic testing is enable During TRNG operation (GE = 1), writing to this bit is prohibited.
1	MOD	TRNG mode selection enable bit Specifies the operation mode of ICUM_TRNGC0RNRR. 0: Normal mode ICUM_TRNGC0RNRR serves as a 32-bit read-only register. 1: Buffer mode ICUM_TRNGC0RNRR serves as a 32-bit x 4 read-only register. This bit must be set up when the GE bit is set to 0.
0	GE	Generate enable bit Specifies whether the ICUM_TRNGC0 starts the random number generation or not. 0: The ICUM_TRNGC0 is stopped. 1: The ICUM_TRNGC0 starts the random number generation.

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9.2.3.5 ICUM_TRNGC0RNRR — TRNG Random Number Result Register

This registers is used for reading out the generated random number. It is made up of a 32-bit × 4 FIFO.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_TRNGC0_base> + $000C_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	RNRR[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	RNRR[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 9.33 ICUM_TRNGC0RNRR Register Contents

Bit Position	Bit Name	Function
31 to 0	RNRR[31:0]	Random number result data The random number generated by the ICUM_TRNGC0 is written into RNRR[31:0]. RNRR[31:0] is valid as a random number when the ICUM_TRNGC0 is running (GE = 1) and the ICUM_TRNGC0SR.RRDY bit is set to 1. A 32-bit random number can be obtained from this register if ICUM_TRNGC0CR.MOD is set to 0. Reading RNRR[31:0] once clears ICUM_TRNGC0SR.RRDY. After that, the generation of a new random number starts. If ICUM_TRNGC0CR.MOD is set to 1, a 128-bit random number can be obtained by reading RNRR[31:0] 4 times. Reading RNRR[31:0] 4 times clears ICUM_TRNGC0SR.RRDY. After that, the generation of a new random number starts.

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9.2.3.6 ICUM_TRNGC0VER — TRNG Version Register

This register indicates TRNG unit version.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_TRNGC0_base> + $001C_H$

Value after reset: $0000\ 0001_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	TRNGVER[3:0]			
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 9.34 ICUM_TRNGC0VER Register Contents

Bit Position	Bit Name	Function
31 to 4	—	Reserved When read, the value after reset is returned.
3 to 0	TRNGVER[3:0]	TRNG unit version bits Indicate TRNG unit version 0001: TRNG unit is the ICUM_TRNGC0

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9.2.4 Operation

9.2.4.1 Basic Operation

(1) Random number generation (Normal mode)

The flow of the random number generation (Normal mode) is shown in **Figure 9.41**.

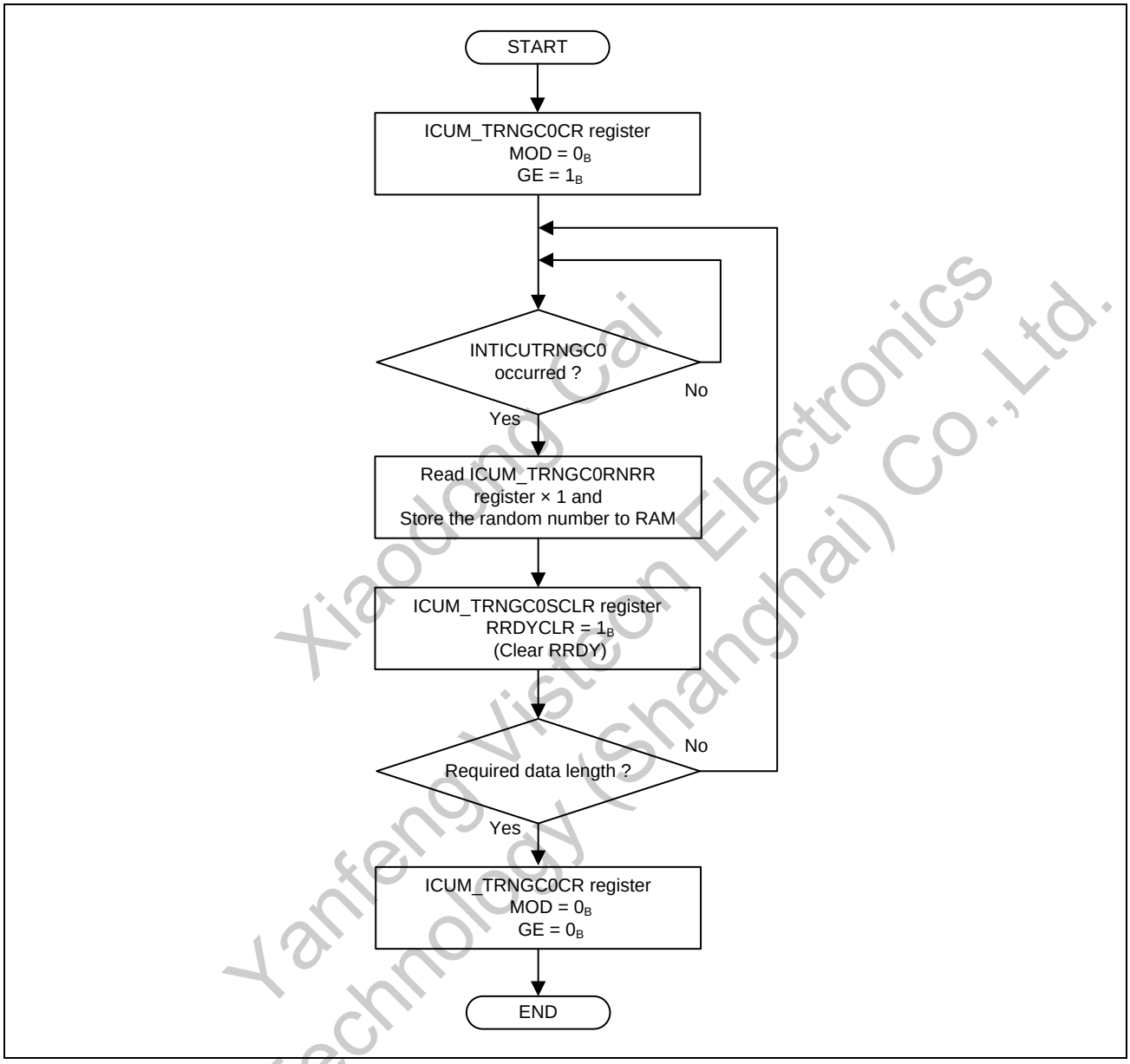


Figure 9.41 Random number generation (Normal mode)

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(2) Random number generation (Buffer mode)

The flow of the random number generation (Buffer mode) is shown in **Figure 9.42**.

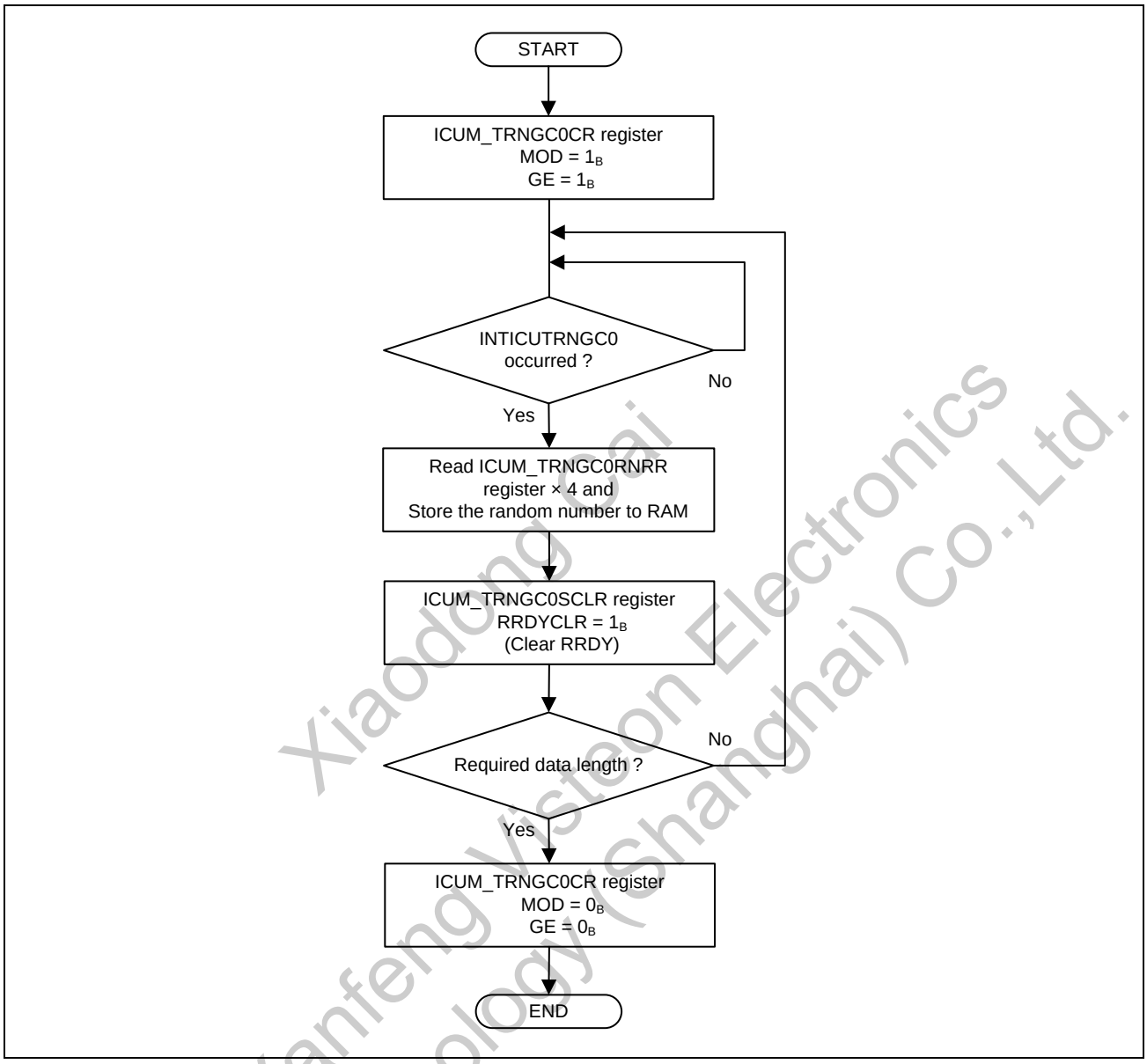


Figure 9.42 Random number generation (Buffer mode)

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9.2.4.2 Self-diagnostic testing mode (AIS31 class P2 criteria)

The flow of the self-diagnostic testing mode (AIS31 class P2 criteria) is shown in **Figure 9.43**.

A description of "Error handling" in Figure 9.43 is described below.
Even if ICUM_TRNGC0 has no failure, there is a possibility that the self-diagnostic test judges the random data generated by ICUM_TRNGC0 to be fail. Therefore, perform the self-diagnostic test again to generate a random number based on AIS31.

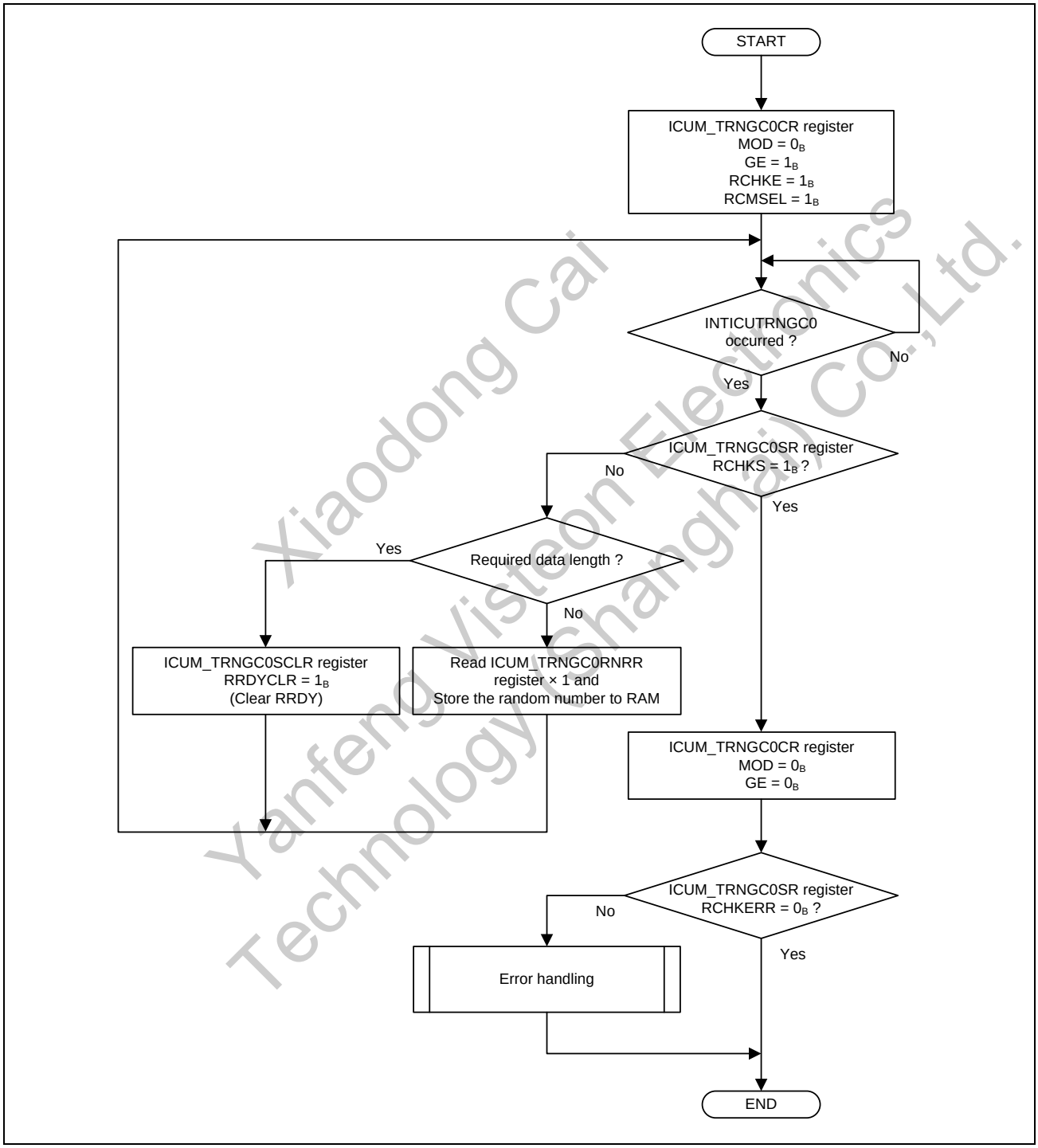


Figure 9.43 Self-diagnostic testing mode (AIS31 class P2 criteria)

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Section 10 ICUMD Control Register (ICUM_CTLREG)

10.1 Features of ICUM_CTLREG

10.1.1 Number of Units and Channels

ICUMD has one ICUM_CTLREG channel.

10.1.2 Register Base Address

The ICUM_CTLREG base address are listed in the table below.

Table 10.1 Register Base Address

Base Address Name	Base Address
<ICUM_CTLREG_base>	FFFE $E200_H$

10.1.3 Clock Supply

The ICUM_CTLREG clock supply is listed in the table below.

Table 10.2 Clock Supply

Unit Name	Unit Clock Name	Supply Clock Name
ICUM_CTLREG	PCLK	CPUCLK_L

10.1.4 Reset Sources

The ICUM_CTLREG reset sources are listed in the table below. ICUM_CTLREG is initialized by these reset sources.

Table 10.3 Reset Sources

Unit Name	Reset Source
ICUM_CTLREG	All reset sources (ISORES)

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10.2 Overview

10.2.1 Functional Overview

The ICUM_CTLREG has the following functions.

- Remap control
- Upgrade protection control
- Dedicated GPIO control
- Challenge & Response authentication control
- Debug I/F control
- STOP mode control
- Handshake control for Non-secure flash programming
- ICUMD-related configuration data reference

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10.3 Registers

10.3.1 List of Registers

The ICUM_CTLREG registers are listed in the table below.

Table 10.4 List of Registers

Module Name	Register Name	Symbol	Address
ICUM_CTLREG	SIC (System Inter Connect) Remap area register	ICUM_SICREMAP	<ICUM_CTLREG_base> + 0000_H
	Upgrade protection register	ICUM_UPGPS	<ICUM_CTLREG_base> + 0004_H
	GPIO control register	ICUM_GPIO	<ICUM_CTLREG_base> + 0008_H
	Challenge data register 0	ICUM_CDAT0	<ICUM_CTLREG_base> + 0010_H
	Challenge data register 1	ICUM_CDAT1	<ICUM_CTLREG_base> + 0014_H
	Challenge data register 2	ICUM_CDAT2	<ICUM_CTLREG_base> + 0018_H
	Challenge data register 3	ICUM_CDAT3	<ICUM_CTLREG_base> + $001C_H$
	Response data register 0	ICUM_RDAT0	<ICUM_CTLREG_base> + 0020_H
	Response data register 1	ICUM_RDAT1	<ICUM_CTLREG_base> + 0024_H
	Response data register 2	ICUM_RDAT2	<ICUM_CTLREG_base> + 0028_H
	Response data register 3	ICUM_RDAT3	<ICUM_CTLREG_base> + $002C_H$
	Debug I/F control register	ICUM_IDAS	<ICUM_CTLREG_base> + 0030_H
	STOP mode register	ICUM_STOP	<ICUM_CTLREG_base> + 0040_H
	ICUMD lock control register	ICUM_LOCKCTRL	<ICUM_CTLREG_base> + 0048_H
	Flash programming request register 0	ICUM_FLPRG0REQ	<ICUM_CTLREG_base> + 0050_H
	Flash programming request acknowledge register 0	ICUM_FLPRG0ACK	<ICUM_CTLREG_base> + 0058_H
	ICUMD configuration register 0	ICUM_CFG0	<ICUM_CTLREG_base> + 0060_H
	ICUMD configuration register 1	ICUM_CFG1	<ICUM_CTLREG_base> + 0064_H
	ICUMD configuration register 2	ICUM_CFG2	<ICUM_CTLREG_base> + 0068_H
	ICUMD configuration register 3	ICUM_CFG3	<ICUM_CTLREG_base> + $006C_H$
	ICUMD configuration register 4	ICUM_CFG4	<ICUM_CTLREG_base> + 0070_H
	Product name storage register 1	ICUM_PRDNAME1	<ICUM_CTLREG_base> + 0078_H
	Product name storage register 2	ICUM_PRDNAME2	<ICUM_CTLREG_base> + $007C_H$
	Product name storage register 3	ICUM_PRDNAME3	<ICUM_CTLREG_base> + 0080_H
	Chip ID register 1 LL	ICUM_CHIPID1_LL	<ICUM_CTLREG_base> + 0088_H
	Chip ID register 1 LH	ICUM_CHIPID1_LH	<ICUM_CTLREG_base> + $008C_H$
	Chip ID register 1 HL	ICUM_CHIPID1_HL	<ICUM_CTLREG_base> + 0090_H
	Chip ID register 1 HH	ICUM_CHIPID1_HH	<ICUM_CTLREG_base> + 0094_H

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10.3.2 ICUM_SICREMAP — SIC Remap Area Register

This register holds the higher-order address of the Remap function. For the details of the Remap area, see **Section 3.3.1, Remap Area**.

The correct write sequence using the ICUM_PROT0PCMD register is required in order to update this register. For details, see **Section 4**, **Write-Protection Registers**.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CTLREG_base> + 0000_H

Value after reset: $FC00\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	SICREMAP[6:0]							—	—	—	—	—	—	—	—	—
Value after reset	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 10.5 ICUM_SICREMAP Register Contents

Bit Position	Bit Name	Function
31 to 25	SICREMAP[6:0]	Specify MSB part of the start address of the memory area to map within the SIC (System Inter Connect) Remap area (see **Table 3.1**, **ICUMD Address Space (F1KH-D8 8-MB 324 Pin Product)** and **Table 3.2, ICUMD Address Space (F1KM-S4 4-MB 272 Pin Product)**).
24 to 0	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.

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10.3.3 ICUM_UPGPS — Upgrade Protection Register

This register temporarily releases upgrade protection.

For information on upgrade protection, see **Section 16.1.1, Functional Overview**.

The correct write sequence using the ICUM_PROT0PCMD register is required in order to update this register.

For details, see **Section 4, Write-Protection Registers**.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CTLREG_base> + 0004_H

Value after reset: $0000\ 0003_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	UP2EN	UP1EN
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R/W	R/W

Table 10.6 ICUM_UPGPS Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
1	UP2EN	Upgrade protection 2 setting*1 0: Disable 1: Enable
0	UP1EN	Upgrade protection 1 setting*2 0: Disable 1: Enable

Note 1. To enable the upgrade protection 2, the UPGP2 bit in ICUMD configuration register 0 must be 0_H.

Note 2. To enable the upgrade protection 1, the UPGP1 bit in ICUMD configuration register 0 must be 0_H.

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10.3.4 ICUM_GPIO — GPIO Control Register

This register is used to change the output function from general-purpose port pin to output pin under ICUMD control (i.e. pin for the security output function). This register is also used to enable or disable output from each pin and to set the level of each pin when it is selected for the security output function.

Available pins depend on each products.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CTLREG_base> + 0008_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	CTLEN3	EOUT3	DOUT3	—	—	—	—	—	CTLEN2	EOUT2	DOUT2
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R/W	R/W	R/W	R	R	R	R	R	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	CTLEN1	EOUT1	DOUT1	—	—	—	—	—	CTLEN0	EOUT0	DOUT0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R/W	R/W	R/W	R	R	R	R	R	R/W	R/W	R/W

Table 10.7 ICUM_GPIO Register Contents (1/2)

Bit Position	Bit Name	Function
31 to 27	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
26	CTLEN3	Selects the general-purpose port pin function or security output pin function. 0: General-purpose port pin function (P2_5) 1: Security output pin function (SECOUT3)
25	EOUT3	Selects the output mode of the SECOUT3 pin. 0: Output is disabled. 1: Output is enabled.
24	DOUT3	Sets the output level of the SECOUT3 pin. 0: Low level 1: High level
23 to 19	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
18	CTLEN2	Selects the general-purpose port pin function or security output pin function. 0: General-purpose port pin function (P2_4) 1: Security output pin function (SECOUT2)
17	EOUT2	Selects the output mode of the SECOUT2 pin. 0: Output is disabled. 1: Output is enabled.
16	DOUT2	Sets the output level of the SECOUT2 pin. 0: Low level 1: High level
15 to 11	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.

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Table 10.7 ICUM_GPIO Register Contents (2/2)

Bit Position	Bit Name	Function
10	CTLEN1	Selects the general-purpose port pin function or security output pin function. 0: General-purpose port pin function (P9_1) 1: Security output pin function (SECOUT1)
9	EOUT1	Selects the output mode of the SECOUT1 pin. 0: Output is disabled. 1: Output is enabled.
8	DOUT1	Sets the output level of the SECOUT1 pin. 0: Low level 1: High level
7 to 3	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
2	CTLEN0	Selects the general-purpose port pin function or security output pin function. 0: General-purpose port pin function (P9_0) 1: Security output pin function (SECOUT0)
1	EOUT0	Selects the output mode of the SECOUT0 pin. 0: Output is disabled. 1: Output is enabled.
0	DOUT0	Sets the output level of the SECOUT0 pin. 0: Low level 1: High level

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10.3.5 ICUM_CDATi — Challenge Data Register i (i = 0 to 3)

These registers are used to store challenge data of challenge & response authentication.

Access: This register can be read or written in 32-bit units.

Address: ICUM_CDAT0: <ICUM_CTLREG_base> + 0010_H

ICUM_CDAT1: <ICUM_CTLREG_base> + 0014_H

ICUM_CDAT2: <ICUM_CTLREG_base> + 0018_H

ICUM_CDAT3: <ICUM_CTLREG_base> + $001C_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICUMCDATi[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICUMCDATi[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Table 10.8 ICUM_CDATi Register Contents

Bit Position	Bit Name	Function
31 to 0	ICUMCDATi[31:0]	32 bits of the 128-bit challenge data to be passed to a debugger in challenge-and-response authentication. ● ICUM_CDAT0: Challenge Data[31:0] ● ICUM_CDAT1: Challenge Data[63:32] ● ICUM_CDAT2: Challenge Data[95:64] ● ICUM_CDAT3: Challenge Data[127:96]

CAUTION

Write these registers in the following order.

ICUM_CDAT0 → ICUM_CDAT1 → ICUM_CDAT2 → ICUM_CDAT3

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10.3.6 ICUM_RDATi — Response Data Register i (i = 0 to 3)

These registers are used to read response data of challenge & response authentication.

Access: This register is a read-only register that can be read in 32-bit units.

Address: ICUM_RDAT0: <ICUM_CTLREG_base> + 0020_H

ICUM_RDAT1: <ICUM_CTLREG_base> + 0024_H

ICUM_RDAT2: <ICUM_CTLREG_base> + 0028_H

ICUM_RDAT3: <ICUM_CTLREG_base> + $002C_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICUMRDATi[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICUMRDATi[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 10.9 ICUM_RDATi Register Contents

Bit Position	Bit Name	Function
31 to 0	ICUMRDATi[31:0]	32 bits of the 128-bit response data to be passed from a debugger in challenge-and-response authentication. ● ICUM_RDAT0: Response Data[31:0] ● ICUM_RDAT1: Response Data[63:32] ● ICUM_RDAT2: Response Data[95:64] ● ICUM_RDAT3: Response Data[127:96]

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10.3.7 ICUM_IDAS — Debug I/F Control Register

This register controls debug interface lock signals to reflect challenge & response authentication result.

The correct write sequence using the ICUM_PROT0PCMD register is required in order to update this register.

For details, see **Section 4, Write-Protection Registers**.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CTLREG_base> + 0030_H

Value after reset: Undefined

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	CDATSEL	—	—	—	—	—	—	—	IDLOCK
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
R/W	R	R	R	R	R	R	R	R/W	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	ICUMCRF2	ICUMCRF1	—	—	—	—	—	—	CRLOCK2	CRLOCK1
Value after reset	0	0	0	0	0	0	X*1	X*1	0	0	0	0	0	0	1	1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R/W	R/W

Table 10.10 ICUM_IDAS Register Contents

Bit Position	Bit Name	Function
31 to 25	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
24	CDATSEL	Challenge Data for C&R select flag 0: Challenge data for CPU (PE) 1: Challenge data for ICUP
23 to 17	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
16	IDLOCK	ID authentication status flag 0: Unlocked 1: Locked
15 to 10	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
9	ICUMCRF2	CPU (PE) C&R enable flag 0: C&R enable 1: C&R disable
8	ICUMCRF1	ICUP C&R enable flag 0: C&R enable 1: C&R disable
7 to 2	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
1	CRLOCK2	CPU (PE) debug interface lock bit*2 0: Unlocked (debugger successfully authenticated) 1: Locked (debugger authentication failed)
0	CRLOCK1	ICUP debug interface lock bit*2 0: Unlocked (debugger successfully authenticated) 1: Locked (debugger authentication failed)

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Note 1. The value after reset of each bit depends on the configuration setting data.

Note 2. This bit can only be changed from 1 to 0.

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10.3.8 ICUM_STOP — STOP Mode Register

This register enables the issuance of STOP mode or DeepSTOP mode requests to the ICUP and enables mode transition.

Setting the STPRQEN bit in this register to 1 enables issuance and of a STOP mode or DeepSTOP mode request to the ICUP.

The correct write sequence using the ICUM_PROT0PCMD register is required in order to update this register.

For details, see **Section 4, Write-Protection Registers**.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CTLREG_base> + 0040_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	STPRQ	STPRQEN
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R/W

Table 10.11 ICUM_STOP Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
1	STPRQ	STOP/DeepSTOP mode request flag 0: STOP or DeepSTOP mode is not requested. 1: STOP or DeepSTOP mode is requested.
0	STPRQEN	STOP/DeepSTOP mode transition enable 0: Requests for STOP or DeepSTOP mode are masked. 1: Requests for STOP or DeepSTOP mode are enabled. This bit can be set to 1 only when the ICUM_STOP.STPRQ is 1, and is cleared to 0 when the ICUM_STOP.STPRQ becomes 0.

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10.3.9 ICUM_LOCKCTRL — ICUMD Lock Control Register

This register controls for the accessibility of flash peripheral register.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CTLREG_base> + 0048_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	EEPRDCYCLWLK	—	—	—	BFASELRWLK
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R/W	R	R	R	R/W

Table 10.12 ICUM_LOCKCTRL Register Contents

Bit Position	Bit Name	Function
31 to 5	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
4	EEPRDCYCLWLK	Lock a write access to data flash wait cycle control register (EEPRDCYCL) 0: Not locked 1: Locked
3 to 1	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
0	BFASELRWLK	Lock a write access to BFA selection register (BFASELR) 0: Not locked 1: Locked

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10.3.10 ICUM_FLPRG0REQ — Flash Programming Request Register 0

This register monitors the issuance of flash programming requests.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CTLREG_base> + 0050_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	FENTDUPREQ	FENTCUPREQ
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 10.13 ICUM_FLPRG0REQ Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When read, the value after reset is returned.
1	FENTDUPREQ0	Data flash programming request 0: A request is not issued. 1: A request is issued.
0	FENTCUPREQ0	Code flash programming request 0: A request is not issued. 1: A request is issued.

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10.3.11 ICUM_FLPRG0ACK — Flash Programming Request Acknowledge Register 0

This register responds to issued flash programming requests.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CTLREG_base> + 0058_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	FENTDUPACKA0	FENTDUPACK0	FENTCUPACK0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R/W	R/W	R/W

Table 10.14 ICUM_FLPRG0ACK Register Contents

Bit Position	Bit Name	Function
31 to 3	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
2	FENTDUPACKA0	Automatic response setting for data flash programming requests If this bit is set to 0, the FENTDUPACK0 bit can be set. 0: No automatic response is set for programming requests. 1: An automatic response is set for programming requests.
1	FENTDUPACK0	Acknowledgment of data flash programming requests This bit is enabled only when the FENTDUPACKA0 bit is 0. 0: Programming requests are disabled. 1: Programming requests are enabled. FENTDUPACK0 can be set to 1 only when the ICUM_FLPRG0REQ.FENTDUPREQ0 is 1, and is cleared to 0 when the ICUM_FLPRG0REQ.FENTDUPREQ0 becomes 0.
0	FENTCUPACK0	Acknowledgment of code flash programming requests 0: Programming requests are disabled. 1: Programming requests are enabled. FENTCUPACK0 can be set to 1 only when the ICUM_FLPRG0REQ.FENTCUPREQ0 is 1, and is cleared to 0 when the ICUM_FLPRG0REQ.FENTCUPREQ0 becomes 0.

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10.3.12 ICUM_CFG0 — ICUMD Configuration Register 0

This register is used to reference the status of upgrade protection 1 and 2. This register references option byte 8 (OPBT8). For details, see **Section 16.8.1, OPBT8 — Option Byte 8**.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CTLREG_base> + 0060_H

Value after reset: Undefined

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	UPGP1[3:0]				—	—	—	—
Value after reset	0	0	0	0	1	1	1	1	X*1	X*1	X*1	X*1	1	1	1	1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	UPGP2[3:0]				—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	X*1	X*1	X*1	X*1	1	1	1	1	1	1	1	1	1	1	1	1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 10.15 ICUM_CFG0 Register Contents

Bit Position	Bit Name	Function
31 to 24	—	Reserved When read, the value after reset is returned.
23 to 20	UPGP1[3:0]	Upgrade protection 1 flag F_H: Protection is disabled 0_H: Protection is enabled Others: Setting is prohibited
19 to 16	—	Reserved When read, the value after reset is returned.
15 to 12	UPGP2[3:0]	Upgrade protection 2 flag F_H: Protection is disabled 0_H: Protection is enabled Others: Setting is prohibited
11 to 0	—	Reserved When read, the value after reset is returned.

Note 1. The value after reset of each bit depends on the configuration setting data.

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10.3.13 ICUM_CFG1 — ICUMD Configuration Register 1

This register is used to see the reset vector of ICUP. This register references option byte 9 (OPBT9). For details, see **Section 16.8.2, OPBT9 — Option Byte 9**.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CTLREG_base> + 0064_H

Value after reset: Undefined

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICUMRESV[31:16]															
Value after reset	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICUMRESV[15:0]															
Value after reset	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 10.16 ICUM_CFG1 Register Contents

Bit Position	Bit Name	Function
31 to 0	ICUMRESV[31:0]	ICUP reset vector address

Note 1. The value after reset of each bit depends on the configuration setting data.

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10.3.14 ICUM_CFG2 — ICUMD Configuration Register 2

This register is used to see the start address of the secure area (code flash). This register references option byte 10 (OPBT10). For details, see **Section 16.8.3, OPBT10 — Option Byte 10**.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CTLREG_base> + 0068_H

Value after reset: Undefined

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICUMRAC[31:16]															
Value after reset	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICUMRAC[15:0]															
Value after reset	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 10.17 ICUM_CFG2 Register Contents

Bit Position	Bit Name	Function
31 to 0	ICUMRAC[31:0]	Start address of the secure code flash area

Note 1. The value after reset of each bit depends on the configuration setting data.

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10.3.15 ICUM_CFG3 — ICUMD Configuration Register 3

This register is used to see the start address of the secure area (data flash). This register references option byte 11 (OPBT11). For details, see **Section 16.8.4, OPBT11 — Option Byte 11**.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CTLREG_base> + $006C_H$

Value after reset: Undefined

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICUMRAD[31:16]															
Value after reset	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICUMRAD[15:0]															
Value after reset	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 10.18 ICUM_CFG3 Register Contents

Bit Position	Bit Name	Function
31 to 0	ICUMRAD[31:0]	Start address of the secure data flash area

Note 1. The value after reset of each bit depends on the configuration setting data.

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10.3.16 ICUM_CFG4 — ICUMD Configuration Register 4

This register is used to see the ICUMD option bytes. This register references option byte 12 (OPBT12). For details, see **Section 16.8.5, OPBT12 — Option Byte 12**.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CTLREG_base> + 0070_H

Value after reset: Undefined

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICUMOPBT[31:16]															
Value after reset	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICUMOPBT[15:0]															
Value after reset	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1	X*1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 10.19 ICUM_CFG4 Register Contents

Bit Position	Bit Name	Function
31 to 0	ICUMOPBT[31:0]	ICUMD option bytes

Note 1. The value after reset of each bit depends on the configuration setting data.

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10.3.17 ICUM_PRDNAMEn — Product Name Storage Register n (n = 1 to 3)

This register is used to see the PRDNAME.

For details, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

The only differences from the RH850/F1KH, RH850/F1KM are register symbol and address.

Delete the prefix "ICUM_" when searching for the register symbol in the manual.

For register symbol and address, see **Table 10.4, List of Registers**.

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10.3.18 ICUM_CHIPID1_XX — Chip ID Register 1 XX (XX = LL, LH, HL, HH)

This register is used to see the chip ID.

For details, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

The only differences from the RH850/F1KH, RH850/F1KM are register symbol and address.

Delete the prefix "ICUM_" when searching for the register symbol in the manual.

For register symbol and address, see **Table 10.4, List of Registers**.

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Section 11 ICUMD Command Register (ICUM_CMDREG)

ICUM_CMDREG is used to share information between ICUP and CPU (PE).

11.1 Features of ICUM_CMDREG

11.1.1 Number of Units and Channels

ICUMD has one ICUM_CMDREG channel.

11.1.2 Register Base Address

The ICUM_CMDREG base addresses are listed in the table below.

Table 11.1 Register Base Address

Base Address Name	Base Address
<ICUM_CMDREG_base>	FF1F 0000$_H$

11.1.3 Clock Supply

The ICUM_CMDREG clock supply is listed in the table below.

Table 11.2 Clock Supply

Unit Name	Unit Clock Name	Supply Clock Name
ICUM_CMDREG	gapb32_pclk	CPUCLK_L

11.1.4 Interrupt Requests

Interruptions between ICUP and CPU (PE) are controlled by ICUM_CMDREG registers.

Interrupt requests from CPU (PE) to ICUP are listed in the table below.

Table 11.3 Interrupt Requests from CPU (PE) to ICUP

Unit Interrupt Signal	Description	Interrupt Number of INTC on ICUP	DMA Trigger Number
ICUM_CMDREG			
INTPES	Single interrupt request from CPU (PE) to ICUP	0	—
INTPE	Interrupt request from CPU (PE) to ICUP	1	—

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Interrupt requests from ICUP to CPU (PE) are listed in the table below.

Table 11.4 Interrupt Requests from ICUP to CPU (PE)

Unit Interrupt Signal	Description	Interrupt Number of INTC on CPU (PE)	DMA Trigger Number
ICUM_CMDREG			
INTICUPS	Single interrupt request from ICUP to CPU (PE)	6	—
INTICUP	Interrupt request from ICUP to CPU (PE)	7	—
INTFICUIF0CFRQPE*1	Code flash programming request for CPU (PE)	358	—
INTFICUIF0DFRQPE*1	Data flash programming request for CPU (PE)	359	—

Note 1. This interrupt is not issued before ICUP allows CPU (PE) operating. When the OPBT12.ICUMFPHH bit is 0, this interrupt is not issued at all.

Table 11.5 lists EI level maskable interrupts for CPU (PE). These registers can be accessed by CPU (PE).

Table 11.5 EI Level Maskable Interrupt Sources for CPU (PE)

Interrupt			Interrupt Request					Handler Address (Offset)*1		
	Control Register							Direct Jumping to an Address		
Channel No.*1	Name	Address	Name	Source	Unit	Detection Type*1	Exception Source code	RINT = 0	RINT = 1	Reference to a Table*1
6	ICICUPS	FFFE EA0C$_H$	INTICUPS	Single interrupt request from ICUP to CPU (PE)	ICUM_CMDREG	Level	1006$_H$	*1	*1	+018$_H$
7	ICICUP	FFFE EA0E$_H$	INTICUP	Interrupt request from ICUP to CPU (PE)	ICUM_CMDREG	Level	1007$_H$			+01C$_H$
358	ICFICUIF0CFRQPE*2	FFFF B2CC$_H$	INTFICUIF0CFRQPE	Code flash programming request for CPU (PE)	ICUM_CMDREG	Edge	1166$_H$			+598$_H$
359	ICFICUIF0DFRQPE*2	FFFF B2CE$_H$	INTFICUIF0DFRQPE	Data flash programming request for CPU (PE)	ICUM_CMDREG	Edge	1167$_H$			+59C$_H$

Note 1. For details, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

Note 2. This register is used under the OPBT12.ICUMFPHH = 1 condition.

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11.1.5 Reset Sources

The ICUM_CMDREG reset sources are listed in the table below. ICUM_CMDREG is initialized by these reset sources.

Table 11.6 Reset Sources

Unit Name	Reset Source
ICUM_CMDREG	All reset sources (ISORES)

11.2 Overview

11.2.1 Functional Overview

The ICUM_CMDREG has the following functions.

- Indicate version and identification information
- Provide ICUP and CPU (PE) internal status information
- Handshake control for secure flash programming
- Handle interruption between ICUP and CPU (PE).
- Semaphore operations between ICUP and CPU (PE).

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11.3 Registers

11.3.1 List of Registers

The ICUM_CMDREG registers are listed in the table below.

Table 11.7 List of Registers (1/2)

Module Name	Register Name	Symbol	Address	Access authority from ICUP	Access authority from CPU (PE)
ICUM_CMDREG	ICU-M ID register	ICUM_ICUID	<ICUM_CMDREG_base> + 0000_H	R	R
	ICU-M version register	ICUM_ICUVER	<ICUM_CMDREG_base> + 0004_H	R	R
	ICUMD internal status information register	ICUM_ICU2PES	<ICUM_CMDREG_base> + 0010_H	R/W	R
	CPU (PE) internal status information register	ICUM_PE2ICUS	<ICUM_CMDREG_base> + 0014_H	R	R/W
	ICUMD activating flag register	ICUM_ACTFLAG	<ICUM_CMDREG_base> + 0018_H	R	R
	CPU (PE) flash programming request register 0	ICUM_PEFLPRG0REQ	<ICUM_CMDREG_base> + 0020_H	R	R
	CPU (PE) flash programming request acknowledge register 0	ICUM_PEFLPRG0ACK	<ICUM_CMDREG_base> + 0028_H	R	R/W
	CPU (PE) interrupt request flag register	ICUM_PE2ICUF	<ICUM_CMDREG_base> + 0040_H	R	R
	ICUM_PE2ICUF set register	ICUM_PE2ICUFS	<ICUM_CMDREG_base> + 0044_H	—	W
	ICUM_PE2ICUF clear register	ICUM_PE2ICUFC	<ICUM_CMDREG_base> + 0048_H	W	—
	ICUM_PE2ICUF enable register	ICUM_PE2ICUIE	<ICUM_CMDREG_base> + $004C_H$	R/W	—
	ICUP interrupt request flag register	ICUM_ICU2PEF	<ICUM_CMDREG_base> + 0050_H	R	R
	ICUM_ICU2PEF set register	ICUM_ICU2PEFS	<ICUM_CMDREG_base> + 0054_H	W	—
	ICUM_ICU2PEF clear register	ICUM_ICU2PEFC	<ICUM_CMDREG_base> + 0058_H	—	W
	ICUM_ICU2PEF enable register	ICUM_ICU2PEIE	<ICUM_CMDREG_base> + $005C_H$	R	R/W
	ICUM_ICU2PEF selection register*1	ICUM_ICU2PEIS	<ICUM_CMDREG_base> + 0060_H	R	R/W
	CPU (PE) interrupt single request flag register	ICUM_PE2ICUSF	<ICUM_CMDREG_base> + 0070_H	R	R
	ICUM_PE2ICUSF set register	ICUM_PE2ICUSFS	<ICUM_CMDREG_base> + 0074_H	—	W
	ICUM_PE2ICUSF clear register	ICUM_PE2ICUSFC	<ICUM_CMDREG_base> + 0078_H	W	—
	ICUM_PE2ICUSF enable register	ICUM_PE2ICUSIE	<ICUM_CMDREG_base> + $007C_H$	R/W	—
	ICUP interrupt single request flag register	ICUM_ICU2PESF	<ICUM_CMDREG_base> + 0080_H	R	R
	ICUM_ICU2PESF set register	ICUM_ICU2PESFS	<ICUM_CMDREG_base> + 0084_H	W	—
	ICUM_ICU2PESF clear register	ICUM_ICU2PESFC	<ICUM_CMDREG_base> + 0088_H	—	W

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Table 11.7 List of Registers (2/2)

Module Name	Register Name	Symbol	Address	Access authority from ICUP	Access authority from CPU (PE)
ICUM_CMDREG	ICUM_ICU2PESF enable register	ICUM_ICU2PESIE	<ICUM_CMDREG_base> + $008C_H$	R	R/W
	ICUM_ICU2PESF selection register*1	ICUM_ICU2PESIS	<ICUM_CMDREG_base> + 0090_H	R	R/W
	CPU1 (PE1) semaphore register	ICUM_SEMAPE1	<ICUM_CMDREG_base> + $00A0_H$	R/W	R/W only by CPU1 (PE1)
	CPU2 (PE2) semaphore register*1	ICUM_SEMAPE2	<ICUM_CMDREG_base> + $00A4_H$	R/W	R/W only by CPU2 (PE2)

Note 1. These registers area supported only inRH850/F1KH.

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11.3.2 ICUM_ICUID — ICU-M ID Register

This register indicates ICUMD identification information.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CMDREG_base> + 0000_H

Value after reset: $0601\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICUID[31:16]															
Value after reset	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	1
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICUID[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 11.8 ICUM_ICUID Register Contents

Bit Position	Bit Name	Function
31 to 0	ICUID[31:0]	ICUMD identification register

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11.3.3 ICUM_ICUVER — ICU-M Version Register

This register indicates ICUMD identification information.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CMDREG_base> + 0004_H

Value after reset: $0303\ 0020_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICUVER[31:16]															
Value after reset	0	0	0	0	0	0	1	1	0	0	0	0	0	0	1	1
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICUVER[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 11.9 ICUM_ICUVER Register Contents

Bit Position	Bit Name	Function
31 to 0	ICUVER[31:0]	ICUMD version register

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11.3.4 ICUM_ICU2PES — ICUMD Internal Status Information Register

This register is a general status register for reporting information from ICUP to CPU (PE).

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CMDREG_base> + 0010_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICU2PES[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICU2PES[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 11.10 ICUM_ICU2PES Register Contents

Bit Position	Bit Name	Function
31 to 0	ICU2PES[31:0]	Information reported to CPU (PE)

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11.3.5 ICUM_PE2ICUS — CPU (PE) Internal Status Information Register

This register is a general status register for reporting information from CPU (PE) to ICUP.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CMDREG_base> + 0014_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	PE2ICUS[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PE2ICUS[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Table 11.11 ICUM_PE2ICUS Register Contents

Bit Position	Bit Name	Function
31 to 0	PE2ICUS[31:0]	Information reported from CPU (PE)

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11.3.6 ICUM_ACTFLAG — ICUMD Activating Flag Register

This register indicates active state of ICUMD.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CMDREG_base> + 0018_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ICUMACT[1:0]	
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 11.12 ICUM_ACTFLAG Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When read, the value after reset is returned.
1, 0	ICUMACT[1:0]	ICUMD active status 00: ICUMD is inactive 11: ICUMD is active

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11.3.7 ICUM_PEFLPRG0REQ — CPU (PE) Flash Programming Request Register 0

This register monitors the issuance of flash programming requests.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CMDREG_base> + 0020_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	FENTDUPREQ 0	FENTCUPREQ 0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 11.13 ICUM_PEFLPRG0REG Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When read, the value after reset is returned.
1	FENTDUPREQ0	Data flash programming request from ICUP 0: A request is not issued. 1: A request is issued.
0	FENTCUPREQ0	Code flash programming request from ICUP 0: A request is not issued. 1: A request is issued.

CAUTION

- This register is used after ICUP allows CPU (PE) operating. (Code flash and data flash can be shifted to P/E mode before ICUP activates CPU (PE).)
- When the OPBT12.ICUMFPHH bit is 0, this register is not used at all.

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11.3.8 ICUM_PEFLPRG0ACK — CPU (PE) Flash Programming Request Acknowledge Register 0

This register responds to issued flash programming requests.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CMDREG_base> + 0028_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	FENTDUPACKA0	FENTDUPACK0	FENTCUPACK0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R/W	R/W	R/W

Table 11.14 ICUM_PEFLPRG0ACK Register Contents

Bit Position	Bit Name	Function
31 to 3	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
2	FENTDUPACKA0	Automatic response setting for data flash programming requests If this bit is set to 0, the FENTDUPACK0 bit can be set. 0: No automatic response is set for programming requests. 1: An automatic response is set for programming requests.
1	FENTDUPACK0	Acknowledgment of data flash programming requests This bit is enabled only when the FENTDUPACKA0 bit is 0. 0: Programming requests are disabled. 1: Programming requests are enabled. FENTDUPACK0 can be set to 1 only when the ICUM_PEFLPRG0REQ.FENTDUPREQ0 is 1, and is cleared to 0 when the ICUM_PEFLPRG0REQ.FENTDUPREQ0 becomes 0.
0	FENTCUPACK0	Acknowledgment of code flash programming requests 0: Programming requests are disabled. 1: Programming requests are enabled. FENTCUPACK0 can be set to 1 only when the ICUM_PEFLPRG0REQ.FENTCUPREQ0 is 1, and is cleared to 0 when the ICUM_PEFLPRG0REQ.FENTCUPREQ0 becomes 0.

CAUTION

- This register is used after ICUP allows CPU (PE) operating. (Code flash and data flash can be shifted to P/E mode before ICUP activates CPU (PE).)
- When the OPBT12.ICUMFPHH bit is 0, this register is not used at all.

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11.3.9 ICUM_PE2ICUF — CPU (PE) Interrupt Request Flag Register

This register is an interrupt status flag register from CPU (PE) to ICUP. Each bit can be set to 1 individually from CPU (PE) by writing the corresponding bit in the register ICUM_PE2ICUFS. Each bit can be set to 0 individually from ICUP by writing the corresponding bit in the register ICUM_PE2ICUFC.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CMDREG_base> + 0040_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	PE2ICUF[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PE2ICUF[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 11.15 ICUM_PE2ICUF Register Contents

Bit Position	Bit Name	Function
31 to 0	PE2ICUF[31:0]	Interrupt request (INTPE) flags from CPU (PE) to ICUP

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11.3.10 ICUM_PE2ICUFS — ICUM_PE2ICUF Set Register

This register is used to set 1 to each bit of ICUM_PE2ICUF by CPU (PE).

Access: This register is a write-only register that can be written in 32-bit units.

Address: <ICUM_CMDREG_base> + 0044_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	PE2ICUFS[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CPU (PE) R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PE2ICUFS[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CPU (PE) R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W

Table 11.16 ICUM_PE2ICUFS Register Contents

Bit Position	Bit Name	Function
31 to 0	PE2ICUFS[31:0]	Set register for ICUM_PE2ICUF When 1 is written to PE2ICUFS[i], ICUM_PE2ICUF.PE2ICUF[i] is set to 1 (i = 0 to 31). Writing 0 to this register is invalid.

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11.3.11 ICUM_PE2ICUFC — ICUM_PE2ICUF Clear Register

This register is used to set 0 to each bit of ICUM_PE2ICUF by ICUP.

Access: This register is a write-only register that can be written in 32-bit units.

Address: <ICUM_CMDREG_base> + 0048_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	PE2ICUFC[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
CPU (PE) R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PE2ICUFC[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
CPU (PE) R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Table 11.17 ICUM_PE2ICUFC Register Contents

Bit Position	Bit Name	Function
31 to 0	PE2ICUFC[31:0]	Clear register for ICUM_PE2ICUF When 1 is written to PE2ICUFC[i], ICUM_PE2ICUF.PE2ICUF[i] is cleared to 0 (i = 0 to 31). Writing 0 to this register is invalid.

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11.3.12 ICUM_PE2ICUIE — ICUM_PE2ICUF Enable Register

This register is used to enable the status of the ICUM_PE2ICUF register by ICUP.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CMDREG_base> + $004C_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	PE2ICUIE[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
CPU (PE) R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PE2ICUIE[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W
CPU (PE) R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Table 11.18 ICUM_PE2ICUIE Register Contents

Bit Position	Bit Name	Function
31 to 0	PE2ICUIE[31:0]	Interrupt (INTPE) enable flags from CPU (PE) to ICUP

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11.3.13 ICUM_ICU2PEF — ICUP Interrupt Request Flag Register

This register is an interrupt status flag register from ICUP to CPU (PE). Each bit can be set to 1 individually from ICUP by writing the corresponding bit in the register ICUM_ICU2PEFS. Each bit can be set to 0 individually from CPU (PE) by writing the corresponding bit in the register ICUM_ICU2PEFC.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CMDREG_base> + 0050_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICU2PEF[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICU2PEF[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 11.19 ICUM_ICU2PEF Register Contents

Bit Position	Bit Name	Function
31 to 0	ICU2PEF[31:0]	Interrupt request (INTICUP) flags from ICUP to CPU (PE)

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11.3.14 ICUM_ICU2PEFS — ICUM_ICU2PEF Set Register

This register is used to set 1 to each bit of ICUM_ICU2PEF by ICUP.

Access: This register is a write-only register that can be written in 32-bit units.

Address: <ICUM_CMDREG_base> + 0054_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICU2PEFS[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
CPU (PE) R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICU2PEFS[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
CPU (PE) R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Table 11.20 ICUM_ICU2PEFS Register Contents

Bit Position	Bit Name	Function
31 to 0	ICU2PEFS[31:0]	Set register for ICUM_ICU2PEF When 1 is written to ICU2PEFS[i], ICUM_ICU2PEF.ICU2PEF[i] is set to 1 (i = 0 to 31). Writing 0 to this register is invalid.

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11.3.15 ICUM_ICU2PEFC — ICUM_ICU2PEF Clear Register

This register is used to set 0 to each bit of ICUM_ICU2PEF by CPU (PE).

Access: This register is a write-only register that can be written in 32-bit units.

Address: <ICUM_CMDREG_base> + 0058_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICU2PEFC[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CPU (PE) R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICU2PEFC[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CPU (PE) R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W

Table 11.21 ICUM_ICU2PEFC Register Contents

Bit Position	Bit Name	Function
31 to 0	ICU2PEFC[31:0]	Clear register for ICUM_ICU2PEF When 1 is written to ICU2PEFC[i], ICUM_ICU2PEF.ICU2PEF[i] is cleared to 0 (i = 0 to 31). Writing 0 to this register is invalid.

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11.3.16 ICUM_ICU2PEIE — ICUM_ICU2PEF Enable Register

This register is used to enable the status of the ICUM_ICU2PEF register by CPU (PE).

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CMDREG_base> + $005C_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICU2PEIE[31:16]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICU2PEIE[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Table 11.22 ICUM_ICU2PEIE Register Contents

Bit Position	Bit Name	Function
31 to 0	ICU2PEIE[31:0]	Interrupt (INTICUP) enable flags from ICUP to CPU (PE)

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11.3.17 ICUM_ICU2PEIS — ICUM_ICU2PEF Selection Register

This register controls the destination selection of INTICUP by CPU (PE). (RH850/F1KH)

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CMDREG_base> + 0060_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	PESEL7[3:0]				PESEL6[3:0]				PESEL5[3:0]				PESEL4[3:0]			
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PESEL3[3:0]				PESEL2[3:0]				PESEL1[3:0]				PESEL0[3:0]			
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Table 11.23 ICUM_ICU2PEIS Register Contents (1/2)

Bit Position	Bit Name	Function
31 to 28	PESEL7[3:0]	INTICUP destination selection for ICU2PEF[31:28] $1xxx_B$: INTICUP to all CPUs (PEs) 0001_B: INTICUP to CPU2 (PE2) 0000_B: INTICUP to CPU1 (PE1) Other than above: Setting prohibited
27 to 24	PESEL6[3:0]	INTICUP destination selection for ICU2PEF[27:24] $1xxx_B$: INTICUP to all CPUs (PEs) 0001_B: INTICUP to CPU2 (PE2) 0000_B: INTICUP to CPU1 (PE1) Other than above: Setting prohibited
23 to 20	PESEL5[3:0]	INTICUP destination selection for ICU2PEF[23:20] $1xxx_B$: INTICUP to all CPUs (PEs) 0001_B: INTICUP to CPU2 (PE2) 0000_B: INTICUP to CPU1 (PE1) Other than above: Setting prohibited
19 to 16	PESEL4[3:0]	INTICUP destination selection for ICU2PEF[19:16] $1xxx_B$: INTICUP to all CPUs (PEs) 0001_B: INTICUP to CPU2 (PE2) 0000_B: INTICUP to CPU1 (PE1) Other than above: Setting prohibited
15 to 12	PESEL3[3:0]	INTICUP destination selection for ICU2PEF[15:12] $1xxx_B$: INTICUP to all CPUs (PEs) 0001_B: INTICUP to CPU2 (PE2) 0000_B: INTICUP to CPU1 (PE1) Other than above: Setting prohibited

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Table 11.23 ICUM_ICU2PEIS Register Contents (2/2)

Bit Position	Bit Name	Function
11 to 8	PESEL2[3:0]	INTICUP destination selection for ICU2PEF[11:8] $1xxx_B$: INTICUP to all CPUs (PEs) 0001_B: INTICUP to CPU2 (PE2) 0000_B: INTICUP to CPU1 (PE1) Other than above: Setting prohibited
7 to 4	PESEL1[3:0]	INTICUP destination selection for ICU2PEF[7:4] $1xxx_B$: INTICUP to all CPUs (PEs) 0001_B: INTICUP to CPU2 (PE2) 0000_B: INTICUP to CPU1 (PE1) Other than above: Setting prohibited
3 to 0	PESEL0[3:0]	INTICUP destination selection for ICU2PEF[3:0] $1xxx_B$: INTICUP to all CPUs (PEs) 0001_B: INTICUP to CPU2 (PE2) 0000_B: INTICUP to CPU1 (PE1) Other than above: Setting prohibited

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11.3.18 ICUM_PE2ICUSF — CPU (PE) Interrupt Single Request Flag Register

This register is an interrupt single status flag register from CPU (PE) to ICUP.

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CMDREG_base> + 0070_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PE2ICUSF0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 11.24 ICUM_PE2ICUSF Register Contents

Bit Position	Bit Name	Function
31 to 1	—	Reserved When read, the value after reset is returned.
0	PE2ICUSF0	Interrupt single request (INTPES) flags from CPU (PE) to ICUP

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11.3.19 ICUM_PE2ICUSFS — ICUM_PE2ICUSF Set Register

This register is used to set 1 to ICUM_PE2ICUSF by CPU (PE).

Access: This register is a write-only register that can be written in 32-bit units.

Address: <ICUM_CMDREG_base> + 0074_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CPU (PE) R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PE2ICUSFS0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CPU (PE) R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W

Table 11.25 ICUM_PE2ICUSFS Register Contents

Bit Position	Bit Name	Function
31 to 1	—	Reserved When writing, write the value after reset.
0	PE2ICUSFS0	Set register for ICUM_PE2ICUSF When 1 is written to PE2ICUSFS0, ICUM_PE2ICUSF.PE2ICUSF0 is set to 1. Writing 0 to this register is invalid.

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11.3.20 ICUM_PE2ICUSFC — ICUM_PE2ICUSF Clear Register

This register is used to set 0 to ICUM_PE2ICUSF by ICUP.

Access: This register is a write-only register that can be written in 32-bit units.

Address: <ICUM_CMDREG_base> + 0078_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
CPU (PE) R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PE2ICUSFC0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
CPU (PE) R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Table 11.26 ICUM_PE2ICUSFC Register Contents

Bit Position	Bit Name	Function
31 to 1	—	Reserved When writing, write the value after reset.
0	PE2ICUSFC0	Clear register for ICUM_PE2ICUSF When 1 is written to PE2ICUSFC0, ICUM_PE2ICUSF.PE2ICUSF0 is set to 0. Writing 0 to this register is invalid.

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11.3.21 ICUM_PE2ICUSIE — ICUM_PE2ICUSF Enable Register

This register is used to enable the status of the ICUM_PE2ICUSF register by ICUP.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CMDREG_base> + $007C_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PE2ICUSIE0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R/W
CPU (PE) R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Table 11.27 ICUM_PE2ICUSIE Register Contents

Bit Position	Bit Name	Function
31 to 1	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
0	PE2ICUSIE0	Interrupt single (INTPES) enable flags from CPU (PE) to ICUP

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11.3.22 ICUM_ICU2PESF — ICUP Interrupt Single Request Flag Register

This register is an interrupt single status flag register from ICUP to CPU (PE).

Access: This register is a read-only register that can be read in 32-bit units.

Address: <ICUM_CMDREG_base> + 0080_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ICU2PESF0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 11.28 ICUM_ICU2PESF Register Contents

Bit Position	Bit Name	Function
31 to 1	—	Reserved When read, the value after reset is returned.
0	ICU2PESF0	Interrupt single request (INTICUPS) flag from ICUP to CPU (PE)

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11.3.23 ICUM_ICU2PESFS — ICUM_ICU2PESF Set Register

This register is used to set 1 to ICUM_ICU2PESF by ICUP.

Access: This register is a write-only register that can be written in 32-bit units.

Address: <ICUM_CMDREG_base> + 0084_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
CPU (PE) R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ICU2PESFS0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W
CPU (PE) R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Table 11.29 ICUM_ICU2PESFS Register Contents

Bit Position	Bit Name	Function
31 to 1	—	Reserved When writing, write the value after reset.
0	ICU2PESFS0	Set register for ICUM_ICU2PESF When 1 is written to ICU2PESFS0, ICUM_ICU2PESF.ICU2PESF0 is set to 1. Writing 0 to this register is invalid.

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11.3.24 ICUM_ICU2PESFC — ICUM_ICU2PESF Clear Register

This register is used to set 0 to ICUM_ICU2PESF by CPU (PE).

Access: This register is a write-only register that can be written in 32-bit units.

Address: <ICUM_CMDREG_base> + 0088_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CPU (PE) R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ICU2PESFC0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
CPU (PE) R/W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W	W

Table 11.30 ICUM_ICU2PESFC Register Contents

Bit Position	Bit Name	Function
31 to 1	—	Reserved When writing, write the value after reset.
0	ICU2PESFC0	Clear register for ICUM_ICU2PESF When 1 is written to ICU2PESFC0, ICUM_ICU2PESF.ICU2PESF0 is cleared to 0. Writing 0 to this register is invalid.

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11.3.25 ICUM_ICU2PESIE — ICUM_ICU2PESF Enable Register

This register is used to enable the status of the ICUM_ICU2PESF register by ICUP.

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CMDREG_base> + $008C_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ICU2PESIE0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R/W

Table 11.31 ICUM_ICU2PESIE Register Contents

Bit Position	Bit Name	Function
31 to 1	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
0	ICU2PESIE0	Interrupt single (INTICUPS) enable flag from ICUP to CPU (PE)

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11.3.26 ICUM_ICU2PESIS — ICUM_ICU2PESF Selection Register

This register controls the destination selection of INTICUPS by CPU (PE). (RH850/F1KH)

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CMDREG_base> + 0090_H

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	PESSEL[3:0]			
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R
CPU (PE) R/W	R	R	R	R	R	R	R	R	R	R	R	R	R/W	R/W	R/W	R/W

Table 11.32 ICUM_ICU2PESIS Register Contents

Bit Position	Bit Name	Function
31 to 4	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
3 to 0	PESSEL[3:0]	INTICUPS destination selectoin $1xxx_B$: INTICUPS to all CPUs (PEs) 0001_B: INTICUPS to CPU2 (PE2) 0000_B: INTICUPS to CPU1 (PE1) Other than above: Setting prohibited

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11.3.27 ICUM_SEMAPE1 — CPU1 (PE1) Semaphore Register

This register is a semaphore register to support synchronize operation between ICUP and CPU1 (PE1).

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CMDREG_base> + $00A0_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	AUTH[1:0]		SEMAPE1FL[29:16]													
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1
CPU1 (PE1) R/W	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	SEMAPE1FL[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1
CPU1 (PE1) R/W	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1

Table 11.33 ICUM_SEMAPE1 Register Contents

Bit Position	Bit Name	Function
31, 30	AUTH[1:0]	Write authority status flag Indicate write-authority of this register. If read when AUTH[1:0]=00, the read value is 00. The value of AUTH[1:0] is updated according to the access master. If read by CPU1 (PE1), AUTH[1:0] is updated to 10. If read by ICUP, AUTH[1:0] is updated to 11. If read when AUTH[1:0]=10, the read value of AUTH[1:0] is 10. The value of AUTH[1:0] is not updated. If read when AUTH[1:0]=11, the read value of AUTH[1:0] is 11. The value of AUTH[1:0] is not updated. If the read vale of AUTH[1:0] is 00, it indicates that the access master has acquired write access authority to this register. If the read vale of AUTH[1:0] is other than 00, it indicates that the access master indicated by the read value has write access authority. Writable when the access master has write authority to this register. By writing 00_B to AUTH[1:0], access authority will be released. 00: ICUP and CPU1 (PE1) are possible to only read. 01: Undefined. 10: CPU1 (PE1) is possible to R/W*2. ICUP is possible only read. 11: ICUP is possible to R/W*3. CPU1 (PE1) is possible only read.
29 to 0	SEMAPE1FL[29:0]	Flag for task manager R/W is available depending on the AUTH[1:0] value. These bits can be used as a flag for task management. When CPU1 (PE1) updates the value of SEMAPE1FL[29:0], AUTH[1:0]=00,10 must be written simultaneously with the value. When ICUP updates the value of SEMAPE1FL[29:0], AUTH[1:0]=00,11must be written simultaneously with the value.

Note 1. Depending on AUTH[1:0] value

Note 2. CPU1 (PE1) can write 00 or 10 to AUTH[1:0]. Other values are invalid.

Note 3. ICUP can write 00 or 11 to AUTH[1:0]. Other values are invalid.

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11.3.28 ICUM_SEMAPE2 — CPU2 (PE2) Semaphore Register

This register is a semaphore register to support synchronize operation between ICUP and CPU2 (PE2) (RH850/F1KH).

Access: This register can be read or written in 32-bit units.

Address: <ICUM_CMDREG_base> + $00A4_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	AUTH[1:0]		SEMAPE2FL[29:16]													
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1
CPU2 (PE2) R/W	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	SEMAPE2FL[15:0]															
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ICUP R/W	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1
CPU2 (PE2) R/W	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1	R/W*1

Table 11.34 ICUM_SEMAPE2 Register Contents

Bit Position	Bit Name	Function
31, 30	AUTH[1:0]	Write authority status flag Indicate write-authority of this register. If read when AUTH[1:0]=00, the read value is 00. The value of AUTH[1:0] is updated according to the access master. If read by CPU2 (PE2), AUTH[1:0] is updated to 10. If read by ICUP, AUTH[1:0] is updated to 11. If read when AUTH[1:0]=10, the read value of AUTH[1:0] is 10. The value of AUTH[1:0] is not updated. If read when AUTH[1:0]=11, the read value of AUTH[1:0] is 11. The value of AUTH[1:0] is not updated. If the read vale of AUTH[1:0] is 00, it indicates that the access master has acquired write access authority to this register. If the read vale of AUTH[1:0] is other than 00, it indicates that the access master indicated by the read value has write access authority. Writable when the access master has write authority to this register. By writing 00_B to AUTH[1:0], access authority will be released. 00: ICUP and CPU2 (PE2) are possible to only read. 01: Undefined. 10: CPU2 (PE2) is possible to R/W*2. ICUP is possible only read. 11: ICUP is possible to R/W*3. CPU2 (PE2) is possible only read.
29 to 0	SEMAPE2FL[29:0]	Flag for task manager R/W is available depending on the AUTH[1:0] value. These bits can be used as a flag for task management. When CPU2 (PE2) updates the value of SEMAPE2FL[29:0], AUTH[1:0]=00,10 must be written simultaneously with the value. When ICUP updates the value of SEMAPE2FL[29:0], AUTH[1:0]=00,11must be written simultaneously with the value.

Note 1. Depending on AUTH[1:0] value

Note 2. CPU2 (PE2) can write 00 or 10 to AUTH[1:0]. Other values are invalid.

Note 3. ICUP can write 00 or 11 to AUTH[1:0]. Other values are invalid.

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Section 12 Window Watchdog Timer (ICUM_WDTA)

This section describes the features of the window watchdog timer (ICUM_WDTA) included in the ICUMD.

12.1 Features of ICUM_WDTA

12.1.1 Number of Units and Channels

The ICUMD has one ICUM_WDTA channel (n = 0).

Table 12.1 Index

Index	Meaning
n	Throughout this section, individual ICUM_WDTA units are identified by the index “n”: for example, ICUM_WDTAn enable register (ICUM_WDTAnWDTE).

12.1.2 Register Base Address

The base address of the ICUM_WDTAn is shown in the table below. ICUM_WDTAn register addresses are given as offsets from the base address.

Table 12.2 Register Base Address

Base Address Name	Base Address
<ICUM_WDTAn_base>	FFFE E080$_H$

12.1.3 Clock Supply

The ICUM_WDTAn clock supply is shown in the following table.

Table 12.3 Clock Supply

Unit Name	Unit Clock Name	Supply Clock Name
ICUM_WDTA0	WDTACKI	LS IntOsc f_{RL}
	Register access clock	CPUCLK_L

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12.1.4 Interrupt Requests

Interrupt requests for ICUM_WDTAn are listed below.

Table 12.4 Interrupt Requests

Unit Interrupt Signal	Description	Interrupt Number	DMA Trigger Number
ICUM_WDTA0			
INTWDTAn	ICUM_WDTA0 75% interrupt	9	—

Table 12.5 Interrupt Requests (FE Level Non-Maskable Interrupt)

Unit Interrupt Signal	Description	Interrupt Name	DMA Trigger Number
ICUM_WDTA0			
WDTAnTNMI	ICUM_WDTA0 FENMI interrupt (in WDTA error detection mode with an NMI request)	ICUWDTA0NMI	—

12.1.5 Reset Sources

The ICUM_WDTAn resets sources are listed in the table below. ICUM_WDTAn is initialized by these reset sources.

Table 12.6 Reset Sources

Unit Name	Reset Source
ICUM_WDTA0	All reset sources (ISORES)

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12.2 Overview

12.2.1 Functional Overview

This section describes the startup options of the ICUM_WDTA.

For functions other than start-up options, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

Table 12.7 ICUM_WDTA Start-Up Options

Startup Operation	Function	Description	Option Byte
ICUMOPWDEN	ICUM_WDTA setting	Enables/disables the ICUM_WDTA. 0: ICUM_WDTA is disabled. 1: ICUM_WDTA is enabled	ICUM_WDTA0: OPBT12.ICUMWDT0_0
ICUMOPWDOVF[2:0]	Overflow interval time reset value setting	Overflow interval time control. Specify the reset value of overflow interval time bits ICUM_WDTAnMD.WDTAnOVF[2:0].	ICUM_WDTA0: OPBT12.ICUMWDT_[2:0]
ICUMOPWDRUN	Start mode setting	Specify the start mode. 0: Software trigger start mode 1: Default start mode	ICUM_WDTA0: OPBT12.ICUMWDT0_1
ICUMOPWDVAC	Variable activation code setting	Specify the trigger register for the generation of the counter restart triggers to keep the counter from overflowing. 0: ICUM_WDTAnWDTE (Fixed) 1: ICUM_WDTAnEVAC (Variable)	ICUM_WDTA0: OPBT12.ICUMWDT0_3

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12.3 Registers

12.3.1 List of Registers

ICUM_WDTA registers are listed in the table below.

For details, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

The only differences from the window watchdog timer incorporated in the RH850/F1KH, RH850/F1KM are the module names and register symbols. Read "WDTAn" as "ICUM_WDTAn".

For details on <ICUM_WDTAn_base>, see **Section 12.1.2, Register Base Address**.

Table 12.8 List of Registers

Module Name	Register Name	Symbol	Address
ICUM_WDTAn	WDTA enable register	ICUM_WDTAnWDTE	<ICUM_WDTAn_base> + 0000_H
	WDTA enable VAC register	ICUM_WDTAnEVAC	<ICUM_WDTAn_base> + 0004_H
	WDTA reference value register	ICUM_WDTAnREF	<ICUM_WDTAn_base> + 0008_H
	WDTA mode register	ICUM_WDTAnMD	<ICUM_WDTAn_base> + $000C_H$

12.4 Interrupt Sources

For details, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

The only differences from the RH850/F1KH, RH850/F1KM are the names of the interrupts.

Read "WDTAn" as "ICUWDTAn".

12.5 Functions

See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

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Section 13 OS Timer (ICUM_OSTM)

This section describes the features of the OS timer (ICUM_OSTM) included in the ICUMD.

13.1 Features of ICUM_OSTM

13.1.1 Number of Units and Channels

The ICUMD includes two ICUM_OSTM channels, ICUM_OSTMn. (n = 0, 1)

Table 13.1 Index

Index	Meaning
n	Throughout this section, individual ICUM_OSTM units are identified by the index “n”: for example, ICUM_OSTMnCNT is the ICUM_OSTM counter register.

13.1.2 Register Base Address

ICUM_OSTMn base addresses are listed in the table below.

ICUM_OSTMn register addresses are given as offsets from the base addresses.

Table 13.2 Register Base Address

Base Address Name	Base Address
<ICUM_OSTM0_base>	FFFE E000$_H$
<ICUM_OSTM1_base>	FFFE E040$_H$

13.1.3 Clock Supply

The clock supply of the ICUM_OSTMn is shown in the table below.

Table 13.3 Clock Supply

Unit Name	Unit Clock Name	Supply Clock Name
ICUM_OSTMn	PCLK	CPUCLK_L

13.1.4 Interrupt Requests

The ICUM_OSTMn interrupt requests are listed in the table below.

Table 13.4 Interrupt Requests

Unit Interrupt Signal	Description	Interrupt Number	DMA Trigger Number
ICUM_OSTM0			
OSTMTINT	ICUM_OSTM0 interrupt	7	—
ICUM_OSTM1			
OSTMTINT	ICUM_OSTM1 interrupt	8	—

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Table 13.5 Interrupt Requests (FE Level Maskable Interrupt Request)

Unit Interrupt Signal	Description	Interrupt Name	DMA Trigger Number
ICUM_OSTM0			
OSTMTINT	ICUM_OSTM0 interrupt	INTICUOSTM0_FE	—
ICUM_OSTM1			
OSTMTINT	ICUM_OSTM1 interrupt	INTICUOSTM1_FE	—

13.1.5 Reset Sources

The ICUM_OSTMn reset sources are listed in the table below. ICUM_OSTMn is initialized by these reset sources.

Table 13.6 Reset Sources

Unit Name	Reset Source
ICUM_OSTMn	All reset sources (ISORES)

13.2 Overview

See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

13.3 Registers

13.3.1 List of Registers

The ICUM_OSTM registers are listed in the table below.

For details, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

The only differences from the OS timer incorporated in the RH850/F1KH, RH850/F1KM are the module names and register symbols.

Read "OSTMn" as "ICUM_OSTMn".

For details on the <ICUM_OSTMn_base>, see **Section 13.1.2, Register Base Address**.

Table 13.7 List of Registers

Module Name	Register Name	Symbol	Address
ICUM_OSTMn	OSTMn compare register	ICUM_OSTMnCMP	<ICUM_OSTMn_base> + 00_H
	OSTMn counter register	ICUM_OSTMnCNT	<ICUM_OSTMn_base> + 04_H
	OSTMn count enable status register	ICUM_OSTMnTE	<ICUM_OSTMn_base> + 10_H
	OSTMn count start trigger register	ICUM_OSTMnTS	<ICUM_OSTMn_base> + 14_H
	OSTMn count stop trigger register	ICUM_OSTMnTT	<ICUM_OSTMn_base> + 18_H
	OSTMn control register	ICUM_OSTMnCTL	<ICUM_OSTMn_base> + 20_H
	OSTMn emulation register	ICUM_OSTMnEMU	<ICUM_OSTMn_base> + 24_H

13.4 Operations

See the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

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Section 14 Pin Function

This section describes the pin functions of the ICUMD.

14.1 Pin Description

The ICUP is able to control output from four pins. These pins are controlled by CPU (PE) as general-purpose port pins after reset is deasserted, but can be changed by register setting from the ICUP to operate as security output pins (SECOUTn) controlled by the ICUP.

Table 14.1 Pin Description

General-purpose Port Pin (Under CPU (PE) Control)*1	Security Output Pin (Under ICUP Control)
P9_0	SECOUT0
P9_1	SECOUT1
P2_4	SECOUT2
P2_5	SECOUT3

Note 1. If a general-purpose port pin is to be used as a security output pin, do not use the pin for input.

14.2 Pin Functions During and After Reset

Table 14.2 Pin Functions During and After Reset

Pins	During Reset	After Reset
P9_0, P9_1, P2_4, P2_5	High impedance	High impedance

14.3 Pin State in Standby Mode

Pin state in standby mode is same as other general pins.

For details, see I/O Buffer Control section in the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

14.4 Registers

14.4.1 ICUM_GPIO

See **Section 10.3.4, ICUM_GPIO — GPIO Control Register**.

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Section 15 On-Chip Debug Unit (OCD)

Executing special authentication using the on-chip debugging function of the RH850/F1KH, RH850/F1KM enables debugging of the ICUMD.

CAUTION

The ICUMD supports the debugging functions described in this section. However, whether they are usable depends on the debugger. For the details of debugging, see the user's manual of the debugger.

15.1 Functional Overview

This section describes break function and challenge & response authentication. For details of other functions, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

15.2 Break Function

The microcontroller supports synchronous type and asynchronous type break function.

When synchronous type break function is enabled and ICUP is stopped by the break function, CPU (PE) stop simultaneously. When asynchronous type break function is enabled, ICUP, CPU (PE) can be stopped by the break function respectively.

15.3 CPU(PE) Independent Debug Function *1,*2

CPU (PE)'s software debug can be supported without disclose ICUP's software. ICUP executes normally.

Note 1. This function is only available when synchronous type break is selected.

Note 2. This function is selected by option byte.

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15.4 Challenge & Response Authentication

The microcontroller supports challenge and response authentication (C&R authentication) to allow debugger access to CPU (PE) in addition to ID authentication.

There are 2 types of C&R authentication. One is C&R authentication to connect CPU (PE). Another is C&R authentication to connect ICUP.

Figure 15.1 shows high-level view of C&R authentication.

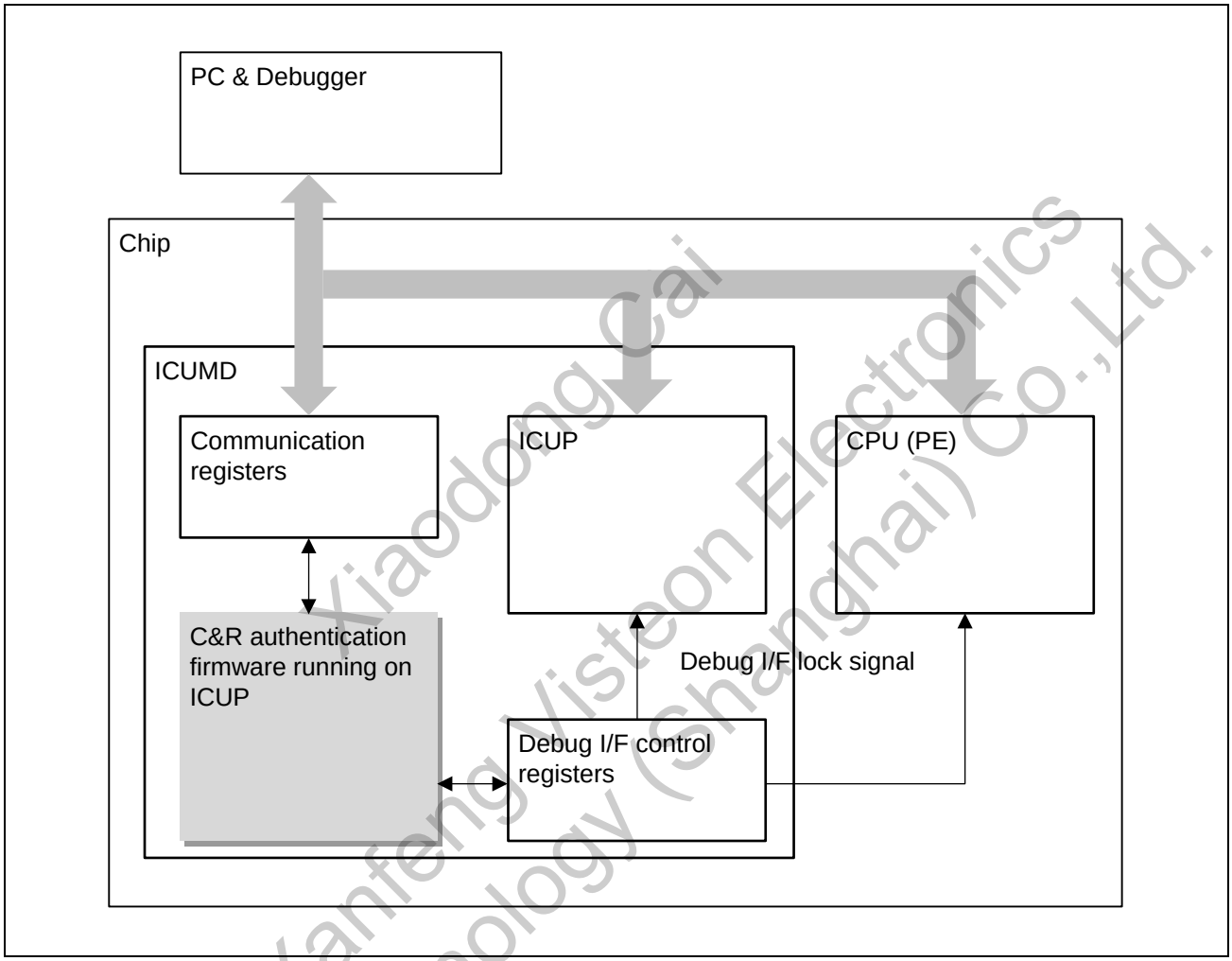


Figure 15.1 C&R Authentication High-Level View

The function of C&R authentication is substantiated by combination of hardware and software. There is a communication interface between ICUMD and the debugger tool. ICUMD has debug allowance registers for CPU (PE) and ICUP. The debugger can issue interruption to ICUP.

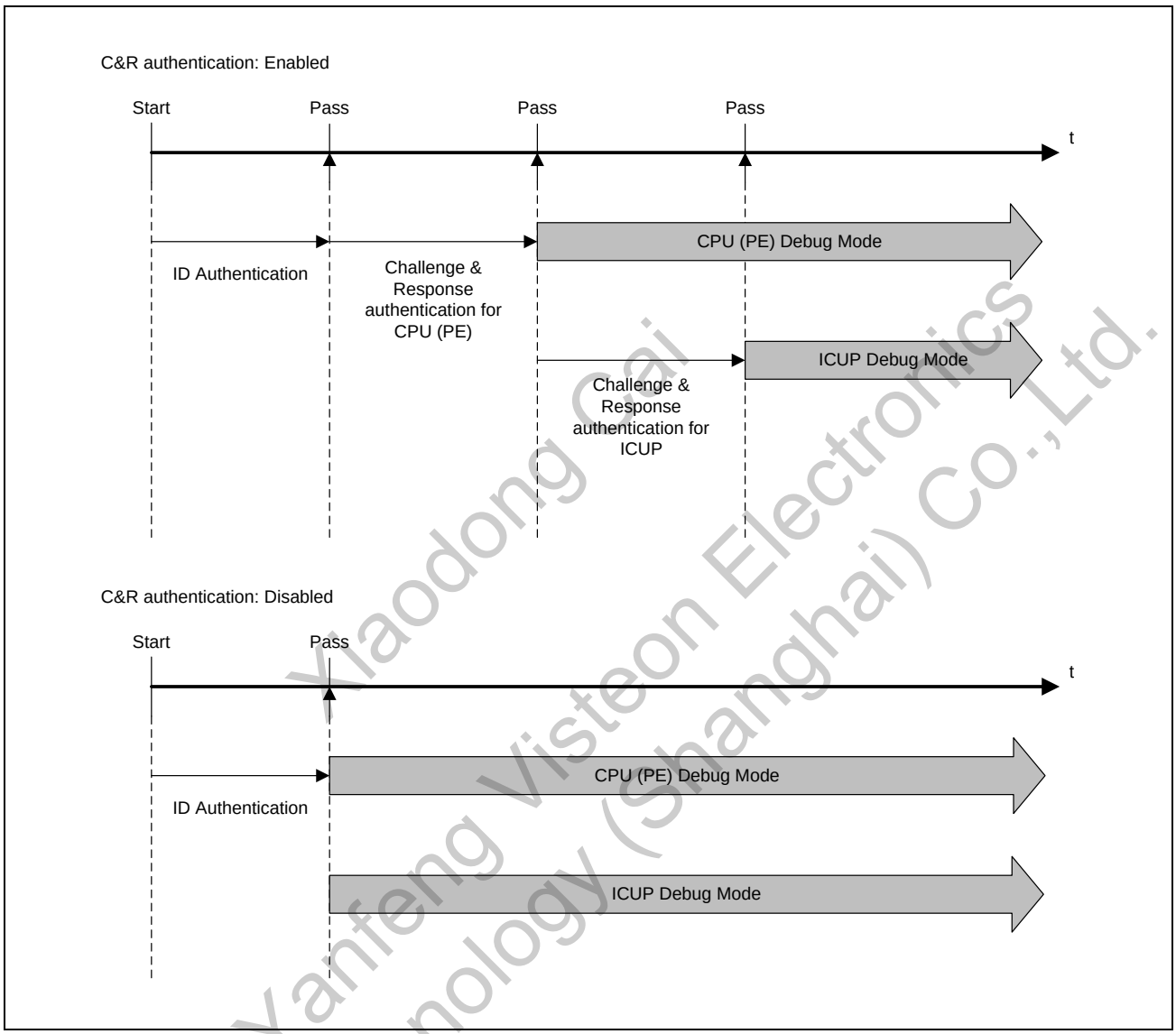
C&R authentication firmware running on ICUP and PC software is needed to handle the sequence. The firmware and the PC software must be prepared by the user. The firmware generates the challenge data by using TRNG. The PC software generates the response data in accordance with a user-defined algorithm. The firmware verifies the response data with user-defined algorithm and control debug interface lock signals on the basis of the verification result.

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15.5 Debug Authentication Sequence

Figure 15.2 shows the conceptual time-sequence diagram of debugger authentication. If both C&R authentications are enabled, time-sequence is shown in upper part. If C&R authentications are disabled, time-sequence is shown in lower part.



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Figure 15.2 Time-sequence Diagram of Debugger Authentication

NOTES

1. When ICUMD is not activated, C&R authentication is disabled.
   When ICUMD is activated, availability of C&R authentication is determined by option byte setting (OPBT12.ICUMCRF1 and OPBT12.ICUMCRF2).
2. Background with skip setting of C&R.

   For C&R authentication is executed by software of ICUMD. Therefore debugging of C&R authentication program is needed. There is C&R authentication skip setting for debugging of C&R authentication program.

Figure 15.3 shows debug authentication protocol when C&R authentications are enabled. For the response time of the C&R authentication firmware, see **Section** 18.7**, Challenge & Response**.

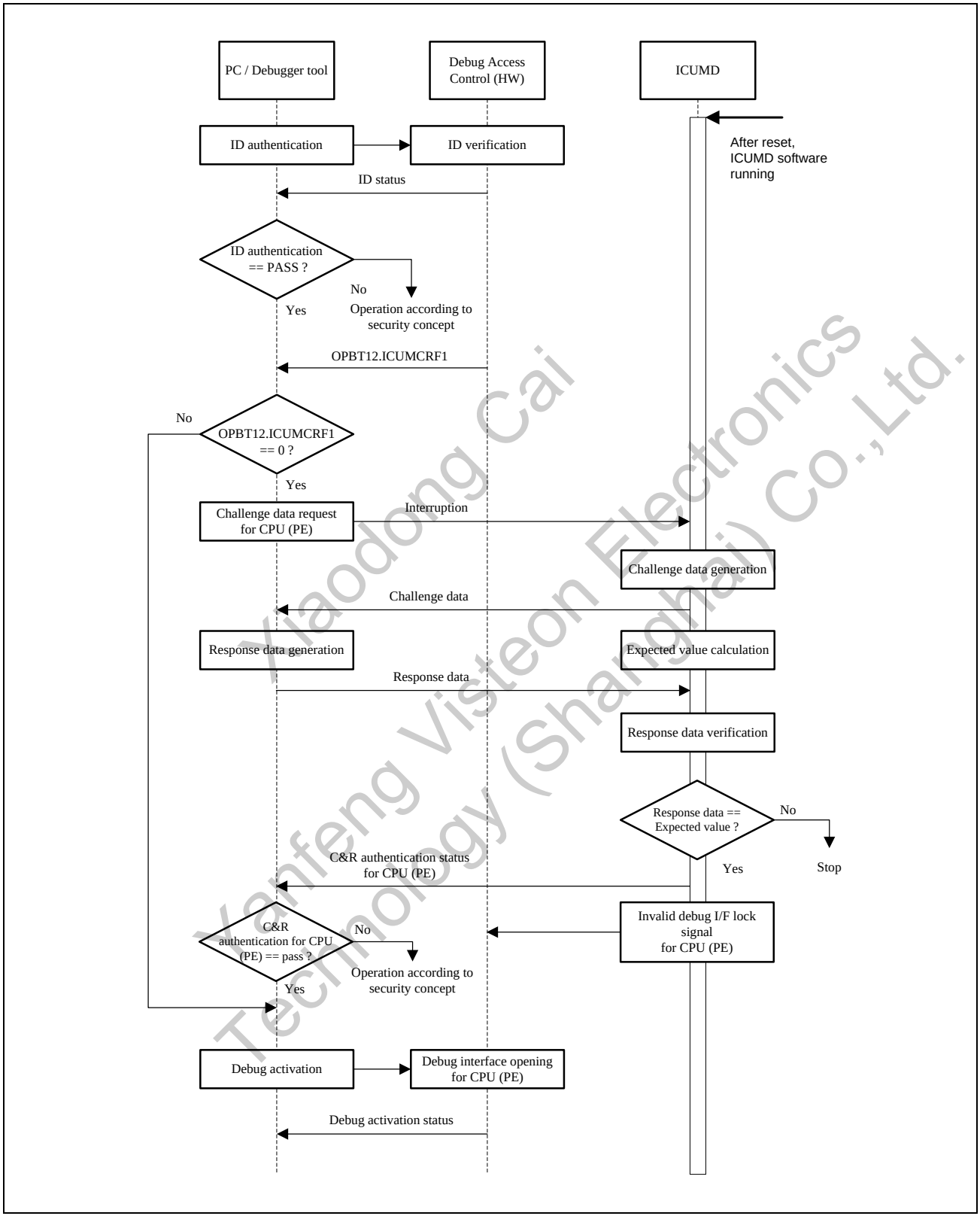
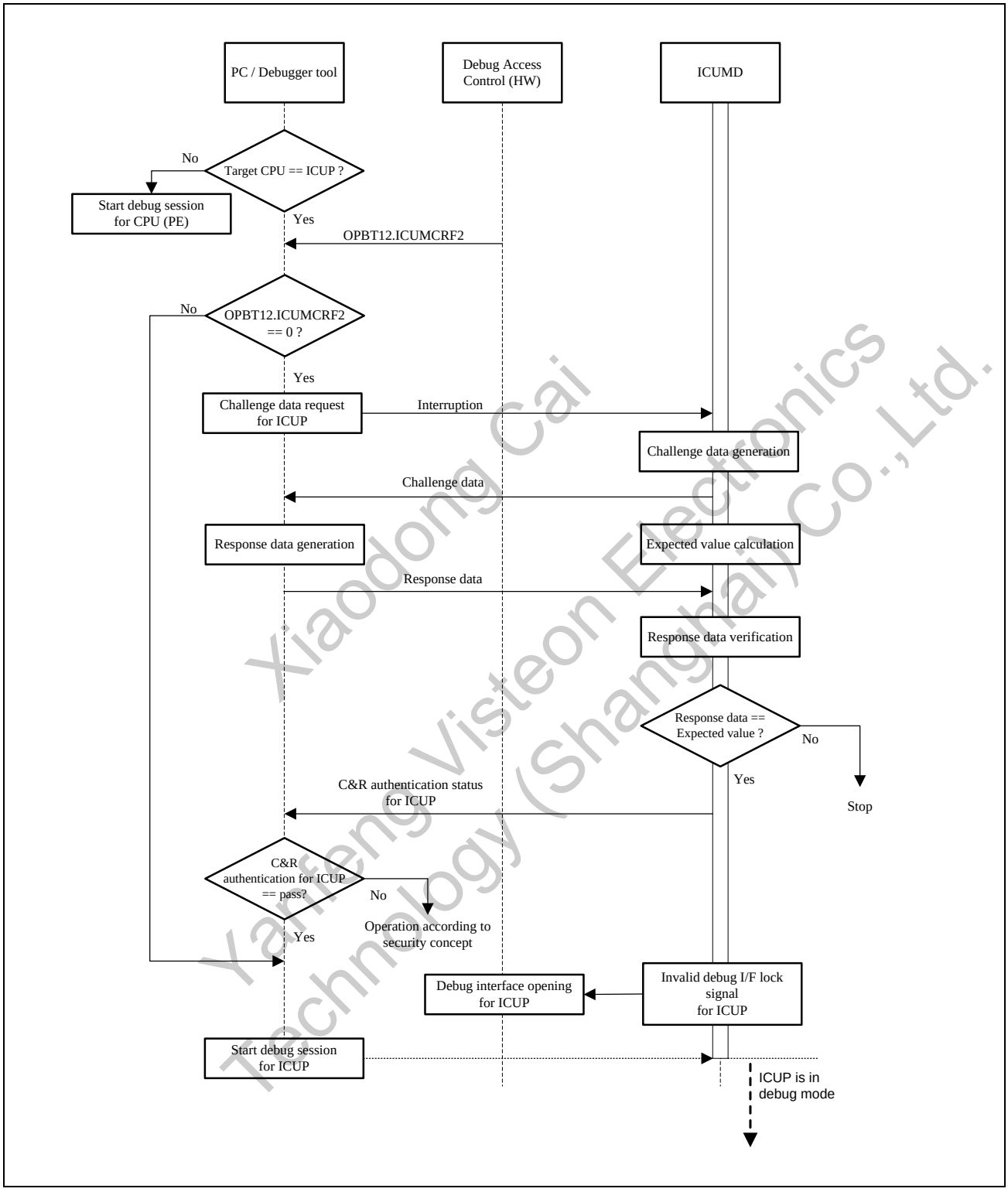


Figure 15.3 Debug Authentication Protocol (1/2)

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Figure 15.3 Debug Authentication Protocol (2/2)

15.6 Peripheral Break Control

The Peripheral Break function generates a stop request to the peripheral modules of the ICUMD when the ICUP program is stopped as in breakpoint hit.

During peripheral break, the peripheral module in the ICUMD operates as follows.

Table 15.1 indicates modules which stop during peripheral break regardless of the debugger-request.

Table 15.1 Modules which Stop Regardless of the Debugger-Request

Module
Window watchdog timer (ICUM_WDTA0)

Table 15.2 indicates modules which can continue to run during peripheral break.

Table 15.2 Modules which can Continue to Run

Module	Control Register/Bit
OS timer (ICUM_OSTMn)	ICUM_OSTMnEMU.OSTMnSVSDIS (n = 0, 1) 0: Stop during break 1: Continue during break

When ICUM_OSTMnEMU.OSTMnSVSDIS is set to 1, the stop request from the debugger is invalid.

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Section 16 Flash Memory

This section describes the flash memory protection and ECC functions supported by the ICUMD. For basic information such as the flash memory capacity, block configuration, and addresses, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

16.1 Security Functions

16.1.1 Functional Overview

To enhance protection against tampering by unauthorized parties, the ICUMD supports security functions to control overwriting by a dedicated flash memory programmer via the serial interface (serial programming) or by a user program (self-programming). These functions are provided in addition to the standard flash memory security functions. **Table 16.1** and **Table 16.2** show the security functions provided by the ICUMD.

Table 16.1 List of Security Functions

Function	Description
Secure area setting	In the code flash and data flash, secure area where only the ICUMD can access can be independently specified. The area specified as secure area enters the state in which fetching, data access, programming, and block erasure from a master other than ICUMD are prohibited. Overwriting by serial programming is also prohibited.
Upgrade protection 1 (UPGP1)	Programming and block erasure in an area other than secure area of the code flash is prohibited. This function is enabled for masters other than the ICUMD and temporary release is possible only from the ICUMD. When UPGP1 is enabled, overwriting by serial programming is prohibited for code flash.
Upgrade protection 2 (UPGP2)	Configuration setting is prohibited. This function is valid for all masters including the ICUMD and temporary release is possible only from the ICUMD. When UPGP2 is enabled, configuration setting by serial programming is prohibited.

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Table 16.2 Summary of Security Functions

Function	Erasure/Write/Read Operation under Security Setting (√: Executable, ×: Inexecutable)		Notes for Security Setting	
	Serial Programming	**Self-Programming**	**Serial Programming**	**Self-Programming**
Secure area setting	● Secure area Block erasure command: × Programming command: × Read command: × ● Areas other than secure area Block erasure command: √ Programming command: √ Read command: √	● Secure area (from other than ICUMD) Block erasure command: × Programming command: × Read command: × ● Secure area (from ICUMD) Block erasure command: √ Programming command: √ Read command: √ ● Area other than secure area Block erasure command: √ Programming command: √ Read command: √	If UPGP2 is valid, a secure area cannot be released by serial programming.	If UPGP2 is valid, a secure area cannot be released by other than the ICUMD.
Upgrade protection 1 (UPGP1)	● Area other than secure area of the code flash Block erasure command: × Programming command: × Read command: √	● Area other than secure area of the code flash (from other than ICUMD) Block erasure command: × Programming command: × Read command: √ ● Area other than secure area of the code flash (from ICUMD) Block erasure command: √ Programming command: √ Read command: √	UPGP1 cannot be released by serial programming	UPGP1 can temporarily be released only by ICUMD.
Upgrade protection 2 (UPGP2)	Configuration setting: × Read command: √	Configuration setting: × Read command: √	UPGP2 cannot be released by serial programming	UPGP2 can temporarily be released only by ICUMD.

16.1.2 Programming Sequence while ICUMD is Enabled

If CPU1 (PE1) starts a programming sequence during access by the ICUP to a secure area, the access by the ICUP is blocked. The ICUP starting a sequence of programming affects CPU1 (PE1) in the same way. If the ICUMD is enabled, arbitration between the ICUP and CPU1 (PE1) by software is required to avoid this problem. This arbitration is realized by the following functions.

- Using the PBUS guard to control access to the flash sequencer
- Processing to handle requests and acknowledgements between the ICUP and CPU1 (PE1) during flash programming sequences.

Examples of the programming sequences are shown in **Figure 16.1** to **Figure 16.4**.

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16.1.2.1 Code Flash Programming from CPU1 (PE1)

The figure below shows a simple flowchart of code flash programming from CPU1 (PE1).

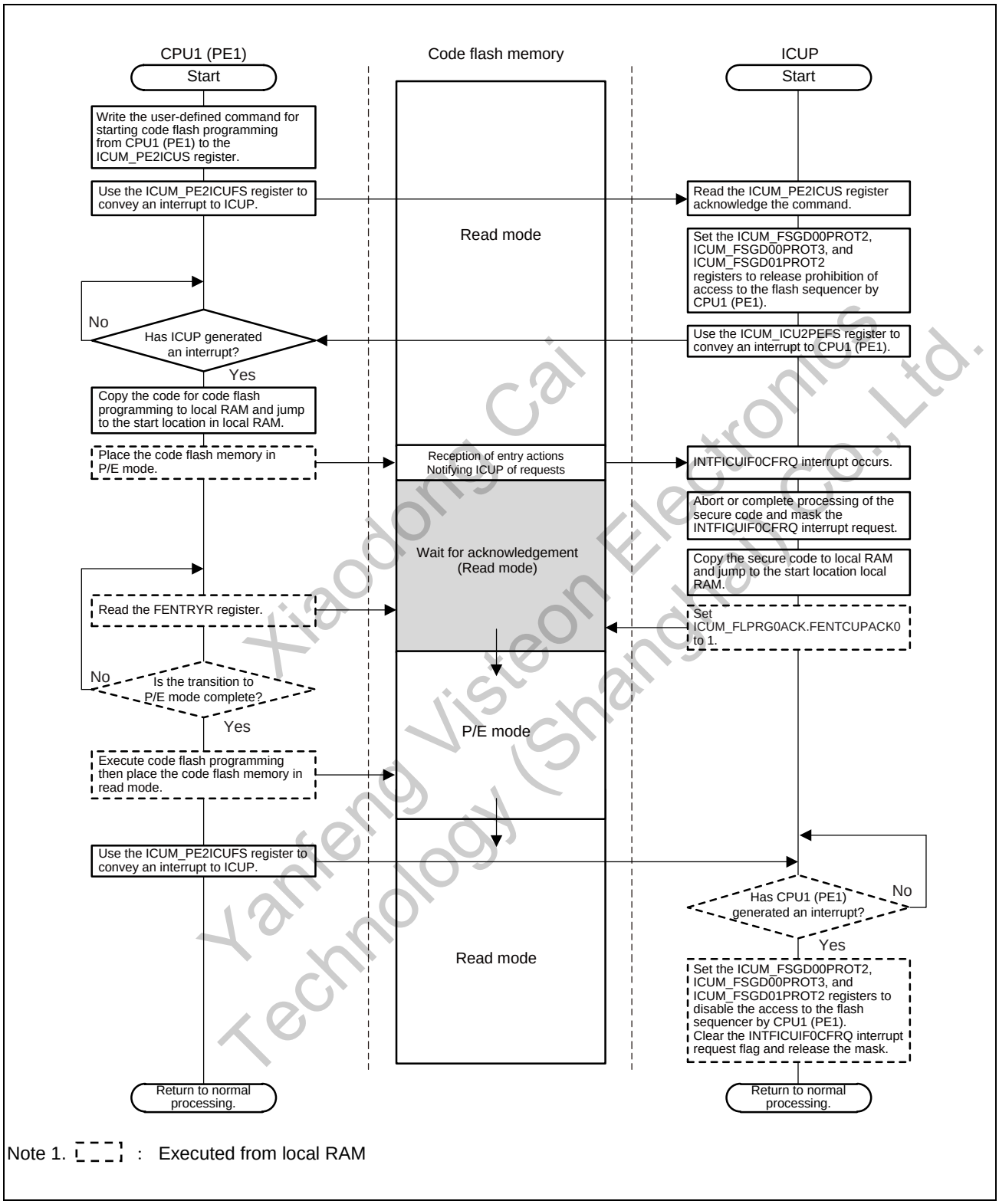


Figure 16.1 Simple Flowchart of Code Flash Programming from CPU1 (PE1)

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16.1.2.2 Code Flash Programming from ICUP

The figure below shows a simple flowchart of code flash programming from ICUP.

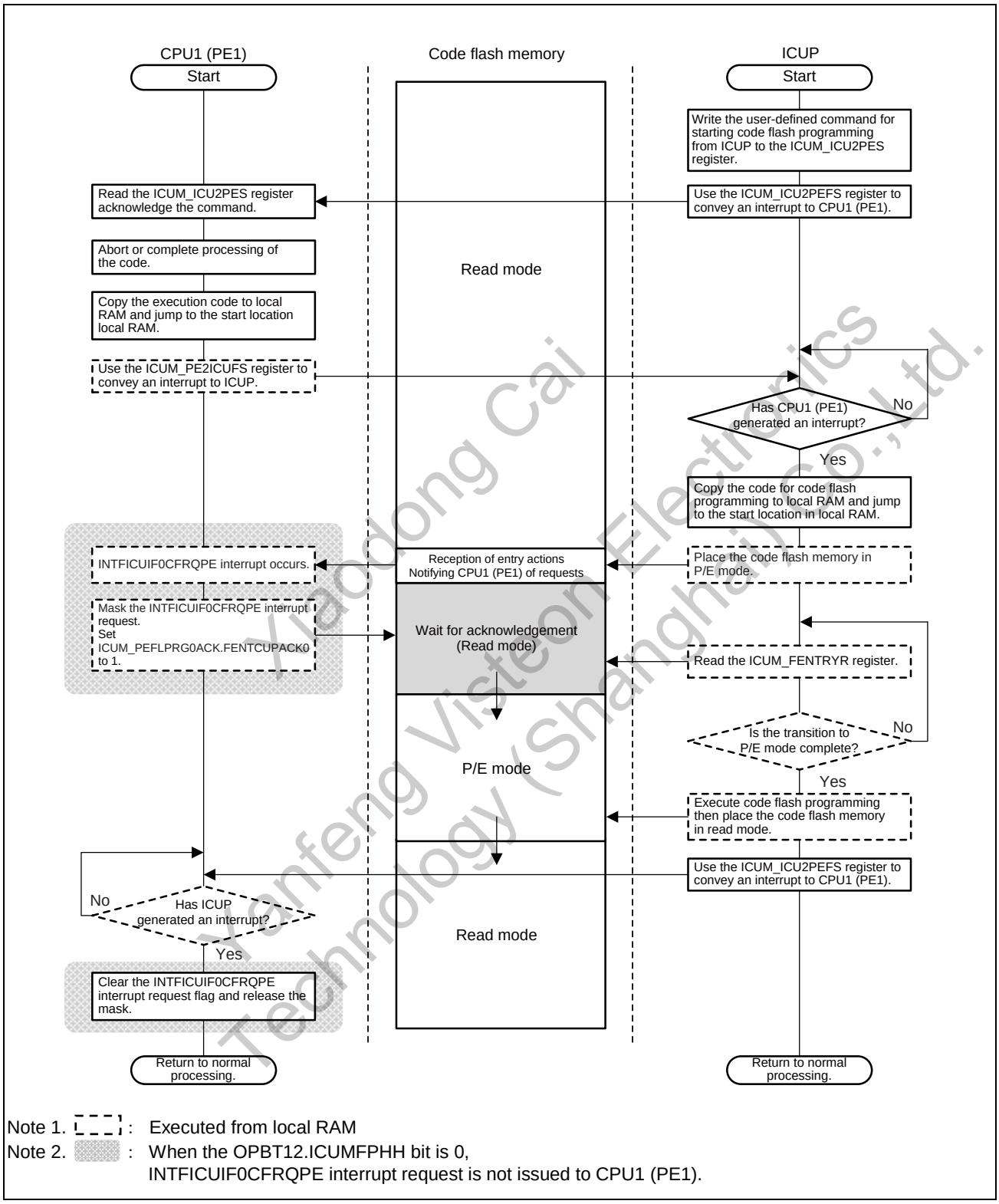


Figure 16.2 Simple Flowchart of Code Flash Programming from ICUP

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16.1.2.3 Data Flash Programming from CPU1 (PE1)

The figure below shows a simple flowchart of data flash programming from CPU1 (PE1).

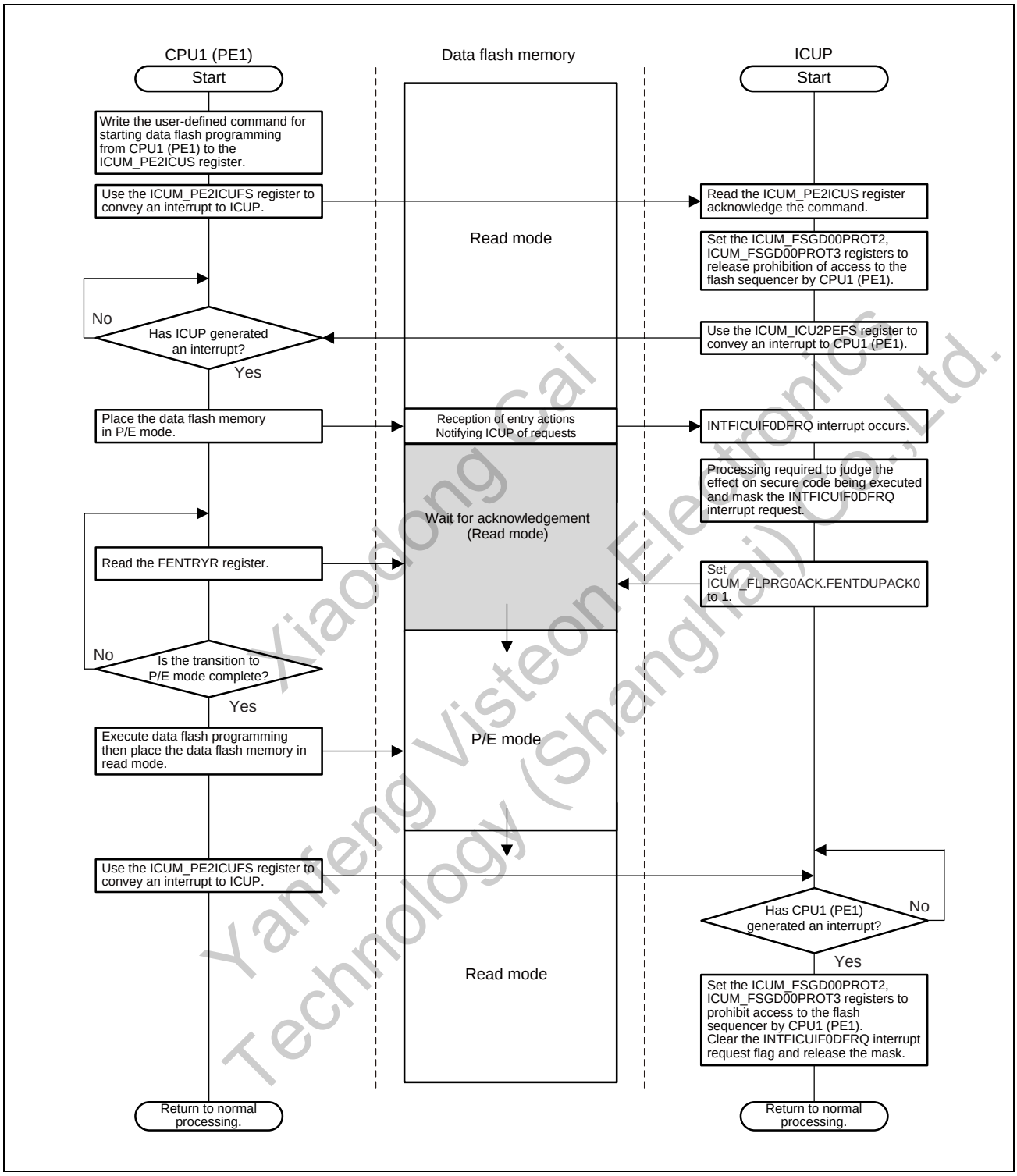
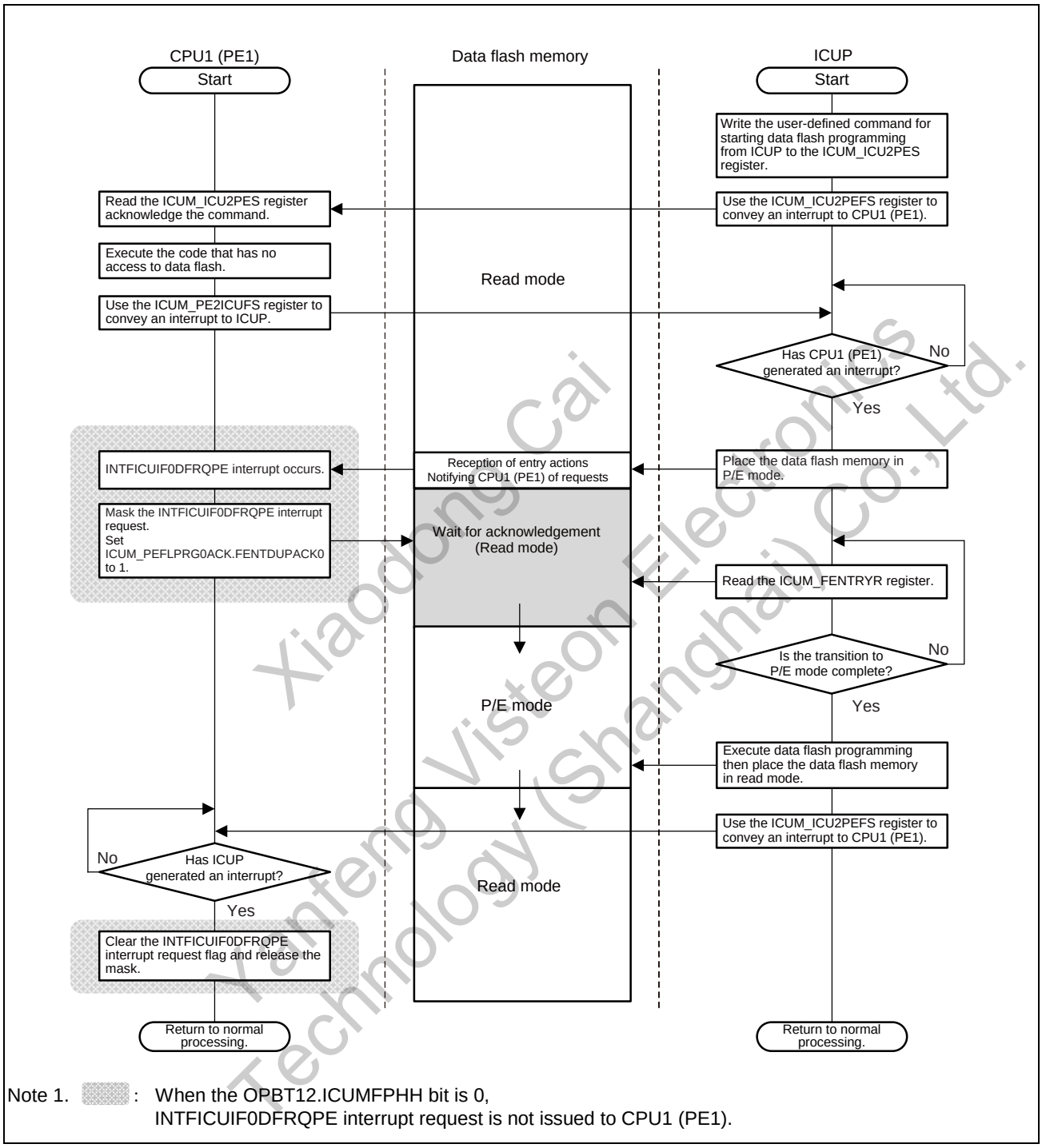


Figure 16.3 Simple Flowchart of Data Flash Programming from CPU1 (PE1)

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16.1.2.4 Data Flash Programming from ICUP

The figure below shows a simple flowchart of data flash programming from ICUP.



Note 1. : When the OPBT12.ICUMFPHH bit is 0, INTFICUIF0DFRQPE interrupt request is not issued to CPU1 (PE1).

Figure 16.4 Simple Flowchart of Data Flash Programming from ICUP

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16.2 ECC Error Detection and Correction of Code Flash Memory

16.2.1 Outline of ECC Functions for Code Flash Memory

Table 16.3 gives an outline of the ECC functions of the ICUMD code flash memory.

Table 16.3 List of ECC Functions for the Code Flash Memory

Item	Description
ECC error detection and correction	ECC error detection and correction can be enabled or disabled. When enabled, either of the following settings can be selected. • 2-bit error detection and 1-bit error detection / correction. • 2-bit error detection and 1-bit error detection. When disabled, neither error detection nor correction is carried out. In the initial state, this function is enabled, and 1-bit errors are detected and corrected, 2-bit errors are detected.
Error notification	A notification is sent when an ECC error occurs. • Enabling or disabling of error notification in case of detection of 2-bit ECC errors is selectable. • Enabling or disabling of error notification in case of detection of 1-bit ECC errors is selectable. In the initial state, 2-bit error notification is enabled, and 1-bit error notification is enabled.
Error status	The detection of 2-bit ECC errors and 1-bit ECC errors is monitored. The 1-bit ECC error status is set only when no error status has been set. The 2-bit ECC error status is set even when the 1-bit ECC error status is set. A register for clearing the error status is provided.
Address capture	When no ECC error status has been set, the address at which the first ECC error occurred is captured. In addition, when the retained address source is a 1-bit ECC error, the address of the 2-bit ECC error is also captured.
Self-diagnosis	The ECC bit can be read directly.
Inhibiting instruction execution	Generating a SYSERR exception in response to the detection of a 2-bit ECC error or a 1-bit ECC error (SECDIS = 1) during instruction fetching prevents the execution of incorrect instructions.

16.2.2 Interrupt Requests

Interrupt requests for code flash ECC are listed below.

Table 16.4 Interrupt Requests for Code Flash ECC

Unit Interrupt Signal	Description	Interrupt Name	DMA Trigger Number
—	1-bit ECC error interrupt of code flash memory	INTICUECCFLI0	—
—	2-bit ECC error interrupt of code flash memory		—

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16.2.3 Registers

Table below is a list of the registers of the code flash ECC.

For details, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

The only differences from the RH850/F1KH, RH850/F1KM ,the registers suffixed with symbols "_VCI", are register symbols, access size, and addresses.

Read "CF" as "ICUM_CF" and delete the suffix "_VCI" when searching for the register symbols in the manual.

All registers are accessed in 32 bits.

Table 16.5 List of Registers

Register Name	Symbol	Address
Code flash ECC control register	ICUM_CFECCCTL	FFFE EE00$_H$
Code flash error notification control register	ICUM_CFERRINT	FFFE EE04$_H$
Code flash error status clear register	ICUM_CFSTCLR	FFFE EE08$_H$
Code flash error overflow status register	ICUM_CFOVFSTR	FFFE EE0C$_H$
Code flash 1st error status register	ICUM_CF1STERSTR	FFFE EE10$_H$
Code flash 1st error address register	ICUM_CF1STEADR0	FFFE EE50$_H$
Code flash sub test control register	ICUM_CFSTSTCTL	FFFE EF50$_H$

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16.3 ECC Error Detection and Correction of Data Flash Memory

16.3.1 Outline of ECC Functions for Data Flash Memory

Table 16.6 gives an outline of the ECC functions of the ICUMD data flash memory.

Table 16.6 List of ECC Functions for the Data Flash Memory

Item	Description
ECC error detection and correction	ECC error detection and correction can be enabled or disabled. When enabled, either of the following settings can be selected. • 2-bit error detection and 1-bit error detection / correction. • 2-bit error detection and 1-bit error detection. When disabled, neither error detection nor correction is carried out. In the initial state, this function is enabled, and 1-bit errors are detected and corrected, 2-bit errors are detected.
Error notification	A notification is sent when an ECC error occurs. • Enabling or disabling of error notification in case of detection of 2-bit ECC errors is selectable. • Enabling or disabling of error notification in case of detection of 1-bit ECC errors is selectable. In the initial state, 2-bit error notification is enabled, and 1-bit error notification is disabled.
Error status	The detection of 2-bit ECC errors and 1-bit ECC errors is monitored. The function is set only while no error status is set. A register for clearing the error status is provided.
Address capture	When no ECC error status has been set, the address at which the first ECC error occurred is captured.
Self-diagnosis	The ECC bit can be read directly.

16.3.2 Interrupt Requests

Interrupt requests for data flash ECC are listed below.

Table 16.7 Interrupt Requests for Data Flash ECC

Unit Interrupt Signal	Description	Interrupt Name	DMA Trigger Number
—	1-bit ECC error interrupt of data flash memory	INTICUECCEEP0	—
—	2-bit ECC error interrupt of data flash memory		—

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16.3.3 Registers

Table below is a list of the registers of the data flash ECC.

For details, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

The only differences from the RH850/F1KH, RH850/F1KM are register symbols, and addresses.

Read “DF” as “ICUM_DF” when searching for the register symbols in the manual.

Table 16.8 List of Registers

Register Name	Symbol	Address
Data flash ECC control register	ICUM_DFECCCTL	FFFE EDC0$_H$
Data flash error status register	ICUM_DFERSTR	FFFE EDC4$_H$
Data flash error status clear register	ICUM_DFERSTC	FFFE EDC8$_H$
Data flash error overflow status register	ICUM_DFOVFSTR	FFFE EDCC$_H$
Data flash error overflow status clear register	ICUM_DFOVFSTC	FFFE EDD0$_H$
Data flash error notification control register	ICUM_DFERRINT	FFFE EDD4$_H$
Data flash 1st error address register	ICUM_DFEADR	FFFE EDD8$_H$
Data flash test control register	ICUM_DFTSTCTL	FFFE EDDC$_H$

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16.4 Flash Guard Status Register

16.4.1 Functional Overview of Flash Guard

Flash Guard protects read access to secure areas from other than the ICUMD. If illegal access occurs in the secure area, error information is stored in this flash guard status register and the access is reported to the ICUMD by an interrupt.

16.4.2 Interrupt Requests

The following table shows the interrupt requests for flash guard.

Table 16.9 Interrupt Requests for Flash Guard

Unit Interrupt Signal	Description	Interrupt Number	DMA Trigger Number
—	Interrupt indicating an illegal read access to the secure area of the code flash	26	—
—	Interrupt indicating an illegal read access to the secure area of the data flash	27	—

16.4.3 List of Registers

The flash guard registers are listed in the table below.

Table 16.10 List of Registers

Register Name	Symbol	Address
Code flash guard control register	ICUM_MGDCFCTL	FFFE ED80$_H$
Code flash guard error status register	ICUM_MGDCFSTAT	FFFE ED84$_H$
Data flash guard control register	ICUM_ERRSLV03CTL	FFFE EDA0$_H$
Data flash guard error status register	ICUM_ERRSLV03STAT	FFFE EDA4$_H$
Data flash guard error access type register	ICUM_ERRSLV03TYPE	FFFE EDAC$_H$

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16.4.4 Details of Registers

16.4.4.1 ICUM_MGDCFCTL — Code Flash Guard Control Register

This register is used to clear the status of the error status register in the code flash guard.

Access: This register is a write-only register that can be written in 32-bit units.

Address: FFFE ED80$_H$

Value after reset: 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CLRO	CLRE
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	W	W

Table 16.11 ICUM_MGDCFCTL Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When writing, write the value after reset.
1	CLRO	Clears an overflow flag 0: Does not clear an overflow flag. 1: Clears an overflow flag.
0	CLRE	Clears an error flag 0: Does not clear an error flag. 1: Clears an error flag.

CLRO	CLRE	Function
0	0	Neither bit is cleared.
0	1	Setting prohibited.
1	0	Clears the OVF bit in the code flash guard error status register.
1	1	Clears the OVF and ERR bits in the code flash guard error status register.

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16.4.4.2 ICUM_MGDCFSTAT — Code Flash Guard Error Status Register

This register is used to retain the status of the access violation to the secure area of the code flash.

Access: This register is a read-only register that can be read in 32-bit units.

Address: FFFE ED84$_H$

Value after reset: 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	OVF	ERR
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 16.12 ICUM_MGDCFSTAT Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When read, the value after reset is returned.
1	OVF	Overflow flag 0: No overflow has occurred 1: An overflow has occurred
0	ERR	Error flag 0: No access violation to the secure area has occurred 1: An access violation to the secure area has occurred

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16.4.4.3 ICUM_ERRSLV03CTL — Data Flash Guard Control Register

This register is used to clear the status of the error status register of the data flash guard.

Access: This register is a write-only register that can be written in 32-bit units.

Address: FFFE EDA0$_H$

Value after reset: 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CLRO	CLRE
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	W	W

Table 16.13 ICUM_ERRSLV03CTL Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When writing, write the value after reset.
1	CLRO	Clears an overflow flag 0: Does not clear an overflow flag. 1: Clears an overflow flag.
0	CLRE	Clears an error flag 0: Does not clear an error flag. 1: Clears an error flag.

CLRO	CLRE	Function
0	0	Neither bit is cleared.
0	1	Setting prohibited.
1	0	Clears the OVF bit in the data flash guard error status register.
1	1	Clears the OVF and ERR bits in the data flash guard error status register.

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16.4.4.4 ICUM_ERRSLV03STAT — Data Flash Guard Error Status Register

This register is used to retain the status when an access violation to the secure area of the data flash occurs.

Access: This register is a read-only register that can be read in 32-bit units.

Address: FFFE EDA4$_H$

Value after reset: 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	OVF	ERR
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 16.14 ICUM_ERRSLV03STAT Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When read, the value after reset is returned.
1	OVF	Overflow flag 0: No overflow has occurred 1: An overflow has occurred
0	ERR	Error flag 0: No access violation to the secure area has occurred 1: An access violation to the secure area has occurred

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16.4.4.5 ICUM_ERRSLV03TYPE — Data Flash Guard Error Access Type Register

This register is used to retain the access type when an access violation to the secure area of the data flash occurs.

Access: This register is a read-only register that can be read in 32-bit units.

Address: FFFE EDAC$_H$

Value after reset: 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PEID[2:0]			—	—	—	SPID[1:0]		—	UM	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 16.15 ICUM_ERRSLV03TYPE Register Contents

Bit Position	Bit Name	Function
31 to 16	—	Reserved When read, the value after reset is returned.
15 to 13	PEID[2:0]	The PEID of the originator of the access which led to a security violation
12 to 10	—	Reserved When read, the value after reset is returned.
9, 8	SPID[1:0]	The SPID of the originator of the access which led to a security violation
7	—	Reserved When read, the value after reset is returned.
6	UM	The UM of the originator of the access which led to a security violation
5 to 0	—	Reserved When read, the value after reset is returned.

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16.5 ICUM_PBG (ICUM_PBUS Guard)

For details about PBG, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

16.5.1 Functional Overview of ICUM_PBG

Table 16.16 lists peripheral guard modules for the ICUMD to control accesses.

Table 16.16 List of Modules for ICUM_PBG

PBG Group	Group Number	PBG Channel Number	Module for Protection Targets
ICUM_PBG00	00	2	FACI register area
		3	FACI command issuance area
ICUM_PBG01	01	2	FLMD

16.5.2 Interrupt Requests

ICUM_PGB interrupt requests are listed in the table below.

Table 16.17 ICUM_PBG Interrupt Requests

Unit Interrupt Signal	Description	Interrupt Number	DMA Trigger Number
—	Access violation detected in the ICUM_PBG00	23	—
	Access violation detected in the ICUM_PBG01		—

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16.5.3 Registers

16.5.3.1 List of Registers

ICUM_PBG registers are listed in the table below.

Table 16.18 lists the registers that are implemented in each ICUM_PBG channel.

Table 16.18 List of Registers (ICUM_FSGDn)

Register Name	Symbol	Address
ICUM_PBGxx protection register n	ICUM_FSGDxxPROTn	<base_addr0>+4*n_H

Table 16.19 lists the registers that are implemented in each ICUM_PBG group.

Table 16.19 List of Registers (ICUM_ERRSLVxx)

Register Name	Symbol	Address
ICUM_PBGxx error control register	ICUM_ERRSLVxxCTL	<base_addr1>+0_H
ICUM_PBGxx error status register	ICUM_ERRSLVxxSTAT	<base_addr1>+4_H
ICUM_PBGxx error type register	ICUM_ERRSLVxxTYPE	<base_addr1>+C_H

“xx” in the register names and symbols indicate the ICUM_PBG group number and “n” indicates the ICUM_PBG channel number. The value of base address <base_addr0> and <base_addr1> corresponding to each ICUM_PBG group number and PBG channel number are shown below.

Table 16.20 List of ICUM_PBG Base Addresses

PBG Group	Group Number xx	PBG Channel Number n	<base_addr0>	<base_addr1>
ICUM_PBG00	00	2, 3	FFFE ED00$_H$	FFFE ED20$_H$
ICUM_PBG01	01	2	FFFE ED40$_H$	FFFE ED60$_H$

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16.5.3.2 ICUM_FSGDxxPROTn — ICUM_PBGxx Protection Register n

This register specifies protection against access to the control registers of peripheral module circuits.

Access which is prohibited by any of the identifiers is treated as illegal and the register is protected.

Access: This register can be read or written in 32-bit units.

Address: See **Table 16.18, List of Registers (ICUM_FSGDn)**.

Value after reset: 0041 FF0F$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	PROTUM	—	—	—	—	—	PROTPEID2*1	PROTPEID1	—	—
Value after reset	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1
R/W	R	R	R	R	R	R	R/W	R	R	R	R	R	R/W	R/W	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	PROTSPID2*1	PROTSPID1	—	—	—	—	—	—
Value after reset	1	1	1	1	1	1	1	1	0	0	0	0	1	1	1	1
R/W	R	R	R	R	R	R	R	R	R/W	R/W	R	R	R	R	R	R

Table 16.21 ICUM_FSGDxxPROTn Register Contents

Bit Position	Bit Name	Function
31 to 26	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
25	PROTUM	User Mode Access 0: Enables access in supervisor mode. 1: Enables access in user mode and supervisor mode.
24 to 20	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
19	PROTPEID2	Access with PEID = 2 (CPU2) 0: Disables access with PEID2. 1: Enables access with PEID2.
18	PROTPEID1	Access with PEID = 1 (CPU1) 0: Disables access with PEID1. 1: Enables access with PEID1.
17 to 8	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
7	PROTSPID2	Access with SPID = 2 (CPU2) 0: Disables access with SPID2. 1: Enables access with SPID2.
6	PROTSPID1	Access with SPID = 1 (CPU1) 0: Disables access with SPID1. 1: Enables access with SPID1.
5 to 0	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.

Note 1. This register is supported only in RH850/F1KH-D8.

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16.5.3.3 ICUM_ERRSLVxxCTL — ICUM_PBGxx Error Control Register

This register is used to clear the status of the error status register of the ICUM_PBGxx.

Access: This register is a write-only register that can be written in 32-bit units.

Address: See **Table 16.19, List of Registers (ICUM_ERRSLVxx)**.

Value after reset: 0000 0000_H

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CLRO	CLRE
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	W	W

Table 16.22 ICUM_ERRSLVxxCTL Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When writing, write the value after reset.
1	CLRO	Clears an overflow flag 0: Does not clear an overflow flag. 1: Clears an overflow flag.
0	CLRE	Clears an error flag 0: Does not clear an error flag. 1: Clears an error flag.

CLRO	CLRE	Function
0	0	Neither bit is cleared.
0	1	Setting prohibited.
1	0	Clears the OVF bit in the ICUM_PBGxx error status register.
1	1	Clears the OVF and ERR bits in the ICUM_PBGxx error status register.

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16.5.3.4 ICUM_ERRSLVxxSTAT — ICUM_PBGxx Error Status Register

This register is used to retain the status when an access violation to the ICUM_PBGxx occurs.

Access: This register is a read-only register that can be read in 32-bit units.

Address: See **Table 16.19, List of Registers (ICUM_ERRSLVxx)**.

Value after reset: 0000 0000_H

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	OVF	ERR
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 16.23 ICUM_ERRSLVxxSTAT Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When read, the value after reset is returned.
1	OVF	Overflow flag 0: No overflow has occurred 1: An overflow has occurred
0	ERR	Error flag 0: No protection violation to PBG has occurred. 1: An protection violation to PBG has occurred.

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16.5.3.5 ICUM_ERRSLVxxTYPE — ICUM_PBGxx Error Type Register

This register is used to retain the access type when an access violation to ICUM_PBGxx occurs.

Access: This register is a read-only register that can be read in 32-bit units.

Address: See **Table 16.19, List of Registers (ICUM_ERRSLVxx)**.

Value after reset: 0000 0000_H

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PEID[2:0]			—	—	—	SPID[1:0]		—	UM	—	—	—	—	—	WRITE
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 16.24 ICUM_ERRSLVxxTYPE Register Contents

Bit Position	Bit Name	Function
31 to 16	—	Reserved When read, the value after reset is returned.
15 to 13	PEID[2:0]	The PEID of the originator of the access which led to a PBG protection violation
12 to 10	—	Reserved When read, the value after reset is returned.
9, 8	SPID[1:0]	The SPID of the originator of the access which led to a PBG protection violation
7	—	Reserved When read, the value after reset is returned.
6	UM	The UM of the originator of the access which led to a PBG protection violation
5	—	Reserved When read, the value after reset is returned.
4 to 1	—	Reserved These bits are read as an undefined value.
0	WRITE	This bit is set to 1 when the PBG protection violation was caused by an write access.

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16.6 PBG for ICUM_CMDREG

16.6.1 Functional Overview

The control register in the peripheral circuits is protected against illegal accesses. For details, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

This register can also be accessed by the CPU (PE).

Table 16.25 lists the peripheral circuits to be protected, the corresponding PBG group name, and the PBG channel number.

Table 16.25 PBG Group, Channel, and Target Module

PBG Group	Group No. (Register Name index)	PBG Channel Number	Protection Target Module	Target Register
PBG21	09	7	ICUMD	All registers in ICUM_CMDREG

16.6.2 Register

16.6.2.1 List of Register

Table 16.26 lists the register provided for PBG group.

For details about PBG, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

Table 16.26 List of Register Provided for PBG Channel

PBG Group	Group No. (Register Name index)	Register Symbol	Register Name	R/W	Value after Reset	Address	Access Size	Power Domain
PBG21	09	FSGD09PROT7	PBG09 Protection Register 7	R/W	0667 FF77$_H$	FFDD D11C$_H$	8/16/32	ISO

In addition to the above registers, there is also a register group used to report errors in the PBG. For the addresses of the PBG error registers, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*. The base address of the PBG error registers is determined by the group number. For details about the PBG error registers, see the *RH850/F1KH, RH850/F1KM User's Manual: Hardware*.

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16.7 Registers and Addresses Related to Flash Programming

The ICUMD has the following dedicated registers for flash programming.

(1) Flash Sequencer Hardware Interface for the ICUMD

The flash sequencer is accessed from the ICUMD via these dedicated registers.
Although access to the sequencer via these registers is only possible from the ICUMD, the registers themselves are accessible from cores other than the ICUMD.

(2) Clearing the Cache for the Code Flash Interface

Updating the code flash memory through self-programming produces a temporary inconsistency between the contents of the code flash memory and of the cache between the ICUMD and the code flash interface, so the ICUMD includes a cache clearing function.
For details, see **Section 16.7.1.3, ICUM_FCBWCTL — Flash Cache Control Register**.

16.7.1 Registers

16.7.1.1 List of Registers

ICUMD flash programming-related registers are listed in the table below.
There are no changes to the BFA selection registers and the registers related to the self-programming ID.

Table 16.27 List of Registers

Module Name	Register Name	Symbol	Address
FACI *1	Flash pin monitor register	ICUM_FPMON	FFBB 0000$_H$
	Flash access status register	ICUM_FASTAT	FFBB 0010$_H$
	Flash sequencer end error interrupt enable register	ICUM_FAEINT	FFBB 0014$_H$
	Code flash memory area select register	ICUM_FAREASELC	FFBB 0020$_H$
	FACI command start address register	ICUM_FSADDR	FFBB 0030$_H$
	FACI command end address register	ICUM_FEADDR	FFBB 0034$_H$
	Flash status register	ICUM_FSTATR	FFBB 0080$_H$
	Flash P/E mode entry register	ICUM_FENTRYR	FFBB 0084$_H$
	Code flash protect register	ICUM_FPROTR	FFBB 0088$_H$
	Flash sequencer set-up initialization register	ICUM_FSUINITR	FFBB 008C$_H$
	Lock bit status register	ICUM_FLKSTAT	FFBB 0090$_H$
	FACI reset transfer status register	ICUM_FRTSTAT	FFBB 0098$_H$
	FACI command register	ICUM_FCMDR	FFBB 00A0$_H$
	Flash P/E status register	ICUM_FPESTAT	FFBB 00C0$_H$
	Data flash blank check control register	ICUM_FBCCNT	FFBB 00D0$_H$
	Data flash blank check status register	ICUM_FBCSTAT	FFBB 00D4$_H$
	Data flash programming start address register	ICUM_FPSADDR	FFBB 00D8$_H$
	Flash sequencer processing switching register	ICUM_FCPSR	FFBB 00E0$_H$
	Flash sequencer processing clock notification register	ICUM_FPCKAR	FFBB 00E4$_H$
	Flash ECC encoder monitor register	ICUM_FECCEMON	FFBB 0100$_H$
	Flash ECC test mode register	ICUM_FECCTMD	FFBB 0104$_H$
	Flash dummy ECC register	ICUM_FDMYECC	FFBB 0108$_H$
—	Flash cache control register	ICUM_FCBWCTL	FFFE E500$_H$
	Flash preload control register	ICUM_FLPLCTL	FFFE E504$_H$

Note 1. These registers are identical to CPU(PE)'s one which has the same symbol when reading registers without the prefix "ICUM_".

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16.7.1.2 Details of Registers

For the flash sequencer functions and details of the registers, see the *RH850/F1KH, RH850/F1KM, RH850/F1K Flash Memory User's Manual: Hardware Interface*.

Delete the prefix "ICUM_" when searching for the register symbols in the manual.

16.7.1.3 ICUM_FCBWCTL — Flash Cache Control Register

This register is enabled for reads and writes in supervisor mode but enabled only for read accesses in user mode; any write accesses are ignored.

Access: This register can be read or written in 32-bit units.

Address: FFFE $E500_H$

Value after reset: 0000 0001_H

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ICHCLR	ICHEN
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	W	R/W

Table 16.28 ICUM_FCBWCTL Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
1	ICHCLR	Setting this bit (to 1) clears the instruction cache and BWC Buffer (Buffer for the code flash Interface). Writing a 0 has no meaning. This bit is always read as 0.
0	ICHEN	Enables or disables the instruction cache function. 1: Enables hit test on cache and fill operations on Cache RAM. 0: Disables hit test on cache and fill operations on Cache RAM.

NOTES

To maintain the state after having cleared cache, ICHCLR have need to be set after first disabling the cache function (ICHEN = 0).

The procedure for enabling/disabling a cache function at intentional timing is described below.

1. Set ICUM_FCBWCTL.ICHEN bit by the instruction placed in the local RAM.
2. Place "SYNCM" instruction immediately after the instruction to set ICUM_FCBWCTL.ICHEN bit, and place a branch instruction to code flash after it.

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16.7.1.4 ICUM_FLPLCTL — Flash Preload Control Register

This register is enabled for reads and writes in supervisor mode but enabled only for read accesses in user mode; any write accesses are ignored.

This register is set the enabled/disabled about pre-load function on Data access. This function can be used to check the code on secure boot. This function can be used only for secure boot.

The access to code flash is prohibited when changing the value of this register.

Access: This register can be read or written in 32-bit units.

Address: FFFE E504$_H$

Value after reset: 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	EXPL
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R/W

Table 16.29 ICUM_FLPLCTL Register Contents

Bit Position	Bit Name	Function
31 to 1	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
0	EXPL	Enabling/disabling the preload function by the BWC during data access. 1: The preload function is enabled during instruction fetch access and data access. 0: The preload function is enabled during instruction fetch access only.

NOTES

When this function is used, secure boot processing need to be executed on RAM.

The procedure for enabling/disabling a cache function at intentional timing is described below.

1. Set ICUM_FCBWCTL.ICHEN bit by the instruction placed in the local RAM.
2. Place “SYNCM” instruction immediately after the instruction to set ICUM_FCBWCTL.ICHEN bit.

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16.7.2 Addresses related to flash programming

When an area related to flash programming is accessed by the ICUMD, the address may differ from when the area is accessed by CPU1 (PE1). The addresses of each area are shown in the table below.

Table 16.30 Addresses of Flash Programming Related Areas

Area	Address	Size	Different from CPU1 (PE1) mapping
Hardware register areas	See **Section 16.7.1.1, List of Registers**	See **Section 16.7.1.1, List of Registers**	Yes
FACI command issuance area	FFBC 0000_H	4 bytes	Yes
Configuration setting area	FF30 0040_H to FF30 $00AF_H$	112 bytes	No
OTP setting area	FF38 0040_H to FF38 $009F_H$	96 bytes	No

In addition, addresses of option byte for ICUMD are shown below. ICUMD option bytes can be set by configuration setting command.

Table 16.31 ICUMD Option Byte Addresses Used by Configuration Setting Command

Address	Setting Data
FF30 0090_H	Option byte 8, Option byte 9, Option byte 10, Option byte 11 *1
FF30 $00A0_H$	Option byte 12 *2

Note 1. The setting data is as follows *3

n = 0 : Bit 15 to 0 of Option byte 8
n = 1 : Bit 31 to 16 of Option byte 8
n = 2 : Bit 15 to 0 of Option byte 9
n = 3 : Bit 31 to 16 of Option byte 9
n = 4 : Bit 15 to 0 of Option byte 10
n = 5 : Bit 31 to 16 of Option byte 10
n = 6 : Bit 15 to 0 of Option byte 11
n = 7 : Bit 31 to 16 of Option byte 11

Note 2. The setting data is as follows *3

n = 0 : Bit 15 to 0 of Option byte 12
n = 1 : Bit 31 to 16 of Option byte 12

Note 3. For the meaning of "n", see the Usage of the Configuration Setting Command in the *RH850/F1KH, RH850/F1KM, RH850/F1K Flash Memory User's Manual: Hardware Interface*.

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16.8 Option Bytes

The flash memory has an expansion area for the user to define the ICUMD operations.

The modification becomes valid after reset is deasserted.

16.8.1 OPBT8 — Option Byte 8

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICUMON[3:0]				—	—	—	—	UPGP1[3:0]				—	—	—	—
Setting value	0/1	0/1	0/1	0/1	1	1	1	1	0/1	0/1	0/1	0/1	1	1	1	1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	UPGP2[3:0]				—	—	—	—	—	—	—	—	—	—	—	—
Setting value	0/1	0/1	0/1	0/1	1	1	1	1	1	1	1	1	1	1	1	1

Table 16.32 OPBT8 Register Contents

Bit Position	Bit Name	Function
31 to 28	ICUMON[3:0]	ICUMD startup flag When ICUMON is 0_H, these flags satisfy one of the conditions for the ICUMD startup. For details, see **Section** 18.1**,** Enabling ICUMD.
27 to 24	—	Reserved When writing, always write 1.
23 to 20	UPGP1[3:0]	Upgrade protection 1 enable flag Setting 0_H during ICUMD startup enables upgrade protection 1. When the upgrade protection 1 is enabled, overwriting the code flash by serial programming mode is prohibited. Specify the settings of these flags before shipment.
19 to 16	—	Reserved When writing, always write 1.
15 to 12	UPGP2[3:0]	Upgrade protection 2 enable flag Setting 0_H during ICUMD startup enables upgrade protection 2. When the upgrade protection 2 is enabled, overwriting option byte by serial programming mode is prohibited. Specify the settings of these flags before shipment.
11 to 0	—	Reserved When writing, always write 1.

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16.8.2 OPBT9 — Option Byte 9

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICUMRESV[31:16]															
Setting value	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICUMRESV[15:0]															
Setting value	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1

Table 16.33 OPBT9 Register Contents

Bit Position	Bit Name	Function
31 to 0	ICUMRESV[31:0]	ICUMD reset vector address These bits set instruction fetch start address of the ICUMD after a reset. Set the fetch start address within the secure area of the code flash.

16.8.3 OPBT10 — Option Byte 10

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICUMRAC[31:16]															
Setting value	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICUMRAC[15:0]															
Setting value	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1

Table 16.34 OPBT10 Register Contents

Bit Position	Bit Name	Function
31 to 0	ICUMRAC[31:0]	Code flash secure area start address Set the address where the secure area of the code flash memory starts. The code flash memory at the start address and higher addresses becomes the secure area (except for the extended user area). The start address can be set in blocks 0020 0000_H (RH850/F1KM-S4) / 0080 0000_H (RH850/F1KH-D8) and beyond. Set the start address of the block to be used as the secure area.

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16.8.4 OPBT11 — Option Byte 11

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	ICUMRAD[31:16]															
Setting value	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	ICUMRAD[15:0]															
Setting value	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1	0/1

Table 16.35 OPBT11 Register Contents

Bit Position	Bit Name	Function
31 to 0	ICUMRAD[31:0]	Data flash secure area start address Set the address where the secure area of the data flash memory starts. The data flash memory at the start address and higher addresses becomes the secure area. The start address can be set in blocks. Set the start address of the block to be used as the secure area.

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16.8.5 OPBT12 — Option Byte 12

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	ICUMPEDBG	—	—	ICUMCRF2	ICUMCRF1	—	ICUMFPHH	—	—	—	—	ICUMOPGPIO	—
Setting value	1	1	1	0/1	1	1	0/1	0/1	1	0/1	1	1	1	1	0/1	1

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	ICUMWDT0_3	ICUMWDT0_1	ICUMWDT0_0	ICUMWDT_[2:0]		
Setting value	1	1	1	1	1	1	1	1	1	1	0/1	0/1	0/1	0/1	0/1	0/1

Table 16.36 OPBT12 Register Contents

Bit Position	Bit Name	Function
31 to 29	—	Reserved When writing, always write 1.
28	ICUMPEDBG	Enables or disables CPU (PE) independent debug. 0: CPU (PE) independent debug is enabled. 1: CPU (PE) independent debug is disabled.
27, 26	—	Reserved When writing, always write 1.
25	ICUMCRF2	Enables or disables challenge and response (C&R) for ICUP. 0: C&R authentication is enabled. 1: C&R authentication is skipped.
24	ICUMCRF1	Enables or disables challenge and response (C&R) for CPU (PE). 0: C&R authentication is enabled. 1: C&R authentication is skipped.
23	—	Reserved When writing, always write 1.
22	ICUMFPHH	Enables or disables hardware handshake of flash programming from ICUP. 0: Hardware handshake is disabled. 1: Hardware handshake is enabled.
21 to 18	—	Reserved When writing, always write 1.
17	ICUMOPGPIO	Enables or disables GPIO control by ICUM_GPIO register. 0: ICUMD GPIO control is disabled. 1: ICUMD GPIO control is enabled.
16 to 6	—	Reserved When writing, always write 1.
5	ICUMWDT0_3	Specify activation code method of the ICUM_WDTA0. 0: Fixed activation code 1: Variable activation code
4	ICUMWDT0_1	Specify start mode of the ICUM_WDTA0. 0: Software trigger start mode 1: Default start mode
3	ICUMWDT0_0	Enables or disables the ICUM_WDTA0. 0: ICUM_WDTA0 is disabled. 1: ICUM_WDTA0 is enabled.
2 to 0	ICUMWDT_[2:0]	Overflow internal time control of the ICUM_WDTA0. Specify the value after reset of the ICUM_WDTAnMD.WDTAnOVF[2:0].

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Section 17 RAM

This section describes the dedicated RAMs of the ICUMD and the error correction function (ECC) of these RAMs.

17.1 Overview

- ICUMD has local RAM (64 KB) and instruction cache RAM (4KB-2way).
- Each RAM has Error detection/correction function (ECC).
- Includes an ECC function that can be used to detect 2-bit errors and detect/correct 1-bit errors.

17.2 Local RAM

The Local RAM has the following characteristics:

- Size: 64 KB

17.3 Instruction Cache

The instruction cache has the following characteristics:

- Size: 4 KB
- 2-way associative
- Enable/disable caching control

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17.4 ECC for Local RAM

17.4.1 Outline of ECC Function for Local RAM

Table 17.1 shows an outline of the ECC functions for the RAM.

Table 17.1 List of the ECC Functions for RAM

Item	Description
ECC error detection and correction	ECC error detection and correction can be enabled or disabled. When enabled, either of the following settings can be selected. • 2-bit error detection and 1-bit error detection / correction. • 2-bit error detection and 1-bit error detection. When disabled, neither error detection nor correction is carried out. In the initial state, this function is enabled, and 1-bit errors are detected and corrected, 2-bit errors are detected.
Error notification	A notification is sent when an ECC error occurs. • Enabling or disabling of error notification in case of detection of 2-bit ECC errors is selectable. • Enabling or disabling of error notification in case of detection of 1-bit ECC errors is selectable. In the initial state, 2-bit error notification is enabled, and 1-bit error notification is enabled.
Error status	The detection of 2-bit ECC errors and 1-bit ECC errors is monitored. The 1-bit ECC error status is set only when no error status has been set. The 2-bit ECC error status is set even when the 1-bit ECC error status is set. A register for clearing the error status is provided.
Address capture	When no ECC error status has been set, the address at which the first ECC error occurred is captured. In addition, when the retained address source is a 1-bit ECC error, the address of the 2-bit ECC error is also captured.
Self-diagnosis	Arbitrary data can be written to RAM data and the ECC bit. The ECC bit can be read.
Inhibiting instruction execution	Generating a SYSERR exception in response to the detection of a 2-bit ECC error or a 1-bit ECC error (SECDIS = 1) during instruction fetching prevents the execution of incorrect instructions.

CAUTION

- When ECC error detection/correction for the local RAM is enabled, initialize the RAM with the access length set to 32 bits before the RAM is used.
- If the RAM before initialization is read, an FE-level maskable interrupt or SYSERR exceptional processing may be generated.
- Moreover, if the RAM is not initialized by using a 32-bit length (and initialized by using a 8- or 16-bit length), an FE-level maskable interrupt or SYSERR exceptional processing may be generated.

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17.4.2 Interrupt Requests

The following table shows interrupt requests for local RAM ECC.

Table 17.2 Interrupt Requests by for Local RAM ECC (FE Level Maskable Interrupt)

Unit Interrupt Signal	Function	Interrupt Name	DMA Trigger Number
—	1-bit ECC error interrupt of local RAM	INTICUECCLRAM	—
	2-bit ECC error interrupt of local RAM		—

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17.4.3 Registers

17.4.3.1 List of Registers

The table below is a list of the ECC-related registers for the local RAM.

Table 17.3 List of Registers

Register Name	Symbol	Address
Local RAM ECC control register	ICUM_LRECCCTL	FFFE E300$_H$
Local RAM 1st error status register	ICUM_LR1STERSTR	FFFE E304$_H$
Local RAM error status clear register	ICUM_LRSTCLR	FFFE E324$_H$
Local RAM error overflow status register	ICUM_LROVFSTR	FFFE E328$_H$
Local RAM error overflow status clear register	ICUM_LROVFSTC	FFFE E32C$_H$
Local RAM 1st error address register 0	ICUM_LR1STEADR0	FFFE E330$_H$
Local RAM error notification control register	ICUM_LRERRINT	FFFE E3B0$_H$
Local RAM test control register	ICUM_LRTSTCTL	FFFE E3B4$_H$
Local RAM test data read buffer 0	ICUM_LRTDATBF0	FFFE E3B8$_H$

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17.4.3.2 ICUM_LRECCCTL — Local RAM ECC Control Register

This register enables or disables the ECC error detection/correction and 1-bit error correction. Set the PROT1 and PROT0 bits to 01_B when writing to this register.

Access: This register can be read or written in 16-bit units.

Address: FFFE $E300_H$

Value after reset: 0000_H

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PROT1	PROT0	—	—	—	—	—	—	—	—	—	—	—	—	SECDIS	ECCDIS
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R/W	R/W	R	R	R	R	R	R	R	R	R	R	R	R	R/W	R/W

Table 17.4 ICUM_LRECCCTL Register Contents

Bit Position	Bit Name	Function
15	PROT1	These bits enable or disable programming of the ECCDIS and SEDDIS bits. Values written are not retained. These bits are always read as 0. Set the PROT1 and PROT0 bits to 0 and 1 when writing to this register.
14	PROT0	
13 to 2	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
1	SECDIS	1-Bit error correction disable This bit enables or disables 1-bit error correction when the ECC error detection/ correction is enabled. When writing to this bit, write 0 and 1 to PROT1 and PROT0 at the same time. 0: 1-bit error correction is performed when a 1-bit error is detected. 1: 1-bit error correction is not performed when a 1-bit error is detected.
0	ECCDIS	ECC disable This bit enables or disables ECC error detection/correction. When writing to this bit, write 0 and 1 to PROT1 and PROT0 at the same time. ECC error detection/correction is enabled in the initial state. 0: ECC error detection/correction is enabled. 1: ECC error detection/correction is disabled. ECC data is still written even if the error detection/correction function is disabled.

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17.4.3.3 ICUM_LR1STERSTR — Local RAM 1st Error Status Register

This register monitors the occurrence of the first error.

Detecting a 1-bit ECC error sets the SEDF0 bit and detecting a 2-bit ECC error sets the DEDF0 bit while ECC error detection/correction is enabled.

Access: This register is a read-only register that can be read in 32-bit units.

Address: FFFE E304$_H$

Value after reset: 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DEDF0	SEDF0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 17.5 ICUM_LR1STERSTR Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When read, the value after reset is returned.
1	DEDF0	Local RAM 2-bit ECC error monitor flag This bit is set to 1 when a 2-bit ECC error occurs, regardless of the SEDF0 or DEDF0 state. [Clearing condition] An internal reset, generation of a pin reset, or setting the 1STERRCLR0 bit in local RAM error status clear register to 1 [Setting condition] When a 2-bit ECC error occurs
0	SEDF0	Local RAM 1-bit ECC error monitor flag This bit is set to 1 when a 1-bit ECC error occurs while SEDF0 and DEDF0 are 0. [Clearing condition] An internal reset, generation of a pin reset, or setting the 1STERRCLR0 bit in local RAM error status clear register to 1 [Setting condition] When a 1-bit ECC error occurs while both SEDF0 and DEDF0 are 0.

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17.4.3.4 ICUM_LRSTCLR — Local RAM Error Status Clear Register

This register clears the error flag in the local RAM 1st error status register.

Access: This register is a write-only register that can be written in 8-bit units.

Address: FFFE E324$_H$

Value after reset: 00$_H$

Bit	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	1STERRCLR0
Value after reset	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	W

Table 17.6 ICUM_LRSTCLR Register Contents

Bit Position	Bit Name	Function
7 to 1	—	Reserved When writing, write the value after reset.
0	1STERRCLR0	First error status register clear Writing 1 to this bit clears the error flag in the local RAM 1st error status register.

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17.4.3.5 ICUM_LROVFSTR — Local RAM Error Overflow Status Register

This register monitors occurrence of a local RAM error overflow. Setting the ERROVFCLR0 bit in the ICUM_LROVFSTC register clears the ERROVF0 bit.

Access: This register is a read-only register that can be read in 8-bit units.

Address: FFFE $E328_H$

Value after reset: 00_H

Bit	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	ERROVF0
Value after reset	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R

Table 17.7 ICUM_LROVFSTR Register Contents

Bit Position	Bit Name	Function
7 to 1	—	Reserved When read, the value after reset is returned.
0	ERROVF0	Error overflow flag This bit is set when an ECC error occurs while the error address register is enabled. ● If the retained address source is SEDF: When a 1-bit error occurs in the same address, no overflow occurs. When a 1-bit error occurs in a different address, an overflow occurs. When a 2-bit error occurs, an overflow occurs. ● If the retained address source is DEDF: When a 1-bit error occurs, an overflow occurs. When a 2-bit error occurs in the same address, no overflow occurs. When a 2-bit error occurs in a different address, an overflow occurs.

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17.4.3.6 ICUM_LROVFSTC — Local RAM Error Overflow Status Clear Register

This register clears the local RAM error overflow flag. Setting 1 to the ERROVFCLR0 bit clears this ERROVF0 bit in the ICUM_LROVFSTR register.

Access: This register is a write-only register that can be written in 8-bit units.

Address: FFFE E32C$_H$

Value after reset: 00$_H$

Bit	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	ERROVFCLR0
Value after reset	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	W

Table 17.8 ICUM_LROVFSTC Register Contents

Bit Position	Bit Name	Function
7 to 1	—	Reserved When writing, write the value after reset.
0	ERROVFCLR0	Error overflow flag clear Writing 1 to this bit clears the ERROVF0 flag.

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17.4.3.7 ICUM_LR1STEADR0 — Local RAM 1st Error Address Register 0

LR1STEADR0 retains the address of an ECC error occurrence. This register retains an internal address. Add the base address FEC0 0000_H to convert it to the actual address.

Access: This register is a read-only register that can be read in 32-bit units.

Address: FFFE $E330_H$

Value after reset: $0000\ 0000_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	LR1STEADR0[21:16]					
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	LR1STEADR0[15:2]														—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 17.9 ICUM_LR1STEADR0 Register Contents

Bit Position	Bit Name	Function
31 to 22	—	Reserved When read, the value after reset is returned.
21 to 2	LR1STEADR0 [21:2]	First error occurrence address These bits are read-only bits to monitor the address where the first error has occurred. If the retained address source is SEDF: When a 1-bit error occurs in the same address, the address is not overwritten. When a 1-bit error occurs in a different address, the address is not overwritten. When a 2-bit error occurs, the address is overwritten. If the retained address source is DEDF: When a 1-bit error occurs, the address is not overwritten. When a 2-bit error occurs in the same address, the address is not overwritten. When a 2-bit error occurs in a different address, the address is not overwritten
1, 0	—	Reserved When read, the value after reset is returned.

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17.4.3.8 ICUM_LRERRINT — Local RAM Error Notification Control Register

This register is used for specifying whether to enable or disable error notification signals generated when a 2-bit ECC error or a 1-bit ECC error is detected.

Access: This register can be read or written in 8-bit units.

Address: FFFE E3B0$_H$

Value after reset: 03$_H$

Bit	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	DEDIE	SEDIE
Value after reset	0	0	0	0	0	0	1	1
R/W	R	R	R	R	R	R	R/W	R/W

Table 17.10 ICUM_LRERRINT Register Contents

Bit Position	Bit Name	Function
7 to 2	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
1	DEDIE	2-bit ECC error notification enable This bit controls whether to report an error when a 2-bit error is detected while ECC error detection/correction is enabled. 0: 2-bit ECC error notification is disabled. 1: 2-bit ECC error notification is enabled.
0	SEDIE	1-bit ECC error notification enable This bit controls whether to report an error when a 1-bit error is detected while ECC error detection/correction is enabled. 0: 1-bit ECC error notification is disabled. 1: 1-bit ECC error notification is enabled.

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17.4.3.9 ICUM_LRTSTCTL — Local RAM Test Control Register

This register is used for an ECC test (self diagnosis). After the ECC test mode is enabled by setting ECCTST = 1, arbitrary data can be written to the ECC bits. The DATSEL bit is used to select the RAM data or the ECC bits.

When writing to the ICUM_LRTSTCTL register, PROT1 and PROT0 must be 01_B.

Access: This register can be read or written in 16-bit units.

Address: FFFE E3B4$_H$

Value after reset: 0000_H

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PROT1	PROT0	—	—	—	—	—	—	—	—	—	—	—	—	ECCTST	DATSEL
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R/W	R/W	R	R	R	R	R	R	R	R	R	R	R	R	R/W	R/W

Table 17.11 ICUM_LRTSTCTL Register Contents

Bit Position	Bit Name	Function
15	PROT1	These two bits specify whether rewriting the ECCTST and DATSEL bits is disabled or enabled. The value written to these bits is not retained. When these bits are read, the read value is always 0. When writing to the LRTSTCTL register, (PROT1, PROT0) must be (0,1).
14	PROT0	
13 to 2	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
1	ECCTST	ECC test mode This bit specifies the ECC test mode. 0: Normal mode 1: ECC test mode
0	DATSEL	Data selection This bit is enabled when ECCTST = 1. This bit specifies the RAM bit which can be accessed when writing. 0: RAM data is selected. 1: ECC bits are selected.

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17.4.3.10 ICUM_LRTDATBF0 — Local RAM Test Data Read Buffer 0

In test mode (ECCTST = 1), the ECC bits can be read. While ECCTST = 1 in the local RAM test control register, reading the RAM reads out the data of the ECC bits stored in the RAM, and the data is stored in this buffer.

Access: This register is a read-only register that can be read in 32-bit units.

Address: FFFE E3B8$_H$

Value after reset: 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	LRTDATBF0[6:0]						
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 17.12 ICUM_LRTDATBF0 Register Contents

Bit Position	Bit Name	Function
31 to 7	—	Reserved When read, the value after reset is returned.
6 to 0	LRTDATBF0[6:0]	These bits are enabled when ECCTST = 1 (test mode) in the local RAM test control register. When reading the RAM, the data of the ECC bits is stored in LRTDATBF06 to LRTDATBF00.

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17.5 ECC for Instruction Cache RAM

17.5.1 Outline of ECC Functions for Instruction Cache RAM

Table 17.13 shows an outline of the ECC functions for the instruction cache RAM.

Table 17.13 List of the ECC Functions for the Instruction Cache RAM

Item	Description
ECC error detection and correction	ECC error detection and correction can be enabled or disabled. When enabled, either of the following settings can be selected. • 2-bit error detection and 1-bit error detection / correction. • 2-bit error detection and 1-bit error detection. When disabled, neither error detection nor correction is carried out. In the initial state, this function is enabled, and 1-bit errors are detected and corrected, 2-bit errors are detected.
Error notification	A notification is sent when an ECC error occurs. • Enabling or disabling of error notification in case of detection of 2-bit ECC errors is selectable. • Enabling or disabling of error notification in case of detection of 1-bit ECC errors is selectable. In the initial state, 2-bit error notification is enabled, and 1-bit error notification is enabled.
Error status	The detection of 2-bit ECC errors and 1-bit ECC errors is monitored. The 1-bit ECC error status is set only when no error status has been set. The 2-bit ECC error status is set even when the 1-bit ECC error status is set. A register for clearing the error status is provided.
Inhibiting instruction execution	Generating a SYSERR exception in response to the detection of a 2-bit ECC error or a 1-bit ECC error (SECDIS = 1) during instruction fetching prevents the execution of incorrect instructions.

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17.5.2 Interrupt Requests

The following table shows interrupt requests for instruction cache RAM ECC.

Table 17.14 Interrupt Requests by for Instruction Cache RAM ECC (FE Level Maskable Interrupt)

Unit Interrupt Signal	Function	Interrupt Name	DMA Trigger Number
—	1-bit ECC error interrupt of instruction cache RAM	INTICUECCCRAM	—
	2-bit ECC error interrupt of instruction cache RAM		—

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17.5.3 Registers

17.5.3.1 List of Registers

The table below is a list of the ECC-related registers for the instruction cache RAM.

Table 17.15 List of Registers

Register Name	Symbol	Address
Instruction cache RAM ECC control register	ICUM_ICECCCTL	FFFE E400$_H$
Instruction cache RAM 1st error status register	ICUM_IC1STERSTR	FFFE E404$_H$
Instruction cache RAM error status clear register	ICUM_ICSTCLR	FFFE E424$_H$
Instruction cache RAM error overflow status register	ICUM_ICOVFSTR	FFFE E428$_H$
Instruction cache RAM error overflow status clear register	ICUM_ICOVFSTC	FFFE E42C$_H$
Instruction cache RAM error notification control register	ICUM_ICERRINT	FFFE E4B0$_H$

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17.5.3.2 ICUM_ICECCCTL — Instruction Cache RAM ECC Control Register

This register enables or disables the ECC error detection/correction and 1-bit error correction. Set the PROT1 and PROT0 bits to 01_B when writing to this register.

Access: This register can be read or written in 16-bit units.

Address: FFFE $E400_H$

Value after reset: 0000_H

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	PROT1	PROT0	—	—	—	—	—	—	—	—	—	—	—	—	SECDIS	ECCDIS
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R/W	R/W	R	R	R	R	R	R	R	R	R	R	R	R	R/W	R/W

Table 17.16 ICUM_ICECCCTL Register Contents

Bit Position	Bit Name	Function
15	PROT1	These bits enable or disable programming of the ECCDIS and SEDDIS bits. Values written are not retained. These bits are always read as 0. Set the PROT1 and PROT0 bits to 0 and 1 when writing to this register.
14	PROT0	
13 to 2	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
1	SECDIS	1-Bit error correction disable This bit enables or disables 1-bit error correction when the ECC error detection/ correction is enabled. When writing to this bit, write 0 and 1 to PROT1 and PROT0 at the same time. 0: 1-bit error correction is performed when a 1-bit error is detected. 1: 1-bit error correction is not performed when a 1-bit error is detected.
0	ECCDIS	ECC disable This bit enables or disables ECC error detection/correction. When writing to this bit, write 0 and 1 to PROT1 and PROT0 at the same time. ECC error detection/correction is enabled in the initial state. 0: ECC error detection/correction is enabled. 1: ECC error detection/correction is disabled. ECC data is still written even if the error detection/correction function is disabled.

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17.5.3.3 ICUM_IC1STERSTR — Instruction Cache RAM 1st Error Status Register

This register monitors the occurrence of the first error.

Detecting a 1-bit ECC error sets the SEDF0 bit and detecting a 2-bit ECC error sets the DEDF0 bit while ECC error detection/correction is enabled.

Access: This register is a read-only register that can be read in 32-bit units.

Address: FFFE E404$_H$

Value after reset: 0000 0000$_H$

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DEDF0	SEDF0
Value after reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R	R

Table 17.17 ICUM_IC1STERSTR Register Contents

Bit Position	Bit Name	Function
31 to 2	—	Reserved When read, the value after reset is returned.
1	DEDF0	Instruction cache RAM 2-bit ECC error monitor flag This bit is set to 1 when a 2-bit ECC error occurs, regardless of the SEDF0 or DEDF0 state. [Clearing condition] ● An internal reset, generation of a pin reset, or setting the 1STERRCLR0 bit in instruction cache RAM error status clear register to 1 [Setting condition] ● When a 2-bit ECC error occurs
0	SEDF0	Instruction cache RAM 1-bit ECC error monitor flag This bit is set to 1 when a 1-bit ECC error occurs while SEDF0 and DEDF0 are 0. [Clearing condition] ● An internal reset, generation of a pin reset, or setting the 1STERRCLR0 bit in instruction cache RAM error status clear register to 1 [Setting condition] ● When a 1-bit ECC error occurs while both SEDF0 and DEDF0 are 0.

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17.5.3.4 ICUM_ICSTCLR — Instruction Cache RAM Error Status Clear Register

This register clears the error flag in the instruction cache RAM 1st error status register.

Access: This register is a write-only register that can be written in 8-bit units.

Address: FFFE E424$_H$

Value after reset: 00$_H$

Bit	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	1STERRCLR0
Value after reset	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	W

Table 17.18 ICUM_ICSTCLR Register Contents

Bit Position	Bit Name	Function
7 to 1	—	Reserved When writing, write the value after reset.
0	1STERRCLR0	First error status register clear Writing 1 to this bit clears the error flag in the instruction cache RAM 1st error status register.

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17.5.3.5 ICUM_ICOVFSTR — Instruction Cache RAM Error Overflow Status Register

This register monitors occurrence of instruction cache RAM error overflow. Setting the ERROVFCLR0 bit in the ICUM_ICOVFSTC register clears the ERROVF0 bit.

Access: This register is a read-only register that can be read in 8-bit units.

Address: FFFE E428$_H$

Value after reset: 00$_H$

Bit	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	ERROVF0
Value after reset	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	R

Table 17.19 ICUM_ICOVFSTR Register Contents

Bit Position	Bit Name	Function
7 to 1	—	Reserved When read, the value after reset is returned.
0	ERROVF0	Error overflow flag This bit is set when an ECC error occurs while the error address register is enabled. ● If the retained address source is SEDF: When a 1-bit error occurs in the same address, no overflow occurs. When a 1-bit error occurs in a different address, an overflow occurs. When a 2-bit error occurs, an overflow occurs. ● If the retained address source is DEDF: When a 1-bit error occurs, an overflow occurs. When a 2-bit error occurs in the same address, no overflow occurs. When a 2-bit error occurs in a different address, an overflow occurs.

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17.5.3.6 ICUM_ICOVFSTC — Instruction Cache RAM Error Overflow Status Clear Register

This register clears the instruction cache RAM error overflow flag. Setting 1 to the ERROVFCLR0 bit clears this ERROVF0 bit in the ICUM_ICOVFSTR register.

Access: This register is a write-only register that can be written in 8-bit units.

Address: FFFE E42C$_H$

Value after reset: 00$_H$

Bit	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	—	ERROVFCLR0
Value after reset	0	0	0	0	0	0	0	0
R/W	R	R	R	R	R	R	R	W

Table 17.20 ICUM_ICOVFSTC Register Contents

Bit Position	Bit Name	Function
7 to 1	—	Reserved When writing, write the value after reset.
0	ERROVFCLR0	Error overflow flag clear Writing 1 to this bit clears the ERROVF0 flag.

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17.5.3.7 ICUM_ICERRINT — Instruction Cache RAM Error Notification Control Register

This register is used for specifying whether to enable or disable error notification signals generated when a 2-bit ECC error or a 1-bit ECC error is detected.

Access: This register can be read or written in 8-bit units.

Address: FFFE E4B0$_H$

Value after reset: 03$_H$

Bit	7	6	5	4	3	2	1	0
	—	—	—	—	—	—	DEDIE	SEDIE
Value after reset	0	0	0	0	0	0	1	1
R/W	R	R	R	R	R	R	R/W	R/W

Table 17.21 ICUM_ICERRINT Register Contents

Bit Position	Bit Name	Function
7 to 2	—	Reserved When read, the value after reset is returned. When writing, write the value after reset.
1	DEDIE	2-bit ECC error notification enable This bit controls whether to report an error when a 2-bit error is detected while ECC error detection/correction is enabled. 0: 2-bit ECC error notification is disabled. 1: 2-bit ECC error notification is enabled.
0	SEDIE	1-bit ECC error notification enable This bit controls whether to report an error when a 1-bit error is detected while ECC error detection/correction is enabled. 0: 1-bit ECC error notification is disabled. 1: 1-bit ECC error notification is enabled.

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Section 18 Operation

18.1 Enabling ICUMD

ICUMD is enabled by setting the configuration setting data shown in the table below.

Table 18.1 Configuration for ICUMD Enabling

ICUMD Configuration Parameter			
ICUMRAC	**ICUMRAD**	**ICUMRESV**	**ICUMON**
Other than FFFF FFFF$_H$	Other than FFFF FFFF$_H$	Other than FFFF FFFF$_H$	0$_H$

For details of the setting, see **Section 16.8, Option Bytes**.

CAUTION

When the ICUMD is used, default start mode (WDTA1/WDTA2*[1]) cannot be used. Always select software trigger start mode by using the option bytes.

Note 1. WDTA2 is supported only in RH850/F1KH.

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18.2 ICUMD Startup

An example of the startup flow of the ICUMD is shown below.

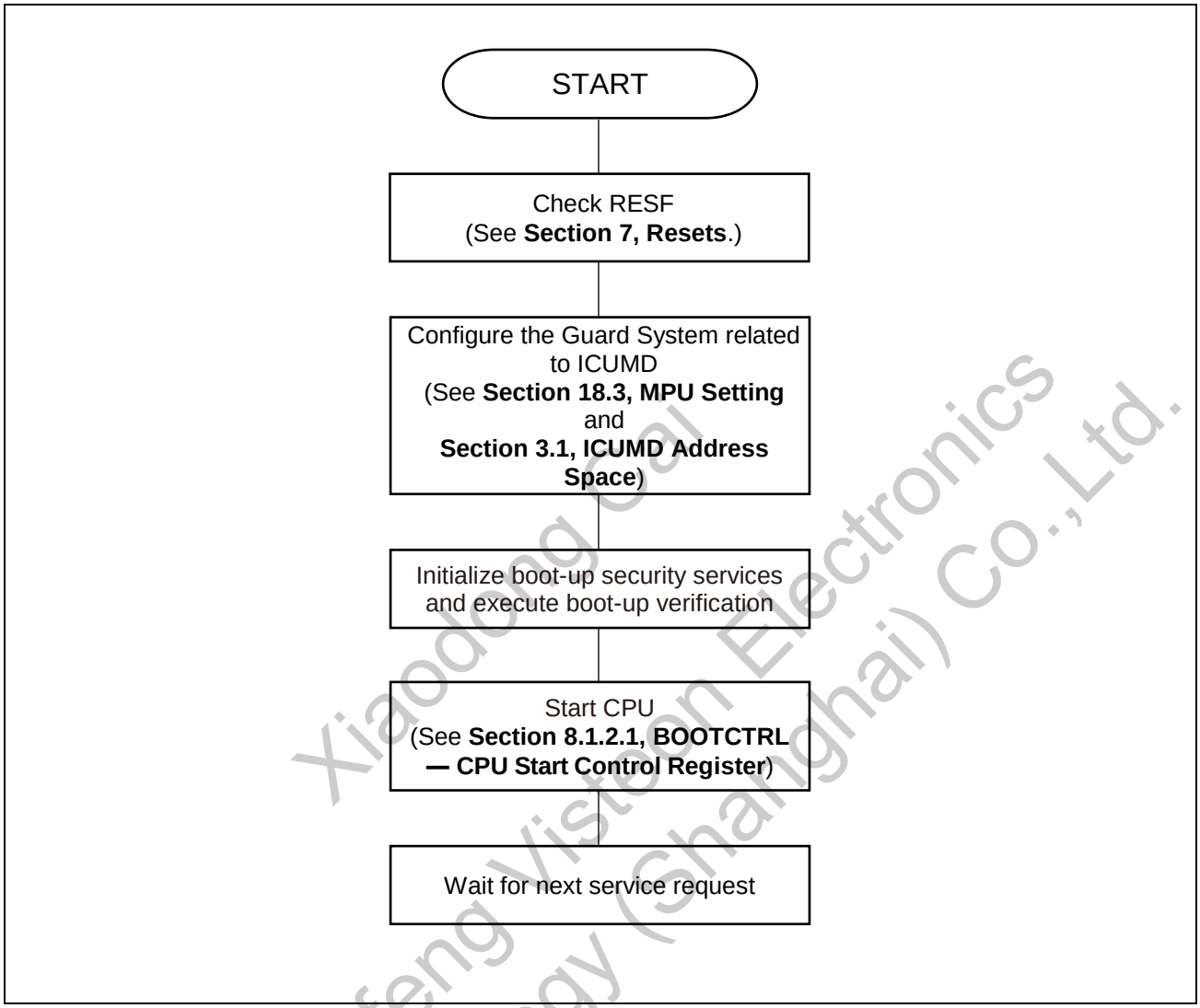


Figure 18.1 Startup Flow

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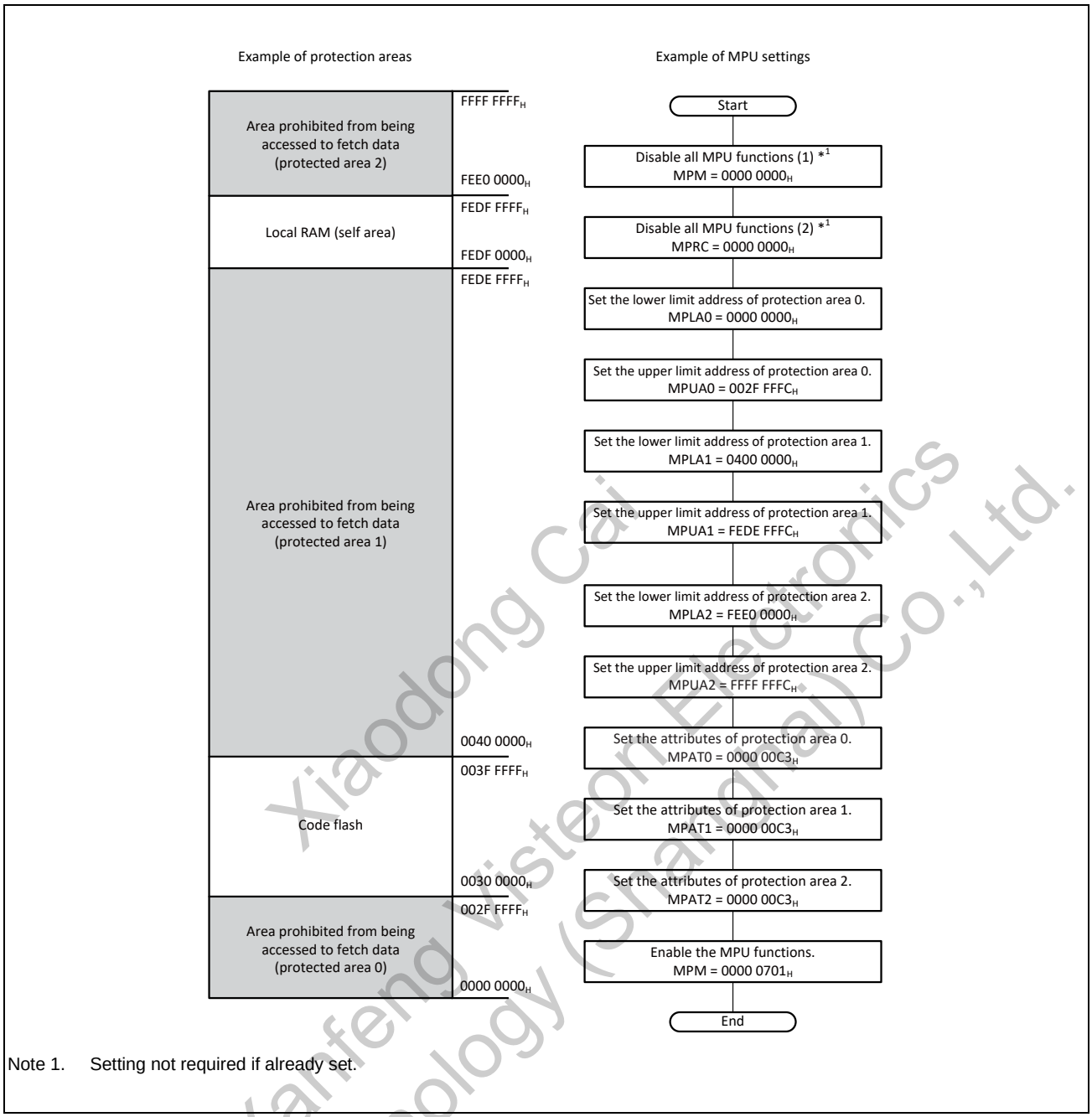
18.3 MPU Setting

An MPU function can be used to guard areas other than the designated secure areas from being accessed by the ICUP for fetching data.

For the MPU, see **Section 2.1.1.1, Memory Management**.

The figure below shows an example of how to configure the MPU to prevent areas other than the code flash (addresses $0030\ 0000_H$ and beyond) and the local RAM (self area) from being accessed to fetch data.

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Note 1. Setting not required if already set.

Figure 18.2 Example of MPU Setting Procedure in RH850/F1KM-S4

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18.4 Communication between CPU (PE) and ICUP

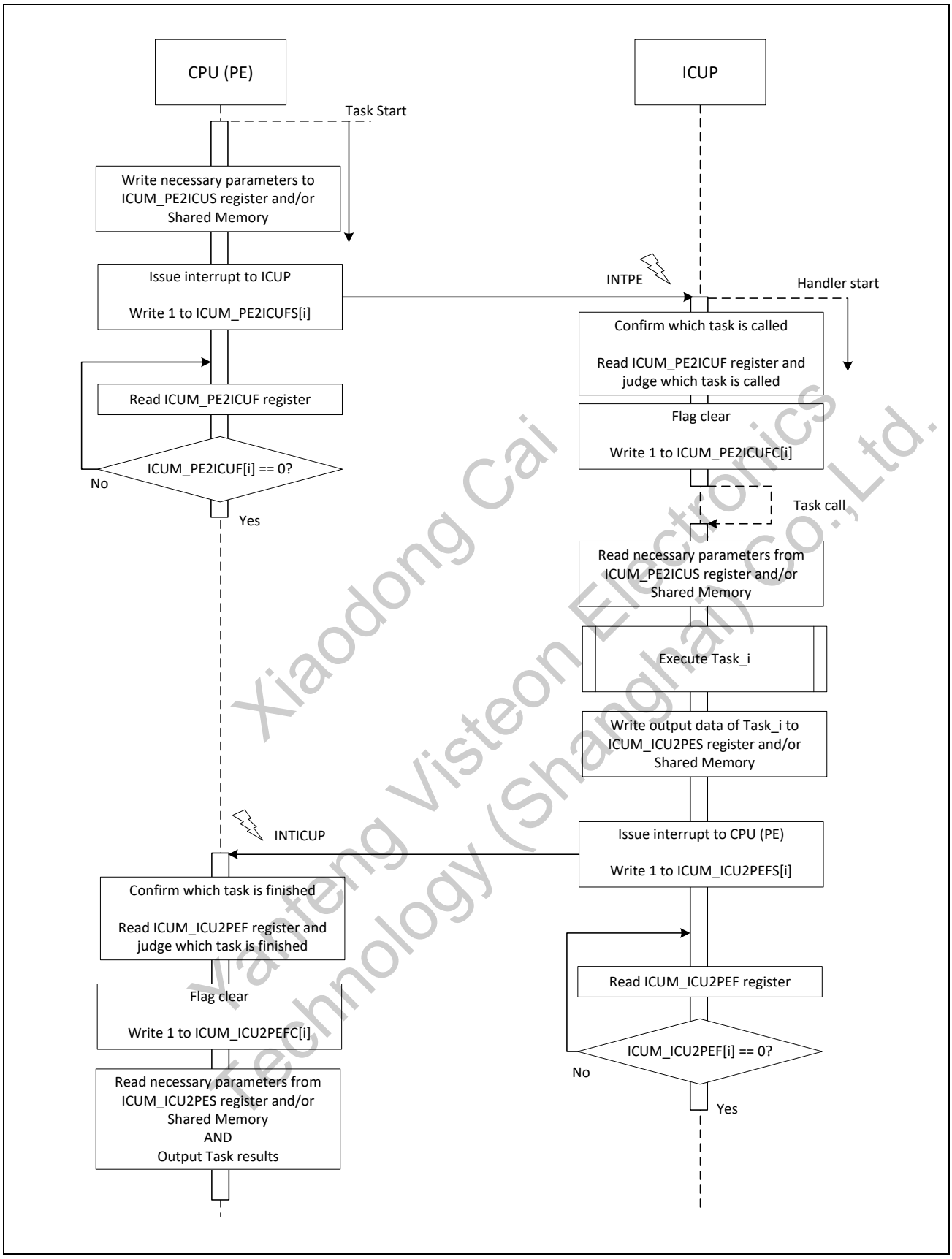
Figure 18.3 shows example procedure of communication between CPU (PE) and ICUP which uses INTPE and INTICUP.

Precondition is as shown below:

- CPU (PE) requests ICUP to execute task by issueing INTPE.
- ICUP notifies CPU (PE) of the task completion by issueing INTICUP.
- Each task allocates bit number of ICUM_PE2ICUF and ICUM_ICU2PEF
  - CPU (PE) write 1 to ICUM_PE2ICUFS[i] to request executing Task_i (i=0~31)
  - ICUP writes 1 to ICUM_PE2ICUFC[i] to notifies CPU (PE) of receiving request of Task_i (i=0~31)
  - ICUP writes 1 to ICUM_ICU2PEFS[i] to notifies CPU (PE) of Task_i's completion (i=0~31)
  - CPU (PE) writes 1 to ICUM_ICU2PEFC[i] to notifies ICUP of receiving notification of Task_i's completion (i=0~31)

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Figure 18.3 Communication procedure between CPU (PE) and ICUP which uses INTPE and INTICUP

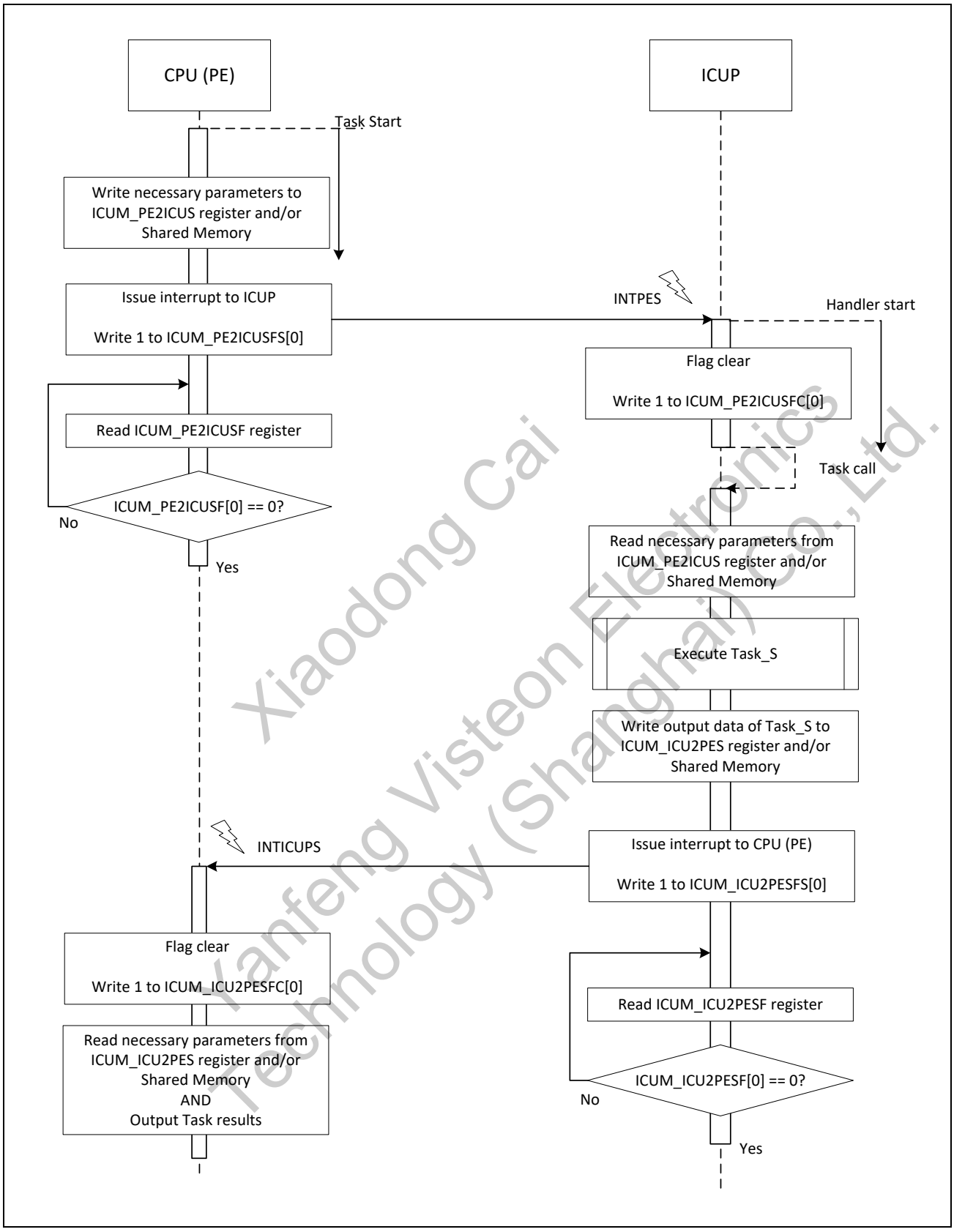
Figure 18.4 shows example procedure of communication between CPU (PE) and ICUP which uses INTPES and INTICUPS.

Precondition is as shown below:

- CPU (PE) requests ICUP to execute task by issueing INTPES.
- ICUP notifies CPU (PE) of the task completion by issueing INTICUPS.
- A higher-priority task (Task_S) allocates interruption of INTPES and INTICUPS.
  - CPU (PE) write 1 to ICUM_PE2ICUSFS[0] to request ICUP to execute Task_S
  - ICUP writes 1 to ICUM_PE2ICUSFC[0] to notifies CPU (PE) of receiving request of Task_S
  - ICUP writes 1 to ICUM_ICU2PESFS[0] to notifies CPU (PE) of Task_S's completion
  - CPU (PE) writes 1 to ICUM_ICU2PESFC[0] to notifies ICUP of receiving notification of Task_S's completion

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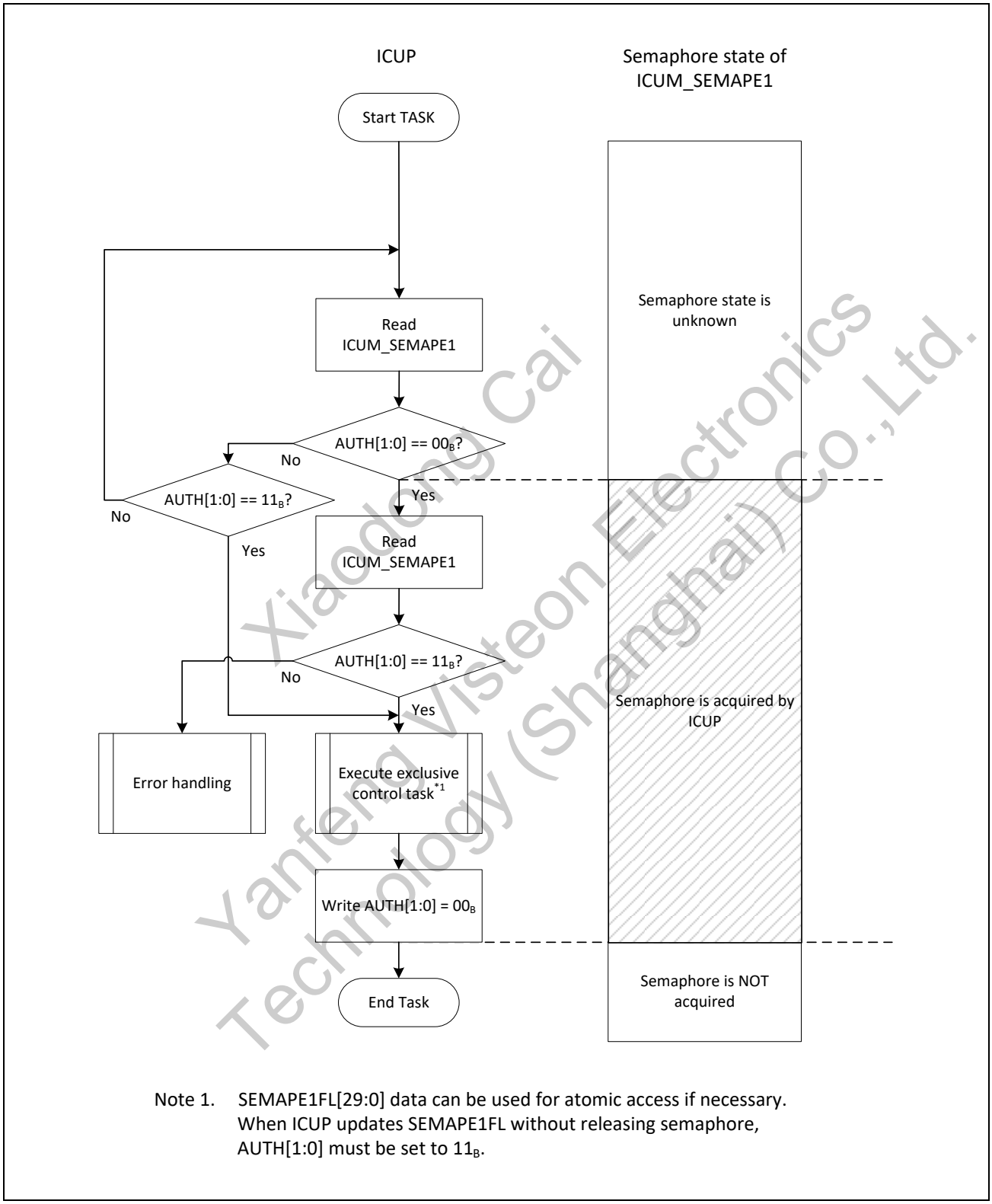


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Figure 18.4 Communication procedure between CPU (PE) and ICUP which uses INTPES and INTICUPS

18.5 Semaphore Register

Figure 18.5 shows example flow of ICUM_SEMAPE1 accessing by ICUP.



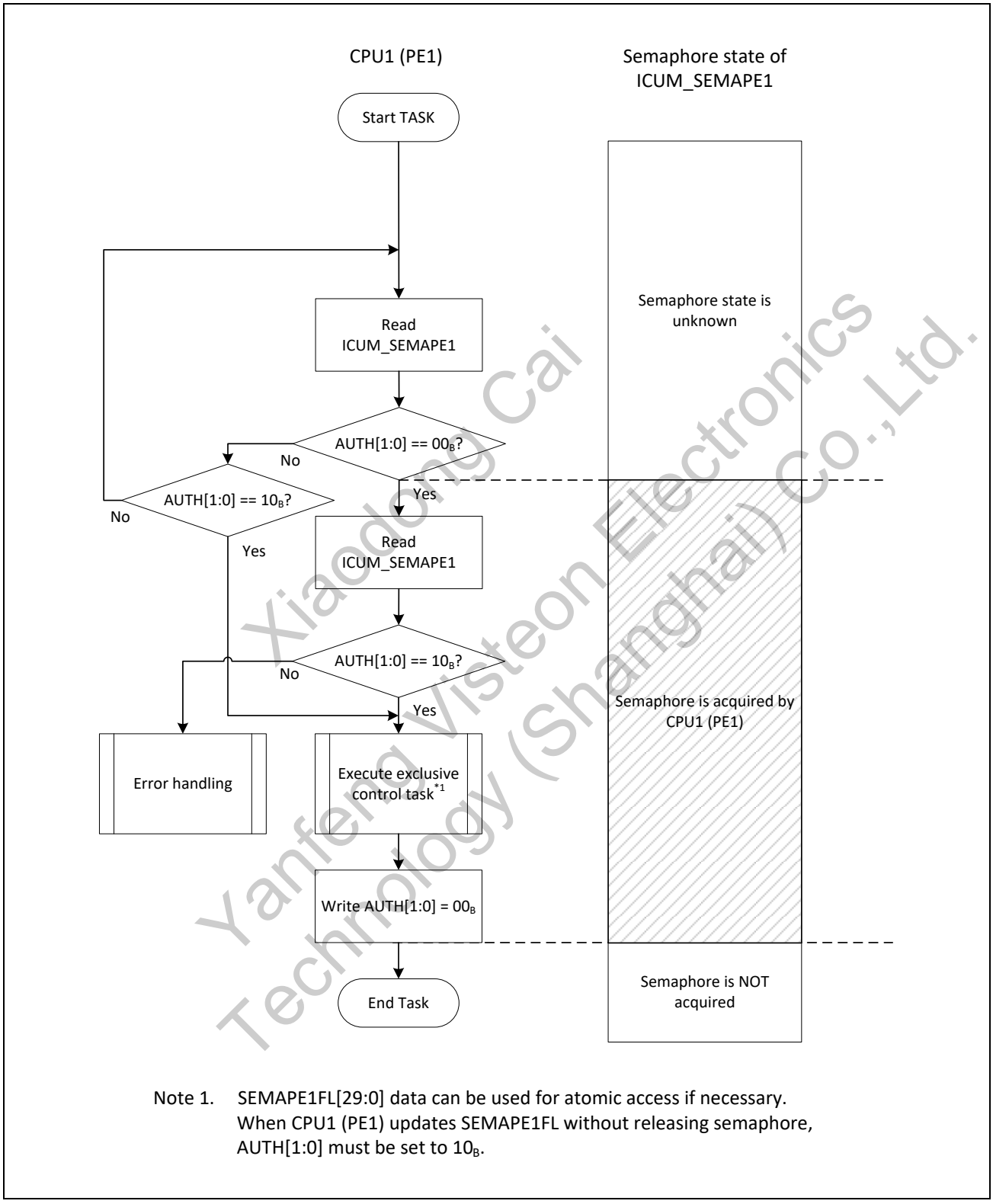
Note 1. SEMAPE1FL[29:0] data can be used for atomic access if necessary. When ICUP updates SEMAPE1FL without releasing semaphore, AUTH[1:0] must be set to 11_B.

Figure 18.5 ICUM_SEMAPE1 accessing procedure by ICUP

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Figure 18.6 shows example flow of ICUM_SEMAPE1 accessing by CPU1 (PE1).



Note 1. SEMAPE1FL[29:0] data can be used for atomic access if necessary. When CPU1 (PE1) updates SEMAPE1FL without releasing semaphore, AUTH[1:0] must be set to 10_B.

Figure 18.6 ICUM_SEMAPE1 accessing procedure by CPU1 (PE1)

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Figure 18.7 shows example flow of accessing ICUM_SEMAPE1 by ICUP and CPU1 (PE1).

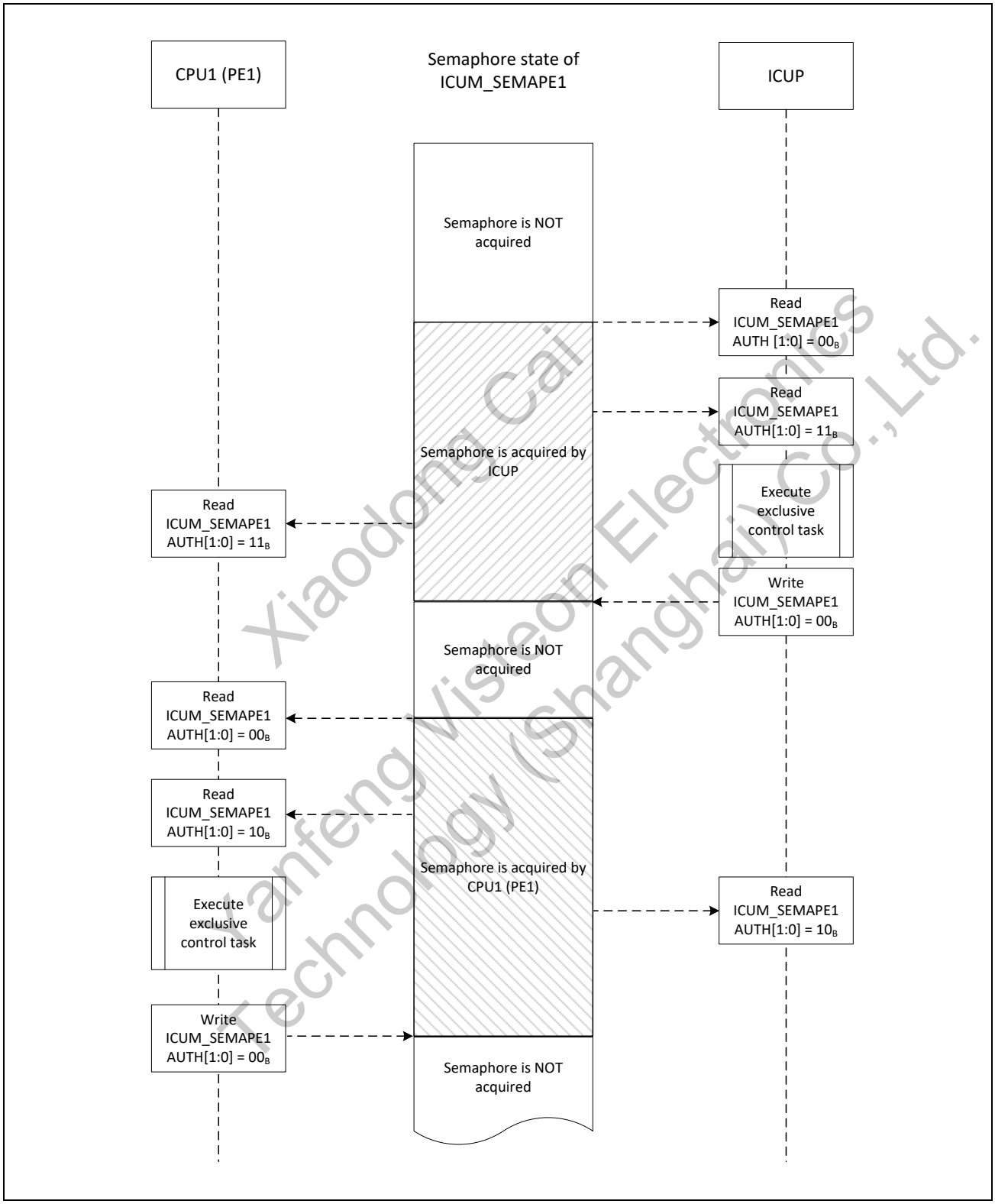


Figure 18.7 ICUM_SEMAPE1 accessing flow by ICUP and CPU1 (PE1)

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18.6 AES operation

Figure 18.8 shows example of ICUM_AESB0 procedure.

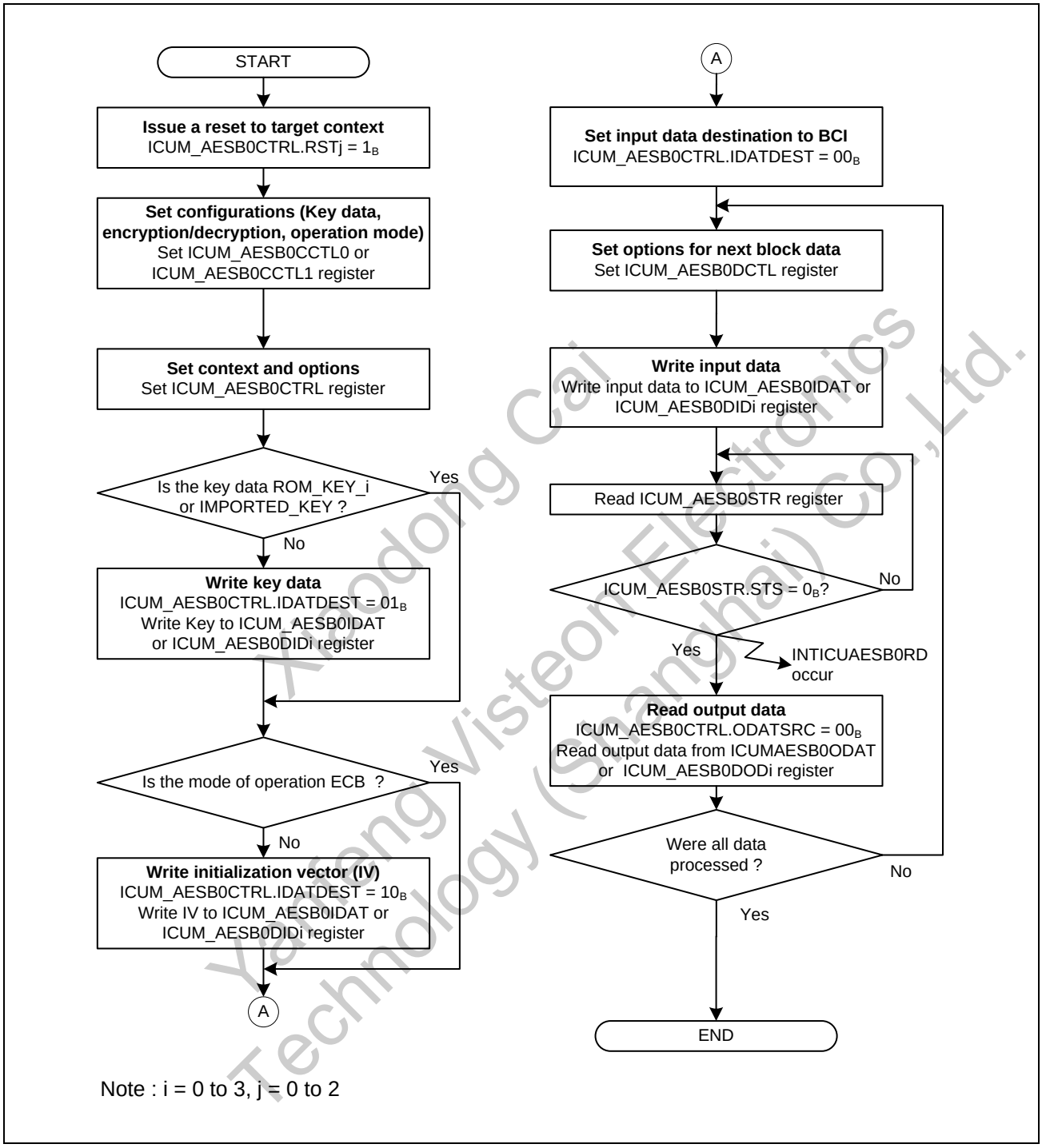


Figure 18.8 ICUM_AESB0 Setting Procedure

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18.7 Challenge & Response

Figure 18.9 and **Figure 18.10** shows example of challenge & response authentication procedure for debugger connection.

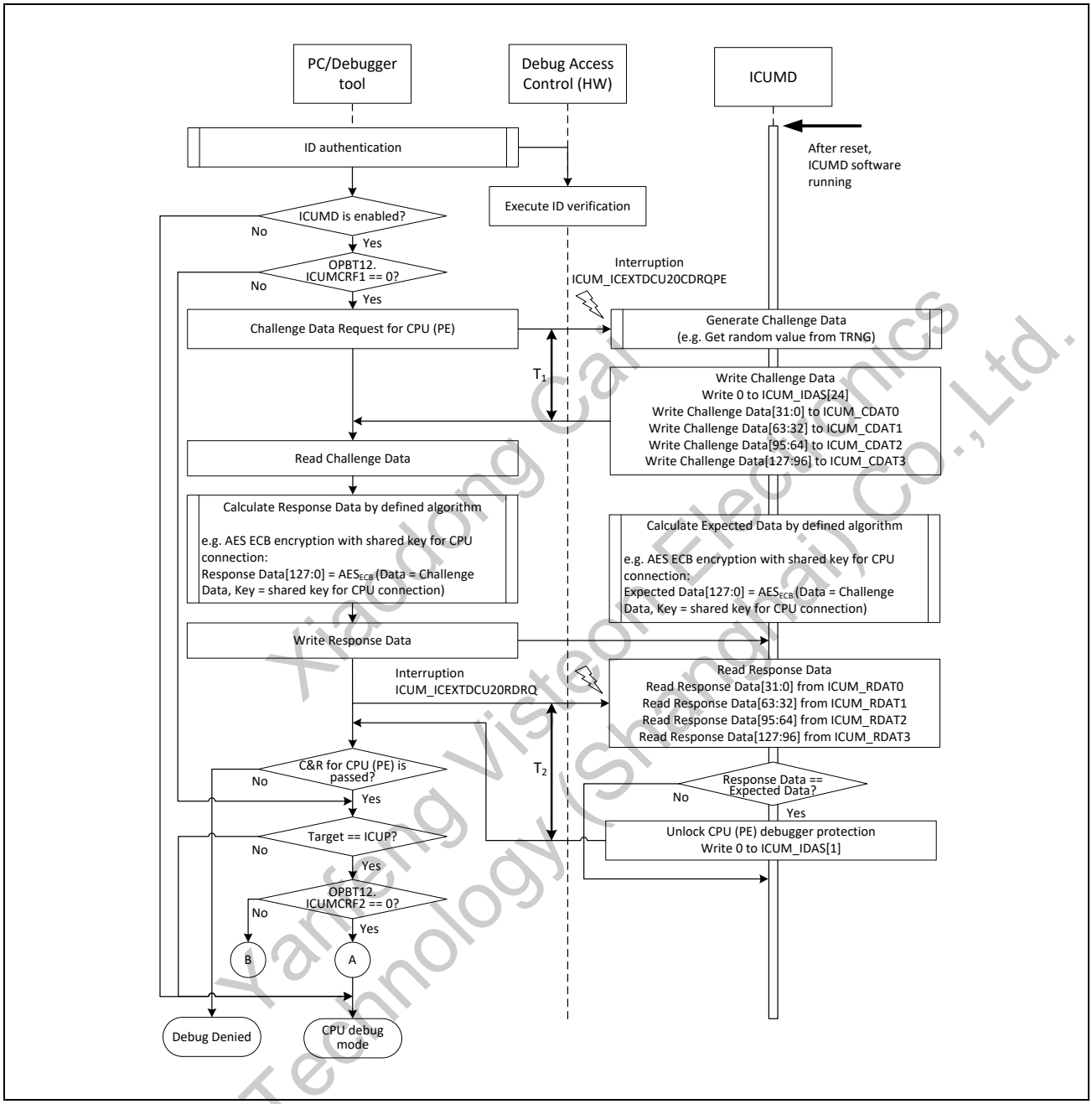


Figure 18.9 Challenge & response authentication procedure for debugger connection (1/2)

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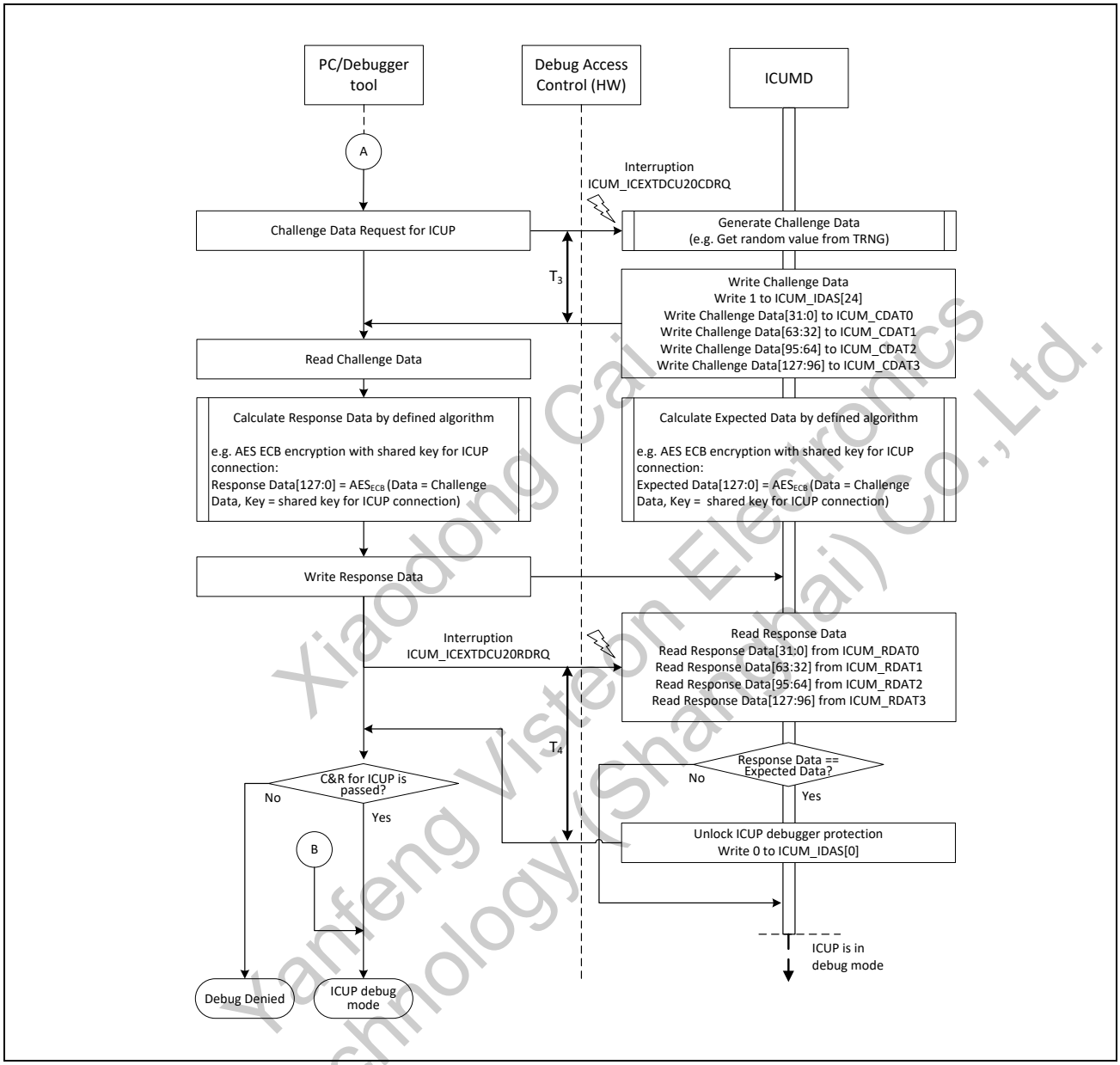


Figure 18.10 Challenge & response authentication procedure for debugger connection (2/2)

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CAUTION

Duration from issuing interruption of challenge data request to writing challenge data (T_1, T_3) and duration from issuing interruption of response data verification to updating debugger protection status (T_2, T_4) depend on C&R firmware implementation. The C&R firmware shall be implemented to satisfy following conditions.

- T_1 , T_3 < 10 sec
- T_2, T_4 < 1 sec

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Section 19 Electrical Specification

19.1 TRNG Characteristics

Condition: REGVCC = EVCC = 3.0 V to 5.5 V, BVCC = 3.0 V to REGVCC, A0VREF = 3.0 V to 5.5 V, A1VREF = 3.0 V to 5.5 V, AWOVSS = ISOVSS = EVSS = BVSS = A0VSS = A1VSS = 0 V, CAWOVCL: 0.1 µF ± 30%, CISOVCL: 0.1 µF ± 30%, Tj = –40 to (depend on the product) °C

Table 19.1 TRNG Characteristics

Item	Symbol	Condition	MIN	TYP	MAX	Unit
True random number generation time	t_{TRNG}	Size: 32-bit	—	—	10.24	ms

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Colophon 5.0

RH850/F1KH, RH850/F1KM ICUMD

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